SSC JE

STRENGTH OF MATERIALS & PRODUCTION ENGINEERING

Topper-Style Concept Notes

SSC JE Mechanical • Paper-I & Paper-II • Concept + Formula + PYQ-Trend + Solved Numericals

Engineering Mechanics • Stress & Strain • Material Testing • Principal Stresses & Failure Theories • SFD & BMD • Bending & Shear Stresses • Deflection • Torsion & Springs • Columns • Cylinders • Engineering Materials & Heat Treatment • Casting • Welding • Metal Forming • Machining & Machine Tools • Non-Traditional Machining • Metrology • CNC & Industrial Engineering

Concept-based notes – no MCQ practice. "PYQ-Trend" boxes list concepts SSC JE has repeatedly tested (based on recurring topics, not copied questions). Cross-check values with a standard textbook.

Contents

How to Use These Notes 11

Why SOM and Production Matter . . . 11

How Each Chapter Is Built . . . 11

1. Engineering Mechanics – Forces, Equilibrium and Friction 12

1.1 Force Systems and Resultants . . . 12

1.2 Equilibrium . . . 12

1.3 Friction . . . 13

1.4 Trusses . . . 13

1.5 Centroid and Moment of Inertia . . . 14

1.6 Dynamics Essentials . . . 14

2. Simple Stresses and Strains 16

2.1 Stress and Strain . . . 16

2.2 Elastic Constants . . . 16

2.3 Bars of Varying Section and Compound Bars . . . 17

2.4 Thermal Stresses . . . 17

2.5 Strain Energy and Impact Loading . . . 17

3. Mechanical Properties and Material Testing 19

3.1 Stress–Strain Diagram (Mild Steel) . . . 19

3.2 Mechanical Properties . . . 20

3.3 Hardness Tests . . . 21

3.4 Impact, Fatigue and Creep Tests . . . 21

3B. Non-Destructive Testing, Composites, Ceramics and Polymers 23

3B.1 Non-Destructive Testing (NDT) . . . 23

3B.2 Composite Materials . . . 24

3B.3 Ceramics and Polymers . . . 24

4. Principal Stresses, Mohr's Circle and Theories of Failure 26

4.1 Stresses on an Inclined Plane . . . 26

4.2 Principal Stresses and Planes . . . 26

4.3 Mohr's Circle . . . 27

4.4 Theories of Failure . . . 27

5. Shear Force and Bending Moment Diagrams 30

5.1 Beams, Supports and Loads . . . 30

5.2 Relations Between Load, SF and BM . . . 30

5.3 Standard Results . . . 31

6. Bending Stresses in Beams 33

6.1 Theory of Simple Bending . . . 33

6.2 Beams of Uniform Strength and Composite Beams . . . 34

6.3 Combined Direct and Bending Stresses . . . 34

6.4 Plastic Bending and Shape Factor . . . 35

6.5 Curved Beams and Unsymmetrical Bending (Basics) . . . 35

7. Shear Stresses in Beams and Deflection of Beams 37

7.1 Shear Stress Distribution . . . 37

7.2 Deflection – Differential Equation and Methods . . . 37

7.3 Standard Slopes and Deflections . . . 38

8. Torsion of Shafts and Springs 40

8.1 Torsion of Circular Shafts . . . 40

8.2 Combined Bending and Torsion; Shaft Design . . . 40

8.3 Close-Coiled Helical Springs . . . 41

8.4 Leaf (Laminated) Springs . . . 41

9. Columns and Struts 43

9.1 Basics . . . 43

9.2 Euler's Theory . . . 43

9.3 Rankine–Gordon and Other Formulae . . . 44

10. Thin and Thick Cylinders and Spheres 45

10.1 Thin Cylinders (t < d/20) . . . 45

10.2 Thin Spheres . . . 45

10.3 Thick Cylinders (Lamé's Theory) . . . 45

11. Riveted, Welded and Bolted Joints (Strength Basics) 47

11.1 Riveted Joints . . . 47

11.2 Welded Joints . . . 47

11.3 Bolted Joints, Keys and Couplings (Quick Facts) . . . 48

11B. Special Topics – Rotating Rings and Discs, Dams, Curved Bars and Contact Stresses 50

11B.1 Rotating Rings and Discs . . . 50

11B.2 Dams and Retaining Walls (Masonry Structures) . . . 50

11B.3 Curved Bars, Springs and Thin Rings . . . 51

11B.4 Contact (Hertz) Stresses and Stress Concentration . . . 51

12. Engineering Materials and Classification of Steels 52

12.1 Classification of Materials and Crystal Structure . . . 52

12.2 Classification of Steels (Paper-I Topic) . . . 52

12.3 Alloying Elements . . . 54

12.4 Cast Irons . . . 55

12.5 Non-Ferrous Metals and Alloys . . . 55

12B. Material Selection, Corrosion and Strengthening Mechanisms 57

12B.1 Strengthening Mechanisms . . . 57

12B.2 Corrosion . . . 57

12B.3 Material Selection for Typical Components . . . 58

13. Iron–Carbon Diagram and Heat Treatment 60

13.1 Iron–Iron Carbide Diagram . . . 60

13.2 TTT Diagram and Hardenability . . . 61
13.3 Heat Treatment Processes . . . 61
13.4 Surface Hardening (Case Hardening) . . . 62
14. Metal Casting – Patterns, Moulds and Gating 64
14.1 Casting Basics . . . 64
14.2 Patterns . . . 64
14.3 Moulding Sand and Cores . . . 65
14.4 Gating and Risering . . . 66
15. Special Casting Processes, Melting and Casting Defects 68
15.1 Special Casting Processes . . . 68
15.2 Melting Furnaces . . . 69
15.3 Casting Defects . . . 69
16. Welding, Brazing and Soldering 72
16.1 Classification . . . 72
16.2 Gas Welding . . . 72
16.3 Arc Welding . . . 72
16.4 Resistance and Solid-State Welding . . . 73
16.5 Welding Metallurgy, Defects and Inspection . . . 74
16.6 Brazing, Soldering and Adhesive Bonding . . . 75
16B. Welding Practice – Symbols, Joint Design, Metals and Residual Stresses 77
16B.1 Weld Symbols and Joint Design . . . 77
16B.2 Distortion and Residual Stresses . . . 77
16B.3 Welding of Different Metals . . . 78
16B.4 Cutting Processes . . . 78
17. Metal Forming – Rolling, Forging, Extrusion, Drawing and Sheet Metal 80
17.1 Fundamentals . . . 80
17.2 Rolling . . . 80
17.3 Forging . . . 81
17.4 Extrusion and Drawing . . . 81
17.5 Sheet Metal Working (Press Work) . . . 82
17B. Presses, Press Tools and High-Energy-Rate Forming 85
17B.1 Presses . . . 85
17B.2 Die Elements and Die Types . . . 85
17B.3 High-Energy-Rate Forming (HERF) and Other Processes . . . 86
18. Theory of Metal Cutting 88
18.1 Cutting Tools and Geometry . . . 88
18.2 Chip Formation . . . 88
18.3 Mechanics of Orthogonal Cutting (Merchant) . . . 89
18.4 Tool Wear, Tool Life and Machinability . . . 90
18.5 Tool Materials and Cutting Fluids . . . 91

19. Machine Tools – Lathe, Drilling, Shaper, Planer, Milling and Others **93**

19.1 Lathe . . . 93

19.2 Drilling, Boring and Reaming . . . 94

19.3 Shaper, Planer and Slotter . . . 94

19.4 Milling . . . 95

19.5 Broaching, Gear Cutting and Other Processes . . . 96

20. Grinding, Finishing and Non-Traditional Machining **98**

20.1 Grinding . . . 98

20.2 Fine Finishing Processes . . . 99

20.3 Non-Traditional (Unconventional) Machining . . . 100

21. Metrology – Limits, Fits, Gauges and Measuring Instruments **103**

21.1 Limits, Fits and Tolerances . . . 103

21.2 Gauges . . . 104

21.3 Linear and Angular Measuring Instruments . . . 104

21.4 Surface Roughness . . . 105

22. CNC Machines, Automation, Jigs and Fixtures **107**

22.1 NC and CNC Fundamentals . . . 107

22.2 Automation and Flexible Manufacturing . . . 108

22.3 Jigs and Fixtures . . . 109

23. Powder Metallurgy, Plastics Processing and Surface Treatments **111**

23.1 Powder Metallurgy (P/M) . . . 111

23.2 Processing of Plastics . . . 111

23.3 Surface Treatments and Coatings . . . 112

24. Industrial Engineering – Work Study, PPC, Inventory and Quality **114**

24.1 Productivity and Work Study . . . 114

24.2 Plant Location and Layout . . . 115

24.3 Production Planning and Control, Forecasting . . . 116

24.4 Inventory Control . . . 116

24.5 Quality Control and Reliability . . . 117

24.6 Project Management (PERT/CPM) and Operations Research Basics . . . 118

24B. Maintenance, Safety, Value Engineering and Costing **120**

24B.1 Maintenance . . . 120

24B.2 Industrial Safety . . . 120

24B.3 Value Engineering, Costing and Ergonomics . . . 121

25. Solved Numericals – Set 1 (Stress, Strain, Principal Stresses, Columns, Cylinders) **123**

25.1 Simple Stress and Strain . . . 123

25.2 Thermal, Composite and Impact Problems . . . 124

25.3 Principal Stresses and Failure Theories . . . 124

25.4 Columns . . . 125

25.5 Cylinders . . . 125

25B. Solved Numericals – Set 2 (SFD/BMD, Bending, Shear, Deflection, Torsion, Springs) 127

25B.1 Shear Force and Bending Moment . . . 127
25B.2 Bending and Shear Stresses . . . 127
25B.3 Deflection . . . 128
25B.4 Torsion and Springs . . . 129

25C. Solved Numericals – Set 3 (Casting, Welding, Forming, Machining) 130

25C.1 Casting . . . 130
25C.2 Welding . . . 130
25C.3 Metal Forming . . . 131
25C.4 Metal Cutting . . . 131

25D. Solved Numericals – Set 4 (Metrology, Industrial Engineering, Mechanics) 134

25D.1 Metrology . . . 134
25D.2 Industrial Engineering . . . 135
25D.3 Engineering Mechanics . . . 136

25E. Solved Numericals – Set 5 (Advanced SOM) 138

25E.1 Bars and Energy . . . 138
25E.2 Stress Transformation . . . 138
25E.3 Beams . . . 139
25E.4 Torsion, Columns, Cylinders . . . 140

25F. Solved Numericals – Set 6 (Advanced Production and Industrial Engineering) 141

25F.1 Casting and Welding . . . 141
25F.2 Forming and Press Work . . . 141
25F.3 Machining Economics . . . 142
25F.4 Industrial Engineering . . . 143

25G. Solved Numericals – Set 7 (Quick One-Concept Drills) 145

25G.1 Mechanics and SOM Drills . . . 145
25G.2 Production Drills . . . 146

26. Concept Recall Bank – Part A (Mechanics and SOM) 148

R1. Engineering Mechanics . . . 148
R2. Stress, Strain and Properties . . . 148
R3. Principal Stresses and Failure . . . 149
R4. Beams . . . 150
R5. Torsion, Springs, Columns, Cylinders . . . 150

26B. Concept Recall Bank – Part B (Production Engineering) 152

R6. Materials and Heat Treatment . . . 152
R7. Casting and Welding . . . 153
R8. Forming . . . 153
R9. Machining and Machine Tools . . . 154
R10. Metrology, CNC and Industrial Engineering . . . 155

26C. Concept Recall Bank – Part C (Special Topics) 157

R11. NDT, Composites and Polymers . . . 157

R12. Special Stresses and Presses . . . 157

R13. Welding Practice and Corrosion . . . 158

R14. Maintenance, Safety and Costing . . . 158

27. Paper-II Model Answers 160

P1. Strength of Materials . . . 160

P2. Production Engineering . . . 161

27B. Paper-II Model Answers – Set 2 164

P3. Strength of Materials . . . 164

P4. Production Engineering . . . 165

27C. Paper-II Model Answers – Set 3 167

P5. Mechanics and SOM . . . 167

P6. Production Engineering . . . 168

28. High-Yield Topic Map 170

M1. Engineering Mechanics and SOM . . . 170

M2. Production Engineering . . . 171

M3. How to Use This Map in the Last Two Weeks . . . 171

28B. Abridged Data Tables 172

T1. Typical Mechanical Properties . . . 172

T2. Coefficients of Thermal Expansion ($\times 10^{-6}$ per °C) . . . 172

T3. Standard Section Properties . . . 172

T4. Hardness Conversion (approximate, steels) . . . 173

T5. Fundamental Tolerance (IT) Values for Common Sizes (µm) . . . 173

T6. Typical Cutting Speeds (HSS tool, turning, m/min) . . . 173

T7. Control Chart Factors . . . 173

29. Last-Minute Revision Sheet 175

29.1 Must-Know Values . . . 175

29.2 Formula Flash Card . . . 175

29.3 Comparison Quick-Hits . . . 176

29.4 Traps Toppers Avoid . . . 176

29.5 Final 10-Minute Checklist . . . 177

Appendix A – One-Place Formula Sheet 178

Engineering Mechanics – Forces, Equilibrium and Friction . . . 178

Simple Stresses and Strains . . . 178

Mechanical Properties and Material Testing . . . 179

Non-Destructive Testing, Composites, Ceramics and Polymers . . . 180

Principal Stresses, Mohr's Circle and Theories of Failure . . . 180

Shear Force and Bending Moment Diagrams . . . 180

Bending Stresses in Beams . . . 180

Shear Stresses in Beams and Deflection of Beams . . . 181
Torsion of Shafts and Springs . . . 181
Columns and Struts . . . 181
Thin and Thick Cylinders and Spheres . . . 181
Riveted, Welded and Bolted Joints (Strength Basics) . . . 182
Special Topics – Rotating Rings and Discs, Dams, Curved Bars and Contact Stresses . . . 182
Metal Casting – Patterns, Moulds and Gating . . . 183
Special Casting Processes, Melting and Casting Defects . . . 183
Welding, Brazing and Soldering . . . 183
Metal Forming – Rolling, Forging, Extrusion, Drawing and Sheet Metal . . . 183
Presses, Press Tools and High-Energy-Rate Forming . . . 184
Theory of Metal Cutting . . . 184
Machine Tools – Lathe, Drilling, Shaper, Planer, Milling and Others . . . 184
Grinding, Finishing and Non-Traditional Machining . . . 185
Metrology – Limits, Fits, Gauges and Measuring Instruments . . . 185
CNC Machines, Automation, Jigs and Fixtures . . . 185
Industrial Engineering – Work Study, PPC, Inventory and Quality . . . 186
Maintenance, Safety, Value Engineering and Costing . . . 186

Appendix B – PYQ-Trend Concept Map 187

Engineering Mechanics – Forces, Equilibrium and Friction . . . 187
Simple Stresses and Strains . . . 187
Mechanical Properties and Material Testing . . . 187
Non-Destructive Testing, Composites, Ceramics and Polymers . . . 188
Principal Stresses, Mohr's Circle and Theories of Failure . . . 188
Shear Force and Bending Moment Diagrams . . . 188
Bending Stresses in Beams . . . 188
Shear Stresses in Beams and Deflection of Beams . . . 189
Torsion of Shafts and Springs . . . 189
Columns and Struts . . . 189
Thin and Thick Cylinders and Spheres . . . 189
Riveted, Welded and Bolted Joints (Strength Basics) . . . 189
Special Topics – Rotating Rings and Discs, Dams, Curved Bars and Contact Stresses . . . 190
Engineering Materials and Classification of Steels . . . 190
Material Selection, Corrosion and Strengthening Mechanisms . . . 190
Iron–Carbon Diagram and Heat Treatment . . . 190
Metal Casting – Patterns, Moulds and Gating . . . 191
Special Casting Processes, Melting and Casting Defects . . . 191
Welding, Brazing and Soldering . . . 191
Welding Practice – Symbols, Joint Design, Metals and Residual Stresses . . . 192
Metal Forming – Rolling, Forging, Extrusion, Drawing and Sheet Metal . . . 192
Presses, Press Tools and High-Energy-Rate Forming . . . 192
Theory of Metal Cutting . . . 193

Machine Tools – Lathe, Drilling, Shaper, Planer, Milling and Others 193
Grinding, Finishing and Non-Traditional Machining 193
Metrology – Limits, Fits, Gauges and Measuring Instruments 194
CNC Machines, Automation, Jigs and Fixtures 194
Powder Metallurgy, Plastics Processing and Surface Treatments 194
Industrial Engineering – Work Study, PPC, Inventory and Quality 194
Maintenance, Safety, Value Engineering and Costing 195

INTRODUCTION

How to Use These Notes

Why SOM and Production Matter

- Strength of Materials (SOM) with Engineering Mechanics is one of the highest-scoring blocks of SSC JE Mechanical Paper-I, and it is the backbone of the numerical questions in Paper-II. Most questions test a formula, a standard result (deflection of a cantilever, Euler load for an end condition, Mohr's circle radius) or a definition.
- Production Engineering appears in Paper-I through 'classification of steels' and engineering materials, and in Paper-II and many state/PSU JE papers through casting, welding, forming, machining, metrology and industrial engineering. The questions are largely factual – the process that suits a job, the defect and its cause, the tool or gauge used – so tables and comparison lists are the fastest way to learn it.
- Part A (Chapters 1–11) covers Engineering Mechanics and SOM. Part B (Chapters 12–24) covers Production Engineering. Solved numericals, recall banks, Paper-II answers and revision sheets follow.

How Each Chapter Is Built

- Concept lines and tables give the theory in exam language.
- Yellow formula boxes collect every formula; they are also gathered automatically in the Formula Appendix at the end.
- ★ PYQ-TREND boxes list the concepts that SSC JE papers have repeatedly tested (based on recurring topics, not copied questions). They are gathered in the PYQ-Trend Appendix for final revision.
- ✔ TOPPER TRICK boxes give shortcuts; ✖ COMMON MISTAKE boxes warn about traps.
- ✎ DESCRIPTIVE Q boxes give model answers for Paper-II.
- Sign conventions used: tensile stress positive; sagging bending moment positive; shear force positive when the left portion tends to move up relative to the right.

✔ **TOPPER TRICK:** For SOM, memorise the 'standard cases' tables (deflection, BM, Euler loads) – most numericals are a direct substitution into one of them.

CHAPTER 1

Engineering Mechanics – Forces, Equilibrium and Friction

1.1 Force Systems and Resultants

- Force: a vector with magnitude, direction, point of application. Units: newton (N); 1 kgf = 9.81 N.
- Principle of transmissibility: a force can be moved along its line of action without changing its external effect on a rigid body (not valid for internal effects or deformable bodies).

System	Description	Resultant
Coplanar concurrent	Lines of action meet at a point, in one plane	Single force through the point
Coplanar parallel	Lines parallel	Single force or a couple
Coplanar non-concurrent	General plane system	Force + couple, or single force
Collinear	Same line of action	Algebraic sum
Non-coplanar	Space forces	Force + couple (wrench)

Parallelogram law: $R = \sqrt{(P^2 + Q^2 + 2PQ\cos\theta)}$; $\tan\alpha = Q\sin\theta/(P + Q\cos\theta)$

Resolution: $R = \sqrt{((\Sigma Fx)^2 + (\Sigma Fy)^2)}$; $\tan\alpha = \Sigma Fy/\Sigma Fx$

Moment of a force M = F × perpendicular distance; Varignon's theorem: moment of resultant = sum of moments of components

Couple: two equal, opposite, parallel non-collinear forces; moment = F × arm, same about every point (free vector)

- A force can be replaced by an equal force at another point plus a couple (force–couple system).
- R is maximum (P + Q) when θ = 0 and minimum (|P − Q|) when θ = 180°; when P = Q, R = 2P cos(θ/2).

★ **PYQ-TREND:** Parallelogram law and resultant of two equal forces (2P cos θ/2) – frequent; Varignon's theorem; couple is a free vector; principle of transmissibility.

1.2 Equilibrium

Coplanar equilibrium: $\Sigma Fx = 0$, $\Sigma Fy = 0$, $\Sigma M = 0$ (three equations)

Lami's theorem (three concurrent forces in equilibrium): $P/\sin\alpha = Q/\sin\beta = R/\sin\gamma$ (α, β, γ = angles opposite each force)

- Triangle law: if three concurrent forces are in equilibrium they can be represented by the sides of a triangle taken in order. Polygon law for more forces.
- Two-force member: forces are equal, opposite and collinear. Three-force member: forces must be concurrent or parallel.
- Free-body diagram (FBD): isolate the body and show all external forces and reactions.

Support	Reactions
Roller / smooth surface	1 (normal to surface)
Hinge / pin	2 (two components)
Fixed (built-in)	3 (two forces + moment)
Link / cable	1 (along its axis; cable only tension)

- Statically determinate: unknown reactions can be found from equilibrium alone (simply supported, cantilever, overhanging beams). Indeterminate: more unknowns (fixed and propped beams, continuous beams) – need compatibility of deformation.

★ **PYQ-TREND:** Lami's theorem (very frequent); number of reactions at hinge/roller/fixed support; conditions of equilibrium; two-force member.

1.3 Friction

- Friction resists relative motion between surfaces in contact. Static friction is self-adjusting up to limiting friction; kinetic (dynamic) friction during motion is slightly less than limiting friction; rolling friction is much smaller still.

```
Limiting friction F = μN; angle of friction φ: tanφ = μ
Angle of repose α = angle of friction φ (for a body on an inclined plane just on the point of sliding)
Cone of friction: semi-vertex angle = φ
Force to pull up an incline (force parallel to plane): P = W(sinα + μcosα); to just prevent sliding down: P = W(sinα − μcosα)
Least force to drag a body on a horizontal plane: P_min = W sinφ, applied at angle φ to the horizontal
Ladder problems: ΣFx = 0, ΣFy = 0, ΣM about foot = 0
Belt friction: T₁/T₂ = e^(μθ) (flat belt); V-belt: T₁/T₂ = e^(μθ cosec β) (β = half groove angle)
Screw jack: effort P = W tan(α + φ); efficiency η = tanα/tan(α + φ); max efficiency = (1 − sinφ)/(1 + sinφ); self-locking if φ > α (η < 50%)
```

- Laws of dry (Coulomb) friction: friction is proportional to normal reaction, independent of apparent contact area, independent of sliding speed (approximately), depends on the nature of surfaces; acts opposite to the direction of impending motion.
- Wedge, screw and self-locking: a self-locking screw jack has efficiency less than 50%.

★ **PYQ-TREND:** Angle of repose = angle of friction (very frequent); μ = tanφ; least pull on horizontal plane at angle φ; belt tension ratio e^(μθ); self-locking screw efficiency < 50%; laws of friction.

✖ **COMMON MISTAKE:** Rolling friction is much less than sliding friction; kinetic friction is less than static friction – not equal to it.

1.4 Trusses

- Truss: framework of members joined at ends by pins, loaded at joints; members are two-force members carrying axial tension or compression.

```
Perfect (just rigid) truss: m = 2j − 3; deficient: m < 2j − 3; redundant: m > 2j − 3 (m = members, j = joints)
```

- Method of joints: equilibrium of each joint (ΣFx = ΣFy = 0) – start at a joint with only two unknowns. Method of sections: cut through not more than three members of unknown force and use ΣM = 0 – quick for a particular member.
- Zero-force members: at a joint with two non-collinear members and no external load, both are zero; at a joint with three members, two collinear and no load, the third is zero.

★ **PYQ-TREND:** m = 2j − 3 condition (very frequent); zero-force member rules; method of sections uses at most three unknown members.

1.5 Centroid and Moment of Inertia

Shape	Centroid location
Rectangle	At mid-height h/2
Triangle	h/3 from base (2h/3 from apex)
Semicircle	4r/(3π) from diameter
Quarter circle	4r/(3π) from each straight edge
Hemisphere (solid)	3r/8 from flat face
Hollow hemisphere (shell)	r/2 from flat face
Right circular cone (solid)	h/4 from base
Semicircular arc (wire)	2r/π from diameter

```
Parallel axis theorem: I = I_G + A h²; perpendicular axis theorem (plane areas): I_z = I_x + I_y
Radius of gyration k = √(I/A)
```

Section	I about centroidal axis
Rectangle b × d (about axis parallel to b)	bd³/12 (about base: bd³/3)
Square side a	a⁴/12
Circle diameter d	πd⁴/64 (polar πd⁴/32)
Hollow circle	π(D⁴ − d⁴)/64
Triangle (base b, height h)	bh³/36 (about base: bh³/12)
Semicircle (about diameter)	πd⁴/128
Semicircle (centroidal, parallel to diameter)	0.11 r⁴
Mass MOI: solid sphere / hollow sphere	(2/5)MR² / (2/3)MR²
Mass MOI: solid disc / ring	MR²/2 / MR²
Mass MOI: rod about centre / about end	ML²/12 / ML²/3

★ **PYQ-TREND:** Centroid of triangle h/3, semicircle 4r/3π, cone h/4, hemisphere 3r/8 (very frequent); I of rectangle bd³/12 and about base bd³/3; triangle bh³/36; parallel and perpendicular axis theorems; polar MOI of circle πd⁴/32.

✓ **TOPPER TRICK:** Square: same I = a⁴/12 about any centroidal axis in its plane (including the diagonal).

1.6 Dynamics Essentials

```
Equations of motion: v = u + at; s = ut + ½at²; v² = u² + 2as
Projectile: range R = u² sin2θ/g (max at 45°: u²/g); max height H = u² sin²θ/(2g); time of flight T = 2u sinθ/g
```

Work–energy: $W = \Delta KE$; impulse–momentum: $F \cdot t = m\Delta v$

Coefficient of restitution $e = (v_2 - v_1)/(u_1 - u_2)$; $e = 1$ perfectly elastic, $e = 0$ perfectly plastic

Circular motion: centripetal acceleration $v^2/r = \omega^2 r$; banking angle $\tan\theta = v^2/(rg)$

Simple pendulum $T = 2\pi\sqrt{(L/g)}$; spring–mass $T = 2\pi\sqrt{(m/k)}$

- D'Alembert's principle: adding the inertia force (−ma) converts a dynamics problem into a static equilibrium problem.
- Conservation of momentum holds in all collisions; kinetic energy is conserved only in perfectly elastic collisions.

★ **PYQ-TREND:** Maximum range at 45°; coefficient of restitution limits; D'Alembert's principle; banking of roads; conservation of momentum vs energy.

✎ **DESCRIPTIVE Q:** State and prove Lami's theorem.

Lami's theorem states that if three coplanar concurrent forces acting on a body keep it in equilibrium, each force is proportional to the sine of the angle between the other two. Let forces P, Q and R act at a point O in equilibrium, with α the angle between Q and R, β between R and P, and γ between P and Q. Since the forces are in equilibrium they can be represented by the sides of a triangle taken in order (triangle law). In this triangle the interior angles are (180° − α), (180° − β) and (180° − γ), opposite the sides representing P, Q and R respectively. Applying the sine rule: P/sin(180° − α) = Q/sin(180° − β) = R/sin(180° − γ). Since sin(180° − θ) = sinθ, P/sinα = Q/sinβ = R/sinγ. The theorem is used to find tensions in strings and reactions in problems of a body suspended by two strings or resting against smooth surfaces.

CHAPTER 2

Simple Stresses and Strains

2.1 Stress and Strain

- Stress: internal resisting force per unit area, $\sigma = P/A$ ($N/m^2 = Pa$; 1 MPa = 1 N/mm^2). Normal stress (tensile or compressive) acts perpendicular to the section; shear stress τ acts parallel to it.
- Strain: deformation per unit original dimension (dimensionless). Linear strain $e = \delta L/L$; shear strain φ = change in a right angle (radians); volumetric strain $e_v = \delta V/V$.

```
σ = P/A; e = δL/L; τ = P/A (shear); bearing stress σ_b = P/(d t)
Hooke's law: σ = E e (within proportional limit); τ = G φ; σ_v = K e_v
Elongation of a bar: δL = PL/(AE)
Poisson's ratio μ = − lateral strain/longitudinal strain (0 ≤ μ ≤ 0.5)
```

- Typical Poisson's ratio: steel 0.25–0.33 (≈ 0.3), cast iron 0.21–0.26, aluminium 0.33, concrete 0.1–0.2, rubber ≈ 0.5, cork ≈ 0, auxetic materials negative.
- Typical E: steel 200–210 GPa, cast iron ≈ 100–150 GPa, copper ≈ 110–120 GPa, aluminium ≈ 70 GPa, brass ≈ 100 GPa, concrete ≈ 25–30 GPa, rubber ≈ 0.01–0.1 GPa; G of steel ≈ 80 GPa.
- Saint-Venant's principle: away from the point of load application (about one width), the stress distribution depends only on the resultant, not on the way the load is applied.
- Principle of superposition: for linear-elastic small deformations, effects of several loads can be added.

★ **PYQ-TREND:** Hooke's law, units of stress; Poisson's ratio range 0–0.5 and value for steel/rubber/cork (very frequent); $\delta L = PL/AE$; Saint-Venant's principle.

2.2 Elastic Constants

```
E = 2G(1 + μ); E = 3K(1 − 2μ); E = 9KG/(3K + G); μ = (3K − 2G)/(6K + 2G)
Volumetric strain of a rectangular bar under axial load: e_v = e(1 − 2μ); for three-dimensional stresses: e_v = (σx + σy + σz)(1 − 2μ)/E
Volumetric strain of a cylindrical rod: e_v = e_L − 2e_d (e_d = magnitude of diametral strain); of a sphere: e_v = 3 × diametral strain
```

- Number of independent elastic constants: isotropic material 2 (any two of E, G, K, μ); orthotropic 9; general anisotropic 21.
- When μ = 0.5 (incompressible), $K \to \infty$ and $E = 3G$. When μ = 0, $E = 2G = 3K$.
- For steel, $G \approx 0.4E$ and $K \approx 0.83E$ (with μ = 0.3: $K = E/1.2$).

★ **PYQ-TREND:** $E = 2G(1 + \mu)$ and $E = 3K(1 - 2\mu)$ (very frequent); $E = 9KG/(3K + G)$; number of independent constants for isotropic material = 2; incompressible material μ = 0.5.

✔ **TOPPER TRICK:** For μ = 0.25: $E = 2.5G$, $K = E/1.5$ → E : G : K = 1 : 0.4 : 0.67.

2.3 Bars of Varying Section and Compound Bars

Stepped bar: $\delta L = (P/E)\ \Sigma(L_i/A_i)$

Uniformly tapering circular bar (diameters d_1, d_2): $\delta L = 4PL/(\pi E d_1 d_2)$

Tapering rectangular bar (thickness t, widths a, b): $\delta L = PL \ln(a/b)/(E t (a - b))$

Bar hanging under self-weight: $\delta L = \rho g L^2/(2E) = WL/(2AE)$ (half that of the same weight applied at the end)

Bar of uniform strength (constant stress σ under self weight + load): $A_2 = A_1$ e^(ρgL/σ)

Conical bar under self-weight: $\delta L = \rho g L^2/(6E)$

- Composite (compound) bar of two materials sharing a load in parallel: equal strains (compatibility) and sum of loads (equilibrium).

Composite bar: $P = \sigma_1 A_1 + \sigma_2 A_2$; $\sigma_1/E_1 = \sigma_2/E_2 \rightarrow \sigma_1 = (E_1/E_2)\ \sigma_2$ (modular ratio $m = E_1/E_2$)

- Statically indeterminate bar fixed at both ends with a load in between: reactions share the load inversely as the lengths on either side: R_A = P b/L, R_B = P a/L.

★ **PYQ-TREND:** Elongation due to self-weight $\rho g L^2/2E$ (very frequent); tapering bar $4PL/(\pi E d_1 d_2)$; composite bar equal strain; modular ratio; bar of uniform strength exponential profile.

✖ **COMMON MISTAKE:** Self-weight elongation is half of the elongation produced by the same total weight applied at the free end.

2.4 Thermal Stresses

Free thermal expansion $\delta = \alpha L \Delta T$ (no stress)

Fully restrained bar: thermal stress $\sigma = E\alpha\Delta T$ (compressive on heating, tensile on cooling) — independent of length and area

Partially restrained (gap δ allowed): $\sigma = E(\alpha L \Delta T - \delta)/L$

Composite bar of two materials heated ($\alpha_1 > \alpha_2$): $\sigma_1 A_1 = \sigma_2 A_2$ (equilibrium); $\sigma_1/E_1 + \sigma_2/E_2 = (\alpha_1 - \alpha_2)\Delta T$ (compatibility) → material with higher α in compression

Tapered circular bar ($d_1 > d_2$) fixed at both ends and heated: $P = E\alpha\Delta T\ \pi\ d_1 d_2/4$; σ_max (at small end) = $E\alpha\Delta T\ d_1/d_2$

- Typical α: steel 12 × 10^{-6} /°C, copper 17 × 10^{-6}, aluminium 23 × 10^{-6}, brass 19 × 10^{-6}, concrete ≈ 10–12 × 10^{-6} (nearly same as steel – hence RCC works), invar ≈ 1 × 10^{-6}.

★ **PYQ-TREND:** Thermal stress $E\alpha\Delta T$ independent of length (very frequent); compound bar – higher-α material in compression on heating; steel and concrete have similar α; gap allowance.

2.5 Strain Energy and Impact Loading

Strain energy (axial) $U = \sigma^2 V/(2E) = P^2 L/(2AE)$; strain energy density = $\sigma^2/(2E)$

Proof resilience = $\sigma_p^2 V/(2E)$ (σ_p = proof/elastic limit stress); modulus of resilience = $\sigma_p^2/(2E)$

Gradually applied load: $\sigma = P/A$; suddenly applied load: $\sigma = 2P/A$ (twice); impact (falling weight W from height h): $\sigma = (W/A)[1 + \sqrt{(1 + 2AEh/(WL))}]$

If h □ δ: $\sigma \approx \sqrt{(2WhE/(AL))}$

Shear strain energy $U = \tau^2 V/(2G)$; torsion of solid shaft $U = (\tau^2/(4G))\ V$; hollow shaft $U = (\tau^2/(4G))\ (D^2 + d^2)/D^2 \times V$

Bending: $U = \int M^2\ dx/(2EI)$; beam of uniform strength stores more energy

- Modulus of toughness: total area under the stress–strain curve up to fracture (energy absorbed per unit volume before fracture).

- Castigliano's first theorem: $\partial U/\partial P = \delta$ (deflection in the direction of load P). Maxwell's reciprocal theorem: deflection at A due to load at B = deflection at B due to the same load at A.

★ **PYQ-TREND:** Suddenly applied load gives double stress (very frequent); strain energy formula $\sigma^2V/2E$; proof resilience vs modulus of resilience; impact stress; Castigliano's theorem; Maxwell's reciprocal theorem; strain energy of solid shaft $\tau^2V/4G$.

✎ **DESCRIPTIVE Q:** Derive the expression for stress produced by a suddenly applied load.

Let a load P be applied suddenly to a bar of area A and length L (at zero height). The load does work $P\cdot\delta L$ as it moves through the full extension δL, since its magnitude is constant throughout. The bar stores strain energy $U = \sigma^2AL/(2E)$. Equating, $P\,\delta L = \sigma^2AL/(2E)$. With $\delta L = \sigma L/E$: $P\,\sigma L/E = \sigma^2AL/(2E)$, giving $\sigma = 2P/A$. Thus the stress under a suddenly applied load is twice the stress under the same load applied gradually (where the work done is $\frac{1}{2}P\delta L$ because the load increases from zero). The maximum extension is also twice; the bar then oscillates about the static equilibrium position and, with damping, settles at $\sigma = P/A$. For a load falling from a height h, the work done is $P(h + \delta L)$, and solving the resulting quadratic gives $\sigma = (P/A)[1 + \sqrt{(1 + 2AEh/(PL))}]$, which reduces to $2P/A$ when $h = 0$.

CHAPTER 3

Mechanical Properties and Material Testing

3.1 Stress–Strain Diagram (Mild Steel)

- Tensile test on a standard specimen (gauge length $L_0 = 5.65\sqrt{A_0}$ in IS/ISO proportional specimens) on a universal testing machine (UTM) gives load–extension data converted to the engineering stress–strain curve.

Point / region	Meaning
Proportional limit (A)	Stress ∝ strain up to here (Hooke's law); slope = E
Elastic limit (B)	Maximum stress without permanent set; slightly above A
Upper yield point (C)	Stress at which yielding starts; depends on loading rate and specimen
Lower yield point (D)	Stress stays nearly constant while strain increases (Lüders bands) – taken as yield strength
Strain hardening (D–E)	Stress rises again as dislocations multiply
Ultimate tensile strength (E)	Maximum engineering stress; necking begins
Fracture point (F)	Breaking stress (engineering) is lower than UTS because area reduces in the neck; true fracture stress is higher

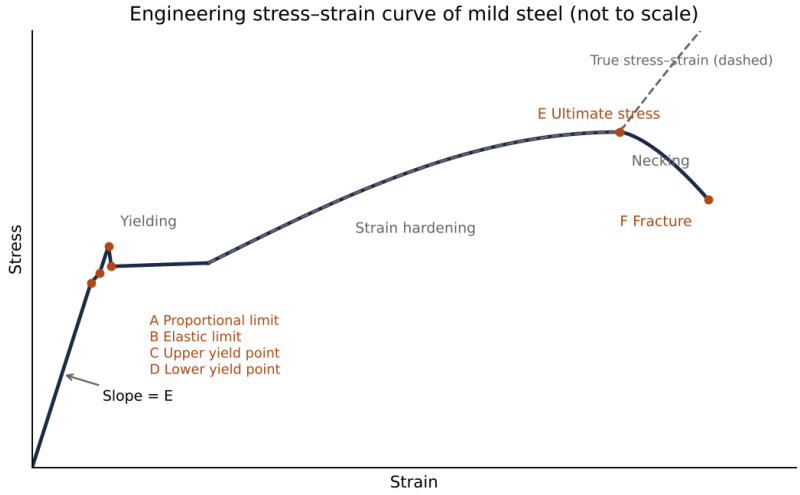
- Materials without a clear yield point (aluminium, high-strength steel, copper) use 0.2% offset (proof) stress.
- True stress $\sigma_t = \sigma(1 + e)$; true strain $\varepsilon = \ln(1 + e)$. The true stress–strain curve keeps rising up to fracture.

Percentage elongation = $(L_f - L_0)/L_0 \times 100$; percentage reduction in area = $(A_0 - A_f)/A_0 \times 100$

Factor of safety = ultimate (or yield) stress/working (allowable) stress

True stress $\sigma_t = \sigma(1 + e)$; true strain $\varepsilon = \ln(1 + e)$; Hollomon: $\sigma_t = K \varepsilon^n$ (n = strain-hardening exponent = true strain at UTS)

- Ductile materials (mild steel, copper, aluminium): large plastic deformation, cup-and-cone fracture at 45° shear; brittle materials (cast iron, glass, concrete): little deformation, flat fracture normal to the load in tension; in compression cast iron fails by shear along an inclined plane (≈ 45–55°).
- Ductility criterion: elongation > 5% → ductile; < 5% → brittle.
- Fracture of a ductile bar in torsion occurs on a plane perpendicular to the axis (maximum shear); brittle (cast iron) bar in torsion breaks along a 45° helix (maximum tensile stress).



★ **PYQ-TREND:** Sequence proportional limit → elastic limit → yield → UTS → fracture (very frequent); necking starts at UTS; 0.2% proof stress; cup-and-cone fracture of ductile materials; percentage elongation and reduction in area; factor of safety; torsion fracture planes of ductile vs brittle materials.

✖ **COMMON MISTAKE:** The breaking (fracture) stress on the engineering curve is less than the ultimate stress only because original area is used; the true fracture stress is the highest.

3.2 Mechanical Properties

Property	Meaning	Example
Elasticity	Regains shape after load removal	Steel, rubber
Plasticity	Permanent deformation without fracture	Lead, clay
Ductility	Can be drawn into wires (tensile plastic deformation)	Gold (most ductile), silver, platinum, copper, aluminium, mild steel
Malleability	Can be hammered/rolled into sheets (compressive plastic deformation)	Gold (most malleable), silver, aluminium, copper, tin, lead
Brittleness	Fractures with little deformation	Cast iron, glass, concrete
Toughness	Ability to absorb energy up to fracture (area under σ–ε curve)	Mild steel, wrought iron
Resilience	Energy absorbed within elastic limit	Spring steel
Hardness	Resistance to indentation, scratching, abrasion	Diamond, tool steel
Stiffness	Resistance to elastic deformation (E)	Steel > aluminium
Fatigue	Failure under repeated/fluctuating stress below UTS	Shafts, springs, aircraft parts
Creep	Slow, time-dependent plastic deformation under constant load (high temperature)	Turbine blades, boiler tubes, lead pipes at room temperature
Machinability	Ease of machining	Free-cutting steel, brass

Property	Meaning	Example
Weldability / castability	Ease of welding/casting	Low-carbon steel / cast iron

★ **PYQ-TREND:** Ductility vs malleability (very frequent); gold most ductile and malleable; toughness = area under curve up to fracture; resilience within elastic limit; creep at high temperature; fatigue below UTS.

3.3 Hardness Tests

Test	Indenter	Hardness number	Remarks
Brinell (BHN)	Hardened steel or tungsten-carbide ball, 10 mm, 3000 kgf for steel	Load/curved area of indentation	Not for very hard or thin materials; large impression – good for castings
Vickers (VHN/HV)	Diamond square pyramid, 136° between faces	Load/surface area of impression (1.854 P/d²)	All materials, thin sections; same scale over wide range
Rockwell (HR)	Diamond cone 120° (scale C, hard steels) or 1/16 in steel ball (scale B, soft)	From depth of penetration (minor load 10 kgf + major load)	Quick, direct reading; most used in production
Knoop	Elongated diamond pyramid	Microhardness	Thin coatings, brittle materials
Shore scleroscope	Diamond-tipped hammer dropped	Rebound height	Portable, non-destructive; large parts, rolls
Mohs scale	Scratch test	1 (talc) to 10 (diamond)	Minerals
Poldi / Moore	Portable impact tests	–	Field testing

$BHN = 2P/[\pi D(D - \sqrt{(D^2 - d^2)})]$ (P in kgf, D ball diameter, d impression diameter, mm)

$VHN = 1.854\ P/d^2$ (d = mean diagonal); approximate: UTS (MPa) ≈ 3.45 × BHN for steels

- Mohs scale order: talc, gypsum, calcite, fluorite, apatite, orthoclase, quartz, topaz, corundum, diamond.

★ **PYQ-TREND:** Vickers indenter 136° diamond pyramid (very frequent); Rockwell C uses 120° diamond cone and measures depth; Brinell uses ball and area; Shore scleroscope rebound; Mohs scale; UTS ≈ 3.45 BHN.

3.4 Impact, Fatigue and Creep Tests

Test	Description
Izod	Cantilever specimen (75 mm × 10 mm square, V-notch 45°, 2 mm deep) struck by pendulum; notch faces the striker; energy absorbed in joules
Charpy	Simply supported specimen (55 mm × 10 × 10, V or U notch) struck at the centre opposite the notch; also used for ductile–brittle transition temperature
Fatigue (rotating beam, R.R. Moore)	Specimen rotated under bending so stress reverses each cycle; S–N curve plotted
Creep test	Constant load at constant high temperature; strain vs time

Test	Description
Torsion test	Gives G and shear strength
Compression test	Brittle materials (cast iron, concrete)
Bend test	Ductility of welds and bars

- Ductile-to-brittle transition: BCC metals (steels) become brittle at low temperature (Titanic hull, Liberty ships); FCC metals (aluminium, copper, austenitic stainless) do not show it.
- Fatigue: S–N (Wöhler) curve; steels show an endurance limit (≈ 0.5 × UTS for polished specimens, at about 10^6–10^7 cycles) below which life is infinite; non-ferrous alloys (aluminium) have no true endurance limit – fatigue strength quoted at a given number of cycles.
- Factors reducing fatigue strength: stress concentration (notches, holes, keyways), poor surface finish, corrosion, size, residual tensile stress, high temperature. Improved by shot peening (compressive residual stress), surface hardening, polishing, fillets.
- Fatigue fracture surface: smooth beach marks (progression zone) and a rough instantaneous fracture zone.

```
Soderberg: σ_a/σ_e + σ_m/σ_y = 1/N; Goodman: σ_a/σ_e + σ_m/σ_u = 1/N; Gerber: σ_a/σ_e + N(σ_m/σ_u)² = 1/N
Stress concentration factor K_t = σ_max/σ_nominal; for a small circular hole in a wide plate K_t = 3; elliptical hole K_t = 1 + 2a/b
```

- Creep curve: primary (decreasing rate – strain hardening), secondary (steady-state, minimum rate – design region), tertiary (accelerating, necking, rupture). Creep becomes significant above ≈ 0.4 T_m (absolute melting temperature). Stress relaxation: stress falls with time at constant strain (bolts, springs at high temperature).

★ **PYQ-TREND:** Izod (cantilever) vs Charpy (simply supported) (very frequent); endurance limit ≈ 0.5 UTS and S–N curve; Goodman/Soderberg lines; stress concentration factor 3 for a circular hole; creep stages – secondary used for design; shot peening improves fatigue; BCC metals ductile–brittle transition.

✓ **TOPPER TRICK:** Izod – 'I' stands upright: specimen held vertically like a cantilever. Charpy – lies like a beam.

✎ **DESCRIPTIVE Q:** Explain the stress–strain diagram for mild steel in tension, indicating all salient points.

When a mild-steel specimen is loaded in tension, stress is at first proportional to strain up to the limit of proportionality, the slope being Young's modulus. Slightly beyond lies the elastic limit, up to which the specimen regains its original length when unloaded. Further loading brings it to the upper yield point, where the stress drops suddenly to the lower yield point; the material then elongates considerably at nearly constant stress (yielding), forming Lüders bands, as dislocations break free from carbon atoms. After yielding, strain hardening raises the stress again with increasing strain until the ultimate tensile strength is reached, the maximum engineering stress. At this point a local neck forms; the load-carrying area decreases rapidly, so the engineering stress (based on original area) falls until fracture at the breaking point, giving a cup-and-cone fracture. Total elongation of about 25–30% and reduction in area of about 60% indicate high ductility. The true stress–strain curve, based on instantaneous area, continues to rise up to fracture. Materials without a distinct yield point use the 0.2% proof stress.

CHAPTER 3B

Non-Destructive Testing, Composites, Ceramics and Polymers

3B.1 Non-Destructive Testing (NDT)

Method	Principle	Detects / remarks
Visual inspection (with borescopes, magnifiers)	Direct observation	Surface defects; first and cheapest step
Dye (liquid) penetrant test (DPT)	Coloured/fluorescent dye drawn into surface-open cracks by capillary action; developer draws it out	Surface-breaking cracks, porosity in any non-porous material (metals, ceramics, plastics); cannot find subsurface defects
Magnetic particle test (MPT)	Magnetised part; flux leaks at defects attract iron particles	Surface and near-surface cracks in ferromagnetic materials only (not aluminium, austenitic stainless); best when crack perpendicular to field
Radiography (X-ray, γ-ray: Co-60, Ir-192)	Differential absorption of penetrating radiation recorded on film/digital detector	Internal volumetric defects – porosity, blowholes, inclusions, shrinkage in castings and welds; permanent record; radiation hazard; planar cracks parallel to beam hard to see
Ultrasonic testing (UT)	High-frequency sound (0.5–25 MHz) pulses reflected from interfaces (pulse-echo) or transmitted	Internal cracks, laminations, thickness measurement; deep penetration; needs couplant; skilled operator; good for planar defects
Eddy current testing (ECT)	Induced eddy currents change near defects (coil impedance)	Surface/near-surface cracks in conductive materials, coating thickness, sorting alloys, tube inspection
Acoustic emission	Stress waves released by growing cracks	Monitoring pressure vessels during loading
Thermography (infrared)	Temperature patterns	Delaminations, electrical hot spots
Leak testing (bubble, helium, pressure decay)	Fluid escape	Leaks in vessels and piping

- Destructive tests: tensile, compression, hardness (slightly destructive), impact, fatigue, creep, bend, torsion, metallography (macro/micro-etch).

★ **PYQ-TREND:** Magnetic particle test only for ferromagnetic materials (very frequent); dye penetrant for surface cracks only; radiography for internal porosity in castings and welds; ultrasonic for internal cracks and thickness; eddy current for conductive materials; Co-60 and Ir-192 gamma sources.

✔ **TOPPER TRICK:** Surface only → DPT; surface + just below, ferrous → MPT; deep inside → UT or radiography.

3B.2 Composite Materials

- Composite: two or more distinct phases combined to obtain properties not possible individually – matrix (continuous, binds and protects, transfers load) and reinforcement (fibres, particles, flakes – carries load).

Classification	Examples
Polymer matrix composites (PMC)	GFRP (glass fibre + polyester/epoxy – boats, tanks, car bodies), CFRP (carbon fibre + epoxy – aircraft, sports goods), aramid (Kevlar) – bullet-proof vests
Metal matrix composites (MMC)	Al–SiC, Al–B; brake rotors, engine parts – high stiffness and wear resistance
Ceramic matrix composites (CMC)	SiC–SiC, C–C (carbon–carbon) – brake discs of aircraft, rocket nozzles, heat shields
Particulate composites	Concrete (aggregate in cement), cermets (WC–Co cemented carbides), filled polymers
Laminated / structural	Plywood, sandwich panels (honeycomb core), clad metals, bimetal strips
Natural composites	Wood (cellulose fibres in lignin), bone

```
Rule of mixtures (longitudinal, iso-strain): E_c = E_f V_f + E_m V_m; σ_c = σ_f V_f + σ_m V_m
Transverse (iso-stress): 1/E_c = V_f/E_f + V_m/E_m
```

- Advantages: high specific strength and stiffness, corrosion resistance, fatigue resistance, tailored directional properties, low thermal expansion; disadvantages: anisotropy, high cost, delamination, difficult repair and recycling.
- Manufacturing: hand lay-up, spray-up, filament winding, pultrusion, resin transfer moulding, autoclave curing of prepregs, compression moulding (SMC/BMC).

★ **PYQ-TREND:** GFRP and CFRP uses; Kevlar for bullet-proof vests; cermets as particulate composites; rule of mixtures (frequent); wood and bone as natural composites; C–C composites for brake discs and rocket nozzles.

3B.3 Ceramics and Polymers

- Ceramics: compounds of metals and non-metals (oxides – alumina Al_2O_3, zirconia ZrO_2; carbides – SiC, WC; nitrides – Si_3N_4, CBN; silicates – glass, porcelain, cement). Ionic/covalent bonds give high hardness, high melting point, chemical inertness, electrical and thermal insulation, high compressive strength; but brittleness and low tensile strength, poor thermal-shock resistance (except some). Uses: refractories, cutting tools, abrasives, spark-plug insulators, bearings, armour, turbine coatings, electronics substrates.
- Refractories: acid (silica, fireclay), basic (magnesite, dolomite), neutral (chromite, alumina, graphite, SiC). Use acid refractories with acid slags and basic with basic slags.
- Glass: amorphous (non-crystalline) silicate – soda-lime (windows, bottles), borosilicate/Pyrex (low expansion, lab ware), lead glass (optical, crystal), fused silica; tempered (toughened) glass has compressive surface stress.

- Polymers: thermoplastics (linear/branched chains, soften on heating, recyclable – polyethylene, polypropylene, PVC, polystyrene, nylon, PTFE (Teflon – lowest friction, chemical inertness), PMMA (acrylic, Perspex), polycarbonate, PET, ABS); thermosets (cross-linked, rigid, cannot be remelted – phenol-formaldehyde (Bakelite), urea- and melamine-formaldehyde, epoxy, polyester, polyurethane (both types)); elastomers (natural rubber – polyisoprene, vulcanised with sulphur; synthetic – neoprene, nitrile, silicone, SBR).
- Polymerisation: addition (chain-growth – polyethylene from ethylene) and condensation (step-growth with by-product – nylon, polyester, Bakelite). Degree of polymerisation = molecular weight of polymer/molecular weight of mer.
- Glass-transition temperature T_g: below it an amorphous polymer is glassy and brittle; above it rubbery.
- Additives: plasticisers (flexibility – PVC), fillers, stabilisers, colourants, flame retardants, lubricants.

★ **PYQ-TREND:** Bakelite is a thermoset; PTFE lowest coefficient of friction; vulcanisation of rubber with sulphur (very frequent); addition vs condensation polymerisation; glass is amorphous; acid/basic/neutral refractories; plasticisers in PVC; T_g.

✎ **DESCRIPTIVE Q:** Compare magnetic particle testing and dye penetrant testing.

Both are surface inspection methods. In dye penetrant testing the cleaned surface is coated with a low-viscosity coloured or fluorescent penetrant that enters surface-breaking cracks by capillary action; after a dwell time, the excess is removed and a developer (white powder) draws the penetrant back out, revealing the defects as coloured or glowing lines (under UV light). It can be applied to any non-porous material – ferrous and non-ferrous metals, ceramics and plastics – is portable and cheap, but detects only defects open to the surface and requires careful cleaning. In magnetic particle testing the part is magnetised by a yoke, coil or direct current; at a crack the magnetic flux leaks out of the surface, and fine iron particles (dry or in a liquid, often fluorescent) sprinkled on the surface gather at the leakage field, outlining the defect. It detects surface and slightly subsurface defects (a few millimetres), is fast and sensitive, but works only on ferromagnetic materials (carbon and low-alloy steels, cast iron, nickel), needs the field roughly perpendicular to the crack (so two directions are tested), and the part may need demagnetising afterwards.

CHAPTER 4

Principal Stresses, Mohr's Circle and Theories of Failure

4.1 Stresses on an Inclined Plane

- Uniaxial stress σ on a plane inclined at θ to the cross-section (normal of plane at θ to the axis):

```
σ_n = σ cos²θ; τ = (σ/2) sin2θ; τ_max = σ/2 at θ = 45°
Two perpendicular direct stresses σx, σy: σ_n = (σx + σy)/2 + ((σx − σy)/2) cos2θ; τ = ((σx − σy)/2) sin2θ
General plane stress (σx, σy, τxy): σ_n = (σx + σy)/2 + ((σx − σy)/2) cos2θ + τxy sin2θ; τ = ((σx − σy)/2) sin2θ − τxy cos2θ
```

- Complementary shear stresses: a shear stress on one plane is always accompanied by an equal shear stress on the perpendicular plane (for moment equilibrium of an element).
- Pure shear state (τ only) is equivalent to equal tensile and compressive stresses of magnitude τ on planes at 45°.
- Sum of normal stresses on any two perpendicular planes is constant: $\sigma x + \sigma y = \sigma_1 + \sigma_2$ (stress invariant).

★ **PYQ-TREND:** Maximum shear in uniaxial tension = σ/2 at 45° (very frequent); complementary shear stresses; pure shear equivalent to ±τ at 45°; invariance of σx + σy.

4.2 Principal Stresses and Planes

- Principal planes: planes on which shear stress is zero; the normal stresses on them (principal stresses) are the maximum and minimum normal stresses. Principal planes are 90° apart; planes of maximum shear are at 45° to them.

```
σ₁,₂ = (σx + σy)/2 ± √[((σx − σy)/2)² + τxy²]
Principal plane: tan2θ_p = 2τxy/(σx − σy)
Maximum in-plane shear τ_max = √[((σx − σy)/2)² + τxy²] = (σ₁ − σ₂)/2; normal stress on that plane = (σx + σy)/2
Absolute maximum shear (3-D): τ_abs = (σ_max − σ_min)/2 — if σ₁ and σ₂ have the same sign, τ_abs = σ₁/2 (taking σ₃ = 0)
Principal strains: ε₁ = (σ₁ − μσ₂)/E; ε₂ = (σ₂ − μσ₁)/E (plane stress)
```

- Combined bending and torsion of a shaft: $\sigma = 32M/(\pi d^3)$, $\tau = 16T/(\pi d^3)$ → $\sigma_1 = (16/\pi d^3)[M + \sqrt{(M^2 + T^2)}]$, $\tau_{max} = (16/\pi d^3)\sqrt{(M^2 + T^2)}$.

```
Equivalent bending moment M_e = ½[M + √(M² + T²)]; equivalent torque T_e = √(M² + T²)
```

- Strain rosettes (rectangular 0–45–90° and delta 0–60–120°) measure strains in three directions to find principal strains on a surface.

★ **PYQ-TREND:** Principal stress formula and shear zero on principal planes (very frequent); principal and maximum shear planes 45° apart; $\tau_{max} = (\sigma_1 - \sigma_2)/2$; equivalent bending moment and torque; strain rosette.

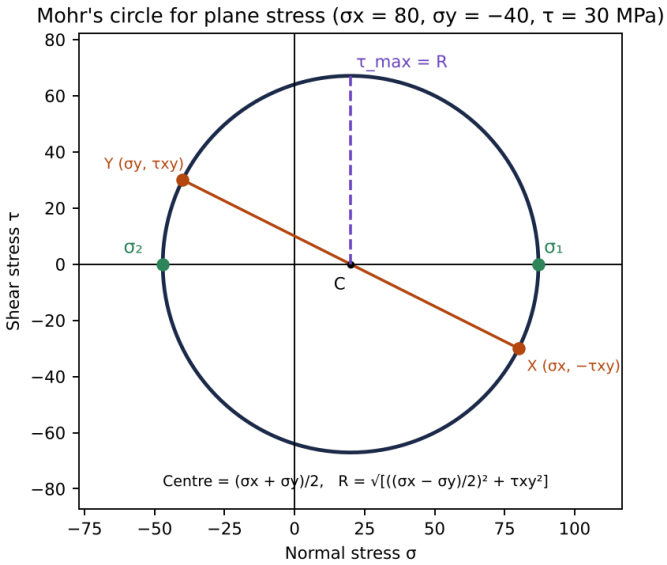
4.3 Mohr's Circle

- Graphical representation of the state of stress at a point: normal stress on the x-axis, shear stress on the y-axis.

Centre $C = ((\sigma x + \sigma y)/2, 0)$; radius $R = \sqrt{[((\sigma x - \sigma y)/2)^2 + \tau xy^2]} = \tau_{max}$
$\sigma_1 = C + R$; $\sigma_2 = C - R$; angle on circle = 2 × angle on element

Stress state	Mohr's circle
Uniaxial tension σ	Circle through origin, centre σ/2, radius σ/2
Equal biaxial tension (σx = σy = σ, τ = 0)	A point (radius zero) – no shear on any plane (hydrostatic in plane)
Pure shear τ	Centre at origin, radius τ ($\sigma_1 = \tau$, $\sigma_2 = -\tau$)
σx = −σy (equal and opposite)	Centre at origin, radius σx – equivalent to pure shear
Hydrostatic stress (3-D)	Point

- Mohr's circle for strain: ε on the x-axis, γ/2 on the y-axis; centre $(\varepsilon x + \varepsilon y)/2$, radius $\sqrt{[((\varepsilon x - \varepsilon y)/2)^2 + (\gamma xy/2)^2]}$.



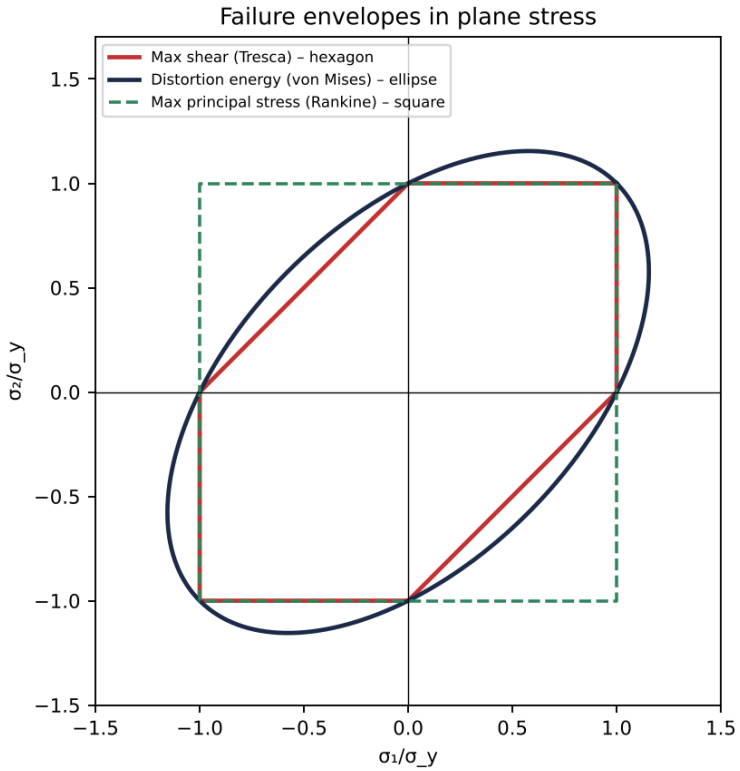
★ **PYQ-TREND:** Mohr's circle centre and radius (very frequent); circle reduces to a point for equal biaxial stresses; pure shear circle centred at origin; angle doubling; radius equals maximum shear stress.

✔ **TOPPER TRICK:** Mohr's circle radius = τ_{max}; its diameter = $\sigma_1 - \sigma_2$. If the circle touches the origin, one principal stress is zero.

4.4 Theories of Failure

Theory	Failure criterion	Equivalent design equation	Suitable for
Maximum principal stress (Rankine)	$\sigma_1 = \sigma_{yt}$ (or σ_u)	$\sigma_1 \le \sigma_y/N$	Brittle materials
Maximum shear stress (Guest–Tresca)	$\tau_{max} = \sigma_y/2$	$\sigma_1 - \sigma_2 \le \sigma_y$ (when signs differ); largest of $\lvert\sigma_1\rvert$, $\lvert\sigma_2\rvert$, $\lvert\sigma_1 - \sigma_2\rvert \le \sigma_y$	Ductile materials – conservative; hexagon locus
Maximum principal strain (St. Venant)	$\varepsilon_1 = \sigma_y/E$	$\sigma_1 - \mu\sigma_2 \le \sigma_y$	Rarely; brittle under some conditions; rhombus locus
Total strain energy (Haigh)	$U = \sigma_y^2/(2E)$	$\sigma_1^2 + \sigma_2^2 - 2\mu\sigma_1\sigma_2 \le \sigma_y^2$	Ductile (ellipse depending on μ)
Distortion energy / shear strain energy (von Mises–Hencky)	$U_d = (1+\mu)\sigma_y^2/(3E)$	$\sigma_1^2 + \sigma_2^2 - \sigma_1\sigma_2 \le \sigma_y^2$	Ductile materials – best agreement with tests; ellipse locus

- Shear yield strength: Tresca $\tau_y = 0.5\sigma_y$; von Mises $\tau_y = \sigma_y/\sqrt{3} = 0.577\sigma_y$.
- Under hydrostatic stress, theories based on shear or distortion predict no failure (no shear and no distortion).
- Shaft design: Tresca → $T_e = \sqrt{(M^2 + T^2)}$; Rankine → $M_e = \frac{1}{2}[M + \sqrt{(M^2 + T^2)}]$; von Mises → $\sqrt{(M^2 + \frac{3}{4}T^2)}$.
- In pure shear the Tresca hexagon lies inside the von Mises ellipse, so Tresca is safer (more conservative).



★ **PYQ-TREND:** Maximum shear stress theory for ductile and principal stress theory for brittle materials (very frequent); von Mises gives $\tau_y = 0.577\sigma_y$; shapes of failure loci (hexagon, ellipse, square, rhombus); Tresca conservative; hydrostatic stress causes no distortion.

✖ **COMMON MISTAKE:** Distortion energy theory is the most accurate for ductile metals, but Tresca (maximum shear) is often preferred in design because it is simpler and conservative.

✎ **DESCRIPTIVE Q:** Explain the maximum shear stress theory and the distortion energy theory, and compare them.

The maximum shear stress theory (Guest–Tresca) states that yielding begins when the maximum shear stress at a point reaches the maximum shear stress at yield in a simple tension test, i.e. $\tau_{max} = \sigma_y/2$. For a biaxial stress state with principal stresses σ_1 and σ_2 the condition is: if they are of like sign, the larger one equals σ_y; if of opposite sign, $\sigma_1 - \sigma_2 = \sigma_y$. On a σ_1–σ_2 plot the safe region is an irregular hexagon. The distortion energy theory (von Mises–Hencky) states that yielding occurs when the strain energy of distortion per unit volume (the part that changes shape, excluding volume change) equals that at yield in simple tension, leading to $\sigma_1^2 + \sigma_2^2 - \sigma_1\sigma_2 = \sigma_y^2$ in two dimensions, an ellipse inscribed around the Tresca hexagon. Both theories are suitable for ductile materials and both predict no yielding under hydrostatic stress. The distortion energy theory agrees best with experiments; it gives shear yield strength $0.577\sigma_y$, while Tresca gives $0.5\sigma_y$, so Tresca is more conservative and simpler, giving up to about 15% heavier designs. For brittle materials, the maximum principal stress theory is used instead.

CHAPTER 5

Shear Force and Bending Moment Diagrams

5.1 Beams, Supports and Loads

Beam	Description	Determinacy
Cantilever	Fixed at one end, free at the other	Determinate
Simply supported	Hinge at one end, roller at the other	Determinate
Overhanging	Simply supported with one or both ends extending beyond supports	Determinate
Fixed (built-in)	Both ends fixed	Indeterminate (degree 3; 2 for vertical loads)
Propped cantilever	Fixed at one end, simply supported at the other	Indeterminate (degree 1)
Continuous	More than two supports	Indeterminate

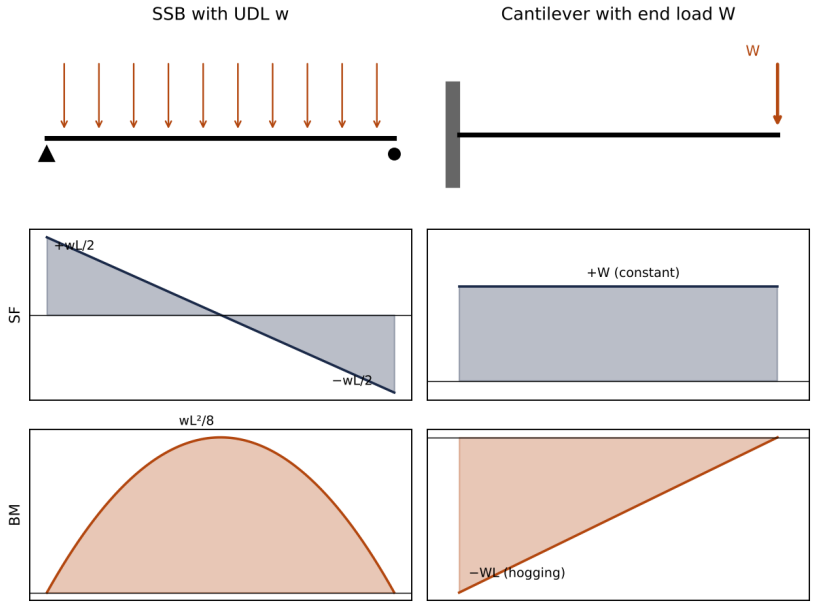
- Loads: point (concentrated) load, uniformly distributed load (UDL, w per unit length), uniformly varying load (UVL – triangular), applied couple (moment).
- Shear force (SF) at a section: algebraic sum of vertical forces on either side of the section. Bending moment (BM): algebraic sum of moments of forces on either side about the section.
- Sign convention: SF positive when the left portion tends to move up relative to right (clockwise shear); BM positive when sagging (concave upward), negative when hogging.

5.2 Relations Between Load, SF and BM

```
dF/dx = −w (rate of change of SF = intensity of load); dM/dx = F (rate of change of BM = SF)
BM is maximum (or minimum) where SF = 0 or changes sign
Change in BM between two sections = area of SFD between them; change in SF = area of load diagram
```

Loading on a span	Shape of SFD	Shape of BMD
No load	Horizontal (constant)	Inclined straight line (linear)
Point load	Vertical jump equal to load	Change in slope (kink)
UDL	Inclined straight line (linear)	Parabola (second degree)
Uniformly varying load (triangular)	Parabola (second degree)	Cubic (third degree)
Applied couple	No change	Vertical jump equal to couple

- Point of contraflexure (inflection): where BM changes sign (zero BM inside the span); occurs in overhanging, fixed and propped beams but not in a simply supported beam with downward loads or a cantilever.



★ **PYQ-TREND:** dM/dx = F and max BM where SF = 0 (very frequent); SFD/BMD shapes for UDL (linear/parabolic) and UVL (parabolic/cubic); point of contraflexure definition; couple causes jump in BMD only.

5.3 Standard Results

Beam and load	Reactions / max SF	Maximum BM
Cantilever, point load W at free end	W (constant)	WL at fixed end (hogging)
Cantilever, UDL w over length L	wL at fixed end	$wL^2/2$ at fixed end
Cantilever, UVL zero at free end to w at fixed end	wL/2	$wL^2/6$ at fixed end
Cantilever, UVL w at free end to zero at fixed end	wL/2	$wL^2/3$ at fixed end
Cantilever, couple M at free end	0	M (constant along length)
SSB, central point load W	W/2 each	WL/4 at centre
SSB, point load W at a from A (b from B)	R_A = Wb/L, R_B = Wa/L	Wab/L under the load
SSB, UDL w over span	wL/2 each	$wL^2/8$ at centre
SSB, UVL zero at one end to w at the other	wL/6 and wL/3	$wL^2/(9\sqrt{3}) \approx 0.064wL^2$ at $L/\sqrt{3}$ from zero end
SSB, symmetric triangular load (peak w at centre)	wL/4 each	$wL^2/12$ at centre
SSB, couple M at a point	±M/L (equal and opposite)	Jump of M at the point
Fixed beam, UDL	wL/2 each	Fixed-end moments $wL^2/12$ (hogging); centre $wL^2/24$ (sagging)
Fixed beam, central point load	W/2 each	WL/8 at ends and centre
Fixed beam, point load at a, b	–	Fixed-end moments Wab^2/L^2 and Wa^2b/L^2
Propped cantilever, UDL	Prop reaction 3wL/8; fixed end 5wL/8	$wL^2/8$ at fixed end; $9wL^2/128$ sagging at 3L/8 from prop

Beam and load	Reactions / max SF	Maximum BM
Propped cantilever, central point load	Prop reaction 5W/16	3WL/16 at fixed end; 5WL/32 under load

- Overhanging beam of total length L with equal overhangs a and UDL w throughout: BM at supports = $-wa^2/2$; BM at centre = $w(L - 2a)^2/8 - wa^2/2$. Equal magnitudes of hogging and sagging BM (most economical section) when a = 0.207L.
- Points of contraflexure: fixed beam with UDL at 0.211L from each end ($L/2 - L/(2\sqrt{3})$); fixed beam with central point load at L/4 from each end; propped cantilever with UDL at L/4 from the fixed end.

★ **PYQ-TREND:** Max BM: SSB UDL $wL^2/8$, central load WL/4, cantilever $wL^2/2$ (very frequent); fixed beam UDL $wL^2/12$ and $wL^2/24$; propped cantilever reaction 3wL/8 and 5W/16; UVL results $wL^2/(9\sqrt{3})$ and $wL^2/6$; contraflexure locations; SSB with couple has constant SF.

✔ **TOPPER TRICK:** For a SSB carrying total load W as UDL, max BM = WL/8 – half the BM of the same W as a central point load (WL/4).

✖ **COMMON MISTAKE:** In a cantilever, the maximum BM is at the fixed end and is hogging (negative), not at the free end.

✎ **DESCRIPTIVE Q:** Draw the SFD and BMD for a simply supported beam carrying a UDL w over its whole span L and locate the maximum bending moment.

By symmetry each reaction is wL/2. At a section x from the left support, SF $F = wL/2 - wx$, which varies linearly from $+wL/2$ at the left support to $-wL/2$ at the right support, passing through zero at mid-span ($x = L/2$). The BM $M = (wL/2)x - wx^2/2$, a parabola that is zero at both supports. Since $dM/dx = F$, the BM is maximum where $F = 0$, i.e. at mid-span, giving $M_{max} = (wL/2)(L/2) - (w/2)(L/2)^2 = wL^2/8$ (sagging). The SFD is therefore a straight inclined line crossing the axis at the centre, and the BMD a symmetrical parabola with its peak $wL^2/8$ at the centre. There is no point of contraflexure since the BM is sagging throughout. If the same total load $W = wL$ were concentrated at the centre, the maximum BM would be WL/4, twice as large, which shows the advantage of spreading loads.

CHAPTER 6

Bending Stresses in Beams

6.1 Theory of Simple Bending

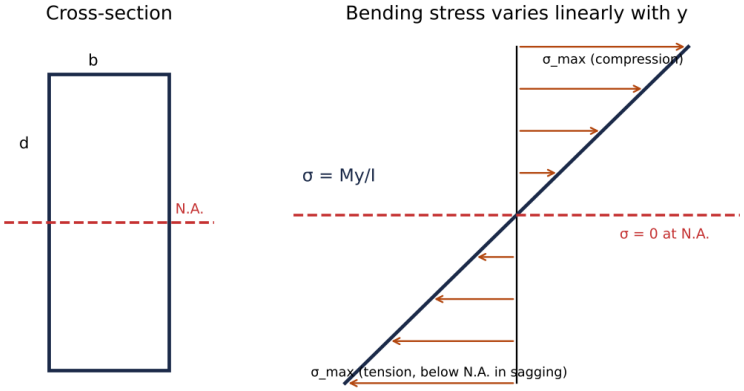
- Assumptions (Euler–Bernoulli): material homogeneous, isotropic and linearly elastic with the same E in tension and compression; beam initially straight with uniform cross-section symmetric about the plane of bending; plane sections remain plane after bending; radius of curvature large compared with depth; stresses within the elastic limit; no resultant axial force; each layer free to expand or contract independently.
- In sagging, fibres above the neutral axis are in compression and below are in tension; the neutral surface has zero stress and zero strain; the neutral axis passes through the centroid of the section (for pure bending with no axial force).

```
Bending equation: M/I = σ/y = E/R
Bending stress σ = M y/I; maximum σ = M/Z; section modulus Z = I/y_max
Moment of resistance M = σ_allow Z; flexural rigidity = EI; curvature 1/R = M/(EI)
```

- Bending stress varies linearly with distance from the neutral axis – zero at the NA, maximum at the extreme fibres.

Section	Section modulus Z
Rectangle b × d	$bd^2/6$
Square side a (side horizontal)	$a^3/6$
Square with diagonal vertical	$a^3/(6\sqrt{2}) = 0.118a^3$
Solid circle diameter d	$\pi d^3/32$
Hollow circle D, d	$\pi(D^4 - d^4)/(32D)$
Triangle (about centroidal axis, base b, height h)	$bh^2/24$ (top fibre) and $bh^2/12$ (bottom fibre)
I-section (approx.)	$[BD^3 - bd^3]/(6D)$ (b = B − web thickness)

- For the same area, the I-section is most efficient, then hollow sections, then rectangle placed with depth vertical, then square, then circle (material near the NA is poorly used).
- Strongest rectangular beam cut from a circular log of diameter D: b = D/√3, d = √(2/3)D (d/b = √2). Stiffest beam (maximum I): d/b = √3.
- Square vs circle of equal area: Z_square/Z_circle ≈ 1.18 – square is stronger. A square beam with its side horizontal is √2 times stronger than with its diagonal vertical.
- Doubling the depth of a rectangular beam increases its strength four times ($Z \propto d^2$) and stiffness eight times ($I \propto d^3$); doubling the width only doubles both.



★ **PYQ-TREND:** M/I = σ/y = E/R (very frequent); section modulus of rectangle $bd^2/6$ and circle $\pi d^3/32$; NA passes through centroid; bending stress zero at NA; strongest beam from log d/b = √2; I-section most economical; effect of doubling depth.

6.2 Beams of Uniform Strength and Composite Beams

- Beam of uniform strength: section varies so that σ_max is the same at every section (σ = M/Z constant). Examples: leaf springs, tapered cantilevers, girders with cover plates.

Cantilever with point load, constant depth: width b ∝ x (triangular plan); constant width: depth d ∝ √x

SSB central load, constant width: d ∝ √x from each support

- Flitched (composite) beam: two materials (e.g. timber with steel plates) bonded so that they bend together – strain is continuous; stresses at the same distance from NA are in the ratio of moduli: σ_s/σ_w = E_s/E_w = m (modular ratio). Use the equivalent (transformed) section: replace steel by m times its width in timber.

Composite beam: $M = M_1 + M_2 = \sigma_1 Z_1 + \sigma_2 Z_2$ with $\sigma_1/\sigma_2 = E_1/E_2$ at equal distance from NA

★ **PYQ-TREND:** Beam of uniform strength (leaf spring example); flitched beams and modular ratio; equivalent section method.

6.3 Combined Direct and Bending Stresses

Eccentric axial load P at eccentricity e: $\sigma = P/A \pm M/Z = P/A \pm P\,e/Z$

No tension condition (rectangle b × d, eccentricity along d): $e \le d/6$ – middle-third rule

Circular section: $e \le d/8$ – load within a central circle of diameter d/4 (middle-quarter rule)

Hollow circle: $e \le (D^2 + d^2)/(8D)$

Core (kern) of rectangular section: a rhombus with diagonals b/3 and d/3

- Applications: columns with eccentric loads, chimneys under wind pressure, dams and retaining walls (no tension in masonry requires the resultant to lie within the middle third of the base).

Chimney/dam base stresses: $\sigma_{max,min} = W/A\,(1 \pm 6e/b)$

★ **PYQ-TREND:** Middle-third rule $e \le d/6$ (very frequent); middle-quarter rule for circular sections d/8; kern of rectangle is a rhombus; σ = P/A ± M/Z; masonry dams no tension.

✔ **TOPPER TRICK:** Kern widths: rectangle d/3 (middle third); circle d/4 (diameter of core) – the core radius is d/8.

6.4 Plastic Bending and Shape Factor

- When the extreme fibres reach yield and loading continues, yielding spreads inward until the whole section is plastic (plastic hinge). In fully plastic state, the neutral axis divides the area into two equal halves (equal-area axis).

Plastic moment $M_p = \sigma_y Z_p$; plastic section modulus $Z_p = (A/2)(\bar{y}_1 + \bar{y}_2)$
Shape factor $S = Z_p/Z = M_p/M_y$

Section	Shape factor
Rectangle	1.5
Solid circle	16/(3π) = 1.70
Thin hollow circular tube	≈ 1.27
I-section (about major axis)	≈ 1.12–1.15
Triangle	2.34
Diamond (square with diagonal vertical)	2.0

- Load factor = collapse load/working load = shape factor × factor of safety.

★ **PYQ-TREND:** Shape factor of rectangle 1.5, circle 1.7, I-section ≈ 1.14, diamond 2 (very frequent); plastic neutral axis divides area equally; plastic hinge.

6.5 Curved Beams and Unsymmetrical Bending (Basics)

- In a curved beam (crane hooks, chain links, C-frames), the stress distribution is hyperbolic, not linear; the neutral axis shifts from the centroidal axis towards the centre of curvature; maximum stress occurs at the inner (concave) fibre. Analysis by Winkler–Bach theory.

Curved beam: $\sigma = (M/(A\,e)) \times (y/(R_n - y))$ where $e = \bar{R} - R_n$ (shift of NA)

- Unsymmetrical bending: load not in a plane of symmetry or section without an axis of symmetry (angles, Z-sections); bending about both principal axes; $\sigma = M_u v/I_u + M_v u/I_v$.
- Shear centre: point through which the load must pass to produce bending without twisting. For doubly symmetric sections it coincides with the centroid; for a channel it lies outside the web on the side opposite the flanges; for an angle section it is at the intersection of the leg centre-lines; for a T-section at the junction of flange and web centre-lines.

★ **PYQ-TREND:** Crane hook – curved beam, maximum stress at inner fibre, NA shifts towards centre of curvature (frequent); shear centre of channel outside the web; shear centre of angle at junction of legs.

✎ **DESCRIPTIVE Q:** State the assumptions of the theory of simple bending and derive the bending equation.

Assumptions: the beam is initially straight and of uniform section with a plane of symmetry in which the loads act; the material is homogeneous, isotropic and obeys Hooke's law with equal moduli in tension and compression; plane transverse sections remain plane after bending; the radius of curvature is large compared with the depth; each layer expands or contracts freely; and there is no resultant axial force. Consider a small length dx bent to radius R about the neutral surface, subtending angle dθ, so dx = R dθ. A layer at distance y from the neutral

surface has length (R + y)dθ; its strain is [(R + y)dθ − R dθ]/(R dθ) = y/R. By Hooke's law, σ = Ey/R, so stress varies linearly with y. For no resultant axial force, ∫σ dA = (E/R)∫y dA = 0, so ∫y dA = 0: the neutral axis passes through the centroid. The moment of the internal forces about the NA must equal the applied moment: M = ∫σ y dA = (E/R)∫y² dA = EI/R. Combining, M/I = σ/y = E/R. The maximum stress is σ = M y_max/I = M/Z.

CHAPTER 7

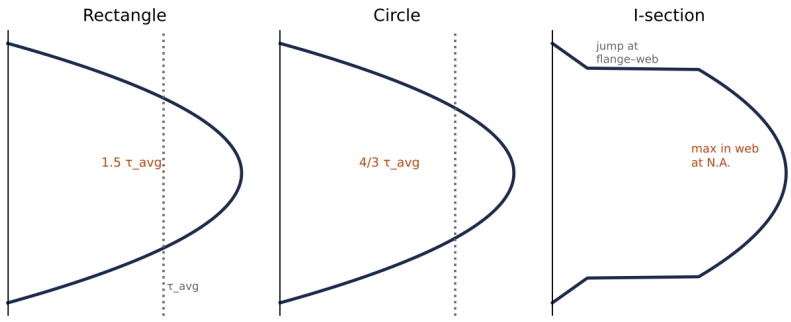
Shear Stresses in Beams and Deflection of Beams

7.1 Shear Stress Distribution

$\tau = V A \bar{y}/(I b)$ (V = shear force, $A\bar{y}$ = first moment of area above the layer about NA, b = width at the layer)

Section	Distribution	τ_{max}/τ_{avg}	Location of τ_{max}
Rectangle	Parabolic	1.5	Neutral axis
Solid circle	Parabolic	4/3 = 1.33	Neutral axis
Thin circular tube	–	2.0	Neutral axis
Triangle	Parabolic	1.5 (at h/2); at NA (h/3 from base) 4/3	Mid-height
Square with diagonal vertical (diamond)	–	9/8 = 1.125	d/8 above and below NA
I-section	Parabolic in flanges and web, sudden jump at the flange–web junction	Web carries ≈ 90–95% of shear	Neutral axis (web)

- Shear stress is zero at the top and bottom fibres (where bending stress is maximum) and maximum near the NA (where bending stress is zero).
- In an I-section, flanges resist most of the bending moment while the web resists most of the shear force; for approximate design, $\tau \approx V/(\text{web area})$.



★ **PYQ-TREND:** $\tau_{max} = 1.5\tau_{avg}$ for rectangle and 4/3 for circle (very frequent); maximum shear at NA (except triangle at h/2); I-section web carries shear and flanges carry bending; jump at web–flange junction; diamond 9/8.

✖ **COMMON MISTAKE:** For a triangular section, the maximum shear stress is at mid-height, not at the neutral axis (which lies at h/3 from the base).

7.2 Deflection – Differential Equation and Methods

Differential equation of the elastic curve: $EI\,d^2y/dx^2 = M$; $EI\,d^3y/dx^3 = F$; $EI\,d^4y/dx^4 = -w$

Method	Feature
Double integration	Integrate M twice; constants from boundary conditions
Macaulay's method	Single equation for the whole beam with discontinuous loads using brackets $\langle x - a \rangle$ (ignored when negative)
Moment-area method (Mohr's theorems)	Change in slope between two points = area of M/EI diagram; deflection of one point relative to tangent at other = moment of M/EI area about that point – convenient for cantilevers and symmetric beams
Conjugate beam method	Slope and deflection of real beam = shear and BM of conjugate beam loaded with M/EI; a fixed end becomes free and a free end becomes fixed; simple supports stay simple supports
Strain energy / Castigliano	$\delta = \partial U/\partial P$; useful for curved members and frames
Unit load method	Virtual work: $\delta = \int Mm\, dx/EI$

- Boundary conditions: fixed end – y = 0 and dy/dx = 0; simple support – y = 0 (and M = 0); free end – M = 0 and F = 0.

★ **PYQ-TREND:** EI $d^2y/dx^2 = M$ (very frequent); Macaulay's method for discontinuous loads; Mohr's moment-area theorems; conjugate beam supports conversion; boundary conditions.

7.3 Standard Slopes and Deflections

Beam and load	Maximum slope	Maximum deflection
Cantilever, point load W at free end	$WL^2/(2EI)$	$WL^3/(3EI)$
Cantilever, UDL w	$wL^3/(6EI)$	$wL^4/(8EI)$
Cantilever, couple M at free end	$ML/(EI)$	$ML^2/(2EI)$
Cantilever, UVL (zero at free end, w at fixed end)	$wL^3/(24EI)$	$wL^4/(30EI)$
Cantilever, point load W at distance a from fixed end	$Wa^2/(2EI)$	Under load $Wa^3/(3EI)$; free end $Wa^3/(3EI) + Wa^2(L - a)/(2EI)$
SSB, central point load W	$WL^2/(16EI)$ at supports	$WL^3/(48EI)$ at centre
SSB, UDL w	$wL^3/(24EI)$ at supports	$5wL^4/(384EI)$ at centre
SSB, eccentric point load (a, b)	–	Under load $Wa^2b^2/(3EIL)$
SSB, couple M at one end	$ML/(3EI)$ at that end, $ML/(6EI)$ at the other	$ML^2/(9\sqrt{3}\ EI)$
SSB, UVL (zero to w)	–	$\approx 0.00652wL^4/EI$
Fixed beam, central point load	0 at ends	$WL^3/(192EI)$
Fixed beam, UDL	0 at ends	$wL^4/(384EI)$
Propped cantilever, UDL	–	$wL^4/(185EI) \approx 0.0054wL^4/EI$

- Same total load W on a SSB: deflection with UDL ($5WL^3/384EI$) is 5/8 of that with a central point load ($WL^3/48EI$).
- Fixed beam with UDL deflects one-fifth as much as a simply supported beam; with central load, one-fourth.
- Deflection $\propto L^3$ for point loads and L^4 for UDL; $\propto 1/E$ and $1/I$. For a rectangle $I \propto bd^3$, so doubling the depth reduces deflection to 1/8.

- Ratio of free-end deflection to slope: cantilever with end load $\delta/\theta = 2L/3$; with UDL $\delta/\theta = 3L/4$.

★ **PYQ-TREND:** Cantilever $WL^3/3EI$ and $wL^4/8EI$; SSB $WL^3/48EI$ and $5wL^4/384EI$ (very frequent); fixed beam $WL^3/192EI$ and $wL^4/384EI$; ratio of deflections UDL vs point load; effect of length and depth on deflection; slope at supports $wL^3/24EI$.

✔ **TOPPER TRICK:** Memory chain for SSB central load: BM $WL/4$, slope $WL^2/16EI$, deflection $WL^3/48EI$ (denominators 4, 16, 48). Cantilever end load: WL, $WL^2/2EI$, $WL^3/3EI$ (1, 2, 3).

✎ **DESCRIPTIVE Q:** State Mohr's theorems (moment-area method) and use them to find the deflection of a cantilever with a point load at the free end.

Mohr's first theorem: the change in slope between any two points A and B on the elastic curve equals the area of the M/EI diagram between them. Mohr's second theorem: the deflection of point B measured from the tangent drawn at point A equals the moment of the M/EI diagram between A and B taken about B. For a cantilever of length L with load W at the free end, the bending moment varies linearly from zero at the free end to WL at the fixed end, so the M/EI diagram is a triangle of height WL/EI and base L. At the fixed end the slope is zero and the tangent is horizontal, so the slope at the free end = area of the M/EI diagram = $\frac{1}{2} \times L \times WL/EI = WL^2/(2EI)$. The deflection of the free end relative to the horizontal tangent at the fixed end = moment of this area about the free end = $(WL^2/2EI) \times (2L/3) = WL^3/(3EI)$, since the centroid of the triangle is 2L/3 from the free end. The method is especially convenient for cantilevers and symmetrically loaded beams where the tangent at one point is known.

CHAPTER 8

Torsion of Shafts and Springs

8.1 Torsion of Circular Shafts

- Assumptions: material homogeneous, isotropic, elastic; plane cross-sections remain plane and radii remain straight; shaft circular and uniform; twist uniform along the length; stresses within the elastic limit.

```
Torsion equation: T/J = τ/r = Gθ/L
Polar moment of inertia: solid J = πd⁴/32; hollow J = π(D⁴ − d⁴)/32
Maximum shear stress: solid τ = 16T/(πd³); hollow τ = 16TD/[π(D⁴ − d⁴)]
Polar modulus Z_p = J/r_max: solid πd³/16; hollow π(D⁴ − d⁴)/(16D)
Power transmitted P = 2πNT/60 (W, with T in N·m, N in rpm)
Torsional rigidity = GJ; torsional stiffness k_t = GJ/L; angle of twist θ = TL/(GJ)
```

- Shear stress varies linearly from zero at the centre to maximum at the surface. Because the core carries little torque, hollow shafts are stronger and stiffer per unit weight.
- Hollow vs solid of the same weight (material) and length: hollow shaft transmits more torque and is stiffer. Hollow vs solid of the same strength: hollow is lighter.
- For a hollow shaft with d = D/2: J_hollow = (15/16) J_solid of the same outer diameter, while material is only 3/4 – a weight saving of 25% for only 6.25% loss of strength.
- Shafts in series (different segments): same torque, angles add: $\theta = \Sigma TL_i/(G_iJ_i)$. Shafts in parallel (composite or fixed at both ends): same angle, torques add.
- Shaft fixed at both ends with torque T at a point: reactions T_A = T b/L, T_B = T a/L.
- Non-circular shafts: sections warp; maximum shear stress in a rectangle occurs at the middle of the longer side.

★ **PYQ-TREND:** T/J = τ/r = Gθ/L (very frequent); τ = 16T/πd³; P = 2πNT/60; hollow shaft advantages; series vs parallel shafts; angle of twist TL/GJ; shear stress zero at centre.

✔ **TOPPER TRICK:** For equal strength, torque ∝ d³; for equal stiffness, torque ∝ d⁴ (per unit twist). Doubling the diameter → 8× strength, 16× stiffness.

8.2 Combined Bending and Torsion; Shaft Design

```
Equivalent torque (Tresca): T_e = √(M² + T²) → τ_max = 16T_e/(πd³)
Equivalent bending moment (Rankine): M_e = ½[M + √(M² + T²)] → σ_max = 32M_e/(πd³)
With shock and fatigue factors (ASME code): T_e = √((K_m M)² + (K_t T)²)
Shaft under axial load + torque: σ = P/A, τ = 16T/πd³ → principal stresses from formula
```

- Strain energy in torsion: solid shaft U = (τ²/4G) × V; hollow shaft U = (τ²/4G) × [(D² + d²)/D²] × V.
- Critical speed (whirling) of shafts – resonance when rotational speed equals natural frequency of transverse vibration (covered in Theory of Machines).

★ **PYQ-TREND:** Equivalent torque $\sqrt{(M^2 + T^2)}$ and equivalent BM $\frac{1}{2}(M + \sqrt{(M^2 + T^2)})$ (very frequent); ASME code factors; strain energy in torsion.

8.3 Close-Coiled Helical Springs

- Wire diameter d, mean coil diameter D, number of active turns n, spring index C = D/d (typically 4–12), helix angle small (< 10°) for close-coiled springs; wire is mainly under torsion.

Shear stress $\tau = 8WD/(\pi d^3)$ (pure torsion); with Wahl factor $K = (4C - 1)/(4C - 4) + 0.615/C$: $\tau = K \times 8WD/(\pi d^3)$

Deflection $\delta = 8WD^3n/(Gd^4) = 64WR^3n/(Gd^4)$

Stiffness $k = W/\delta = Gd^4/(8D^3n)$

Strain energy $U = \frac{1}{2}W\delta$; springs in series: $1/k = 1/k_1 + 1/k_2$ (same load); springs in parallel: $k = k_1 + k_2$ (same deflection)

Cutting a spring into two halves doubles the stiffness of each half

- Wahl factor accounts for direct shear and curvature effects; stress is maximum at the inner side of the coil.
- Open-coiled helical springs: larger helix angle; wire under both bending and torsion.
- Spring materials: high-carbon steel, oil-tempered wire, music wire, chrome-vanadium, silicon-manganese steel, phosphor bronze, stainless steel.

★ **PYQ-TREND:** Spring stiffness $Gd^4/(8D^3n)$ and deflection $8WD^3n/Gd^4$ (very frequent); series and parallel springs; Wahl factor; spring index; cutting spring in half doubles stiffness.

✖ **COMMON MISTAKE:** Springs in series behave like resistances in parallel (reciprocals add) – the combination is softer, not stiffer.

8.4 Leaf (Laminated) Springs

- Built of plates of equal width and thickness but decreasing lengths, behaving approximately like a beam of uniform strength. Used in vehicle suspensions (semi-elliptic), the central band clamping the leaves. The leaves are given an initial curvature (camber) so that they straighten under proof load. Nipping: giving the full-length leaves a greater radius of curvature than graduated leaves to equalise stresses.

Semi-elliptic leaf spring (span L, n leaves, width b, thickness t, central load W): $\sigma = 3WL/(2nbt^2)$; $\delta = 3WL^3/(8Enbt^3)$

Quarter-elliptic (cantilever) leaf spring: $\sigma = 6WL/(nbt^2)$; $\delta = 6WL^3/(Enbt^3)$

Proof load: load that straightens the spring; camber = δ at proof load

- Torsion bar springs and spiral springs (clocks) are other types; flat spiral spring stores energy by bending.

★ **PYQ-TREND:** Leaf spring is a beam of uniform strength; nipping; stress $3WL/2nbt^2$ and deflection $3WL^3/8Enbt^3$; semi-elliptic springs in vehicles.

✎ **DESCRIPTIVE Q:** Derive the torsion equation for a solid circular shaft.

Consider a shaft of length L and radius R fixed at one end, subjected to torque T at the other. A line AB on the surface parallel to the axis moves to AB' after twisting; the angle of twist of the end section is θ, and the shear strain at the surface is $\varphi = BB'/L = R\theta/L$. At radius r, the shear strain is $r\theta/L$, so, by Hooke's law, the shear stress $\tau_r = G\theta r/L$, i.e. $\tau/r = G\theta/L$ – shear stress

varies linearly with radius. The torque resisted by an elemental ring of radius r and thickness dr is $dT = \tau_r \times 2\pi r\,dr \times r = (G\theta/L)\, r^2\, dA$. Integrating over the section: $T = (G\theta/L)\int r^2\, dA = (G\theta/L)\, J$, where $J = \pi d^4/32$ is the polar moment of inertia. Combining, $T/J = \tau/r = G\theta/L$. The maximum shear stress at the surface is $\tau_{max} = TR/J = 16T/(\pi d^3)$, and the angle of twist $\theta = TL/(GJ)$. For a hollow shaft $J = \pi(D^4 - d^4)/32$.

CHAPTER 9

Columns and Struts

9.1 Basics

- Column: vertical compression member; strut: compression member in any position (e.g. in trusses). Short columns fail by crushing (σ = P/A reaches crushing strength); long columns fail by buckling (lateral instability) at stresses well below the crushing stress; intermediate columns fail by a combination.

```
Slenderness ratio λ = L_e/k_min (k_min = least radius of gyration = √(I_min/A))
```

- Classification (mild steel, roughly): short λ < 30–32; medium 32–120; long λ > 120. Euler's formula is valid only when the Euler stress is below the proportional limit – for mild steel λ > about 80–100.
- Buckling always occurs about the axis of least moment of inertia (least radius of gyration).

★ **PYQ-TREND:** Slenderness ratio uses least radius of gyration (very frequent); short columns crush, long columns buckle; buckling about axis of minimum I.

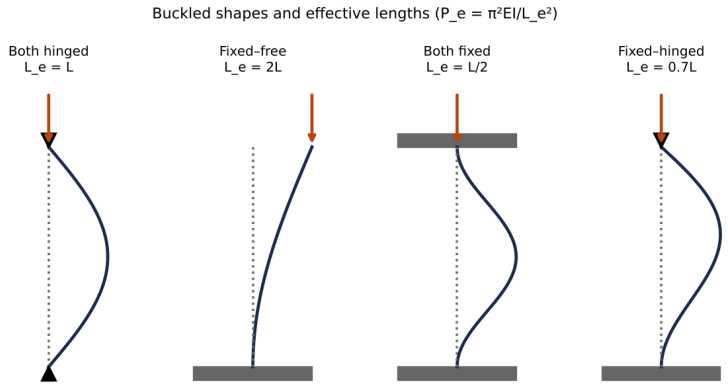
9.2 Euler's Theory

- Assumptions: column initially perfectly straight, load exactly axial, material homogeneous, isotropic and elastic, self-weight neglected, column fails by buckling only, cross-section uniform, length large compared with cross-section.

```
Euler crippling load P_e = π² E I_min/L_e²
Euler critical stress σ_e = π²E/λ²
```

End conditions	Effective length L_e	P_e (in terms of π²EI/L²)
Both ends hinged (pinned)	L	1
One end fixed, other free	2L	1/4
Both ends fixed	L/2	4
One end fixed, other hinged	L/√2 ≈ 0.707L	2

- Ratio of crippling loads (fixed-free : hinged-hinged : fixed-hinged : fixed-fixed) = 1/4 : 1 : 2 : 4 = 1 : 4 : 8 : 16.
- IS 800 practical effective lengths differ slightly (e.g. fixed–fixed 0.65L, fixed–hinged 0.8L, fixed–free 2L) to allow for imperfect fixity.
- Limitations: ignores direct (crushing) stress, initial crookedness and eccentricity; overestimates the load for short and medium columns.



★ **PYQ-TREND:** Euler load π^2EI/L_e^2 and effective lengths for four end conditions (very frequent); ratio 1 : 4 : 8 : 16; Euler valid for long columns only; critical stress π^2E/λ^2; $P \propto 1/L^2$.

✔ **TOPPER TRICK:** Doubling the length of a column reduces the Euler load to one-fourth; doubling the diameter of a solid circular column increases it 16 times ($I \propto d^4$).

9.3 Rankine–Gordon and Other Formulae

```
Rankine: 1/P_R = 1/P_c + 1/P_e → P_R = σ_c A/(1 + a λ²)
Rankine constant a = σ_c/(π²E) (a ≈ 1/7500 for mild steel, 1/1600 for cast iron, 1/750 for timber, for hinged ends)
Johnson's parabolic formula: σ = σ_y[1 − σ_y λ²/(4π²E)] (intermediate columns)
Straight-line formula: P = σ_c A(1 − n λ)
Secant formula (eccentric load e): σ_max = (P/A)[1 + (e y_c/k²) sec((L_e/2k)√(P/(EA)))]
Perry–Robertson formula: allows for initial curvature; basis of IS code curves
```

- Rankine's formula is empirical and valid for all slenderness ratios: for short columns P_e is very large and $P_R \approx P_c$; for long columns P_c is large relative to P_e and $P_R \approx P_e$.
- Column with eccentric load: maximum stress = $P/A + P\,e/Z$ (short column); long columns use secant formula.
- Beam-columns: members with both axial load and lateral load.

★ **PYQ-TREND:** Rankine formula combines crushing and buckling (very frequent); Rankine constant σ_c/π^2E; Johnson formula for intermediate columns; secant formula for eccentric loads.

✎ **DESCRIPTIVE Q:** Derive Euler's formula for a column with both ends hinged.

Consider a long column AB of length L hinged at both ends and subjected to an axial compressive load P. Let it buckle into a curve, with lateral deflection y at a distance x from end A. The bending moment at the section is $M = -Py$ (it tends to reduce curvature). Using $EI\,d^2y/dx^2 = M$: $EI\,d^2y/dx^2 + Py = 0$, or $d^2y/dx^2 + \alpha^2 y = 0$ with $\alpha^2 = P/(EI)$. The general solution is $y = C_1 \cos \alpha x + C_2 \sin \alpha x$. At $x = 0$, $y = 0$, so $C_1 = 0$. At $x = L$, $y = 0$, so $C_2 \sin \alpha L = 0$. Since $C_2 \neq 0$ for a buckled shape, $\sin \alpha L = 0$, giving $\alpha L = n\pi$. The smallest non-zero value ($n = 1$) gives the crippling load: $\sqrt{(P/EI)}\,L = \pi$, i.e. $P_e = \pi^2EI/L^2$. The column buckles in a half sine wave, $y = C_2 \sin(\pi x/L)$, about the axis of least moment of inertia. For other end conditions the same analysis gives $P_e = \pi^2EI/L_e^2$ with the appropriate effective length.

CHAPTER 10

Thin and Thick Cylinders and Spheres

10.1 Thin Cylinders (t < d/20)

- In a thin shell the hoop (circumferential) stress is assumed uniform across the thickness; radial stress is neglected. Failure may occur along a longitudinal joint (by hoop stress) or a circumferential joint (by longitudinal stress).

Hoop (circumferential) stress $\sigma_c = pd/(2t)$

Longitudinal stress $\sigma_l = pd/(4t) = \sigma_c/2$

Maximum in-plane shear $\tau = (\sigma_c - \sigma_l)/2 = pd/(8t)$; absolute maximum shear (including radial ≈ 0) $= \sigma_c/2 = pd/(4t)$

With joint efficiencies: $\sigma_c = pd/(2t\eta_l)$ (longitudinal joint); $\sigma_l = pd/(4t\eta_c)$ (circumferential joint)

Hoop strain $e_c = (\sigma_c - \mu\sigma_l)/E = (pd/(4tE))(2 - \mu)$; longitudinal strain $e_l = (pd/(4tE))(1 - 2\mu)$

Volumetric strain $e_v = 2e_c + e_l = (pd/(4tE))(5 - 4\mu)$; change in volume $\delta V = e_v V$

Wire-wound cylinders: initial wire tension puts the cylinder in compression, increasing its pressure capacity

- Since $\sigma_c = 2\sigma_l$, longitudinal (seam) joints must be stronger; thickness is designed from hoop stress. Hoop stress is the larger, so a pipe bursting under pressure splits lengthwise.
- Ratio of hoop to longitudinal strain ($\mu = 0.3$): $(2 - \mu)/(1 - 2\mu) = 1.7/0.4 = 4.25$.

★ **PYQ-TREND:** $\sigma_c = pd/2t$ and $\sigma_l = pd/4t$ (very frequent); hoop stress twice longitudinal; volumetric strain $(pd/4tE)(5 - 4\mu)$; thin cylinder bursts along longitudinal seam; joint efficiencies.

✖ **COMMON MISTAKE:** Hoop stress uses the longitudinal joint efficiency and longitudinal stress uses the circumferential joint efficiency – students often swap them.

10.2 Thin Spheres

Stress (all directions in the wall) $\sigma = pd/(4t)$

Strain $e = (pd/(4tE))(1 - \mu)$; volumetric strain $e_v = 3e = (3pd/(4tE))(1 - \mu)$

- For the same diameter, pressure and allowable stress, a sphere needs half the thickness of a cylinder – spheres are used for large gas holders and LPG storage (Horton spheres).

★ **PYQ-TREND:** Thin sphere stress pd/4t (same as longitudinal stress of cylinder); volumetric strain $3pd(1 - \mu)/4tE$; sphere needs half the thickness.

10.3 Thick Cylinders (Lamé's Theory)

- When t > d/20, hoop stress varies across the thickness and radial stress is significant.

Lamé's equations: $\sigma_r = a/r^2 - b$ (compressive radial stress, taken positive); $\sigma_c = a/r^2 + b$ (hoop tensile)

Internal pressure p only (radii r_1 inner, r_2 outer): $\sigma_{c,max}$ (at inner surface) $= p(r_2^2 + r_1^2)/(r_2^2 - r_1^2)$

Hoop stress at outer surface = $2p\ r_1^2/(r_2^2 - r_1^2)$; $\sigma_r = p$ at inner surface, 0 at outer surface

Maximum shear stress (inner surface) = $p\ r_2^2/(r_2^2 - r_1^2)$

- Hoop stress is maximum at the inner surface and decreases towards the outside (hyperbolic distribution); increasing thickness gives diminishing returns – the maximum hoop stress cannot fall below p no matter how thick the wall.
- Compound cylinders: a jacket shrunk onto an inner tube creates initial compressive hoop stress in the inner tube, making the stress distribution more uniform under working pressure (gun barrels, high-pressure vessels). Autofrettage achieves a similar effect by over-pressurising to cause partial yielding.
- Thick spheres: $\sigma_r = 2a/r^3 - b$ form (Lamé for spheres).

★ **PYQ-TREND:** Lamé's theory for thick cylinders; maximum hoop stress at inner surface (very frequent); compound cylinders by shrink fit; autofrettage; thick cylinder criterion t > d/20.

✎ **DESCRIPTIVE Q:** Derive the expressions for hoop and longitudinal stresses in a thin cylinder.

Consider a thin cylinder of internal diameter d, length L and wall thickness t subjected to internal pressure p. Hoop stress: imagine the cylinder cut along a diametral plane. The bursting force acting on one half is p multiplied by the projected area, p × d × L. This is resisted by the hoop stress σ_c acting on the two cut walls of area 2tL. Equilibrium gives $\sigma_c \times 2tL = p\ d\ L$, so $\sigma_c = pd/(2t)$. Longitudinal stress: imagine the cylinder cut by a plane perpendicular to its axis. The force tending to separate the two parts is $p \times \pi d^2/4$, resisted by the longitudinal stress σ_l on the wall area πdt (approximately). Hence $\sigma_l \times \pi dt = p\ \pi d^2/4$, giving $\sigma_l = pd/(4t)$. Thus the hoop stress is twice the longitudinal stress, so a thin cylinder tends to fail along a longitudinal line, and the thickness is determined by hoop stress: $t = pd/(2\sigma_{allow}\ \eta_l)$. Radial stress, varying from p at the inner surface to zero at the outer surface, is small compared with these and is neglected.

CHAPTER 11

Riveted, Welded and Bolted Joints (Strength Basics)

11.1 Riveted Joints

- Rivet: a short cylindrical bar with a head, inserted through holes and the second head formed by hammering (hot riveting for steel structures, cold for small rivets). Permanent joint; used in boilers (older), bridges, aircraft (aluminium rivets), ship hulls.

Type	Description
Lap joint	Plates overlap; single or double riveted; eccentric load path
Butt joint	Plates butt end to end with one (single strap) or two cover plates (double strap); rivets in double shear with two straps
Chain riveting	Rivets of adjacent rows directly opposite
Zig-zag riveting	Rivets staggered
Diamond riveting	Number of rivets increases from outer row inward – used in structural butt joints for high efficiency

- Terminology: pitch p (distance between centres of adjacent rivets in a row), back pitch (between rows), margin m (centre of rivet to plate edge, usually 1.5d), diagonal pitch. Unwin's formula for rivet diameter: $d = 6\sqrt{t}$ (mm).

Tearing strength of plate (per pitch) $P_t = (p - d)\, t\, \sigma_t$

Shearing strength of rivets $P_s = n\, (\pi/4)\, d^2\, \tau$ (single shear); $= n \times 2 \times (\pi/4) d^2\, \tau$ (double shear; IS/boiler practice may use 1.75 or 1.875 instead of 2)

Crushing (bearing) strength $P_c = n\, d\, t\, \sigma_c$

Efficiency η = least of (P_t, P_s, P_c)/strength of solid plate $(p\, t\, \sigma_t)$

Efficiency of joint (tearing criterion) $= (p - d)/p$

- Failure modes: tearing of plate between rivets, tearing at the margin, shearing of rivets, crushing of rivets or plate. Rivets are designed so that shearing and crushing strengths roughly equal tearing strength.
- Caulking and fullering make riveted joints in boilers leak-proof (fullering uses a tool as wide as the plate edge – better).

★ **PYQ-TREND:** Joint efficiency = least strength/solid plate strength (very frequent); Unwin's formula $d = 6\sqrt{t}$; double cover butt joint – rivets in double shear; caulking and fullering; diamond riveting most efficient for structures; margin 1.5d.

11.2 Welded Joints

Weld	Strength formula
Butt weld (tension)	P = σ_t × throat (= plate thickness t) × length L

Weld	Strength formula
Fillet weld	Throat thickness = leg size s × cos45° = 0.707s; strength P = 0.707 s L τ
Double transverse fillet	P = 1.414 s L σ_t
Parallel (longitudinal) fillet	P = 1.414 s L τ (two welds)
Circular fillet weld around a shaft under torsion	τ = 2T/(π t d²) (t = throat)

```
Throat of fillet weld t = 0.707 s; weld strength = τ × 0.707 s × L
```

- Parallel fillet welds are loaded mainly in shear; transverse fillets are stronger by about 30% but are usually designed on the same shear basis.
- Welded joints compared with riveted: lighter (no cover plates/heads), leak-proof, faster, can join complex shapes, nearly 100% efficiency possible; but residual stresses, distortion, heat-affected zone, difficult inspection, brittle failures under fatigue if poorly made.

★ **PYQ-TREND:** Throat thickness 0.707 × leg (very frequent); fillet weld strength 0.707 s L τ; welded vs riveted joints; butt weld strength uses plate thickness.

11.3 Bolted Joints, Keys and Couplings (Quick Facts)

- Bolt designated by nominal (major) diameter, e.g. M12 × 1.75 (pitch). Tensile stress area based on mean of pitch and minor diameters. Initial tightening load (empirical) P_i = 2840d N (d in mm) for fluid-tight joints.

```
Bolt in tension: P = (π/4) d_c² σ_t (d_c = core/minor diameter); bolt in shear: P = (π/4) d² τ
Eccentrically loaded bolted bracket: direct shear P/n + torsional shear P e r_i/Σr²
```

- Keys transmit torque between shaft and hub: sunk keys (rectangular – width d/4, square), saddle keys (friction only, light duty), feather keys (axial movement), Woodruff key (semicircular, self-aligning, automotive), splines (many teeth, large torque). Kennedy key (two square keys).

```
Rectangular sunk key: width w = d/4, thickness = 2w/3; shear: T = w l τ d/2; crushing: T = (t/2) l σ_c d/2
Key equally strong in shear and crushing (square key): σ_c = 2τ
```

- Couplings: rigid (sleeve/muff, split-muff, flange) for aligned shafts; flexible (bushed-pin, Oldham's for parallel offset shafts, universal/Hooke's joint for intersecting shafts) to allow misalignment.
- Cotter joint (tapered cotter, taper ≈ 1 in 30) and knuckle joint (pin; tie rods of roof trusses, valve rods) connect rods under axial load.

★ **PYQ-TREND:** Woodruff key is semicircular and self-aligning; saddle key works by friction; Oldham coupling for parallel shafts with lateral offset; Hooke's joint for intersecting shafts; key width d/4; cotter taper 1 in 30; knuckle joint uses a pin.

✎ **DESCRIPTIVE Q:** Explain the different modes of failure of a riveted joint and define its efficiency.

A riveted joint can fail in several ways. (1) Tearing of the plate between rivet holes along a row: resisted area per pitch is (p − d)t, so strength P_t = (p − d)tσ_t. (2) Tearing of the plate at the edge (margin) in front of the rivet, prevented by keeping the margin at least 1.5 times the rivet diameter. (3) Shearing of the rivets: in a lap joint or single-cover butt joint each rivet is in single

shear, $P_s = n(\pi/4)d^2\tau$; in a double-cover butt joint each rivet is in double shear, approximately doubling the strength. (4) Crushing of the rivet or the plate around the hole (bearing failure), with resisting area $d \times t$ per rivet, $P_c = n\,d\,t\,\sigma_c$. The efficiency of the joint is the ratio of the least of these strengths to the strength of the unpunched (solid) plate over the same pitch, $p\,t\,\sigma_t$. Typical efficiencies are 50–60% for single-riveted lap joints and 75–90% for double-riveted double-strap butt joints. A good design makes the tearing, shearing and crushing strengths nearly equal.

CHAPTER 11B

Special Topics – Rotating Rings and Discs, Dams, Curved Bars and Contact Stresses

11B.1 Rotating Rings and Discs

- A thin ring (e.g. flywheel rim, pulley rim) rotating about its axis develops hoop tension due to centrifugal force.

Thin rotating ring: hoop stress $\sigma = \rho v^2 = \rho \omega^2 r^2$ (independent of cross-sectional area)

Maximum peripheral speed for allowable stress: $v_{max} = \sqrt{(\sigma/\rho)}$ (steel σ = 100 MPa → $v \approx$ 113 m/s)

Solid disc of uniform thickness: $\sigma_r = \sigma_\theta = (3 + \mu)\rho\omega^2 R^2/8$ at the centre (maximum)

Disc with central hole (inner radius small): $\sigma_{\theta,max}$ at hole $\approx (3 + \mu)\rho\omega^2 R^2/4$ — twice the solid-disc value

Rotating long cylinder (μ effects similar) and disc of uniform strength: thickness $t = t_0 e^{(-\rho\omega^2 r^2/(2\sigma))}$

- Because a small central hole doubles the maximum stress, turbine discs are made solid or with thick hubs, and discs of uniform strength are used for high-speed turbines (profile thinning outward).

★ **PYQ-TREND:** Hoop stress in rotating ring ρv^2 (frequent); limiting rim speed of flywheels; small hole doubles the stress in a rotating disc; disc of uniform strength.

11B.2 Dams and Retaining Walls (Masonry Structures)

- A masonry dam is checked for no tension at the base: the resultant of self-weight and water thrust must cut the base within the middle third. Other stability checks: against overturning (factor of safety = restoring moment/overturning moment, usually ≥ 1.5), sliding ($\mu\Sigma V \geq \Sigma H$), and crushing (maximum compressive stress ≤ allowable).

Water pressure on vertical face of height h: $P = \rho g h^2/2$ per unit length, acting at h/3 above the base

Base stresses (width b, total vertical load W, eccentricity e): $\sigma_{max,min} = (W/b)(1 \pm 6e/b)$

Earth pressure (Rankine, active, dry cohesionless soil): $p = k_a w h$ with $k_a = (1 - \sin\varphi)/(1 + \sin\varphi)$; total $P = k_a w h^2/2$ at h/3

- Chimneys: wind pressure P = k p A (k = 0.6–0.7 for circular sections, 1 for square) causes bending at the base combined with self-weight: $\sigma = W/A \pm M/Z$.

★ **PYQ-TREND:** Water thrust $\rho g h^2/2$ acting at h/3 (very frequent); no tension in dam base requires e ≤ b/6; Rankine active earth pressure coefficient; chimney wind load coefficient for circular sections.

11B.3 Curved Bars, Springs and Thin Rings

- Curved beam (Winkler–Bach): stress distribution hyperbolic; neutral axis shifted towards the centre of curvature by e = R − R_n; maximum stress at the inner fibre. For a rectangular section R_n = h/ln(R_o/R_i); for a circular section R_n = (√R_o + √R_i)²/4.
- Crane hooks are designed with trapezoidal or T sections, thicker on the inner side, to balance inner tensile and outer compressive stresses.
- Chain links and rings: a closed circular ring under diametral load W has maximum BM 0.182WR at the load points (by strain energy/Castigliano) – used to design chain links.

```
Closed ring under diametral pull W: M at load = 0.182WR (= WR(1/2 − 1/π)); M at 90° = −0.137WR... (sign opposite)
Curved bar with small curvature (R/h > 10): use straight-beam formula with little error
```

★ **PYQ-TREND:** Maximum stress in crane hook at inner fibre; neutral axis shifts towards centre of curvature; Winkler–Bach theory; curved-beam correction negligible for large R/h.

11B.4 Contact (Hertz) Stresses and Stress Concentration

- When two curved elastic bodies (ball on race, gear teeth, cam and follower, wheel on rail) are pressed together, contact occurs over a small area with very high compressive stress (Hertzian). Maximum shear stress occurs slightly below the surface (≈ 0.47–0.78 of contact half-width), which is why pitting (surface fatigue) starts as subsurface cracks.

```
Two spheres: contact radius a ∝ (F R)^(1/3); maximum pressure p_max = 1.5F/(πa²)
Two cylinders (line contact): half-width b = √(2F/(πL) × [(1 − μ₁²)/E₁ + (1 − μ₂²)/E₂]/(1/d₁ + 1/d₂)); p_max = 2F/(πbL)
```

- Stress concentration: abrupt changes of section (holes, notches, keyways, fillets, threads) raise local stress. Theoretical factor K_t (from elasticity/photoelasticity) and fatigue notch factor K_f = 1 + q(K_t − 1), where q is notch sensitivity (0 to 1; ductile materials under static load are notch-insensitive, brittle materials and fatigue loading are sensitive). Remedies: generous fillets, relief grooves, gradual transitions, removing material near notches, smooth finish, shot peening.

```
Fatigue stress concentration K_f = 1 + q(K_t − 1)
```

★ **PYQ-TREND:** Hertz contact stress in bearings and gears; subsurface maximum shear causes pitting; K_f = 1 + q(K_t − 1); notch sensitivity; reducing stress concentration with fillets and relief grooves; photoelasticity to find K_t.

CHAPTER 12

Engineering Materials and Classification of Steels

12.1 Classification of Materials and Crystal Structure

Class	Examples	Typical features
Ferrous metals	Steels, cast irons, wrought iron	Iron-based; strong, cheap, magnetic (mostly)
Non-ferrous metals	Al, Cu, Zn, Pb, Sn, Ni, Ti, Mg and alloys	Light or corrosion resistant, good conductors
Ceramics	Alumina, SiC, glass, cement	Hard, brittle, refractory, insulating
Polymers	Thermoplastics, thermosets, elastomers	Light, corrosion resistant, low strength and temperature
Composites	GFRP, CFRP, concrete, cermets, wood	Tailored properties

Crystal structure	Atoms per unit cell	Coordination number	Packing factor	Examples
Simple cubic	1	6	0.52	Polonium
BCC	2	8	0.68	α-iron (ferrite), Cr, Mo, W, V, Na
FCC	4	12	0.74	γ-iron (austenite), Al, Cu, Ni, Ag, Au, Pb
HCP	6	12	0.74	Zn, Mg, Ti (α), Cd, Co

- FCC metals are the most ductile (many slip systems, close-packed planes); HCP metals have limited ductility at room temperature; BCC metals show a ductile–brittle transition.
- Allotropy (polymorphism): iron – α (BCC, up to 912 °C), γ (FCC, 912–1394 °C), δ (BCC, 1394–1539 °C). Iron is magnetic below the Curie temperature 768 °C (β-iron is the old name for non-magnetic BCC between 768 and 912 °C).
- Defects: point (vacancies, interstitials, substitutional), line (dislocations – edge and screw; plastic deformation occurs by dislocation motion), surface (grain boundaries, twin boundaries), volume (voids, inclusions). Grain refinement increases strength and toughness (Hall–Petch: $\sigma_y = \sigma_0 + k\, d^{(-1/2)}$).

★ **PYQ-TREND:** APF of BCC 0.68 and FCC/HCP 0.74; coordination numbers 8 and 12 (very frequent); allotropic forms of iron and temperatures 912 °C and 1394 °C; Curie temperature 768 °C; plastic deformation by dislocation movement; fine grains increase strength (Hall–Petch).

12.2 Classification of Steels (Paper-I Topic)

Basis	Classes
Carbon content	Low-carbon (mild) steel < 0.3% C; medium-carbon 0.3–0.6% C; high-carbon 0.6–1.5% C (tool steels ≈ 0.8–1.5%)
Alloy content	Plain carbon steel; low-alloy (< 5% total alloying); high-alloy (> 5–10%, e.g. stainless, HSS)
Use	Structural, tool and die steels, spring steels, bearing steels, stainless, heat-resisting, free-cutting
Deoxidation	Killed (fully deoxidised by Si/Al – no gas, used for forging, sound), semi-killed, rimmed (not deoxidised – good surface, for sheets), capped
Microstructure	Hypo-eutectoid (< 0.8% C: ferrite + pearlite), eutectoid (0.8% C: pearlite), hyper-eutectoid (> 0.8% C: pearlite + cementite)

Plain carbon steel	% C	Uses
Dead mild steel	0.05–0.15	Wires, sheets, rivets, nails, chains, tubes
Mild steel	0.15–0.3	Structural sections, bolts, shafts, boiler plates, machine parts
Medium carbon steel	0.3–0.6	Axles, crankshafts, connecting rods, rails, gears, forgings
High carbon steel	0.6–0.8	Springs, hammers, wire ropes, die blocks
Tool steel	0.8–1.5	Chisels, drills, files, razors, taps, dies, cutting tools

- Effect of carbon: as C increases, hardness and tensile strength increase (UTS maximum near 0.8–1.0% C), while ductility, toughness, weldability and machinability decrease. Steels with more than about 0.3% C need preheating for welding.

Designations

System	Example	Meaning
IS (old)	C40 / 40C8	Plain carbon steel with 0.40% C (40C8: 0.8% Mn)
IS alloy	40Cr1, 35Mn6Mo3	0.40% C, 1% Cr; numbers after alloy symbols are multiplied factors
IS cast iron	FG 200, SG 500/7	Grey CI with UTS 200 MPa; SG iron 500 MPa, 7% elongation
AISI–SAE	1040	10 = plain carbon, 40 = 0.40% C
AISI–SAE	4140	41 = Cr–Mo steel, 0.40% C
AISI	304, 316	Austenitic stainless steels (18-8 type)
SAE prefix	2xxx Ni, 3xxx Ni–Cr, 4xxx Mo, 5xxx Cr, 6xxx Cr–V, 9xxx Si–Mn	Main alloying element

★ **PYQ-TREND:** Carbon ranges of low/medium/high carbon steel (very frequent); uses of each grade (rails, springs, tools); 40C8 designation; AISI 1040; killed vs rimmed steel; effect of carbon on properties; eutectoid 0.8% C.

12.3 Alloying Elements

Element	Main effects
Nickel	Increases toughness, strength, corrosion resistance; austenite stabiliser; low-temperature toughness
Chromium	Hardenability, wear and corrosion resistance (> 12% stainless); carbide former
Molybdenum	Hardenability, high-temperature strength, reduces temper brittleness, creep resistance
Tungsten	Hot hardness (red hardness) – high-speed steels; carbide former
Vanadium	Grain refinement, hardness, wear resistance, fatigue strength; strong carbide former
Manganese	Deoxidiser, combines with S (MnS) to prevent hot shortness, hardenability; Hadfield steel (12–14% Mn) – work-hardening, wear-resistant
Silicon	Deoxidiser; electrical (transformer) steels – reduces hysteresis loss; spring steels (Si–Mn)
Cobalt	Increases red hardness of HSS; only element that reduces hardenability (increases critical cooling rate)
Titanium, niobium	Stabilise stainless steels against intergranular corrosion (carbide formers)
Sulphur	Free-cutting (machinability) as MnS; otherwise causes hot shortness (FeS)
Phosphorus	Increases strength but causes cold shortness (brittleness at low temperature)
Lead	Free-machining steels and brasses
Boron	Very small amounts greatly increase hardenability
Aluminium	Deoxidiser, grain refiner; nitriding steels (Nitralloy)

- Special steels: stainless – ferritic (Cr 12–27%, magnetic), martensitic (Cr 12–18% + higher C, hardenable – cutlery, surgical tools), austenitic (18% Cr–8% Ni, non-magnetic, best corrosion resistance – 304); duplex; precipitation hardening. Weld decay (sensitisation): chromium carbide precipitation at grain boundaries at 450–850 °C – prevented by low carbon (304L) or stabilising with Ti/Nb.
- High-speed steel (HSS): 18-4-1 (18% W, 4% Cr, 1% V, ≈ 0.7% C); molybdenum HSS (M2: 6% W, 5% Mo, 4% Cr, 2% V); cobalt HSS (super HSS). Retains hardness up to about 600 °C.
- Hadfield manganese steel: 12–14% Mn, 1–1.4% C; austenitic, work-hardens under impact – rock crushers, railway crossings, excavator teeth.
- Maraging steel (18% Ni, Co, Mo, very low C) – ultra-high strength by age hardening. Invar (36% Ni) – near-zero thermal expansion (measuring tapes, clock pendulums).

★ **PYQ-TREND:** Element effects – Ni toughness, Cr corrosion, W red hardness, Mn prevents hot shortness, S free cutting, P cold shortness, Co reduces hardenability (very frequent); 18-4-1 HSS composition; 18-8 stainless; Hadfield steel; invar; weld decay in stainless steel.

✔ **TOPPER TRICK:** 'Sulphur – hot short; Phosphorus – cold short' (S for 'scorching', P for 'polar').

12.4 Cast Irons

Type	Carbon form	Properties / uses
Grey CI	Graphite flakes	Fracture grey; good castability, machinability, damping, compressive strength; brittle in tension – machine beds, engine blocks, pipes
White CI	Cementite (Fe_3C), no free graphite	Very hard, brittle, white fracture; wear parts, raw material for malleable iron
Malleable CI	Temper carbon (rosettes)	Made by annealing white CI (blackheart/whiteheart); ductile – pipe fittings, brackets, small parts
Nodular (SG, ductile) CI	Spheroidal graphite nodules (Mg or Ce added)	High strength and ductility – crankshafts, gears, pipes
Chilled CI	White surface, grey core (rapid surface cooling)	Rolls, wheels of railway wagons
Mottled CI	Mixed grey and white	–
Alloy CI (Ni-hard, Ni-resist)	–	Wear/corrosion/heat resistance
Compacted graphite iron	Vermicular graphite	Engine blocks (between grey and SG)

- Cast irons contain 2–4 (up to 6.67) % C. Silicon promotes graphite formation (grey); manganese and rapid cooling promote cementite (white).
- Wrought iron: almost pure iron (< 0.1% C) with slag fibres; tough, ductile, corrosion resistant; chains, hooks.

★ **PYQ-TREND:** Grey CI flakes and good damping/machine beds; SG iron by Mg/Ce; malleable iron from white CI by annealing (very frequent); white CI hardest; chilled CI for rolls; carbon range 2–4%.

12.5 Non-Ferrous Metals and Alloys

Alloy	Composition	Uses
Brass	Cu + Zn (cartridge 70/30, Muntz 60/40, admiralty brass with Sn)	Cartridges, radiators, condenser tubes, fittings
Bronze	Cu + Sn (phosphor bronze adds P; aluminium bronze; gun metal Cu–Sn–Zn 88-10-2)	Bearings, gears, springs, valves, statues
Monel	Ni ≈ 67% + Cu ≈ 30%	Corrosion-resistant – marine, chemical, pumps
Inconel	Ni–Cr–Fe	High-temperature parts – jet engines, furnaces
Nichrome	Ni 80% + Cr 20%	Heating elements
Duralumin	Al + 4% Cu + Mg + Mn	Aircraft structures (age-hardenable)
Y-alloy	Al + Cu + Ni + Mg	Pistons, cylinder heads (high-temperature strength)
Babbitt (white metal)	Sn + Sb + Cu (or Pb base)	Bearing linings

Alloy	Composition	Uses
German silver	Cu + Zn + Ni (no silver)	Utensils, decorative items
Solder (soft)	Pb + Sn (now lead-free Sn–Ag–Cu)	Electrical joints
Constantan	Cu 55% + Ni 45%	Thermocouples, resistors
Magnalium	Al + Mg	Light instruments
Stellite	Co + Cr + W + C	Cutting tools, valve seats, hardfacing

- Aluminium: low density (2.7), high conductivity, corrosion resistance by oxide film; precipitation (age) hardening of Al–Cu alloys: solution treatment, quenching, ageing (natural or artificial) forming GP zones.
- Copper: highest conductivity after silver; used for electrical wires; hardened only by cold working (no heat treatment hardening except alloys like beryllium copper).
- Titanium: high strength-to-weight ratio, corrosion resistance; aerospace, implants. Magnesium: lightest structural metal (1.74).
- Polymers: thermoplastics (soften on heating, recyclable – PE, PP, PVC, PS, nylon, PTFE, acrylic) vs thermosets (cross-linked, set permanently – bakelite, epoxy, urea- and melamine-formaldehyde, polyester); elastomers (rubber – vulcanised with sulphur).

★ **PYQ-TREND:** Brass = Cu + Zn, bronze = Cu + Sn (very frequent); gun metal composition; duralumin for aircraft; Monel Ni–Cu; babbitt for bearings; German silver has no silver; age hardening of duralumin; thermoplastic vs thermoset; bakelite thermoset; vulcanisation with sulphur.

CHAPTER 12B

Material Selection, Corrosion and Strengthening Mechanisms

12B.1 Strengthening Mechanisms

Mechanism	How it works	Example
Strain (work) hardening	Cold working multiplies dislocations that hinder each other	Cold-drawn wire, cold-rolled sheet
Grain refinement	Grain boundaries block dislocation motion (Hall–Petch)	Normalising, controlled rolling, microalloying with Nb/V/Ti
Solid-solution strengthening	Solute atoms distort the lattice	Brass (Zn in Cu), Mn and Si in ferrite
Precipitation (age) hardening	Fine coherent precipitates from supersaturated solution	Duralumin, maraging steel, 17-4 PH stainless
Dispersion strengthening	Stable insoluble particles	Oxide-dispersion strengthened alloys, SAP (Al–Al_2O_3)
Martensitic transformation	Supersaturated BCT structure	Hardened steels
Second-phase strengthening	Hard phase in soft matrix	Pearlite in ferrite

- Recovery, recrystallisation and grain growth occur on annealing cold-worked metal: recovery relieves internal stress (properties little changed), recrystallisation forms new strain-free grains (strength falls, ductility restored), and grain growth coarsens grains if heating continues.
- Precipitation hardening steps: solution treatment (single phase) → quenching (supersaturated solid solution) → ageing (natural at room temperature or artificial at elevated temperature); over-ageing coarsens precipitates and reduces strength.

★ **PYQ-TREND:** Age hardening steps for duralumin (very frequent); Hall–Petch grain-size strengthening; recrystallisation restores ductility; strain hardening by cold work; over-ageing reduces strength.

12B.2 Corrosion

- Corrosion: electrochemical deterioration of metals with their environment. Anode corrodes (oxidation); cathode is protected. Rusting of iron requires both oxygen and moisture (Fe → Fe^{2+}; ultimately $Fe_2O_3 \cdot xH_2O$).
- Galvanic series in sea water (more noble/cathodic to more active/anodic): platinum, gold, graphite, titanium, silver, stainless steel (passive), nickel, copper, brass, tin, lead, stainless steel (active), cast iron, mild steel, aluminium, cadmium, zinc, magnesium. When two metals are coupled, the more active one corrodes – e.g. zinc protects steel, steel bolts in copper plates corrode.

Type	Description / example
Uniform (general)	Even attack over surface – rusting of unprotected steel
Galvanic	Dissimilar metals in electrolyte – steel fasteners on copper; worse with small anode/large cathode
Pitting	Localised deep pits – stainless steel in chlorides
Crevice	In gaps where stagnant solution forms – under gaskets, washers
Intergranular	Along grain boundaries – sensitised stainless steel (weld decay)
Stress-corrosion cracking	Tensile stress + specific environment – brass in ammonia ('season cracking'), austenitic stainless in chlorides, caustic embrittlement of boiler steel
Erosion–corrosion	Fluid velocity removes protective film – pump impellers, pipe bends
Fretting	Small oscillatory slip between contacting surfaces
Dezincification (selective leaching)	Zinc removed from brass leaving porous copper – prevented by adding As/Sb/Sn (admiralty brass)
Hydrogen embrittlement	Atomic hydrogen diffuses into high-strength steel – from pickling, plating, welding

- Prevention: proper material selection (stainless, Cu–Ni, titanium), design (avoid crevices, dissimilar contact, ensure drainage; large anode area), protective coatings (paint, galvanising, plating, anodising), cathodic protection (sacrificial Zn/Mg/Al anodes or impressed current), inhibitors (chromates, phosphates, amines in boiler water), environment control (deaeration, pH control, dehumidification).

★ **PYQ-TREND:** Galvanic series – zinc/magnesium most active, used as sacrificial anodes (very frequent); rusting needs O_2 and water; season cracking of brass in ammonia; dezincification of brass; caustic embrittlement in boilers; pitting of stainless steel by chlorides; small anode–large cathode is worst.

✔ **TOPPER TRICK:** In a galvanic couple the metal lower in the series (more active) is sacrificed – 'galvanised' iron stays protected even when the zinc coating is scratched.

12B.3 Material Selection for Typical Components

Component	Typical material and reason
Machine tool beds, engine blocks	Grey cast iron – damping, castability, compressive strength, machinability
Crankshafts	Forged medium-carbon/alloy steel (40C8, 40Cr1) or SG iron – fatigue strength
Connecting rods	Forged medium-carbon or alloy steel; aluminium in small engines
Pistons	Aluminium alloy (Y-alloy, Lo-Ex Al–Si) – light, good conductivity
Cylinder liners	Alloy cast iron – wear resistance
Gears	Case-hardened alloy steel (carburised), cast iron for low duty, nylon/phenolic for quiet light duty

Component	Typical material and reason
Springs	High-carbon or Si–Mn/Cr–V spring steel, phosphor bronze for corrosion
Bearings (plain)	White metal (babbitt), bronze, PTFE, porous sintered bronze
Ball/roller bearings	High-carbon chromium steel (52100 / 103Cr1)
Cutting tools	HSS, carbides, ceramics, CBN, diamond
Dies (cold work)	High-carbon high-chromium steel (D2)
Turbine blades	Nickel superalloys (Inconel, Nimonic), single-crystal castings
Boiler tubes	Low-carbon/Cr–Mo steels (creep resistance)
Aircraft structures	Duralumin, Al–Li alloys, titanium, CFRP
Electrical conductors	Copper, aluminium
Heating elements	Nichrome
Surgical implants	Titanium alloys, 316L stainless steel, Co–Cr alloys
Chemical plant / marine	Stainless steel, Monel, Cu–Ni, titanium
Rails	Medium/high-carbon (≈ 0.6–0.8% C) pearlitic steel; Hadfield Mn steel at crossings
Files, razors, drills (hand)	High-carbon tool steel (≈ 1–1.3% C)

- Selection criteria: required mechanical properties (strength, stiffness, toughness, fatigue, creep), physical properties (density, conductivity, expansion), environmental resistance, manufacturability (castability, formability, weldability, machinability), availability and cost; material indices such as specific strength (σ/ρ) and specific stiffness (E/ρ) guide lightweight design (Ashby charts).

★ **PYQ-TREND:** Machine bed of grey CI because of damping (very frequent); pistons of aluminium alloy; bearings of babbitt/bronze; ball bearings of high-carbon chromium steel; turbine blades of nickel superalloys; rails high-carbon steel; heating elements nichrome; specific strength for aerospace.

CHAPTER 13

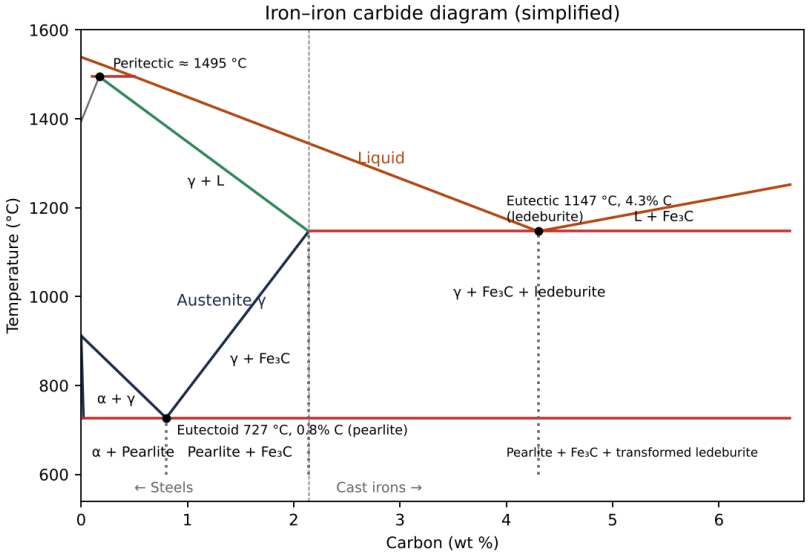
Iron–Carbon Diagram and Heat Treatment

13.1 Iron–Iron Carbide Diagram

Phase / constituent	Description
Ferrite (α)	Interstitial solid solution of C in BCC iron; max 0.022% C at 727 °C (≈ 0.008% at room temperature); soft, ductile, magnetic
Austenite (γ)	Solution of C in FCC iron; max 2.14% C at 1147 °C; non-magnetic, ductile; not stable below 727 °C in plain carbon steel
Cementite (Fe_3C)	Iron carbide, 6.67% C; very hard and brittle
Pearlite	Eutectoid mixture (lamellae) of ferrite and cementite, 0.8% C (0.76–0.8%); forms at 727 °C
Ledeburite	Eutectic mixture of austenite and cementite, 4.3% C, forms at 1147 °C
δ-ferrite	BCC at high temperature (1394–1539 °C)
Martensite	Supersaturated BCT solution of C in iron from rapid quenching; very hard, brittle (not on equilibrium diagram)
Bainite	Fine ferrite + carbide from isothermal transformation between pearlite and martensite ranges

Invariant reaction	Temperature	Composition	Reaction
Peritectic	1493–1495 °C	0.16–0.18% C	δ + liquid → γ
Eutectic	1147 °C	4.3% C	Liquid → γ + Fe_3C (ledeburite)
Eutectoid	727 °C	0.8% C	γ → α + Fe_3C (pearlite)

- Critical lines: A_1 (727 °C, eutectoid line, lower critical temperature), A_3 (upper critical line for hypo-eutectoid steels – γ/γ + α boundary), A_cm (γ/γ + Fe_3C boundary for hyper-eutectoid steels). Steels: up to 2.14% C; cast irons: 2.14–6.67% C.
- Lever rule gives phase fractions: e.g. for a 0.4% C steel just below 727 °C, pearlite ≈ (0.4 − 0.02)/(0.8 − 0.02) ≈ 49%, rest pro-eutectoid ferrite.



★ **PYQ-TREND:** Eutectoid 727 °C, 0.8% C – pearlite; eutectic 1147 °C, 4.3% C – ledeburite; peritectic 1495 °C (very frequent); cementite 6.67% C; max solubility of C in austenite 2.14% and in ferrite 0.022%; martensite BCT not on diagram; lever rule.

✔ **TOPPER TRICK:** Remember the three invariant reactions by temperature: 'PEE' – Peritectic 1495, Eutectic 1147, Eutectoid 727 (high to low).

13.2 TTT Diagram and Hardenability

- Time–temperature–transformation (TTT, isothermal transformation) diagram: C-shaped curves showing start and finish of transformation of austenite at constant temperature. The 'nose' (≈ 550 °C for eutectoid steel) is where transformation is fastest. Above the nose: coarse/fine pearlite; below the nose: upper and lower bainite; below M_s (≈ 220 °C for eutectoid steel): martensite (athermal, diffusionless, depends only on temperature, ends at M_f).
- Critical cooling rate: minimum rate that avoids the nose and gives fully martensitic structure. Alloying elements (except Co) shift the C-curves to the right, lowering the critical cooling rate and increasing hardenability.
- Continuous cooling transformation (CCT) diagram is used for practical continuous cooling.
- Hardenability: depth to which a steel can be hardened (not the maximum hardness, which depends mainly on carbon). Measured by the Jominy end-quench test (water jet on one end of a 25 mm × 100 mm bar; hardness measured along the length). Grossmann's ideal critical diameter.
- Retained austenite: occurs with high carbon/alloy when M_f is below room temperature; removed by sub-zero (cryogenic) treatment.

★ **PYQ-TREND:** TTT diagram nose and critical cooling rate; martensite diffusionless BCT; Jominy end-quench test for hardenability (very frequent); hardness depends on carbon, hardenability on alloying; sub-zero treatment for retained austenite.

13.3 Heat Treatment Processes

Process	Procedure	Purpose / result
Full annealing	Heat 30–50 °C above A_3 (hypo) or A_1 (hyper), hold, furnace cool	Softens, coarse pearlite, relieves stresses, improves machinability and ductility
Process (sub-critical) annealing	Heat below A_1 (550–650 °C), cool	Restores ductility of cold-worked low-carbon steel between cold-working stages (recrystallisation)
Spheroidising	Prolonged heating just below A_1 (or cycling)	Spheroidal cementite – maximum softness and machinability of high-carbon steels
Stress relieving	Heat to ≈ 500–650 °C, slow cool	Removes residual stresses (after welding, casting, machining) without changing structure
Normalising	Heat 30–50 °C above A_3/A_cm, cool in still air	Fine pearlite, refined grains; stronger and harder than annealed; uniform structure in forgings and castings
Hardening	Heat above A_3 (hypo) or A_1 (hyper), quench in water/brine/oil	Martensite – maximum hardness; brittle, internal stresses
Tempering	Reheat hardened steel below A_1 and cool	Reduces brittleness, relieves stresses, increases toughness; low (150–250 °C, tools), medium (350–450 °C, springs), high (500–650 °C, shafts – 'sorbite')
Austempering	Quench into bath above M_s, hold until bainite forms	Bainite – toughness with less distortion and cracking
Martempering (marquenching)	Quench to just above M_s, equalise, then air cool through martensite range	Martensite with minimal distortion and cracking; then temper
Patenting	Austenitise and cool in lead bath (≈ 500 °C)	Fine pearlite for cold drawing of high-carbon wire

- Quenching media in decreasing severity: brine > water > oil > molten salt > air. Severe quenching gives greater hardness but more distortion and quench cracks.
- Hardening + high-temperature tempering = 'quenching and tempering' (Q&T) – best combination of strength and toughness for shafts, gears and bolts.
- Temper brittleness: loss of toughness on tempering in 250–400 °C or slow cooling through 450–600 °C (Ni–Cr steels) – reduced by Mo.

★ **PYQ-TREND:** Annealing (furnace cool) vs normalising (air cool) (very frequent); hardening by quenching gives martensite; tempering reduces brittleness; spheroidising for machinability of high-carbon steel; austempering gives bainite; martempering reduces distortion; quenching media severity order.

✖ **COMMON MISTAKE:** Normalising does not make steel softer than annealing – normalised steel is slightly harder and stronger because air cooling is faster and gives finer pearlite.

13.4 Surface Hardening (Case Hardening)

- Purpose: hard, wear-resistant surface with a tough, ductile core – gears, cams, camshafts, crankshaft journals, piston pins.

Process	Medium / temperature	Case depth / remarks
Pack carburising	Low-carbon steel packed in charcoal + $BaCO_3$ energiser, 900–950 °C	0.5–2 mm; followed by hardening; slow
Gas carburising	CH_4/propane/endothermic gas, ≈ 925 °C	Controllable, mass production
Liquid carburising / cyaniding	Molten NaCN salt bath (adds C and N), 800–870 °C	Thin case 0.1–0.3 mm; toxic salts
Carbonitriding	Gas with NH_3 + hydrocarbon, 800–870 °C	Thin case; less distortion
Nitriding	NH_3 gas at 500–550 °C (below A_1); steels with Al, Cr, Mo (Nitralloy)	Very hard (≈ 1000 HV) thin case; no quenching needed – minimum distortion; long time (up to 100 h)
Flame hardening	Oxy-acetylene flame heating of surface, then water spray	Medium-carbon steel (0.4–0.6% C); large parts
Induction hardening	High-frequency induced current heats surface, then quench	Fast, localised; case depth ∝ 1/√frequency (skin effect); crankshafts, gears
Laser / electron-beam hardening	Focused beam	Precise, very thin case

- Carburising, cyaniding, carbonitriding and nitriding change surface chemistry (thermochemical); flame and induction hardening do not – they require steels with enough carbon (≥ 0.35–0.4%).

★ **PYQ-TREND:** Nitriding needs no quenching and is done below A_1 (very frequent); carburising for low-carbon steels; flame and induction hardening for medium-carbon steels; cyaniding adds C and N; induction case depth decreases with frequency; pack carburising energiser $BaCO_3$.

✎ **DESCRIPTIVE Q:** Distinguish between annealing, normalising, hardening and tempering.

All four are heat-treatment processes involving heating, holding and cooling, but they differ in temperature and cooling rate and therefore in the structure produced. In full annealing the steel is heated about 30–50 °C above the upper critical temperature, held to form uniform austenite and cooled very slowly in the furnace; it gives coarse pearlite and ferrite, maximum softness and ductility, relieves stresses and improves machinability. Normalising uses a similar or slightly higher temperature but the steel is cooled in still air; the faster cooling gives fine pearlite and refined grain size, so normalised steel is somewhat harder and stronger than annealed steel and has a more uniform structure – it is used for forgings, castings and before hardening. Hardening heats the steel into the austenite range and quenches it in water or oil so that austenite transforms to martensite, giving maximum hardness and wear resistance but high brittleness and internal stresses. Tempering always follows hardening: the steel is reheated to a temperature below the lower critical (150–650 °C depending on the required properties) and cooled; it relieves stresses and converts martensite partly into tempered martensite, trading some hardness for much greater toughness.

CHAPTER 14

Metal Casting – Patterns, Moulds and Gating

14.1 Casting Basics

- Casting: pouring molten metal into a mould cavity of the desired shape and allowing it to solidify. Suitable for complex shapes, internal cavities, very large parts and materials difficult to work (cast iron). Sand casting is the most widely used and versatile process.
- Steps in sand casting: pattern making → mould and core making → melting → pouring → solidification and cooling → shakeout → fettling/cleaning → inspection.

14.2 Patterns

Pattern type	Use
Single-piece (solid)	Simple shapes, small quantities
Split (two-piece)	Complex shapes – parting at the split line (e.g. pipes, spindles)
Loose-piece	Projections that would prevent withdrawal
Match-plate	Cope and drag halves mounted on a plate – machine moulding, mass production of small castings
Cope and drag	Large castings – separate patterns
Gated	Several patterns joined with gating – mass production of small parts
Sweep	Symmetrical shapes (bells, large wheels) – template rotated about a spindle
Skeleton	Very large castings – frame filled with loam
Segmental	Circular parts made section by section
Follow-board	Supports thin/weak patterns during moulding

- Pattern materials: wood (teak, pine, mahogany – cheap, easy, for few castings), metal (aluminium, cast iron, brass – long life, mass production), plaster, plastics (epoxy), wax (investment casting), expanded polystyrene (lost foam / full-mould).

Allowance	Purpose / typical value
Shrinkage (contraction)	Pattern made larger to compensate solid contraction: cast iron ≈ 10 mm/m (1%), steel ≈ 20 mm/m (2%), aluminium ≈ 13 mm/m, brass ≈ 15 mm/m
Machining (finish)	Extra metal on surfaces to be machined; more for cast iron/steel and large parts
Draft (taper)	Taper on vertical faces (≈ 1–3° external; more on internal) for easy withdrawal

Allowance	Purpose / typical value
Distortion (camber)	Shape deliberately distorted opposite to expected warping (U- and V-shapes, long thin parts)
Shake (rapping)	Negative allowance – pattern made slightly smaller as rapping enlarges the cavity

- Only shrinkage of the solid (patternmaker's contraction) is compensated by the pattern allowance; liquid and solidification shrinkage are fed by risers. Grey cast iron has low shrinkage because graphite expansion offsets contraction.
- Core prints: projections on the pattern that make seats in the mould to locate and support cores.
- Colour coding of patterns (common practice): red/orange – surfaces to be machined; black – surfaces left unmachined; yellow – core prints; red stripes on yellow – loose pieces.

★ **PYQ-TREND:** Pattern allowances – shake allowance is negative (very frequent); shrinkage allowance for cast iron 1% and steel 2%; match-plate patterns for mass production; sweep pattern for symmetrical shapes; draft for withdrawal; core prints.

✖ **COMMON MISTAKE:** Shake (rapping) allowance is the only negative allowance – the pattern is made smaller, not larger.

14.3 Moulding Sand and Cores

- Green sand: silica sand (SiO_2, 85–91%) + clay binder (bentonite/fireclay, 5–20%) + water (2–8%) + additives; 'green' means moist, not baked. Dry sand moulds are baked for better strength and finish (larger castings). Loam sand: high clay content (≈ 50%) for large castings with sweeps.

Sand property	Meaning
Permeability (porosity)	Allows gases and steam to escape – measured by permeability number (air flow through standard specimen)
Green strength	Strength of moist sand to hold shape
Dry strength	Strength after drying – resist metal erosion
Refractoriness	Ability to withstand high temperatures without fusing
Collapsibility	Ability to break down as casting contracts – avoids hot tears
Flowability / plasticity	Ability to fill pattern details under ramming
Cohesiveness / adhesiveness	Sand grains sticking to each other / to the flask

- Additives: sea coal/coal dust (better surface finish, prevents metal penetration in cast iron), cereal binders, wood flour (collapsibility), silica flour (hot strength), molasses, dextrin.
- Facing sand: fine sand next to the pattern; backing (floor) sand fills the rest; parting sand (dry, clay-free) prevents cope and drag sticking.
- Sand testing: grain fineness number (AFS sieve test), clay content, moisture content, permeability test, strength tests (universal sand testing machine), mould hardness tester.
- Cores form internal cavities: made of sand with binders (oil, sodium silicate–CO_2 process, resin shell cores, cold-box process); baked to give strength. Requirements: strength, permeability (vent), collapsibility, refractoriness. Chaplets are metal supports that hold cores in position and fuse into the casting. Core boxes produce cores.

- Moulding methods: hand moulding (bench, floor, pit), machine moulding (jolt, squeeze, jolt-squeeze, sand slinger – for large moulds).
- CO_2 process: sand mixed with sodium silicate is hardened by passing CO_2 gas – quick, strong moulds and cores without baking.

★ **PYQ-TREND:** Moulding sand properties – permeability, collapsibility, refractoriness (very frequent); bentonite clay binder; CO_2 process uses sodium silicate; chaplets support cores; sea coal for cast iron finish; grain fineness number; parting sand clay-free.

14.4 Gating and Risering

- Gating system: pouring basin (cup) → sprue (tapered downward to avoid aspiration) → sprue well (base) → runner → ingates (gates) → mould cavity. Skim bob and strainer/filter trap slag and dross.

Gate	Feature
Top gate	Metal falls directly into cavity – fast filling, favourable temperature gradient, but erosion and turbulence; simple shapes, ferrous metals
Bottom gate	Fills from below – minimal turbulence and erosion, but unfavourable thermal gradient (hottest metal at bottom); non-ferrous, tall castings
Parting-line (side) gate	Metal enters at parting plane – most common; compromise
Step gate	Several ingates at different heights – large, tall castings

Gating ratio = sprue area : runner area : ingate area — pressurised (e.g. 1 : 0.75 : 0.5, choke at gates; ferrous) or unpressurised (e.g. 1 : 2 : 2 to 1 : 4 : 4, choke at sprue; aluminium, magnesium)

Sprue taper (avoid aspiration): $A_1/A_2 = \sqrt{(h_2/h_1)}$ (continuity + Bernoulli; velocity at bottom $V = \sqrt{(2gh)}$)

Filling time (top gate): $t = V/(A_g \sqrt{(2gH)})$; bottom gate: $t = (A_m/A_g)(2/\sqrt{(2g)})(\sqrt{H} - \sqrt{(H - h_m)})$

- Riser (feeder): reservoir of molten metal that feeds the casting as it shrinks during solidification. It must solidify after the casting and be placed at the heaviest section (last to freeze). Open riser (top open to atmosphere) and blind riser (enclosed); exothermic or insulating sleeves keep riser liquid longer.

Chvorinov's rule: solidification time $t_s = B\,(V/A)^n$ ($n \approx 2$) — modulus $M = V/A$

Riser design: riser modulus ≥ 1.2 × casting modulus (solidification time of riser ≈ 1.2–1.44 × casting)

Caine's method: freezing ratio $X = (A/V)_{casting}/(A/V)_{riser}$; most efficient riser shape is a sphere, cylinder with H = D commonly used

- Directional solidification: solidification should progress from the farthest point towards the riser; achieved by chills (metal inserts that speed cooling locally), padding, riser placement, insulating sleeves.
- Shrinkage stages: liquid contraction (fed by riser), solidification shrinkage (fed by riser – causes shrinkage cavities/porosity if not fed), solid contraction (compensated by pattern allowance).

★ **PYQ-TREND:** Chvorinov's rule $t \propto (V/A)^2$ (very frequent); riser should freeze last – placed at thickest section; sprue tapered to prevent aspiration; chills promote directional solidification; pressurised vs unpressurised gating; bottom gating minimises turbulence; spherical riser most efficient; three stages of shrinkage.

✔ **TOPPER TRICK:** For the same volume, a sphere has the least surface area, so a spherical riser stays liquid the longest. Among a sphere, cylinder and cube of equal volume, the cube solidifies fastest.

CHAPTER 15

Special Casting Processes, Melting and Casting Defects

15.1 Special Casting Processes

Process	Principle	Typical products / features
Shell moulding (Croning)	Heated metal pattern (≈ 200–250 °C) covered with sand + thermosetting resin (phenol-formaldehyde) – thin shell forms; two shells clamped	Good finish and accuracy, thin sections; crankshafts, cylinder heads, small parts in mass production
Investment (lost-wax) casting	Wax pattern coated with refractory slurry (ceramic), wax melted out, metal poured into hot shell	Very complex, precise parts; excellent finish; turbine blades, jewellery, dental, surgical, small intricate parts
Gravity die (permanent mould) casting	Metal poured by gravity into reusable metal mould	Non-ferrous: pistons, gear blanks; better finish and properties than sand
Pressure die casting – hot chamber	Injection mechanism immersed in molten metal; plunger forces metal into die	Low-melting alloys (Zn, Sn, Pb, Mg) – fast cycle
Pressure die casting – cold chamber	Metal ladled into a cold chamber, then injected at high pressure	Higher-melting alloys (Al, Cu/brass, Mg)
Centrifugal casting (true)	Mould rotated about its axis; metal forced outward; no core needed	Hollow cylindrical parts – pipes, liners, bushes; dense, impurity-free outer layer (impurities to the inner surface)
Semi-centrifugal	Rotation about vertical axis, central core may be used	Symmetrical solids – wheels, pulleys, gear blanks
Centrifuging (centrifuge casting)	Several moulds around a central sprue, rotated	Small irregular parts (not symmetrical)
Continuous casting	Molten metal poured into water-cooled open-ended mould and withdrawn continuously	Billets, blooms, slabs – steel plants
Slush casting	Metal poured and dumped after a thin shell solidifies	Hollow ornamental items, toys (low-melting alloys)
Squeeze casting	Metal solidifies under high pressure	Near-net shape, fine grain
Plaster mould casting	Gypsum mould	Non-ferrous, good finish
Ceramic mould casting	Ceramic slurry mould	High-temperature alloys, precise
Full-mould / lost-foam (evaporative pattern) casting	Expanded polystyrene pattern left in sand; vaporises when metal is poured	No parting line, no cores for complex shapes; engine blocks
Vacuum (V-process) moulding	Dry sand held by vacuum with plastic film	Good finish, no binders

- Die casting advantages: high production rate, thin sections, excellent finish and accuracy, minimal machining. Limitations: high die cost (economic only for large quantities), limited to non-ferrous alloys of lower melting point, porosity from entrapped air, size limits.
- Centrifugal force in true centrifugal casting: speed chosen so that G-factor (centrifugal/gravity) ≈ 60–80 for horizontal moulds.

`Rotational speed for G-factor GF: N = (30/π)√(2 g GF/D) (D = inside diameter)`

★ **PYQ-TREND:** Investment casting for turbine blades and jewellery (very frequent); hot chamber die casting for zinc and cold chamber for aluminium; true centrifugal casting for pipes without core; shell moulding uses thermosetting resin; slush casting for hollow toys; continuous casting for billets; lost-foam uses polystyrene; die casting suits large quantities of non-ferrous parts.

✔ **TOPPER TRICK:** Hot chamber → low melting point metals (Zn, Sn, Pb); cold chamber → aluminium and brass (they would attack the immersed hot-chamber parts).

15.2 Melting Furnaces

Furnace	Use
Cupola	Melting cast iron (most common in foundries); vertical shaft charged with coke, pig iron, scrap and limestone flux; air through tuyeres; continuous operation
Crucible furnace (pit, tilting, lift-out)	Small quantities of non-ferrous metals (Al, brass, bronze)
Electric arc furnace (direct arc)	Steel melting from scrap; large capacity
Induction furnace (coreless, channel)	Steel, cast iron, non-ferrous; clean, good temperature control, stirring action
Reverberatory furnace	Large quantities of non-ferrous metals (Cu, Al); flame reflected from roof
Open hearth / converters (Bessemer, LD/BOF)	Steel making from pig iron (LD uses pure oxygen)
Blast furnace	Iron ore to pig iron (coke, limestone, hot blast)

- Cupola zones (bottom to top): well (crucible), combustion (oxidising) zone, reducing zone, melting zone, preheating zone, stack. Coke-to-iron ratio ≈ 1:8 to 1:10. Limestone forms slag with impurities and ash.

★ **PYQ-TREND:** Cupola for cast iron (very frequent); cupola zones; limestone as flux; induction furnace clean melting; crucible furnace for non-ferrous; LD converter uses oxygen.

15.3 Casting Defects

Defect	Description	Main causes	Remedies
Blowholes	Smooth rounded cavities at or below the surface	Gas from moist sand, low permeability, excess moisture, hard ramming	Control moisture, improve permeability and venting, moderate ramming
Pin holes	Many tiny gas holes	Hydrogen dissolved in metal, wet charge/sand	Degassing, dry materials
Shrinkage cavity	Irregular cavity in heavy section	Inadequate feeding, poor riser design	Proper risers, chills, directional solidification

Defect	Description	Main causes	Remedies
Porosity (gas/shrinkage)	Fine voids throughout	Dissolved gases or interdendritic shrinkage	Degassing, better feeding
Misrun	Mould not completely filled	Low pouring temperature, slow pouring, thin sections, low fluidity	Raise pouring temperature and speed, better gating
Cold shut	Two metal streams meet without fusing – visible line	Low temperature, interrupted pouring, poor gating	Higher temperature, continuous pouring, better gating
Hot tear (hot crack)	Ragged crack while metal is hot and weak	Hindered contraction, poor collapsibility of mould/core, abrupt section changes	Improve collapsibility, uniform sections, fillets
Mould shift (shift)	Mismatch at parting line	Misaligned cope and drag, worn pins	Proper alignment, dowel pins
Core shift	Internal cavity displaced	Core not secured, buoyancy, wrong core prints	Chaplets, proper core prints
Swell	Enlargement of casting	Soft ramming, metallostatic pressure	Firm ramming
Fin / flash	Thin projection at parting line	Poor fit of mould halves, insufficient weighting	Proper clamping
Scab	Rough patches where sand layer broke and metal flowed behind	Sand expansion, low hot strength, uneven ramming	Additives (wood flour), uniform ramming
Rat tail / buckle	Thin lines/depressions from sand expansion	Expansion of silica sand	Additives, lower pouring temperature
Drop	Sand falls from cope into cavity	Low green strength, soft ramming	Better sand strength and ramming
Metal penetration	Metal enters sand pores – rough surface	Coarse sand, high temperature/pressure	Fine facing sand, sea coal, coatings
Fusion	Sand fused to casting	Low refractoriness, high pouring temperature	Better sand refractoriness
Wash / cut	Sand eroded by metal flow	Low sand strength, high metal velocity	Stronger sand, better gating
Slag inclusion	Slag particles inside casting	Poor skimming, no strainer	Skimming, strainers, skim bobs
Hard spots	Local chilled areas in cast iron	Rapid cooling, composition	Uniform cooling
Warpage	Distortion	Uneven cooling, residual stress	Uniform sections, stress relief

- Inspection: visual, dimensional, pressure tests, dye penetrant (surface cracks), magnetic particle (ferromagnetic, surface/near-surface), radiography (X-ray/γ-ray – internal porosity, shrinkage), ultrasonic testing (internal flaws, thickness), eddy current.

★ **PYQ-TREND:** Blowholes from moisture and low permeability; misrun vs cold shut (very frequent); hot tear from poor collapsibility; swell from soft ramming; scab and rat tail from sand expansion; core shift prevented by chaplets; radiography for internal porosity; dye penetrant for surface cracks.

✖ **COMMON MISTAKE:** Misrun – metal freezes before filling the mould; cold shut – the mould fills but two streams fail to fuse. Both come from low fluidity or low temperature, but the result is different.

✎ **DESCRIPTIVE Q:** Explain investment casting with its advantages and applications.

Investment (lost-wax) casting begins with making a wax (or plastic) pattern for each casting, usually by injecting wax into a metal die. Several patterns are attached to a wax sprue to form a 'tree'. The assembly is dipped repeatedly in a fine refractory slurry (silica or zircon flour with a binder such as ethyl silicate) and coated with stucco sand, building a ceramic shell; alternatively it is placed in a flask and surrounded by investment slurry. After drying, the mould is heated so that the wax melts and runs out (hence 'lost wax'), and then fired at high temperature to harden it and burn out residues. Molten metal is poured into the hot mould, often by gravity, vacuum or centrifugal force. After solidification the ceramic is broken away and the castings are cut from the tree. Advantages: very intricate shapes and thin sections, no parting line, excellent surface finish and dimensional accuracy, little or no machining, and any metal including high-temperature alloys can be cast. Limitations: expensive, size limited (usually below a few kilograms, though larger is possible), long process cycle. Applications: gas-turbine blades and vanes, jewellery, dental crowns, surgical implants, small precision machine parts, firearm components and golf-club heads.

CHAPTER 16

Welding, Brazing and Soldering

16.1 Classification

- Welding: joining metals by heat and/or pressure, with or without filler, producing coalescence. Fusion welding melts the joint edges (gas, arc, thermit, electron beam, laser). Solid-state (pressure) welding joins without melting (forge, friction, ultrasonic, explosive, diffusion, cold, resistance welding partly).
- Welding positions: flat (easiest, 1G/1F), horizontal, vertical, overhead (most difficult). Joints: butt, lap, T, corner, edge; edge preparations: square, single-V, double-V, U, J, bevel.

16.2 Gas Welding

Flame	Oxygen : acetylene	Temperature	Use
Neutral	≈ 1 : 1	≈ 3100–3200 °C (hottest inner cone tip)	Most metals – mild steel, cast iron, copper, aluminium
Carburising (reducing)	Excess acetylene	Lower (≈ 3000 °C)	High-carbon steel, hardfacing, non-ferrous like nickel/Monel; three zones visible (feather)
Oxidising	Excess oxygen	Highest (≈ 3300–3500 °C)	Brass, bronze (zinc fumes reduced by oxide film)

- Oxy-acetylene: acetylene (C_2H_2) stored dissolved in acetone in a cylinder filled with porous mass (maroon colour cylinder in India); oxygen cylinder black. Acetylene pressure should not exceed 1 bar (15 psi) free gas – explosive. Combustion: primary (inner cone) $C_2H_2 + O_2 \rightarrow 2CO + H_2$; secondary (outer envelope) with atmospheric $O_2 \rightarrow CO_2 + H_2O$.
- Leftward (forehand) welding: thin plates (< 5 mm); rightward (backhand): thicker plates, faster, better view of weld pool, less filler.
- Other fuel gases: oxy-hydrogen (lower temperature, Al/Pb), oxy-LPG/propane (cutting, heating).
- Oxy-fuel cutting (gas cutting): steel is preheated to ignition temperature (≈ 870 °C) and burnt in a jet of pure oxygen ($Fe \rightarrow Fe_3O_4$); works for ferrous metals whose oxide melts below the metal and which burns exothermically. Not suitable for stainless steel, cast iron, aluminium, copper (use plasma or powder cutting).

★ **PYQ-TREND:** Neutral flame for most metals, oxidising for brass, carburising for high-carbon steel (very frequent); neutral flame temperature ≈ 3200 °C; acetylene dissolved in acetone; cylinder colours (oxygen black, acetylene maroon); oxy-fuel cutting only for ferrous; leftward vs rightward welding.

16.3 Arc Welding

Process	Electrode / shielding	Features / uses
Shielded metal arc (SMAW, MMA, stick)	Consumable flux-coated electrode	Most versatile, field work, all positions; slag protection
Submerged arc (SAW)	Bare wire under a blanket of granular flux	Very high deposition rate, deep penetration, no visible arc; flat position only; thick plates, ships, pipes, pressure vessels
Gas tungsten arc (GTAW, TIG)	Non-consumable tungsten electrode, inert gas (Ar/He), separate filler	High quality, thin sheets, Al, Mg, stainless, Ti; AC for Al (cleaning action)
Gas metal arc (GMAW, MIG/MAG)	Consumable wire fed continuously, inert (MIG) or active CO_2 (MAG) gas	Fast, semi-automatic; aluminium (MIG), steels (CO_2)
Flux-cored arc (FCAW)	Tubular wire with flux core	Outdoor, high deposition
Plasma arc (PAW)	Constricted arc through nozzle – very high temperature (up to ≈ 30,000 °C)	Precision welding, keyhole mode; plasma cutting of all metals
Electroslag (ESW)	Heat from resistance of molten slag; vertical position	Very thick plates in one pass
Atomic hydrogen	Arc between two tungsten electrodes in hydrogen; H_2 dissociates and recombines	Very hot, thin and special alloys
Stud welding	Arc between stud and plate, then pressed	Fixing studs

- Polarity (DC): straight polarity DCEN/DCSP – electrode negative, work positive; about two-thirds of heat at the work (anode) → deeper penetration, thick plates, TIG on steel. Reverse polarity DCEP/DCRP – electrode positive; more heat at electrode, shallow penetration, faster melting of electrode – thin sheets, MIG, and cleaning action (removes oxide) useful for aluminium. AC: balanced heat, used in TIG for Al and Mg.
- Electrode coating (flux) functions: stabilise the arc (ionising elements – K, Na), form slag to protect and slow cooling, provide shielding gas (cellulose), add alloying elements and deoxidisers, control bead shape. Electrode coding IS 815/AWS E6013: E = electrode, 60 = 60 ksi UTS, 1 = all positions, 3 = coating/current type.
- Arc length ≈ electrode diameter; open-circuit voltage 50–100 V; arc voltage 20–40 V; current 50–500 A. Power sources: transformer (AC), rectifier and generator (DC). Constant-current (drooping) characteristic for SMAW/TIG; constant-voltage for MIG/SAW (self-adjusting arc).

Heat input per unit length $H = \eta V I/v$ (v = welding speed); arc power = $V \times I$

Arc characteristic (Ayrton): $V = A + B \cdot l$ (voltage increases linearly with arc length)

★ **PYQ-TREND:** DCSP gives deeper penetration – two-thirds heat at the work; DCRP for thin sheets and aluminium (cleaning action) (very frequent); TIG uses non-consumable tungsten; MIG consumable wire; SAW flat position, thick plates; functions of electrode coating; plasma arc highest temperature; heat input VI/v.

✖ **COMMON MISTAKE:** In TIG welding the tungsten electrode is not consumed – filler is added separately; in MIG the wire itself is the filler.

16.4 Resistance and Solid-State Welding

Heat generated in resistance welding $H = I^2 R t$ (current most influential – heat $\propto I^2$)

Process	Description / use
Spot welding	Overlapping sheets pressed between copper electrodes, current passed – car bodies, sheet metal
Seam welding	Rotating wheel electrodes produce overlapping spots – leak-tight seams in fuel tanks, cans
Projection welding	Current concentrated at projections – several welds at once; nuts, studs
Flash butt welding	Ends brought together, flashing, then upset – rails, chains, pipes
Upset butt welding	Ends pressed and heated by resistance without flashing – wires, rods
Percussion welding	Rapid discharge of stored energy
Friction welding	One part rotated against another under pressure; heat by friction; then forging pressure – dissimilar metals, shafts, axles
Friction stir welding	Rotating non-consumable tool stirs plasticised material – aluminium alloys, aerospace
Ultrasonic welding	High-frequency vibration under light pressure – foils, wires, plastics
Explosive welding	Controlled detonation – cladding dissimilar plates
Diffusion bonding	Pressure and temperature over time – aerospace, titanium
Forge welding	Heating and hammering – oldest method (blacksmith)
Thermit welding	Exothermic reaction of Al powder with iron oxide (≈ 2500–3000 °C) – rails, heavy sections, repair of castings
Electron beam welding	Focused electron beam in vacuum – deep narrow welds, reactive metals
Laser beam welding	Focused laser – precise, small HAZ, no vacuum needed

- Electrodes in resistance welding: copper alloys (Cu–Cr, Cu–Be) of high conductivity and hardness. Spot welding cycle: squeeze, weld, hold, off.

★ **PYQ-TREND:** Resistance welding heat I^2Rt (very frequent); spot welding for car bodies; seam welding for leak-tight tanks; thermit welding for rails (aluminium + iron oxide); friction welding for dissimilar metals; electron beam welding in vacuum; flash butt welding for rails.

16.5 Welding Metallurgy, Defects and Inspection

- Weld zones: fusion zone (melted and solidified, cast structure), heat-affected zone (HAZ – not melted but microstructure changed: grain growth, hardening in high-carbon/alloy steels), unaffected base metal. Preheating and post-weld heat treatment (stress relieving) reduce cracking in medium/high-carbon and alloy steels. Carbon equivalent $CE = C + Mn/6 + (Cr + Mo + V)/5 + (Ni + Cu)/15$ – $CE < 0.4$ indicates good weldability.

Defect	Cause
Porosity / blowholes	Gas entrapment – moisture, contamination, poor shielding

Defect	Cause
Slag inclusion	Incomplete slag removal between passes
Lack of fusion	Low current, fast travel, poor manipulation
Incomplete penetration	Low current, improper joint preparation, fast speed
Undercut	Groove along weld toe – excessive current, long arc, wrong angle
Overlap	Weld metal flows over surface without fusing – low current, slow speed
Cracks (hot/cold)	Hot: high sulphur, restraint; cold (hydrogen-induced): hydrogen + hard HAZ + stress – use low-hydrogen electrodes and preheat
Spatter	High current, long arc
Distortion	Non-uniform heating and contraction
Arc blow	Magnetic deflection of arc in DC welding – use AC or change earth position

- Inspection: visual, dye penetrant, magnetic particle, radiography, ultrasonic; destructive – tensile, bend, nick-break, macro-etch tests.

16.6 Brazing, Soldering and Adhesive Bonding

Process	Filler melting point	Base metal melts?	Examples
Welding	Same as base	Yes	–
Brazing	Above 450 °C (below base metal)	No	Brass (spelter), silver alloys, copper; capillary action; carbide tool tips, pipes, radiators
Braze welding	Above 450 °C	No	Filler deposited like welding (no capillary) – cast iron repair
Soldering	Below 450 °C	No	Sn–Pb (60/40 eutectic 183 °C), lead-free Sn–Ag–Cu; electronics, plumbing

- Fluxes: borax for brazing; zinc chloride/ammonium chloride (acid flux) or rosin (non-corrosive, for electronics) for soldering.
- Adhesive bonding: epoxies, cyanoacrylates, structural adhesives; joins dissimilar materials, distributes stress, no heat damage.

★ **PYQ-TREND:** Brazing above 450 °C, soldering below 450 °C, base metal not melted (very frequent); brazing uses capillary action; soft solder Sn–Pb; borax flux; HAZ; arc blow in DC; undercut from excess current; carbon equivalent and weldability.

✎ **DESCRIPTIVE Q:** Compare TIG and MIG welding.

Both are gas-shielded arc welding processes in which an inert or active gas protects the arc and weld pool from the atmosphere, so no flux or slag is involved. In TIG (gas tungsten arc) welding the arc is struck between a non-consumable tungsten electrode and the workpiece in a shield of argon or helium; filler, when required, is added separately by hand as a rod. It gives very high quality, clean, precise welds with excellent control, is suitable for thin sheets and for aluminium, magnesium, stainless steel, titanium and root passes in pipes; AC is used for

aluminium for the oxide-cleaning action, DCEN for steels. It is slow and needs skilled operators. In MIG (gas metal arc) welding, a consumable bare wire electrode is fed continuously from a spool through the torch, acting both as electrode and filler, using reverse polarity DC and argon (MIG) or CO_2 and mixtures (MAG). It has a much higher deposition rate and welding speed, is easily semi-automated or robotised, and suits medium and thick sections of steel and aluminium in fabrication and automobile industries. MIG has more spatter and somewhat lower precision than TIG, and its equipment (wire feeder) is more complex, but it is more productive.

CHAPTER 16B

Welding Practice – Symbols, Joint Design, Metals and Residual Stresses

16B.1 Weld Symbols and Joint Design

- Welding symbol (IS 813 / AWS A2.4): reference line, arrow pointing to the joint, basic weld symbol placed below the line for the arrow side and above for the other side; supplementary symbols – weld-all-round (circle at the bend), field weld (flag), flush/convex/concave contour; dimensions – size on the left, length and pitch on the right of the symbol; tail for process reference.

Basic symbol	Weld
Triangle	Fillet weld
Two parallel lines (‖)	Square butt weld
V	Single-V butt weld
Half V (bevel)	Single-bevel butt weld
U / J	Single-U / single-J butt weld
Circle	Spot / seam weld
Rectangle (slot)	Plug or slot weld

- Edge preparation by thickness (manual arc welding of steel, typical): up to ≈ 5 mm – square butt; 5–20 mm – single-V (included angle 60–70°, root face and gap ≈ 1.5–3 mm); above ≈ 20 mm – double-V, U or J (less weld metal and distortion). Backing strips and root runs ensure full penetration.
- Fillet weld size: leg length s; throat 0.707s; minimum size depends on the thicker plate (to ensure fusion), maximum ≈ plate thickness − 1.5 mm at square edges. Intermittent (chain/staggered) welds for light loads.

★ **PYQ-TREND:** Weld symbol placement – below reference line for arrow side (frequent); triangle symbol for fillet weld; edge preparation by plate thickness; double-V reduces distortion and filler; field-weld flag.

16B.2 Distortion and Residual Stresses

- Non-uniform heating and cooling cause the weld and HAZ to contract while restrained by cooler metal, leaving tensile residual stresses (up to yield) in the weld and compressive stresses in the surrounding plate, and producing distortion.

Distortion	Description
Longitudinal shrinkage	Shortening along the weld
Transverse shrinkage	Shrinkage across the weld
Angular distortion	Rotation about the weld line (V-welds)
Bowing / buckling	Thin plates under compressive residual stress

- Control: minimise weld metal (correct joint design, avoid over-welding), balanced welding about the neutral axis (double-V, alternating sides), back-step and skip welding sequences, presetting parts opposite to expected distortion, clamping/jigs, tack welds, intermittent welds, preheating, peening of weld layers.
- Relief of residual stresses: post-weld heat treatment (stress relieving at ≈ 600–650 °C for steels), vibratory stress relief, peening.

★ **PYQ-TREND:** Weld region left in residual tension (frequent); back-step and skip welding reduce distortion; presetting; stress-relieving heat treatment after welding; angular distortion in single-V welds.

16B.3 Welding of Different Metals

Material	Key points
Low-carbon steel	Excellent weldability; any process
Medium/high-carbon and alloy steels	Hardenable HAZ – risk of cold (hydrogen) cracking; preheat, low-hydrogen electrodes, controlled cooling, PWHT; carbon equivalent CE > 0.45 needs preheat
Austenitic stainless steel	Low thermal conductivity, high expansion – distortion; sensitisation (weld decay) – use low-carbon (L) or stabilised grades; TIG/MIG/SMAW
Cast iron	Brittle, high carbon – preheat (≈ 500–600 °C) and slow cool, nickel-based electrodes, or braze welding; cold welding with short beads and peening
Aluminium	Tenacious oxide (Al_2O_3 melts ≈ 2050 °C vs Al 660 °C), high conductivity, hydrogen porosity – AC TIG or DCEP MIG with argon, thorough cleaning; no colour change before melting
Copper	High conductivity needs preheat and high heat input; TIG/MIG
Titanium	Very reactive – inert gas shielding including trailing shield or chamber; TIG, EBW, LBW
Dissimilar metals	Friction welding, explosive welding, diffusion bonding, brazing

- Weldability factors: chemical composition (C, S, P), thermal conductivity and expansion, oxide formation, hardenability, melting point, and susceptibility to cracking.

★ **PYQ-TREND:** Aluminium welded with AC TIG due to oxide film (very frequent); preheating for high-carbon steel and cast iron; nickel electrodes for cast iron; low-hydrogen electrodes prevent cold cracking; stainless steel weld decay; titanium needs full inert shielding.

16B.4 Cutting Processes

Process	Suitable materials / features
Oxy-fuel cutting	Carbon and low-alloy steels only; thick sections, cheap
Powder (flux) cutting	Iron powder/flux added to oxygen stream – stainless steel, cast iron
Oxygen lance cutting	Very thick sections, concrete

Process	Suitable materials / features
Plasma arc cutting	All conductive metals (stainless, Al, Cu); fast, narrow kerf
Air-carbon arc gouging	Removing defective welds, back gouging – carbon electrode + compressed air jet
Laser cutting	Sheet metals and non-metals; precise, small HAZ
Abrasive water jet	Any material, no HAZ

★ **PYQ-TREND:** Gas cutting not suitable for stainless steel and aluminium; plasma cutting for non-ferrous; air-carbon arc gouging for back gouging; kerf.

CHAPTER 17

Metal Forming – Rolling, Forging, Extrusion, Drawing and Sheet Metal

17.1 Fundamentals

- Metal forming changes shape by plastic deformation without removing material. Advantages: no material waste, improved mechanical properties (grain flow, work hardening), high production rates.
- Hot working: deformation above the recrystallisation temperature (≈ 0.4–0.6 T_m in kelvin; for steel ≈ 1000–1200 °C working range) – low forces, large deformation, refined grains, no strain hardening, but oxidation/scale, poorer finish and dimensional accuracy. Cold working: below recrystallisation temperature (usually room temperature) – strain hardening increases strength and hardness, better finish and accuracy, but higher forces, lower ductility, residual stresses; may need intermediate annealing.
- Recrystallisation temperature of some metals: lead and tin below room temperature (so working them at room temperature is hot working), zinc ≈ room temperature, aluminium ≈ 150 °C, copper ≈ 200 °C, iron ≈ 450 °C, tungsten ≈ 1200 °C.
- Warm working: intermediate temperatures (≈ 0.3–0.5 T_m).

True strain $\varepsilon = \ln(L/L_0) = \ln(A_0/A)$; flow stress $\sigma_f = K\,\varepsilon^n$; average flow stress $= K\,\varepsilon^n/(n+1)$

★ **PYQ-TREND:** Hot vs cold working defined by recrystallisation temperature (very frequent); lead worked at room temperature is hot working; cold working improves finish and strength by strain hardening; recrystallisation temperature ≈ 0.4 T_m.

17.2 Rolling

- Metal passed between rotating rolls; friction draws it into the gap. Primary rolling: ingots/continuous-cast slabs into blooms, billets, slabs; then plates, sheets, strips, structural shapes, rails.

Rolling mill	Use
Two-high (reversing or non-reversing)	Blooming, slabbing mills
Three-high	Reversal by lifting/lowering stock, no roll reversal
Four-high	Small work rolls backed by large backup rolls – reduces deflection; plates, sheets
Cluster (Sendzimir, 12- or 20-high)	Very thin, hard strips (stainless) with tiny work rolls
Tandem (continuous)	Several stands in series – high production of strip
Planetary	Large reduction in single pass

Draft $\Delta h = h_0 - h_f$; maximum draft $\Delta h_{max} = \mu^2 R$ (R = roll radius)

Angle of bite (contact angle) α: $\cos\alpha = 1 - \Delta h/(2R)$; condition for bite: $\mu \geq \tan\alpha$

Projected contact length $L \approx \sqrt{(R\,\Delta h)}$; roll force $F \approx \sigma_{avg} \times w \times \sqrt{(R\Delta h)}$; power per roll $P = F\,(L/2)\,\omega$ (approx.)

`Volume constancy: h₀w₀v₀ = h_f w_f v_f → exit speed greater than entry speed`

- Neutral (no-slip) point: where the roll surface speed equals strip speed; before it the strip moves slower than the rolls, after it faster. Forward slip = (v_f − v_r)/v_r.
- Rolling defects: edge cracks, alligatoring (splitting of slab ends), wavy edges (roll bending), centre cracks, laminations, scale pits. Thread rolling and ring rolling are other rolling processes – thread rolling gives stronger threads (unbroken grain flow) than cutting.

★ **PYQ-TREND:** Maximum draft μ²R and angle of bite (very frequent); neutral point; four-high mill uses backup rolls; cluster mill for thin hard strips; thread rolling stronger than cut threads; volume constancy.

17.3 Forging

Type	Description
Open-die (smith/hand) forging	Flat dies, simple shapes; upsetting, drawing out, fullering, edging, punching
Closed-die (impression-die) forging	Shaped dies; flash forms in gutter; good accuracy
Flashless (precision) forging	Closed die, no flash; exact volume of billet
Drop forging	Hammer dropped by gravity on die (closed die) – gear blanks, crankshafts, connecting rods, spanners
Press forging	Slow squeezing action by hydraulic/mechanical press – deep, uniform deformation; large parts
Upset forging (heading)	Increasing diameter by pressing end – bolt heads, rivets, valves (on headers)
Roll forging	Shaped rolls reduce cross-section – axles, tapered levers
Swaging	Rotating dies hammer a rod or tube to reduce diameter – tapering, pointing
Coining	Closed-die cold forging giving fine surface detail – coins, medals
Hobbing / isothermal forging	Die cavities / superalloys at uniform temperature

- Smith forging operations: drawing down (reducing cross-section, increasing length), upsetting (increasing cross-section, reducing length), fullering (reducing section between, with fullers), edging, swaging, bending, punching, cutting, welding.
- Forging advantages: directional grain flow following part contour gives high strength, toughness and fatigue resistance (crankshafts, connecting rods, gears).

`Forging load (open-die, cylinder): F = σ_f π r²(1 + 2μr/(3h))`

- Defects: cold shut (lap/fold), unfilled sections, scale pits, die shift, flakes (internal cracks due to hydrogen), surface cracks, improper grain flow, burnt metal.

★ **PYQ-TREND:** Drop forging for crankshafts and connecting rods; upsetting increases cross-section; fullering; coining is closed-die cold forming (very frequent); flash in closed-die forging; forging gives favourable grain flow; swaging.

17.4 Extrusion and Drawing

Process	Description
Direct (forward) extrusion	Ram pushes billet through die in the same direction; friction between billet and container – higher force
Indirect (backward/inverted) extrusion	Die moves with hollow ram, billet stationary relative to container – less friction and force
Impact extrusion	Punch strikes a slug in a die – metal flows back around punch; collapsible tubes (toothpaste), cans, cartridge cases (soft metals: Al, Pb, Sn, Zn)
Hydrostatic extrusion	Billet pressurised by fluid – no container friction; brittle materials
Tube extrusion	Mandrel used to form hollow sections
Wire drawing	Pulling rod through progressively smaller dies (tungsten carbide/diamond) – cold process; force applied at exit (tensile)
Tube drawing	With or without mandrel/plug

Extrusion ratio $R = A_0/A_f$; extrusion pressure $p = \sigma_f \ln R$ (ideal) or $k(a + b \ln R)$ (Johnson)

Drawing stress (ideal) $\sigma_d = \sigma_f \ln(A_0/A_f)$; maximum reduction per pass (ideal, perfect plasticity) $= 1 - 1/e \approx 63\%$

- In drawing the force is tensile at the exit, so the drawing stress must remain below the yield strength of the drawn product, which limits reduction per pass; in extrusion the force is compressive, allowing larger reductions.
- Extrusion defects: centre burst (chevron cracking) at low extrusion ratios and high die angles, piping (tailpipe/fishtailing) from surface oxides drawn inward, surface cracking (speed/temperature too high – 'fir-tree' cracking).
- Products: extrusion – aluminium window sections, tubes, rods, complex constant cross-sections; drawing – wires, rods, seamless tubes of accurate size.

★ **PYQ-TREND:** Impact extrusion for collapsible toothpaste tubes (very frequent); indirect extrusion needs less force; wire drawing is a tensile process with maximum reduction 63% (ideal); centre burst in extrusion; aluminium sections by extrusion; extrusion ratio.

17.5 Sheet Metal Working (Press Work)

Operation	Description
Blanking	Cutting a piece of desired shape – removed piece is the product
Punching (piercing)	Making a hole – removed piece is scrap
Notching, lancing, nibbling, slitting, trimming, shaving	Cutting variants
Bending (V, U, edge, roll)	Plastic deformation about a straight axis; springback
Deep drawing (cupping)	Flat blank drawn into cup by punch through die with blank holder
Spinning	Sheet pressed over rotating mandrel by tool – cones, reflectors, utensils
Stretch forming	Sheet stretched over a form block – aircraft skins

Operation	Description
Embossing	Raised/depressed design, thickness nearly unchanged
Coining	Squeezing with different patterns on each side – thickness changes
Hydroforming, rubber-pad forming (Guerin), explosive forming, electromagnetic forming	High-energy-rate or flexible-die processes

Blanking/punching force $F = L\ t\ \tau_s$ (L = perimeter, t = thickness, τ_s = shear strength)

Clearance $c \approx (0.03 \text{ to } 0.1)\ t$ per side ($\approx 0.0032\ t\ \sqrt{\tau}$ for steel by empirical rule)

Blanking: die size = blank size, punch = blank size − $2c$; punching: punch size = hole size, die = hole size + $2c$

Shear on punch or die reduces peak force: $F' = F_{max} \times (p\ t)/S$ (p = fractional penetration at fracture, S = amount of shear, valid for $S \geq p\ t$)

Deep drawing blank diameter (cup without flange): $D = \sqrt{(d^2 + 4dh)}$; drawing ratio $D/d \leq \approx 2$ (limiting drawing ratio); percentage reduction $(D - d)/D \leq \approx 50\%$

Bend allowance $BA = \alpha(R + k\ t)$ (α in radians; $k \approx 0.33$ for $R < 2t$, 0.5 for $R \geq 2t$); minimum bend radius $\propto t$

Press tonnage = cutting force × factor of safety; work done = $F \times t \times$ penetration fraction

- Springback: elastic recovery after bending – compensated by overbending, bottoming or coining. Greater for high yield strength, low E and larger R/t.
- Deep-drawing defects: wrinkling in flange (insufficient blank-holder force), tearing/fracture at the cup bottom (excessive blank-holder force, small die radius), earing (planar anisotropy), surface scratches.
- Dies: progressive die (several operations at successive stations as strip advances), compound die (several cutting operations at one station in one stroke, e.g. blanking and punching), combination die (cutting and forming at one station), transfer die.

★ **PYQ-TREND:** Blanking vs punching – which part is the product (very frequent); clearance applied on punch for blanking and on die for punching; punching force $L\ t\ \tau$; deep drawing blank diameter $\sqrt{(d^2 + 4dh)}$; limiting drawing ratio ≈ 2; springback; wrinkling and earing; progressive vs compound dies; spinning for utensils.

✔ **TOPPER TRICK:** 'Blank = Die' (die gets the exact blank size); 'Punch = Hole' (punch gets the exact hole size). Clearance always goes to the other tool.

✎ **DESCRIPTIVE Q:** Distinguish between hot working and cold working of metals.

Hot working is plastic deformation carried out above the recrystallisation temperature, so that new strain-free grains form continuously during deformation; cold working is carried out below it, usually at room temperature, so the deformed grains remain and the metal strain-hardens. In hot working the flow stress is low, so large deformations are possible with smaller forces and equipment, porosity is closed up, coarse cast grains are refined and directional grain flow is obtained; however, surface oxidation (scale) occurs, finish and dimensional accuracy are poor, and handling of hot metal is difficult. Cold working requires greater forces and gives limited deformation before the metal must be annealed, but it produces good surface finish, close tolerances and higher strength and hardness through strain hardening, at the expense of ductility; residual stresses and anisotropy may result. Examples: hot rolling of billets and hot forging of crankshafts versus cold rolling of sheets, wire drawing, cold heading of bolts and

coining. Note that whether a process is hot or cold depends on the material – working lead at room temperature is hot working because its recrystallisation temperature is below room temperature.

CHAPTER 17B

Presses, Press Tools and High-Energy-Rate Forming

17B.1 Presses

Classification	Types / remarks
Source of power	Manual (fly press), mechanical (crank, eccentric, toggle, knuckle joint, screw/friction), hydraulic (slow, full force throughout stroke – deep drawing, large forging), pneumatic, servo-electric
Frame	C-frame / gap-frame (open on three sides, easy access; deflects under load), straight-side (high rigidity), open-back inclinable (OBI – scrap falls behind), arch, horn
Number of slides	Single action (one slide), double action (blank holder + punch – deep drawing), triple action
Drive	Flywheel-driven crank presses (fast, force maximum near bottom of stroke); toggle and knuckle-joint presses (very high force near the bottom – coining, embossing)

- Press specification: capacity (tonnage at specified distance above bottom dead centre), stroke length, strokes per minute, shut height (distance between bed and slide at bottom of stroke with stroke down and adjustment up), bed size, throat depth.
- Mechanical crank presses are faster and suited to blanking and piercing; hydraulic presses give controllable speed and full force over the whole stroke, suited to deep drawing and forming.

Flywheel energy released: $E = \frac{1}{2} I(\omega_1^2 - \omega_2^2) = I\,\omega^2\,C_s$ (energy for one stroke = $F \times t \times$ penetration)

★ **PYQ-TREND:** Hydraulic press for deep drawing (full force over stroke); knuckle-joint press for coining; shut height definition; C-frame vs straight-side presses; double-action press for deep drawing.

17B.2 Die Elements and Die Types

Element	Function
Punch and die	Cutting/forming members (tool steel, carbide)
Stripper	Strips the sheet off the punch on the return stroke (fixed or spring-loaded)
Stock guide and stop	Locate and position the strip
Pilots	Locate the strip precisely in progressive dies using pierced holes
Die set (punch holder, die shoe, guide posts and bushes)	Maintains alignment

Element	Function
Knock-out / ejector	Removes the blank from the die
Blank holder (pressure pad)	Controls metal flow and prevents wrinkling in drawing
Die type	Description
Simple die	One operation per stroke
Progressive (follow-on) die	Several operations at successive stations as the strip advances; part completed at the last station each stroke; pilots for accuracy; high production
Compound die	Two or more cutting operations at one station in one stroke (e.g. blanking + piercing of a washer) – high accuracy and flatness
Combination die	Cutting and non-cutting (e.g. blanking + drawing) at one station in one stroke
Transfer die	Separate stations; part transferred mechanically between them
Inverted die	Die on the ram and punch on the bed

- Strip layout: arrange blanks to minimise scrap; scrap bridge ≈ 1–1.5 t; material utilisation = blank area × number/strip area.
- Centre of pressure: the point where the resultant cutting force acts; the press ram axis should pass through it to avoid tilting (found like a centroid of cut perimeters).

```
Strip utilisation η = (area of blanks)/(area of strip consumed) × 100
```

★ **PYQ-TREND:** Progressive vs compound vs combination dies (very frequent); stripper function; pilots in progressive dies; centre of pressure; compound die for washers; strip layout to reduce scrap.

17B.3 High-Energy-Rate Forming (HERF) and Other Processes

Process	Principle	Uses
Explosive forming	Shock wave from explosive (in water) forces sheet into die – standoff or contact method	Large parts in small quantities – dish ends, aerospace panels
Electrohydraulic forming	Spark discharge between electrodes in water creates shock wave	Small parts, tubes
Electromagnetic (magnetic pulse) forming	Capacitor discharge through coil induces eddy currents in conductive workpiece; repulsive magnetic force deforms it	Crimping, swaging of tubes onto fittings, joining (Al, Cu)
Rubber-pad (Guerin) forming	Rubber pad acts as female die	Shallow aircraft parts, low tooling cost
Hydroforming (sheet and tube)	High-pressure fluid forms sheet/tube into die	Automobile frames, exhaust parts, bicycle frames
Stretch forming	Sheet gripped and stretched over form block	Aircraft skins, curved panels
Spinning and flow forming	Rotating blank pressed over mandrel	Utensils, reflectors, rocket nose cones

Process	Principle	Uses
Roll forming (cold roll forming)	Strip passed through successive roll stands	Long profiles – channels, gutters, door frames
Incremental sheet forming	CNC tool locally deforms sheet	Prototypes
Superplastic forming	Fine-grained alloys at elevated temperature elongate > 200%	Complex aerospace parts (Ti, Al alloys)

- Advantages of HERF: very high deformation velocities (up to ≈ 300 m/s), forming of difficult or large parts with low-cost single dies, reduced springback; limitations: safety, skilled handling, low production rates.

★ **PYQ-TREND:** Explosive forming for large parts; electromagnetic forming for conductive tubes (crimping); hydroforming for auto frames; roll forming for long profiles; superplastic forming; Guerin process uses rubber pad.

✎ **DESCRIPTIVE Q:** Distinguish between progressive, compound and combination dies.

A progressive die has a series of stations arranged along the strip; at each stroke of the press the strip advances by one pitch and a different operation – piercing, notching, bending, blanking – is performed at each station simultaneously, so a finished part is produced every stroke after the first few. Pilots entering pre-pierced holes register the strip accurately. It suits high-volume production of small parts, but accumulated pitch errors can affect accuracy and the die is long and costly. A compound die performs two or more cutting operations, such as blanking the outer profile and piercing the holes of a washer, at a single station in one stroke; because all features are cut simultaneously, concentricity and flatness are excellent, but the die is complex and the part must be ejected from within the die. A combination die also works at a single station in one stroke but combines a cutting operation with a non-cutting operation, typically blanking followed immediately by drawing or bending (e.g. a shallow cup). In short, progressive dies spread operations over several stations; compound dies combine cutting operations at one station; combination dies combine cutting and forming at one station.

CHAPTER 18

Theory of Metal Cutting

18.1 Cutting Tools and Geometry

- Single-point tools (lathe, shaper, planer, boring) and multi-point tools (milling cutters, drills, broaches, grinding wheels).
- Orthogonal cutting: cutting edge perpendicular to the cutting velocity (2-D, two force components; e.g. parting-off, tube end turning). Oblique cutting: edge inclined at an inclination angle (3-D, three forces; most practical turning) – chip flows sideways, longer tool life.
- Tool signature (ASA/ANSI system) of a single-point tool, in order: back rake – side rake – end relief – side relief – end cutting edge angle – side cutting edge angle – nose radius, e.g. 8-14-6-6-6-15-1 (angles in degrees, nose radius in mm or inch). ORS (orthogonal rake system, ISO): inclination angle – orthogonal rake – orthogonal clearance – auxiliary clearance – auxiliary cutting edge angle – approach angle – nose radius.

Angle	Function
Rake angle (positive/negative)	Positive rake: lower cutting force, better finish, weaker edge – soft/ductile materials, HSS. Negative rake: strong edge, higher force – carbide/ceramic tools, hard materials, interrupted cuts
Clearance (relief) angle	Prevents rubbing of flank on work (≈ 5–8°)
Side cutting edge (approach) angle	Spreads the cut over a longer edge, thinner chip; increases radial force
End cutting edge angle	Clearance between end edge and machined surface
Nose radius	Strengthens tip and improves surface finish; too large causes chatter

Ideal surface roughness (sharp tool): $h = f/(\tan \psi + \cot \kappa)$; with nose radius: $h_{max} = f^2/(8R)$

★ **PYQ-TREND:** Tool signature order (very frequent); positive rake for soft materials and negative rake for carbides/hard materials; orthogonal vs oblique cutting; roughness $f^2/8R$ – larger nose radius and smaller feed improve finish.

18.2 Chip Formation

Chip type	Conditions	Remarks
Continuous	Ductile materials, high speed, large positive rake, small feed, sharp tool, good cutting fluid	Best finish; long ribbons need chip breakers
Continuous with built-up edge (BUE)	Ductile materials at low speed, high friction, small rake	Welded work material on tool tip – poor finish, changes geometry; reduced by higher speed, larger rake, coolant

Chip type	Conditions	Remarks
Discontinuous (segmental)	Brittle materials (cast iron, brass), very low speed, small rake, large feed	Easy disposal, acceptable finish in brittle materials
Serrated (non-homogeneous, saw-tooth)	Low-thermal-conductivity metals (titanium, hardened steel) at high speed	Cyclic shear localisation

- Chip breakers (groove or step on rake face, or clamped obstruction) curl and break continuous chips for safety and disposal.
- Shear zones: primary (shear plane – deformation), secondary (chip–tool interface – friction), tertiary (flank–work interface).

★ **PYQ-TREND:** Continuous chips from ductile materials at high speed; discontinuous chips in cast iron; BUE at low speed (very frequent); chip breakers; BUE reduced by higher speed and rake.

18.3 Mechanics of Orthogonal Cutting (Merchant)

Chip thickness ratio $r = t/t_c = L_c/L = V_c/V$ (< 1); cutting ratio $1/r$ = chip reduction coefficient ζ

Shear angle: $\tan\varphi = r\cos\alpha/(1 - r\sin\alpha)$

Shear strain $\gamma = \cot\varphi + \tan(\varphi - \alpha)$; shear strain rate $= V\cos\alpha/(\cos(\varphi - \alpha)\,\Delta y)$

Velocities: chip velocity $V_c = V\sin\varphi/\cos(\varphi - \alpha) = rV$; shear velocity $V_s = V\cos\alpha/\cos(\varphi - \alpha)$

Merchant's circle: $F = F_c\sin\alpha + F_t\cos\alpha$ (friction force); $N = F_c\cos\alpha - F_t\sin\alpha$; $\mu = \tan\beta = F/N = (F_t + F_c\tan\alpha)/(F_c - F_t\tan\alpha)$

Shear force $F_s = F_c\cos\varphi - F_t\sin\varphi$; normal on shear plane $N_s = F_c\sin\varphi + F_t\cos\varphi$; shear stress $\tau_s = F_s\sin\varphi/(b\,t)$

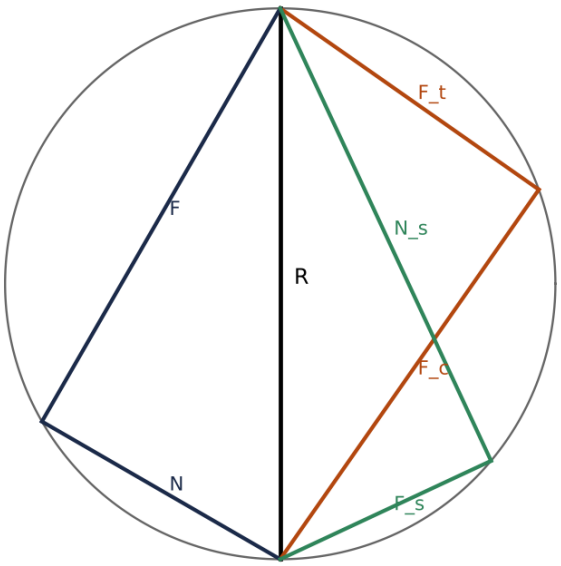
Merchant's first solution (minimum energy): $2\varphi + \beta - \alpha = 90^\circ \rightarrow \varphi = 45^\circ + \alpha/2 - \beta/2$

Lee–Shaffer: $\varphi = 45^\circ + \alpha - \beta$

Cutting power $P = F_c V$; specific cutting energy $u = F_c/(b\,t)$ (J/mm³); material removal rate $MRR = V\,f\,d$ (turning)

- Merchant's circle diagram represents the cutting force system: the resultant R is the diameter; components F_c (cutting/tangential, along velocity), F_t (thrust/feed), F_s and N_s (on shear plane), F and N (on rake face). Assumptions: tool sharp, no side flow (plane strain), continuous chip without BUE, shear on a single plane, uniform stresses, constant speed, orthogonal cutting.
- Cutting force components in turning (dynamometer): tangential (cutting) force F_c (largest, determines power), axial (feed) force, radial (thrust) force. Measured by strain-gauge or piezoelectric dynamometers.
- Heat in metal cutting: shear zone (≈ 60–80%), tool–chip interface (≈ 15–35%), flank–work (≈ 5%). Most heat (≈ 80%) is carried away by the chip; temperature is maximum on the rake face a little away from the cutting edge (where crater wear forms).

Merchant's circle diagram (schematic)



Resultant R is the diameter; F_c–F_t (measured), F–N (rake face), F_s–N_s (shear plane) are perpendicular pairs. μ = F/N = tanβ

★ **PYQ-TREND:** Chip thickness ratio and shear angle formula (very frequent); Merchant's relation 2φ + β − α = 90°; Merchant circle force components; F_c largest force; most heat goes into the chip; specific energy; MRR = V f d; assumptions of Merchant theory.

✔ **TOPPER TRICK:** Chip thickness ratio r is always less than 1 (chip is thicker than uncut layer); higher r means thinner chip, larger shear angle, lower force – better machining.

18.4 Tool Wear, Tool Life and Machinability

- Tool wear mechanisms: abrasion (hard particles), adhesion (welding and tearing), diffusion (atoms migrate at high temperature – crater wear on carbides), oxidation, chemical, plastic deformation of the edge, chipping/fracture.

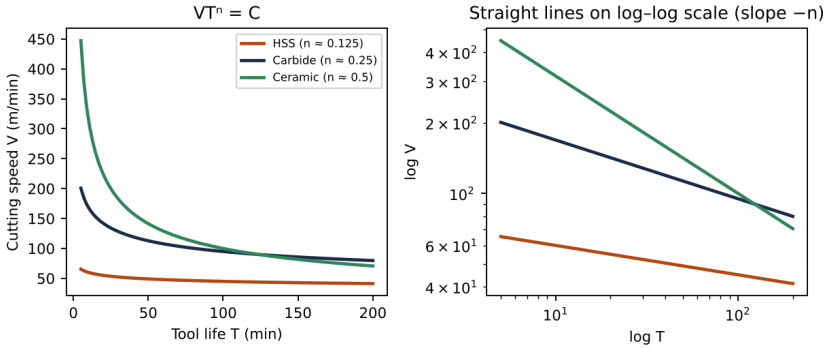
Wear type	Location / cause
Flank wear (wear land VB)	On flank due to rubbing on machined surface; abrasion; used as tool-life criterion (VB ≈ 0.3 mm uniform or 0.6 mm max for carbide, ISO)
Crater wear	On rake face, slightly away from edge; high temperature diffusion; at high speeds
Notch wear	At depth-of-cut line – work-hardened surface, scale
Nose wear, edge chipping	Interrupted cuts, brittle tools

```
Taylor's tool life equation: V Tⁿ = C (V in m/min, T in min)
Extended Taylor: V Tⁿ f^a d^b = K (feed has more effect than depth on tool life)
```

- Typical Taylor exponent n: HSS ≈ 0.08–0.2 (0.125), carbide ≈ 0.2–0.5 (0.25), ceramics ≈ 0.5–0.7. Larger n means tool life is less sensitive to speed.
- Speed affects tool life most, then feed, then depth of cut.

Speed for minimum cost: T_opt = (1/n − 1)(C_t/C_m + t_c) (C_t = tool cost per edge, C_m = machine + labour cost per minute, t_c = tool change time); maximum production rate: T_opt = (1/n − 1) t_c

- Tool life for maximum production rate is shorter (higher speed) than for minimum cost.
- Machinability: ease of machining measured by tool life, surface finish, cutting forces/power, chip form; machinability index = (cutting speed for 20 min tool life of material/that of free-cutting steel AISI 1112) × 100. Improved by adding S, Pb (free-cutting steels and brasses), annealing/spheroidising; reduced by hardness, work hardening (austenitic stainless), abrasive inclusions.



★ **PYQ-TREND:** Taylor's equation $VT^n = C$ and typical n values (very frequent); flank wear criterion and crater wear on rake face; speed most influential on tool life; diffusion wear in carbides; machinability index base AISI 1112; tool life for minimum cost vs maximum production.

18.5 Tool Materials and Cutting Fluids

Tool material	Hot hardness (approx. max temp.)	Remarks
Carbon tool steel	≈ 200–250 °C	Cheap; hand tools, low speeds
High-speed steel (HSS)	≈ 600 °C	Tough; drills, taps, milling cutters, form tools; can be reground
Cast cobalt alloys (Stellite)	≈ 800 °C	Co–Cr–W; between HSS and carbides
Cemented carbides (WC + Co binder)	≈ 1000 °C	Sintered inserts; P (steel, blue), M (stainless, yellow), K (cast iron, red) grades; WC-Co for cast iron; TiC/TaC added for steel to resist cratering
Coated carbides (TiN, TiC, Al_2O_3, TiCN)	≈ 1000+ °C	Longer life; CVD/PVD coatings; TiN gold colour
Cermets (TiC/TiN + Ni/Mo)	≈ 1100 °C	Finishing of steels
Ceramics (Al_2O_3, Si_3N_4, SiAlON)	≈ 1200 °C	Very hard, brittle; high-speed finishing of cast iron and hardened steels; negative rake
Cubic boron nitride (CBN)	≈ 1400 °C	Second hardest; hardened steels, superalloys (does not react with iron)
Diamond (PCD, natural)	≈ 700–800 °C in air (reacts with iron)	Hardest; non-ferrous metals, plastics, composites, ceramics; not for steel

- Order of hardness: diamond > CBN > ceramics > cermets > carbides > cast alloys > HSS > carbon steel. Order of toughness is the reverse.
- Cutting fluids: functions – cooling, lubrication, flushing chips, preventing BUE and corrosion, improving finish and tool life. Types – water-based soluble oils (emulsions, mainly cooling, high-speed cutting), straight cutting oils (mineral/fatty oils, lubrication at low speed – threading, broaching, gear cutting), synthetic/semi-synthetic fluids, extreme-pressure (EP) additives (S, Cl, P) for heavy cuts. Cast iron and brass are usually machined dry (or with air blast); minimum quantity lubrication (MQL) and cryogenic cooling are modern methods.

★ **PYQ-TREND:** Order of hot hardness of tool materials (very frequent); diamond not used for ferrous metals (graphitisation/reaction with iron); CBN for hardened steels; HSS 18-4-1; carbide grades P, M, K; ceramics need negative rake; cast iron machined dry; soluble oils for cooling.

✎ **DESCRIPTIVE Q:** Explain Merchant's circle diagram and state its assumptions.

In orthogonal cutting with a continuous chip, the chip is in equilibrium under two equal and opposite resultant forces: one from the tool along the rake face and one from the workpiece along the shear plane. Merchant represented these forces on a circle whose diameter is the resultant force R. The resultant can be resolved into three pairs of components, all of which can be shown in the same circle: (1) the cutting force F_c (along the cutting velocity) and the thrust force F_t (perpendicular to it), which are measured with a dynamometer; (2) the friction force F along the rake face and the normal force N perpendicular to it, giving the coefficient of friction $\mu = F/N = \tan\beta$; and (3) the shear force F_s along the shear plane and the normal force N_s on it, from which the shear stress $\tau = F_s/A_s$ is obtained. The angles involved are the rake angle α, the shear angle φ and the friction angle β. Using the minimum-energy principle, Merchant derived $2\varphi + \beta - \alpha = 90°$, which predicts that a larger rake angle and lower friction increase the shear angle, giving thinner chips and lower forces. Assumptions: sharp tool with no flank contact, plane-strain (no side flow), continuous chip without built-up edge, shear occurring on a single thin plane, uniform stress distribution on the shear plane, constant cutting velocity and a perfectly plastic work material.

CHAPTER 19

Machine Tools – Lathe, Drilling, Shaper, Planer, Milling and Others

19.1 Lathe

- Main parts: bed (cast iron, with V and flat guideways), headstock (spindle, speed gearbox, live centre), tailstock (dead centre, drilling, reaming; set over for taper turning), carriage (saddle, cross-slide, compound rest, tool post, apron), feed rod (turning feeds), lead screw (thread cutting only, Acme thread, engaged by half-nut/split nut), quick-change gearbox.
- Specification: swing over bed (maximum diameter that can be turned), distance between centres (maximum length), bed length, spindle bore, power, speed range.
- Lathe types: engine (centre) lathe, bench lathe, speed lathe (wood/polishing), tool-room lathe (precision), capstan and turret lathes (semi-automatic – multiple tools, batch production), automatic lathes (single and multi-spindle; Swiss-type for small precise parts), CNC lathe (turning centre).
- Capstan vs turret: capstan has a ram-type tool head moving on a saddle that is clamped – light, bar work, smaller jobs; turret head mounted directly on a saddle that moves on the bed – heavy, chuck work, longer travel.
- Work-holding: three-jaw self-centring chuck (round/hexagonal), four-jaw independent chuck (irregular/eccentric), collet (bar stock, accurate), face plate (irregular shapes), between centres with lathe dog/carrier, mandrel (hollow parts on centres), steady rest (fixed to bed – long slender jobs) and follower rest (moves with carriage).

Operation	Remarks
Facing	Tool moved across the end (cross feed)
Plain/step turning	Tool moved parallel to axis
Taper turning	Compound rest swivel (short steep tapers), tailstock set-over (long gentle tapers, between centres), taper-turning attachment, form tool, combined feeds
Thread cutting	Lead screw with change gears; single-point tool ground to thread form
Knurling	Diamond/straight pattern rolled (not cut) on handles – low speed
Chamfering, grooving, parting-off, drilling, boring, reaming, tapping, spinning, eccentric turning	–

Cutting speed $V = \pi DN/1000$ (m/min, D in mm, N in rpm)

Machining time (turning) $t = L/(f\,N)$ per pass (L = length including approach and overtravel)

MRR (turning) $= \pi\, D_{avg}\, d\, f\, N \approx V\, f\, d$

Taper: $\tan(\alpha/2) = (D - d)/(2L)$; tailstock set-over $S = (D - d)\, L_{total}/(2L_{taper})$ (whole job length L_{total})

Thread cutting: (driver teeth/driven teeth) = pitch of work/pitch of lead screw

Depth of cut in turning $d = (D_1 - D_2)/2$

★ **PYQ-TREND:** Lead screw used only for thread cutting (very frequent); lathe specification (swing, distance between centres); tailstock set-over for long tapers; compound rest for short steep tapers; capstan vs turret; steady rest vs follower rest; four-jaw chuck for irregular work; knurling is forming not cutting; change gear ratio = work pitch/lead screw pitch.

✖ **COMMON MISTAKE:** Feed in turning is in mm per revolution, while feed in milling is usually expressed per tooth or per minute – do not mix them in time calculations.

19.2 Drilling, Boring and Reaming

- Drilling machines: portable, sensitive (bench – hand feed, small holes), upright (pillar/column), radial (arm swings and head moves – large heavy work, holes at different locations without moving the job), gang (several spindles in a row – sequence of operations), multi-spindle (many holes simultaneously), deep-hole (gun drilling).
- Twist drill: point angle 118° for general work (larger for hard materials, ≈ 90° or smaller for soft materials/plastics); helix angle ≈ 24–30°; lip clearance 8–12°; chisel edge (zero/negative rake, does not cut but extrudes – responsible for most thrust); web thickness increases towards shank; flutes remove chips. Tapered shank (Morse taper) or straight shank.

Operation	Purpose
Drilling	Producing a hole
Reaming	Finishing and sizing a drilled hole accurately (small stock removal ≈ 0.1–0.4 mm)
Boring	Enlarging and correcting hole location/straightness with a single-point tool
Counterboring	Enlarging the end of a hole cylindrically for bolt heads
Countersinking	Conical enlargement for flat-head screws
Spot facing	Smoothing the surface around a hole for seating
Tapping	Internal threads with a tap (taper, plug, bottoming taps)
Trepanning	Cutting an annular groove to produce a large hole leaving a core

Drilling time $t = (L + 0.3D)/(f\,N)$; $MRR = (\pi D^2/4)\,f\,N$; cutting speed at periphery $V = \pi DN/1000$

Tap drill size ≈ major diameter − pitch (metric threads)

- Boring machines: horizontal, vertical (vertical turret lathe for large heavy parts), jig boring (precise hole location for jigs and dies).

★ **PYQ-TREND:** Point angle of twist drill 118° (very frequent); radial drilling machine for large jobs; reaming finishes holes; counterboring vs countersinking; chisel edge produces thrust; trepanning; drilling time formula; tap drill size.

19.3 Shaper, Planer and Slotter

Machine	Motion	Use
Shaper	Tool reciprocates (ram), work fed intermittently	Small flat surfaces, slots, keyways; quick-return mechanism (crank and slotted lever, Whitworth, hydraulic)
Planer	Work (table) reciprocates, tool fed	Large heavy flat surfaces – machine beds, guideways; double-housing and open-side planers
Slotter	Vertical shaper – ram moves vertically	Internal keyways, splines, internal profiles

Quick return ratio = time of cutting stroke/time of return stroke = (360° − θ)/θ (θ = crank angle for return) – typically 3 : 2

Average cutting speed (shaper) V = L N (1 + m)/1000 m/min (m = return/cutting time ratio, N = strokes/min)

Machining time = B/(f N) (B = width of job, f = feed per stroke)

- Shaper is specified by maximum stroke length; planer by table size (length × width) and height under the cross-rail.
- Clapper box on the shaper tool head allows the tool to lift on the return stroke to avoid rubbing.

★ **PYQ-TREND:** Quick return mechanism in shaper (crank and slotted lever) (very frequent); planer for large work – work reciprocates; slotter for internal keyways; clapper box; shaper specified by stroke length; quick return ratio.

19.4 Milling

- Milling: multi-point rotating cutter removes material while the work is fed. Machines: column-and-knee (horizontal, vertical, universal – table swivels for helical milling), bed-type (production), planer-type (plano-miller, large work), special (thread, gear, profile), CNC machining centres.
- Up (conventional) milling: cutter rotation opposes feed direction – chip thickness starts at zero and increases; tends to lift the work; rubbing at start causes more tool wear; safer on machines with backlash; preferred for castings/forgings with hard scale. Down (climb) milling: cutter rotation in the feed direction – chip starts thick; presses work down onto the table; better finish and tool life, less power; requires backlash eliminator; preferred on CNC machines.

Cutter	Use
Plain (slab)	Flat horizontal surfaces
Side and face; straddle and gang milling	Steps, slots; two/multiple surfaces at once
End mill, slot drill	Slots, pockets, profiles (vertical milling)
Face mill	Large flat surfaces
Form cutters (convex, concave, gear)	Profiles; involute gear cutters
T-slot, dovetail, Woodruff key cutter	Special slots
Slitting saw	Cutting off, narrow slots
Fly cutter	Single-point cutter in an arbour – economical

Cutting speed V = πDN/1000; table feed $f_m = f_z \times Z \times N$ (mm/min)

Milling time t = (L + approach)/f_m; approach (slab milling) = √(d(D − d)); face milling (centred) = D/2 approx.

MRR = w d f_m; maximum uncut chip thickness (slab) t_max = 2 f_z √(d/D)

Indexing (dividing head, 40 : 1 worm ratio)

Simple indexing: crank turns = 40/N (N = number of divisions); fractional turns using holes in index plate circles (Brown & Sharpe/Cincinnati plates)

Direct (rapid) indexing: 24-hole plate on spindle — divisions that are factors of 24

Angular indexing: crank turns = θ°/9 (one crank turn = 9°)

Differential indexing for numbers not possible by simple indexing (e.g. 61, 127): gear train between spindle and index plate

★ **PYQ-TREND:** Up vs down (climb) milling (very frequent); simple indexing 40/N; angular indexing θ/9; straddle and gang milling; universal milling machine for helical gears; table feed = f_z Z N; T-slot and Woodruff cutters; down milling needs backlash eliminator.

✔ **TOPPER TRICK:** Indexing example: N = 24 → 40/24 = 1 2/3 turns = 1 full turn + 12 holes on an 18-hole circle.

19.5 Broaching, Gear Cutting and Other Processes

- Broaching: multi-tooth tool (broach) with progressively increasing tooth height (rise per tooth) pulled or pushed through or over the work; roughing, semi-finishing and finishing teeth in one pass. Internal (keyways, splines, polygonal holes) and surface broaching. Very fast and accurate for mass production; broach costly and specific to one shape.

Gear manufacturing	Method / remarks
Form milling (gear milling)	Form cutter on milling machine with indexing; set of 8 cutters for each module; low accuracy, jobbing
Gear shaping (generating)	Pinion-shaped cutter reciprocates and rotates with blank; internal gears, cluster gears
Gear hobbing (generating)	Hob (worm-like cutter) rotates in mesh with rotating blank – fastest and most accurate generating method for spur and helical gears, worm wheels; cannot cut internal gears
Gear planing (rack cutter)	Sunderland/Maag process
Bevel gear generators	Gleason machines
Gear finishing	Shaving (before hardening), grinding and lapping/honing (after hardening), burnishing
Gear forming without cutting	Rolling, casting, sintering, extrusion, stamping

- Thread production: single-point turning, die-head chasers, taps and dies, thread milling, thread rolling (strongest, mass production, no chip), thread grinding (precise, hardened).

★ **PYQ-TREND:** Hobbing is the fastest generating process and cannot cut internal gears (very frequent); gear shaper cuts internal gears; form milling uses 8 cutters per module; broaching for internal keyways and splines – mass production; gear shaving before hardening, grinding after; thread rolling strongest threads.

✎ **DESCRIPTIVE Q:** Compare up milling and down milling.

In up (conventional) milling the cutter rotates against the direction of the table feed. Each tooth begins cutting at zero chip thickness, rubs and slides before it bites, and the chip thickness

increases to a maximum at exit. The cutting force tends to lift the workpiece off the table, so stronger clamping is needed; the rubbing causes more tool wear and a poorer finish, and chips are carried into the cutting zone. However, because the feed force opposes the table motion, backlash in the feed screw is taken up, making it safe on conventional machines, and it is preferred for work with hard scale (castings, forgings) since the tooth starts under the surface. In down (climb) milling the cutter rotates in the direction of the feed. Each tooth starts with maximum chip thickness and leaves at zero, so there is no rubbing, giving longer tool life, better finish and lower power; the cutting force presses the work down onto the table, simplifying fixtures, and chips are thrown behind the cutter. But the cutter tends to pull the work into itself, so any backlash in the lead screw causes jerky feed and possible damage – a backlash eliminator (or ball screw as on CNC machines) is essential. Down milling is unsuitable for work with hard surface scale.

CHAPTER 20

Grinding, Finishing and Non-Traditional Machining

20.1 Grinding

- Grinding: material removal by abrasive grains bonded in a wheel; each grain acts as a tiny cutting tool with large negative rake. Gives high accuracy and fine finish; can machine hardened materials; specific energy is very high (most energy goes into heat) – risk of burns and residual stress.

Abrasive	Symbol	Use
Aluminium oxide (alundum)	A	Tough – steels and high-tensile materials
Silicon carbide (carborundum)	C	Harder, brittle – cast iron, brass, aluminium, ceramics, carbides (green SiC)
Cubic boron nitride (CBN)	B	Hardened steels, HSS
Diamond	D	Cemented carbides, glass, stone, ceramics

Wheel specification (IS 551 / ANSI marking order)

Position	Symbol	Meaning
1 (optional)	Prefix number	Manufacturer's abrasive type
2	A / C	Abrasive type
3	8–24 coarse, 30–60 medium, 70–180 fine, 220–600 very fine	Grain (grit) size – mesh number
4	A (softest) to Z (hardest); soft E–G, medium H–P, hard Q–Z	Grade – strength of bond holding grains
5	1 (dense) to 15+ (open)	Structure – spacing of grains
6	V vitrified, B resinoid, R rubber, E shellac, S silicate, M metal	Bond type
7 (optional)	Manufacturer's code	–

- Example: 51 A 46 K 5 V 23 = aluminium oxide, grit 46 (medium), grade K (medium), structure 5, vitrified bond.
- Selection rules: hard material → soft grade wheel (blunt grains drop off and fresh ones appear – self-sharpening); soft material → hard grade wheel. Fine grit for finishing and hard brittle materials; coarse grit for fast removal and soft ductile materials. Large contact area → softer wheel and more open structure. Vitrified bond most common (rigid, porous, unaffected by water); resinoid and rubber for high-speed, cut-off and thin wheels.
- Dressing: restores sharpness (removes glazed/loaded layer) with a diamond dresser or star dresser. Truing: restores the correct geometry/concentricity. Glazing: blunt grains not released (wheel too hard, speed too high) – shiny surface. Loading: pores clogged by metal

chips (soft ductile work, wheel too fine/dense).

Grinding process	Use
Surface grinding	Flat surfaces (horizontal/vertical spindle, reciprocating/rotary table)
Cylindrical grinding (between centres)	External cylindrical surfaces, traverse or plunge
Internal grinding	Bores
Centreless grinding	No centres – work supported on a work-rest blade between grinding wheel and regulating wheel (rubber-bonded, inclined 0–8° for through-feed); pins, rollers, valve stems in mass production; through-feed, in-feed and end-feed methods
Tool and cutter grinding	Sharpening cutters
Creep-feed grinding	Very deep cut with slow feed

`Wheel speed (surface) ≈ 30–35 m/s (vitrified); centreless through-feed rate f = π d_r N_r sin α`

`Grinding ratio G = volume of work removed/volume of wheel wear; specific energy much higher than turning`

★ **PYQ-TREND:** Wheel marking order (very frequent); hard material needs soft wheel; SiC for cast iron and non-ferrous, Al_2O_3 for steels; vitrified bond most common; dressing vs truing; glazing vs loading; centreless grinding with regulating wheel; diamond for carbides.

✔ **TOPPER TRICK:** 'Hard – Soft, Soft – Hard': grind hard work with a soft-grade wheel, soft work with a hard-grade wheel.

20.2 Fine Finishing Processes

Process	Description	Typical roughness Ra (µm)
Honing	Bonded abrasive stones expand and rotate–reciprocate inside a bore – cross-hatch pattern; corrects geometry (roundness, taper); engine cylinder bores	0.1–0.8
Lapping	Loose fine abrasive in a vehicle between work and soft lap (cast iron) – very flat surfaces; gauge blocks, optical flats, valve seats	0.05–0.4
Superfinishing	Fine abrasive stone with short oscillatory strokes under light pressure – removes amorphous surface layer; bearing races	0.01–0.2
Polishing / buffing	Flexible wheels with abrasive (polishing) or fine compound (buffing) – lustre, not accuracy	0.05–0.4
Burnishing	Hard rollers/balls plastically smooth surface – no material removal, improves fatigue strength	0.05–0.4
Electropolishing	Reverse electroplating smooths surfaces	–

- Approximate Ra values: sand casting 12.5–25; forging 3.2–12.5; turning/milling 0.8–6.3; grinding 0.1–1.6; lapping 0.05–0.4.

★ **PYQ-TREND:** Honing for cylinder bores (cross-hatch) (very frequent); lapping uses loose abrasive and gives best flatness (gauge blocks); superfinishing for bearing races; burnishing removes no material; order of finish casting < machining < grinding < honing < lapping.

20.3 Non-Traditional (Unconventional) Machining

- Used when the material is very hard/brittle, the shape complex, the hole very small or deep, or when stress-free, burr-free machining is required. Energy used: mechanical, electrical/thermal, chemical/electrochemical.

Process	Energy / principle	Tool / medium	Materials and applications
Abrasive jet machining (AJM)	Mechanical – high-velocity (150–300 m/s) jet of fine abrasive (Al_2O_3, SiC, 10–50 µm) in gas (air, N_2, CO_2)	Nozzle (tungsten carbide/sapphire)	Hard brittle materials – glass, ceramics; deburring, frosting, cutting thin sections; not for ductile metals
Water jet / abrasive water jet (WJM/AWJM)	Very high-pressure water (≈ 200–400 MPa), abrasive (garnet) added for metals	Sapphire/diamond orifice	Cutting soft materials (paper, food, rubber, plastics, composites), with abrasive – metals, stone; no heat-affected zone
Ultrasonic machining (USM)	Mechanical – tool vibrates at 15–30 kHz with amplitude 10–50 µm; abrasive slurry (B_4C, SiC) hammers the work	Soft, tough tool (mild steel, brass) shaped as cavity; magnetostrictive /piezoelectric transducer	Hard brittle non-conductive – glass, ceramics, gems, carbides; cavities, holes of any shape
Electro-discharge machining (EDM, spark erosion)	Thermal – repetitive sparks between tool and work in dielectric (kerosene, paraffin, deionised water); material melted and vaporised	Tool electrode – graphite, copper, brass, Cu–W; tool usually negative (−)	Any electrically conductive material regardless of hardness – dies, moulds, small holes; produces recast layer and HAZ; tool wears (wear ratio)
Wire EDM (WEDM)	EDM with travelling thin wire (brass, 0.05–0.3 mm), deionised water	Wire	Through-cutting complex profiles in dies, punches, extrusion dies
Electrochemical machining (ECM)	Electrolysis (reverse electroplating, Faraday's laws); work is anode (+), tool cathode (−); electrolyte NaCl/$NaNO_3$ flows at high speed	Copper/brass/stainless tool – no wear	Conductive hard materials – turbine blades, complex cavities, deburring; high MRR, stress-free, no tool wear; expensive, electrolyte disposal
Electrochemical grinding (ECG)	ECM + abrasive grinding (≈ 90% material by electrolysis)	Metal-bonded conductive wheel	Carbide tools, burr-free grinding

Process	Energy / principle	Tool / medium	Materials and applications
Chemical machining / chemical milling, photochemical	Chemical etching through mask (maskant) with etchant (acids/alkalis)	No tool	Aircraft panels (weight reduction), printed circuit boards, thin parts
Laser beam machining (LBM)	Thermal – focused coherent light (CO_2, Nd:YAG, fibre) melts/vaporises	No tool, no vacuum needed	Very small holes, cutting sheets, any material including non-metals; marking, welding; poor efficiency; reflective materials difficult
Electron beam machining (EBM)	Thermal – high-velocity electrons (≈ 0.5–0.8 c) focused in vacuum	Electron gun; needs vacuum	Very small holes (25–100 µm) in any material, high aspect ratio; welding; X-ray hazard
Plasma arc machining (PAM)	Thermal – plasma at 10,000–30,000 °C	Plasma torch	Cutting thick metal plates including stainless steel and aluminium; rough finish
Ion beam machining	Ion sputtering	–	Atomic-level finishing

- Material removal mechanism in ECM: anodic dissolution (Faraday's law: m = A I t/(nF)); in EDM: melting and vaporisation by sparks; in USM: brittle fracture by abrasive impact; in AJM: erosion.

ECM MRR (volume/time) = A I/(ρ n F) per unit (F = 96,500 C) — depends on current, not on hardness

EDM: spark energy $E = \frac{1}{2}CV^2$ (RC relaxation circuit); MRR ∝ energy × frequency; surface finish improves with lower energy per spark

★ **PYQ-TREND:** EDM tool materials (graphite, copper) and dielectric (kerosene) (very frequent); ECM based on Faraday's laws, work is anode, no tool wear; USM for hard brittle non-conductors with 20 kHz vibration; AJM for glass/ceramics; LBM and EBM for micro-holes – EBM needs vacuum; PAM for cutting thick plates; chemical milling for aircraft panels; ECM MRR independent of hardness; wire EDM for dies.

✖ **COMMON MISTAKE:** EDM and ECM need an electrically conductive workpiece; USM and AJM are preferred for non-conductive brittle materials such as glass and ceramics.

✎ **DESCRIPTIVE Q:** Explain the working principle of electric discharge machining (EDM) with its advantages and limitations.

In EDM, the workpiece and a shaped tool electrode, both electrically conductive, are submerged in a dielectric fluid (kerosene, paraffin oil or deionised water) and separated by a small gap of about 0.01–0.5 mm. A pulsed DC power supply (RC relaxation circuit or transistorised pulse generator) applies a voltage across the gap; when the voltage exceeds the breakdown strength of the dielectric, a spark discharges through an ionised channel, producing a local temperature of about 8,000–12,000 °C. A tiny amount of work material melts and vaporises, the plasma channel collapses when the pulse ends, and the dielectric flushes away the debris and re-insulates the gap. Thousands of sparks per second gradually erode a cavity that is a mirror image of the tool, which is fed by a servo mechanism to maintain a constant gap. The tool also wears, so tool materials with high melting point and conductivity – graphite, copper,

copper–tungsten – are used. Advantages: any conductive material can be machined regardless of hardness (hardened steels, carbides); complex shapes, small and deep holes and thin sections can be produced with no mechanical forces, burrs or cutting stresses. Limitations: only conductive materials; low MRR; tool wear and need to make shaped electrodes; recast (white) layer and heat-affected zone with micro-cracks on the surface; slight taper and overcut; fire hazard with oil dielectric. Typical uses are press tool dies, forging dies, injection moulds, cooling holes and wire-EDM cutting of punches and dies.

CHAPTER 21

Metrology – Limits, Fits, Gauges and Measuring Instruments

21.1 Limits, Fits and Tolerances

- Nominal (basic) size: size from which limits are derived. Actual size: measured size. Limits: maximum and minimum permissible sizes. Tolerance: difference between upper and lower limits (always positive). Deviation: algebraic difference between a limit and basic size – upper deviation (ES for holes, es for shafts), lower deviation (EI, ei); fundamental deviation is the one nearer the zero line. Allowance: intentional difference between maximum material limits of mating parts (minimum clearance or maximum interference).
- Unilateral tolerance: variation on one side of basic size (e.g. 25 +0.02/+0.00) – preferred for fits; bilateral: on both sides (25 ± 0.02) – used for locational dimensions.

Fit	Condition	Examples
Clearance	Hole always larger than shaft (minimum clearance > 0)	H7/g6 (sliding, close running), H8/f7 (running), H11/c11 (loose)
Transition	May give clearance or interference	H7/k6, H7/n6 – locational; gears, pulleys on shafts with keys
Interference	Shaft always larger than hole	H7/p6 (press fit), H7/s6 (drive/heavy press), H7/u6 (force/shrink fit)

- Hole-basis system: hole lower deviation zero (H hole); shaft varied – preferred because holes are made with standard fixed-size tools (drills, reamers) and checked with fixed plug gauges. Shaft-basis system: shaft upper deviation zero (h shaft) – used when one shaft carries several parts with different fits (e.g. bright drawn bars, line shafts).
- IS 919 / ISO system: 18 grades of tolerance IT01, IT0, IT1 … IT16 (now IT18); 25–28 fundamental deviations designated by letters – capital for holes (A to ZC), small for shafts (a to zc). H hole and h shaft have zero fundamental deviation. Letters a–h give clearance shafts, j–n transition, p–zc interference.

```
Standard tolerance unit i (μm) = 0.45 ∛D + 0.001D (D = geometric mean of diameter step, mm)
IT grades: IT5 = 7i, IT6 = 10i, IT7 = 16i, IT8 = 25i, IT9 = 40i, IT10 = 64i, IT11 = 100i (multiplying by ≈ 1.6 each grade from IT6)
Maximum clearance = upper limit of hole − lower limit of shaft; minimum clearance (allowance) = lower limit of hole − upper limit of shaft
```

- Typical IT grades by process: lapping/honing IT4–IT5; grinding IT5–IT7; reaming/fine turning IT6–IT8; turning/boring IT7–IT10; milling/shaping IT9–IT11; drilling IT10–IT13; casting/forging IT14–IT16.
- Selective assembly: parts sorted into groups and matched to obtain closer fits economically.
- Geometric dimensioning and tolerancing (GD&T): form (straightness, flatness, circularity, cylindricity), orientation (parallelism, perpendicularity, angularity), location (position,

concentricity, symmetry), runout (circular, total), profile. Maximum material condition (MMC) – hole at its smallest, shaft at its largest.

★ **PYQ-TREND:** Hole-basis system preferred and why (very frequent); clearance/transition/interference fit examples (H7/g6, H7/k6, H7/p6); tolerance vs allowance; 18 IT grades; tolerance unit $0.45\sqrt[3]{D} + 0.001D$; fundamental deviation letters; unilateral vs bilateral tolerances; MMC.

✖ **COMMON MISTAKE:** Allowance is the minimum clearance (or maximum interference) between mating parts – it is not the same as tolerance, which belongs to one part.

21.2 Gauges

- Limit gauges (go/no-go): check whether a part is within limits, not its actual size – fast inspection in mass production. Plug gauge – holes (GO end checks the lower limit, NO-GO the upper limit; NO-GO end is shorter). Ring and snap (gap) gauges – shafts (GO checks upper limit, NO-GO lower limit).
- Taylor's principle of gauging: the GO gauge should check all related dimensions (form and size) simultaneously – full form and length of the mating part; the NO-GO gauge should check one dimension at a time (point contact).
- Gauge tolerance ≈ 10% of work tolerance; wear allowance (≈ 5–10% of work tolerance) is provided on GO gauges only, since they wear in use.

Gauge	Use
Slip (gauge) blocks (Johansson)	Length standards; wrung together; grades 00, 0, I, II; sets of 103/112 pieces; made of hardened steel/carbide/ceramic; lapped
Feeler gauge	Thin strips to measure small gaps (valve clearance)
Radius / fillet gauge	Check radii
Screw pitch gauge	Thread pitch
Wire gauge (SWG)	Diameter of wires, thickness of sheets
Ring, plug, snap, taper, thread gauges	Limit gauging
Sine bar	Angles, with slip gauges: $\sin\theta = h/L$ (accurate up to ≈ 45°; L = 100, 200 or 250 mm between roller centres)
Angle gauges	13 blocks combined to build any angle to 3″
Combination set	Square head, protractor head, centre head with rule
Bevel protractor	Angles to 5′ with vernier
Autocollimator	Small angular deviations, straightness, flatness

★ **PYQ-TREND:** Taylor's principle of gauging (very frequent); GO gauge checks maximum material limit and gets wear allowance; plug gauge for holes, snap gauge for shafts; slip gauges wringing; sine bar for angles below 45°; feeler gauge for gaps.

21.3 Linear and Angular Measuring Instruments

Instrument	Least count (typical)	Remarks
Steel rule	0.5 mm	Line measurement

Instrument	Least count (typical)	Remarks
Vernier caliper	0.02 mm (50 divisions)	Outside, inside, depth; LC = 1 MSD − 1 VSD
Micrometer (outside)	0.01 mm (pitch 0.5 mm, 50 divisions)	Ratchet for constant pressure; Abbe's principle satisfied
Vernier height gauge, depth gauge	0.02 mm	Marking, heights
Dial indicator (dial gauge)	0.01 mm	Comparative measurement, runout, alignment
Bore gauge, telescopic gauge	–	Internal diameters
Comparators (mechanical, optical, pneumatic, electrical)	Up to 0.001 mm	Compare with standard (slip gauges)
Optical flat (interferometry)	Half wavelength per fringe (≈ 0.3 µm with He/monochromatic light)	Flatness
Tool maker's microscope, profile projector	–	Thread profiles, small parts, angles
Coordinate measuring machine (CMM)	≈ 1 µm	3-D measurement, touch-trigger probes
Laser interferometer	Nanometre	Machine tool calibration

```
Vernier least count = value of 1 main scale division/number of vernier divisions; micrometer LC = pitch/number of thimble divisions
Optical flat: height difference = n λ/2 (n = number of fringes)
```

- Abbe's principle: the measuring axis should be in line with the axis of the scale; the micrometer obeys it, the vernier caliper does not (hence more error).
- Errors: systematic (instrument, environmental – temperature: standard 20 °C; parallax; zero error) and random. Accuracy: closeness to true value; precision (repeatability): closeness among repeated readings; sensitivity: output change per input change; resolution: smallest detectable change; calibration traceable to national standards.
- Screw thread measurement: effective (pitch) diameter by the two-wire or three-wire method using best-size wire d = (p/2) sec(θ/2) (= 0.577p for metric 60° threads); major diameter by micrometer; pitch by pitch gauge/microscope.

```
Best wire size d_w = (p/2) sec(θ/2); for 60° thread d_w = 0.577p
```

- Gear measurement: tooth thickness by gear tooth vernier caliper; base tangent method; Parkinson gear tester for composite errors.

★ **PYQ-TREND:** Vernier and micrometer least counts (very frequent); micrometer obeys Abbe's principle; optical flat for flatness λ/2 per fringe; three-wire method and best wire size 0.577p; CMM; accuracy vs precision; standard temperature 20 °C; comparators; gear tooth vernier.

21.4 Surface Roughness

- Surface texture: roughness (fine irregularities from the process), waviness (larger spacing – vibration, deflection), lay (direction of dominant pattern), flaws.

```
Ra (centre-line average, CLA) = (1/L)∫|y| dx – most commonly used (µm)
Rq (RMS) = √((1/L)∫y² dx) ≈ 1.11 Ra for sinusoidal profile; Rz = average of five highest peaks and five deepest valleys (ten-point height); Rt / Rmax = peak-to-valley height
```

- Measured by stylus instruments (Talysurf – diamond stylus, electronic amplification; profilometer), optical methods, comparison specimens. Sampling (cut-off) length typically 0.8 mm.
- Roughness grade numbers N1 (0.025 µm) to N12 (50 µm); each step doubles Ra (N6 = 0.8, N7 = 1.6, N8 = 3.2 µm).
- Lay symbols: = parallel, ⊥ perpendicular, X crossed, M multidirectional, C circular, R radial.

★ **PYQ-TREND:** Ra (CLA) definition and unit µm (very frequent); Talysurf stylus instrument; roughness vs waviness vs lay; N-grade numbers; Rq ≈ 1.11 Ra.

✎ **DESCRIPTIVE Q:** Explain the hole-basis and shaft-basis systems of fits and state why the hole-basis system is preferred.

In a system of fits, the variety of fits needed between mating parts is obtained by keeping the limits of one member fixed and varying the other. In the hole-basis system the hole is the basic member: its lower deviation is zero (H hole, e.g. H7), so its minimum size equals the basic size, and different clearance, transition or interference fits are obtained by choosing shafts with different fundamental deviations (e.g. H7/g6 clearance, H7/k6 transition, H7/p6 interference). In the shaft-basis system the shaft is fixed with zero upper deviation (h shaft), and holes of different deviations are used (e.g. G7/h6, K7/h6, P7/h6). The hole-basis system is generally preferred because holes are produced with fixed-size tools such as drills, reamers and broaches and checked with fixed plug gauges, whereas shafts can be turned or ground to any size with the same tool and measured with adjustable instruments; fewer standard tools and gauges are therefore needed. The shaft-basis system is used where a single shaft of uniform diameter (bright drawn bar, line shaft, piston pin) must carry several components with different fits, or when standard bought-out shafts are used.

CHAPTER 22

CNC Machines, Automation, Jigs and Fixtures

22.1 NC and CNC Fundamentals

- Numerical control (NC): machine controlled by coded instructions (originally punched tape). CNC: dedicated computer stores and edits part programs. DNC: direct/distributed numerical control – several CNC machines linked to a central computer. Adaptive control: parameters adjusted automatically in response to measured conditions (forces, temperature).

Classification	Types
Motion control	Point-to-point (positioning – drilling, spot welding, punching), straight-cut, contouring/continuous path (milling, turning of profiles – interpolation)
Control loop	Open loop (stepper motors, no feedback) and closed loop (servo motors with feedback from encoders/linear scales)
Programming dimension	Absolute (G90 – from program zero) and incremental (G91 – from previous point)
Axes	Z-axis along the main spindle; X horizontal (longest travel, perpendicular to Z); Y completes right-hand system; A, B, C rotations about X, Y, Z

- Machine elements: recirculating ball screws (low friction, no backlash, high efficiency ≈ 90%+), linear motion guideways, servo/stepper drives, encoders (rotary, linear scales), automatic tool changer (ATC), automatic pallet changer (APC), tool magazines (drum/chain).
- Machining centre: CNC milling-based machine with ATC performing milling, drilling, boring, tapping in one set-up; turning centre: CNC lathe with turret and live tools.
- Interpolation: linear (G01), circular (G02 clockwise, G03 counter-clockwise), helical, parabolic.

Common G and M codes (ISO 6983 / RS-274)

Code	Function	Code	Function
G00	Rapid positioning	M00	Program stop
G01	Linear interpolation (feed)	M01	Optional stop
G02	Circular interpolation CW	M02	End of program
G03	Circular interpolation CCW	M03	Spindle on CW
G04	Dwell	M04	Spindle on CCW
G17/G18/G19	XY / ZX / YZ plane selection	M05	Spindle stop
G20/G21	Inch / metric input	M06	Tool change

Code	Function	Code	Function
G28	Return to reference point	M07/M08	Coolant on (mist/flood)
G40/G41/G42	Cutter radius compensation cancel/left/right	M09	Coolant off
G43/G49	Tool length compensation +/cancel	M30	End of program and rewind
G90/G91	Absolute / incremental	M98/M99	Subprogram call / return
G94/G95	Feed per minute / per revolution	–	–

- Program words: N (sequence number), G (preparatory), X/Y/Z (coordinates), I/J/K (arc centre offsets), F (feed), S (spindle speed), T (tool), M (miscellaneous). Canned cycles (G81 drilling, G83 peck drilling, G84 tapping) simplify repetitive operations.
- CAD/CAM: part programs generated automatically from CAD models; APT (Automatically Programmed Tools) was an early high-level NC language.

```
Stepper motor: step angle = 360°/steps per revolution; table movement per pulse (BLU) = lead of screw × step angle/360° (with gear ratio)
Number of pulses = distance/BLU; feed rate = pulse frequency × BLU
```

★ **PYQ-TREND:** G00 rapid, G01 linear, G02 CW, G03 CCW (very frequent); M03 spindle CW, M06 tool change, M30 end and rewind; Z-axis along spindle; point-to-point vs contouring; open vs closed loop; ball screws in CNC; G90/G91; DNC; machining centre with ATC.

✔ **TOPPER TRICK:** Spindle codes in order: M03 (on, CW) → M04 (on, CCW) → M05 (stop). Coolant: M08 on → M09 off.

22.2 Automation and Flexible Manufacturing

Concept	Description
Fixed (hard) automation	Dedicated equipment for very high volumes – transfer lines (automobile engine blocks)
Programmable automation	Batch production with reprogramming – CNC machines, robots
Flexible manufacturing system (FMS)	CNC machines + automated material handling (AGVs, conveyors) + central computer – variety of parts in mid-volume
Computer-integrated manufacturing (CIM)	Integration of design, planning, manufacturing and business by computers
Group technology (GT)	Grouping parts with similar design/process features into families (Opitz, MICLASS coding); cellular manufacturing
CAPP	Computer-aided process planning – variant (retrieval) and generative
Industrial robots	Configurations: Cartesian, cylindrical, polar (spherical), articulated (jointed-arm, most common), SCARA (assembly); end effectors – grippers and tools; degrees of freedom (6 for full positioning and orientation)

Concept	Description
Additive manufacturing (3-D printing)	Layer-by-layer building: stereolithography (SLA – photopolymer resin, UV laser), fused deposition modelling (FDM – thermoplastic filament), selective laser sintering (SLS – powder), direct metal laser sintering/melting (DMLS/SLM), laminated object manufacturing (LOM), 3DP (binder jetting)
Industry 4.0	IoT, cyber-physical systems, digital twins, big data in manufacturing

★ **PYQ-TREND:** FMS components; group technology and Opitz coding; robot configurations – SCARA for assembly; SLA uses photopolymer, FDM uses thermoplastic filament (very frequent); transfer lines for mass production; CAPP types; CIM.

22.3 Jigs and Fixtures

- Jig: holds and locates the work and guides the cutting tool (through hardened drill bushes) – used mainly for drilling, reaming, tapping. Fixture: holds and locates the work but does not guide the tool; tool is set by setting blocks/feeler gauges – milling, turning, grinding, welding, assembly, inspection.
- 3-2-1 principle of location: a block has 12 degrees of freedom (6 translations ±X, ±Y, ±Z and 6 rotations); three locating points on the primary face, two on the secondary and one on the tertiary restrict 9 degrees of freedom, and clamping restricts the remaining 3.
- Locators: flat/plane locators, pins (cylindrical, diamond/relieved pins to avoid redundant location), V-blocks (cylindrical work, centres the work), nesting locators, adjustable locators (for rough castings). Locating surfaces should be machined surfaces where possible; avoid redundant location; allow for burrs and chip clearance (relief grooves).
- Clamps: strap clamps, screw clamps, cam clamps (quick action), toggle clamps, hinged, equalising, pneumatic/hydraulic power clamping. Clamping force should act against locators/solid support, not on unsupported areas, and must not distort the work.
- Drill bushes: press-fit (fixed), renewable, slip (quick change for drilling and reaming in sequence), liner bushes.
- Drill jig types: template, plate, leaf (latch), channel, box (tumble – several faces), indexing, pot jig, sandwich, angle-plate.
- Fixture types: milling (with setting block and tenons/keys), turning (on face plate/chuck), welding, assembly, inspection fixtures.

★ **PYQ-TREND:** Jig guides the tool, fixture only holds (very frequent); 3-2-1 principle restricts 9 DOF by locators and 3 by clamps; 12 degrees of freedom; diamond pin avoids redundant location; slip bushes for successive operations; V-block for cylindrical work; setting block in milling fixtures.

✎ **DESCRIPTIVE Q:** Explain the 3-2-1 principle of location used in jig and fixture design.

A workpiece free in space can move in twelve ways – translation in the positive and negative directions along the X, Y and Z axes (6) and rotation clockwise and counter-clockwise about each axis (6). To machine it accurately and repeatedly these freedoms must be removed. In the 3-2-1 principle, the largest machined face (primary datum) rests on three locating points that are as far apart as possible and not in a straight line; they define a plane and restrict five freedoms – downward translation along Z and rotations about the X and Y axes in both directions. Next, the work is pushed against two locators on an adjacent side (secondary datum), which restrict translation in one direction along X (or Y) and the remaining rotations

about Z – three more. Finally, one locator on the third face (tertiary datum) restricts translation in one direction along the third axis. Together the six locating points restrict nine degrees of freedom; the remaining three (translations away from the locators) are restricted by clamping forces that press the work against the locators. Using more points than required (redundant location) on the same surface can cause rocking or distortion and should be avoided, while locators should be positioned on accurate, machined surfaces and placed to resist cutting forces.

CHAPTER 23

Powder Metallurgy, Plastics Processing and Surface Treatments

23.1 Powder Metallurgy (P/M)

- Process sequence: powder production → blending/mixing (with lubricants like zinc stearate) → compaction (pressing in a die to form a 'green' compact) → sintering (heating below melting point of the main constituent in a protective atmosphere, ≈ 0.7–0.9 T_m, to bond particles by diffusion) → secondary operations (sizing, coining, repressing, infiltration, impregnation with oil, heat treatment, machining).

Powder production method	Feature
Atomisation (gas/water)	Molten metal stream broken by high-pressure jets – most common; spherical/irregular particles
Reduction	Metal oxides reduced by H_2 or CO – iron, tungsten, copper powders (spongy)
Electrolytic deposition	High-purity powders (copper, iron)
Carbonyl process	Ni and Fe carbonyls decomposed – very fine, pure powders
Mechanical comminution (milling, crushing)	Brittle materials
Chemical precipitation, condensation	Fine powders

- Powder characteristics: particle size and distribution, shape, flowability, apparent density, compressibility, purity.
- Compaction: uniaxial die pressing (density varies along height due to wall friction – double-action pressing improves uniformity), isostatic pressing (cold CIP and hot HIP – uniform density), powder rolling, powder injection moulding (MIM), powder forging.
- Products and advantages: self-lubricating porous bronze/iron bearings (oil-impregnated), cemented carbide tools (WC–Co), tungsten filaments, metal filters, magnets, gears, cams and small structural parts, friction materials (brake linings combining metals and non-metals), diamond tools, electrical contacts (Cu–W, Ag–W). Advantages: near-net shape, little waste, controlled porosity, combining materials that cannot be alloyed by melting, high-melting metals (W, Mo); limitations: high powder and die cost, size and shape limits (no undercuts), lower strength and ductility due to porosity, uniform density difficult.

★ **PYQ-TREND:** P/M steps compaction → sintering below melting point (very frequent); atomisation most common for powder production; self-lubricating bearings and cemented carbides by P/M; tungsten filaments; HIP gives full density; green compact; infiltration and impregnation.

23.2 Processing of Plastics

Process	Materials / products
Injection moulding	Thermoplastics (also thermosets) – screw plasticises and injects into closed mould; buckets, housings, toys, bottle caps; most widely used for complex parts in mass production
Extrusion	Continuous profiles – pipes, rods, sheets, films (blown film), wire coating
Blow moulding	Hollow thin-walled containers – bottles (PET, HDPE); extrusion and injection-blow, stretch-blow
Compression moulding	Thermosets (bakelite, melamine) – charge placed in heated open mould and compressed; electrical fittings, dinnerware
Transfer moulding	Thermosets – material heated in chamber, forced into closed mould; parts with inserts, electronic encapsulation
Rotational moulding	Large hollow parts – water tanks, kayaks; powder in rotating heated mould
Thermoforming (vacuum/pressure forming)	Heated sheet formed over mould – packaging, trays, refrigerator liners
Calendering	Thick sheets/films through heated rolls – PVC sheets, flooring
Casting, foam moulding	Acrylic sheets; expanded polystyrene, polyurethane foam
Composite processing	Hand lay-up, spray-up, filament winding (pipes, pressure vessels), pultrusion (constant-section profiles), resin transfer moulding

- Thermoplastics can be remelted (injection moulding, extrusion, blow moulding); thermosets cure irreversibly (compression and transfer moulding).
- Injection-moulding defects: short shot (incomplete filling), flash, sink marks (thick sections shrink), warpage, weld lines, voids, burn marks.

★ **PYQ-TREND:** Compression moulding for thermosets (bakelite) and injection moulding for thermoplastics (very frequent); blow moulding for bottles; rotational moulding for water tanks; calendering for sheets; filament winding for pressure vessels; pultrusion; sink marks.

23.3 Surface Treatments and Coatings

Process	Purpose
Electroplating	Thin metal coating (Cr, Ni, Zn, Cu, Sn, Au) by electrolysis – corrosion protection, appearance, wear; work is the cathode
Galvanising (hot-dip)	Zinc coating on steel by dipping in molten zinc – sacrificial protection; buckets, sheets, wires
Electro-galvanising	Zinc by electroplating
Sherardising	Zinc diffused into steel by heating in zinc dust
Tinning	Tin coating – food cans (tin plate)
Anodising	Thick oxide layer on aluminium by electrolysis (work is anode) – corrosion resistance, dyeable
Phosphating (Parkerising), chromating	Conversion coatings – base for paint

Process	Purpose
Metal spraying (thermal spray – flame, arc, plasma, HVOF)	Coating worn parts, wear/heat resistant layers
Hardfacing	Deposit hard wear-resistant alloy (Stellite) by welding
CVD / PVD	Thin hard coatings (TiN, TiC, Al_2O_3) on tools
Calorising	Aluminium diffusion – oxidation resistance at high temperature
Chromising	Chromium diffusion
Shot peening	Compressive residual stress – fatigue strength
Painting, powder coating, enamelling (vitreous)	Protection and appearance

- Corrosion protection principles: barrier coatings, sacrificial (cathodic) protection (zinc, magnesium anodes for pipelines and ships), impressed-current cathodic protection, inhibitors, proper material selection and design.

★ **PYQ-TREND:** Galvanising is zinc coating; anodising for aluminium (work is anode) (very frequent); tin plate for food cans; sherardising zinc diffusion; calorising aluminium diffusion; sacrificial anodes of zinc/magnesium; PVD/CVD TiN coatings on tools.

✎ **DESCRIPTIVE Q:** Describe the powder metallurgy process and state its advantages and applications.

Powder metallurgy produces parts from metal powders instead of molten metal. First, powders are produced, most commonly by atomisation of molten metal by gas or water jets, or by reduction of oxides, electrolysis or the carbonyl process. The powders are blended with alloying additions and a lubricant to give a uniform mix. The mix is then compacted in a precision die by punches at high pressure (100–800 MPa) to form a green compact that has enough strength for handling. The compact is sintered in a furnace with a protective atmosphere at a temperature below the melting point of the main constituent (about 70–90% of the melting temperature), when diffusion bonds the particles together, increasing strength and density. Secondary operations such as sizing or coining (for accuracy), repressing, infiltration with a lower-melting metal, oil impregnation, heat treatment or machining may follow. Advantages: near-net shapes with minimal waste and machining; controlled porosity (self-lubricating bearings, filters); production of parts from refractory metals and from combinations of metals and non-metals that cannot be made by melting (cemented carbides, electrical contacts, friction materials); good dimensional accuracy and high production rate for small parts. Limitations include costly powders and dies, limits on size and shape, and lower strength because of porosity. Typical applications are oil-impregnated bronze bushes, carbide tool inserts, tungsten filaments, gears and cams for automobiles, magnets and brake linings.

CHAPTER 24

Industrial Engineering – Work Study, PPC, Inventory and Quality

24.1 Productivity and Work Study

```
Productivity = output/input; labour productivity = output/man-hours
```

- Work study = method study + work measurement. Its aim is to improve productivity by better methods and to set time standards.

Method study

- Steps (SREDIM): Select – Record – Examine – Develop – Install – Maintain (Define and Evaluate are added in some texts).

Recording technique	Use
Outline process chart	Only operations and inspections
Flow process chart (man/material/equipment type)	All five activities with symbols
Two-handed (operator) chart	Movements of left and right hands
Multiple activity (man–machine) chart	Simultaneous activities on a common time scale – idle time
SIMO chart	Simultaneous motion cycle chart from micromotion (therbligs) film analysis
Flow diagram	Movement plotted on a layout plan
String diagram	Thread on a scale layout – distance travelled for irregular movements
Travel chart	Movements between departments (tabular)
Cyclegraph / chronocyclegraph	Light on hand photographed – path / direction and speed

Symbol (ASME)	Activity
○ (circle)	Operation
⇨ (arrow)	Transport
□ (square)	Inspection
D	Delay (temporary storage)
▽ (triangle)	Permanent storage

- Therbligs (Gilbreth): 17–18 elementary motions (search, select, grasp, transport empty, transport loaded, hold, release load, position, pre-position, inspect, assemble, disassemble, use, unavoidable delay, avoidable delay, plan, rest to overcome fatigue, find). Principles of motion economy relate to use of the human body, workplace arrangement and design of tools.
- Critical examination questions: purpose, place, sequence, person, means (what, why, where, when, who, how) – leading to eliminate, combine, rearrange, simplify.

★ **PYQ-TREND:** Work study = method study + work measurement (very frequent); flow process chart symbols; therbligs by Gilbreth; string diagram for distance; man–machine chart; SIMO chart uses micromotion study; SREDIM steps.

Work measurement

Normal time = observed time × rating factor (%/100)

Standard time = normal time + allowances = normal time × (1 + allowance fraction)

Number of cycles for stop-watch study (95% confidence, ±5% accuracy): $N = [40\sqrt{(N\Sigma x^2 - (\Sigma x)^2)}/\Sigma x]^2$

Work sampling: number of observations $N = 4p(1 - p)/L^2$ (95% confidence, L = accuracy as a fraction)

Technique	Use
Stop-watch time study (Taylor)	Repetitive short-cycle jobs; rating and allowances
Work sampling (activity/ratio delay, Tippett)	Long or non-repetitive jobs, machine utilisation, based on random observations and probability
Predetermined motion time systems (PMTS) – MTM, Work Factor, MOST	Standard times from basic motion data (MTM unit TMU = 0.00001 h = 0.036 s)
Synthesis from standard data	Times built from element data
Analytical estimating	Non-repetitive maintenance work

- Allowances: personal, fatigue, delay (contingency), special (policy). Rating systems: speed rating, Westinghouse (skill, effort, conditions, consistency), synthetic, objective rating. Normal performance = 100 (British Standard scale 0–100).
- Ergonomics (human engineering): designing work to fit the worker – anthropometry, workplace layout, lighting, noise, climate.

★ **PYQ-TREND:** Standard time = normal time + allowances (very frequent); normal time = observed × rating; work sampling for non-repetitive jobs (Tippett); MTM and TMU = 0.036 s; Westinghouse rating; work sampling sample size $4p(1 - p)/L^2$.

24.2 Plant Location and Layout

Layout	Features	Suitable for
Product (line) layout	Machines arranged in sequence of operations	Mass production of standard products – low handling, low WIP, but inflexible; breakdown stops line
Process (functional) layout	Similar machines grouped together	Job-shop/batch production, variety – flexible but more handling and WIP
Fixed-position layout	Product stays, men and machines move to it	Ships, aircraft, locomotives, large turbines, construction
Cellular (group technology) layout	Machine cells for part families	Medium variety and volume
Combination (hybrid)	Mix	–

- Line balancing: assigning tasks to stations so that idle time is minimum. Cycle time C = available time/required output; minimum number of stations = Σt/C; balance efficiency (line efficiency) = Σt/(n × C) × 100; balance delay = 100 − efficiency. Methods: largest candidate rule, Kilbridge–Wester, ranked positional weight (Helgeson–Birnie).

- Layout tools: travel chart, CRAFT, ALDEP, CORELAP (computerised layout).

```
Line efficiency = Σtᵢ/(n × C); smoothness index = √(Σ(C − tᵢ)²)
```

- Plant location factors: raw materials, market, labour, power, water, transport, climate, government policies; methods – factor rating, break-even (locational cost–volume) analysis, centre of gravity method.

★ **PYQ-TREND:** Product layout for mass production, process layout for job shop, fixed-position for ships/aircraft (very frequent); line balancing efficiency Σt/(nC); cycle time = available time/output; CRAFT.

24.3 Production Planning and Control, Forecasting

- PPC functions: planning (forecasting, routing, estimating, loading), scheduling (when), dispatching (issue of orders – release of work), expediting/follow-up (progress chasing), inspection, corrective action. Routing determines the path (sequence of operations); scheduling fixes the time; dispatching gives the go-ahead.
- Production types: job (unit) production, batch production, mass (flow) production, continuous (process) production.

Forecasting method	Description
Qualitative	Delphi method (panel of experts, anonymous rounds), market survey, executive opinion, sales-force composite
Simple moving average	Average of last n periods
Weighted moving average	Recent data given more weight
Exponential smoothing	F_t+1 = F_t + α(D_t − F_t) (α = smoothing constant 0–1; larger α responds faster)
Regression / least squares	Trend line y = a + bx
Errors	MAD (mean absolute deviation), MSE, MAPE, tracking signal = running sum of forecast errors/MAD

- Scheduling rules for n jobs on one machine: SPT (shortest processing time) minimises mean flow time and average number of jobs in system; EDD (earliest due date) minimises maximum lateness; FCFS; critical ratio. Johnson's rule: n jobs on two machines (same order) minimises makespan – select smallest time; if on machine 1 schedule first, if on machine 2 schedule last.
- Gantt chart: bar chart for scheduling and progress. Master production schedule (MPS), MRP (material requirements planning – uses BOM, MPS, inventory status; dependent demand), MRP-II, ERP, JIT (just-in-time – Toyota, kanban, zero inventory), lean manufacturing, Six Sigma (3.4 defects per million opportunities), TQM, Kaizen (continuous improvement), 5S (Sort, Set in order, Shine, Standardise, Sustain), Poka-yoke (mistake proofing), SMED.

★ **PYQ-TREND:** Routing vs scheduling vs dispatching (very frequent); exponential smoothing formula; Delphi method; Johnson's rule for two machines; SPT minimises mean flow time; MRP uses BOM; JIT and kanban; Six Sigma 3.4 DPMO; 5S; Gantt chart.

24.4 Inventory Control

```
EOQ (Wilson/Harris): Q* = √(2DS/H) (D = annual demand, S = ordering cost per order, H = holding cost per unit per year)
```

At EOQ: annual ordering cost = annual holding cost; total cost = √(2DSH); number of orders = D/Q*; cycle time = Q*/D

Economic production (batch) quantity with production rate p and usage d: Q* = √(2DS/(H(1 − d/p)))

Reorder point ROP = demand during lead time + safety stock = d × L + SS

Safety stock = z σ_L (z for desired service level)

Quantity discount: compare total cost at EOQ and at price-break quantities

- ABC analysis (Pareto, always-better control): A items ≈ 10–20% of items, 70–80% of annual value – tight control; B ≈ 30% items, 15–25% value; C ≈ 50–60% items, 5–10% value – loose control. VED (vital, essential, desirable – spares), FSN (fast, slow, non-moving), XYZ (value of stock), HML (unit price), SDE (availability).
- Inventory systems: fixed-order-quantity (Q system, continuous review, reorder point) and fixed-period (P system, periodic review); two-bin system (C items).
- Doubling demand increases EOQ by √2 (≈ 41%).

★ **PYQ-TREND:** EOQ formula and ordering cost = holding cost at EOQ (very frequent); ABC analysis based on annual consumption value; VED for spare parts; reorder point; safety stock; EBQ with production rate; EOQ ∝ √D.

✔ **TOPPER TRICK:** If both ordering cost and demand double, EOQ doubles (√4); if holding cost doubles, EOQ falls to 1/√2.

24.5 Quality Control and Reliability

Control chart	Type of data	Use
$\bar{X}$ chart (mean)	Variables	Central tendency of process
R chart (range) / s chart	Variables	Dispersion/variability
p chart	Attributes – fraction defective	Variable sample size possible
np chart	Attributes – number defective	Constant sample size
c chart	Attributes – number of defects per unit (Poisson)	Defects on a unit – castings, cloth, painted surface
u chart	Defects per unit, variable unit size	–

$\bar{X}$ chart limits: $\bar{\bar{X}} \pm A_2 \bar{R}$; R chart: UCL = $D_4 \bar{R}$, LCL = $D_3 \bar{R}$; control limits normally ±3σ (99.73% of points within)

p chart: $\bar{p} \pm 3\sqrt{\bar{p}(1-\bar{p})/n}$; c chart: $\bar{c} \pm 3\sqrt{\bar{c}}$

Process capability: $C_p = (USL - LSL)/(6\sigma)$; $C_{pk} = \min[(USL - \mu)/3\sigma, (\mu - LSL)/3\sigma]$; $C_p \geq 1.33$ usually required

- Chance (common, random) causes vs assignable (special) causes of variation; a process is in statistical control when only chance causes act. Run of seven points on one side of the centre line or trends signal assignable causes.
- Acceptance sampling: single, double, multiple, sequential sampling plans; operating characteristic (OC) curve shows probability of acceptance vs lot quality; AQL (acceptable quality level), LTPD (lot tolerance percent defective), producer's risk α (good lot rejected, type I error ≈ 5%), consumer's risk β (bad lot accepted, type II error ≈ 10%); AOQ and AOQL (average outgoing quality limit).
- Reliability: R(t) = e^(−λt) for constant failure rate; MTBF = 1/λ; series system R = R_1R_2…; parallel system R = 1 − (1 − R_1)(1 − R_2)…; bath-tub curve – infant mortality (decreasing

failure rate), useful life (constant, random failures), wear-out (increasing). Availability = MTBF/(MTBF + MTTR).

- Quality gurus: Deming (PDCA cycle, 14 points), Juran (quality trilogy, fitness for use), Crosby (zero defects, cost of quality), Ishikawa (fishbone/cause-and-effect diagram, quality circles), Taguchi (loss function, robust design), Shewhart (control charts). Seven QC tools: check sheet, histogram, Pareto chart, cause-and-effect diagram, scatter diagram, control chart, stratification/flow chart. ISO 9001 – quality management systems.

★ **PYQ-TREND:** $\bar{X}$ and R charts for variables, p and c charts for attributes (very frequent); c chart for defects per unit (Poisson); ±3σ limits; producer's risk vs consumer's risk; OC curve; C_p formula; bath-tub curve; series vs parallel reliability; Ishikawa diagram; Deming PDCA; Pareto principle.

24.6 Project Management (PERT/CPM) and Operations Research Basics

Feature	CPM	PERT
Nature	Deterministic (one time estimate)	Probabilistic (three time estimates)
Orientation	Activity-oriented	Event-oriented
Use	Repetitive construction/maintenance projects – time–cost trade-off (crashing)	Research and development, new projects with uncertain times
Distribution	–	Beta distribution for activity time; normal for project time

PERT expected time $t_e = (t_o + 4t_m + t_p)/6$; variance $\sigma^2 = ((t_p - t_o)/6)^2$

Project duration variance = sum of variances of critical activities; $Z = (T_s - T_e)/\sigma_{project}$

Total float = LS − ES = LF − EF; free float = ES of successor − EF of activity; independent float = ES(succ) − LF(pred) − duration

Critical path: longest path through the network; activities with zero total float

Crash cost slope = (crash cost − normal cost)/(normal time − crash time)

- Network rules: dummy activities (zero time, dashed) show dependency or avoid duplicate numbering; no loops; one start and one end event (Fulkerson's rule for numbering).
- Operations research: linear programming (graphical and simplex methods; feasible region; slack/surplus/artificial variables; duality), transportation problem (NWCR, least-cost, Vogel's approximation for initial solution; MODI/stepping-stone for optimality; m + n − 1 allocations for non-degenerate solution), assignment problem (Hungarian method), queuing (M/M/1: utilisation $\rho = \lambda/\mu$, $L_q = \rho^2/(1-\rho)$, $L_s = \rho/(1-\rho)$, $W_s = 1/(\mu - \lambda)$), sequencing, simulation (Monte Carlo), game theory (saddle point), break-even analysis (BEP = fixed cost/(price − variable cost per unit)).

Break-even quantity $Q_{BE} = F/(p - v)$; margin of safety = (actual sales − BEP sales)/actual sales

M/M/1 queue: $L_s = \lambda/(\mu - \lambda)$; $W_q = \lambda/(\mu(\mu - \lambda))$

★ **PYQ-TREND:** PERT three time estimates and beta distribution, CPM deterministic (very frequent); $t_e = (t_o + 4t_m + t_p)/6$; critical path has zero float; crashing in CPM; Vogel's approximation; Hungarian method for assignment; $m + n - 1$ allocations; break-even point; M/M/1 formulas; dummy activity.

✎ **DESCRIPTIVE Q:** Explain ABC analysis in inventory control.

ABC analysis classifies inventory items according to their annual consumption value (annual usage × unit cost), applying the Pareto principle that a small proportion of items accounts for most of the money spent. All items are listed in descending order of annual consumption value and cumulative percentages of items and value are calculated. Class A items are typically 10–20% of the number of items but account for 70–80% of the total consumption value; class B items are about 30% of items and 15–25% of value; class C items are about 50–60% of items but only 5–10% of value. Different degrees of control are then applied: A items receive strict control – accurate records, frequent review, small and frequent orders, low safety stocks, attention of senior staff and close follow-up with suppliers; B items receive moderate control with periodic review; C items receive simple control – bulk ordering, larger safety stocks and methods like the two-bin system. ABC analysis concentrates management effort where it gives the greatest saving, reduces capital locked in inventory and clerical work. Its limitation is that it considers only value, not criticality – so it is often combined with VED analysis for spare parts.

CHAPTER 24B

Maintenance, Safety, Value Engineering and Costing

24B.1 Maintenance

Type	Description
Breakdown (corrective) maintenance	Repair after failure – suitable for cheap, non-critical items
Preventive (time-based) maintenance	Scheduled inspection, lubrication, replacement at fixed intervals – reduces breakdowns
Predictive (condition-based) maintenance	Monitoring condition – vibration analysis, oil analysis, thermography, ultrasonic, motor current – repair when indicators show deterioration
Proactive / reliability-centred maintenance (RCM)	Analyses failure modes and selects best maintenance for each function
Total productive maintenance (TPM)	Operators involved in routine maintenance; aims at zero breakdowns, zero defects; overall equipment effectiveness (OEE)
Opportunistic, shutdown, design-out maintenance	Work during planned stops; redesign to eliminate recurring failure

```
OEE = availability × performance rate × quality rate (world class ≈ 85%)
Availability = MTBF/(MTBF + MTTR)
```

- Replacement analysis: items that deteriorate (machines) are replaced when the average annual cost (capital + running) is minimum; items that fail suddenly (bulbs) – group vs individual replacement.
- Depreciation methods: straight-line (D = (P − S)/n), declining balance (fixed percentage of book value), sum-of-years' digits, sinking fund.

★ **PYQ-TREND:** Preventive vs predictive maintenance (frequent); vibration analysis for condition monitoring; TPM and OEE; straight-line depreciation formula; replacement when average annual cost is minimum.

24B.2 Industrial Safety

- Causes of accidents: unsafe acts (about 85–88% per Heinrich – carelessness, not using guards or PPE, improper lifting) and unsafe conditions (unguarded machines, poor lighting, slippery floors, defective tools). Heinrich's ratio 1 : 29 : 300 (major injury : minor injuries : no-injury accidents).

```
Frequency rate = (number of lost-time injuries × 10⁶)/man-hours worked
Severity rate = (man-days lost × 10⁶)/man-hours worked (in India per IS 3786; some texts use 1000 man-hours)
```

- Personal protective equipment (PPE): helmets, safety goggles and face shields (welding filters), ear plugs/muffs, gloves, safety shoes, aprons, respirators, safety belts/harness.

- Machine guarding: fixed, interlocked, automatic, adjustable guards; emergency stops; lock-out/tag-out during maintenance.

Fire class	Fuel	Extinguisher
A	Ordinary solids – wood, paper, cloth	Water, foam, ABC powder
B	Flammable liquids – oil, petrol	Foam, CO_2, dry chemical powder (never water jet)
C	Flammable gases	Dry chemical powder; isolate supply
D	Metals – Mg, Na, K	Special dry powder (graphite, sand)
Electrical (E in some standards)	Live electrical equipment	CO_2, dry chemical powder (non-conducting)
F/K	Cooking oils/fats	Wet chemical

- Colour codes (IS 9457 typical): red – stop/fire/danger; yellow – caution; green – safety/first aid; blue – mandatory instruction.
- Statutes in India: Factories Act 1948 (now consolidated in the Occupational Safety, Health and Working Conditions Code 2020), Indian Boilers Act 1923 (a new Boilers Act was passed in 2025), Workmen's Compensation Act.

★ **PYQ-TREND:** Fire classes and extinguishers (water not for oil and electrical fires) (very frequent); unsafe acts cause most accidents; Heinrich's ratio; frequency and severity rates; PPE for welding; Factories Act 1948; colour codes.

24B.3 Value Engineering, Costing and Ergonomics

- Value = function/cost (or worth/cost). Value engineering/analysis (L. D. Miles, General Electric) systematically identifies unnecessary cost without reducing function. Phases: information, function analysis, creative (brainstorming), evaluation, development/recommendation, implementation. Types of value: use, esteem, cost, exchange.

Cost element	Meaning
Prime cost	Direct material + direct labour + direct expenses
Factory (works) cost	Prime cost + factory overheads
Cost of production	Factory cost + administrative overheads
Total cost (cost of sales)	Cost of production + selling and distribution overheads
Selling price	Total cost + profit

- Fixed costs (rent, depreciation, salaries) do not vary with output; variable costs (material, direct labour) vary proportionally. Contribution = selling price − variable cost per unit. Profit–volume ratio = contribution/sales.
- Overhead allocation methods: percentage of direct labour cost, direct material cost, prime cost, labour-hour rate, machine-hour rate (most accurate for machine-intensive shops).
- Ergonomics: fitting the job to the worker – anthropometric data (5th to 95th percentile design), workplace and display/control design, environmental factors (illumination ≈ 300–500 lux for general work, noise limit 90 dB(A) for 8 h in India, thermal comfort), reduction of fatigue.

- Wage incentive plans: Halsey (50% of time saved paid), Rowan (bonus = time saved/time allowed × time taken × rate), Taylor's differential piece rate, Gantt task and bonus, Emerson efficiency plan, Bedaux point system.

```
Halsey bonus = (time saved × rate)/2; Rowan bonus = (time saved/time allowed) × time taken × rate
```

★ **PYQ-TREND:** Value = function/cost (very frequent); prime cost = direct material + labour + expenses; machine-hour rate; Halsey and Rowan plans; fixed vs variable cost; noise limit 90 dB; anthropometry 5th–95th percentile.

✔ **TOPPER TRICK:** Cost ladder: Prime → +Factory OH → Works cost → +Admin OH → Cost of production → +Selling OH → Total cost → +Profit → Selling price.

CHAPTER 25

Solved Numericals – Set 1 (Stress, Strain, Principal Stresses, Columns, Cylinders)

SSC-style problems solved step by step (no options). E = 200 GPa = 2 × 10^5 N/mm² for steel unless stated; 1 MPa = 1 N/mm².

25.1 Simple Stress and Strain

Problem 1 – Axial stress and elongation

A steel rod of 20 mm diameter and 2 m length carries 50 kN tension. Find stress and elongation.

$A = \pi/4 \times 20^2 = 314.16$ mm²; $\sigma = 50{,}000/314.16 = 159.2$ MPa

$\delta L = PL/(AE) = 50{,}000 \times 2000/(314.16 \times 2 \times 10^5) = 1.59$ mm

Problem 2 – Stepped bar

A bar has two portions: A_1 = 400 mm², L_1 = 300 mm and A_2 = 200 mm², L_2 = 500 mm. It carries 40 kN. Find total elongation.

$\delta = (P/E)(L_1/A_1 + L_2/A_2) = (40{,}000/2 \times 10^5)(0.75 + 2.5) = 0.2 \times 3.25 = 0.65$ mm

Problem 3 – Tapering circular bar

A bar tapers from 40 mm to 20 mm diameter over 1 m and carries 60 kN. Find elongation.

$\delta = 4PL/(\pi E d_1 d_2) = 4 \times 60{,}000 \times 1000/(\pi \times 2 \times 10^5 \times 40 \times 20) = 2.4 \times 10^8/5.027 \times 10^8 = 0.477$ mm

Problem 4 – Self-weight elongation

A steel bar 100 m long hangs vertically (ρ = 7850 kg/m³). Find elongation due to self-weight.

$\delta = \rho g L^2/(2E) = 7850 \times 9.81 \times 100^2/(2 \times 2 \times 10^{11}) = 7.70 \times 10^8/4 \times 10^{11} = 1.93 \times 10^{-3}$ m = 1.93 mm

Problem 5 – Poisson's ratio from a test

A 30 mm rod under 100 kN extends 0.212 mm on 300 mm gauge length and its diameter decreases by 0.0064 mm. Find E and μ.

$\sigma = 100{,}000/706.86 = 141.5$ MPa; $e = 0.212/300 = 7.07 \times 10^{-4} \rightarrow E = 141.5/7.07 \times 10^{-4} = 200$ GPa

Lateral strain = $0.0064/30 = 2.13 \times 10^{-4}$; $\mu = 2.13/7.07 = 0.30$

Problem 6 – Elastic constants

For a material E = 200 GPa and μ = 0.25. Find G and K.

$G = E/(2(1 + \mu)) = 200/2.5 = 80$ GPa; $K = E/(3(1 - 2\mu)) = 200/1.5 = 133.3$ GPa

Problem 7 – Poisson's ratio from E and G

E = 210 GPa and G = 84 GPa. Find μ and K.

$\mu = E/(2G) - 1 = 1.25 - 1 = 0.25$; $K = 210/(3 \times 0.5) = 140$ GPa

Problem 8 – Volume change under hydrostatic pressure

A 100 mm steel cube is subjected to hydrostatic pressure 50 MPa (K = 166.7 GPa). Find the change in volume.

$e_v = p/K = 50/166{,}700 = 3 \times 10^{-4}$; $\Delta V = 3 \times 10^{-4} \times 10^6 = 300$ mm³ (decrease)

25.2 Thermal, Composite and Impact Problems

Problem 9 – Thermal stress in a rail

Steel rails ($\alpha = 12 \times 10^{-6}$ /°C) are laid 12 m long. Find the stress for a 40 °C rise if (a) fully restrained (b) a 3 mm gap is left.

(a) $\sigma = E\alpha\Delta T = 2 \times 10^5 \times 12 \times 10^{-6} \times 40 = 96$ MPa (compressive)

(b) free expansion $= 12 \times 10^{-6} \times 12{,}000 \times 40 = 5.76$ mm; $\sigma = E(5.76 - 3)/L = 2 \times 10^5 \times 2.76/12{,}000 = 46$ MPa

Problem 10 – Composite bar

A steel rod (A = 314.16 mm², E = 200 GPa) inside a copper tube (A = 600 mm², E = 100 GPa) shares a load of 100 kN. Find stresses.

Equal strain: $\sigma_s = (E_s/E_c)\sigma_c = 2\sigma_c$; $2\sigma_c \times 314.16 + \sigma_c \times 600 = 100{,}000 \rightarrow \sigma_c = 81.4$ MPa, $\sigma_s = 162.8$ MPa

Problem 11 – Suddenly applied load

A load of 50 kN is applied suddenly to a bar of area 500 mm². Find the maximum stress.

$\sigma = 2P/A = 2 \times 50{,}000/500 = 200$ MPa (gradual load gives 100 MPa)

Problem 12 – Impact load

A 1 kN weight falls 100 mm onto a collar at the end of a bar (A = 1000 mm², L = 2 m). Find the maximum stress and extension.

$\sigma = (W/A)[1 + \sqrt{(1 + 2AEh/(WL))}] = 1 \times [1 + \sqrt{(1 + 2 \times 1000 \times 2 \times 10^5 \times 100/(1000 \times 2000))}] = 1 + \sqrt{20{,}001} = 142.4$ MPa

$\delta = \sigma L/E = 142.4 \times 2000/2 \times 10^5 = 1.42$ mm

Problem 13 – Strain energy

A steel bar 25 mm diameter and 1.5 m long is stressed to 150 MPa. Find the strain energy stored.

$V = 490.9 \times 1500 = 7.36 \times 10^5$ mm³; $U = \sigma^2 V/(2E) = 150^2 \times 7.36 \times 10^5/(4 \times 10^5) = 41{,}420$ N·mm = 41.4 J

25.3 Principal Stresses and Failure Theories

Problem 14 – Principal stresses

$\sigma x = 80$ MPa, $\sigma y = -40$ MPa, $\tau xy = 30$ MPa. Find principal stresses, maximum shear and principal plane.

Centre $= (80 - 40)/2 = 20$; $R = \sqrt{(60^2 + 30^2)} = 67.08$ MPa

$\sigma_1 = 87.08$ MPa; $\sigma_2 = -47.08$ MPa; $\tau_{max} = 67.08$ MPa

$\tan 2\theta = 2 \times 30/(80 + 40) = 0.5 \rightarrow \theta = 13.3°$

Problem 15 – Stresses on an oblique plane

A bar carries 100 MPa axial stress. Find σ_n and τ on a plane whose normal makes 30° with the axis.

$\sigma_n = 100 \cos^2 30° = 75$ MPa; $\tau = 50 \sin 60° = 43.3$ MPa

Problem 16 – Combined bending and torsion

A 60 mm shaft carries M = 1.5 kN·m and T = 2 kN·m. Find maximum shear and maximum normal stress.

$T_e = \sqrt{(1.5^2 + 2^2)} = 2.5$ kN·m → $\tau_{max} = 16 \times 2.5 \times 10^6/(\pi \times 60^3) = 58.9$ MPa

$M_e = \frac{1}{2}(1.5 + 2.5) = 2$ kN·m → $\sigma_{max} = 32 \times 2 \times 10^6/(\pi \times 60^3) = 94.3$ MPa

Problem 17 – Factor of safety by two theories

At a critical point $\sigma_1 = 100$ MPa, $\sigma_2 = -50$ MPa; $\sigma_y = 250$ MPa. Find FS by Tresca and von Mises.

Tresca: $\sigma_1 - \sigma_2 = 150$ → FS = 250/150 = 1.67

von Mises: $\sqrt{(100^2 + 50^2 + 100 \times 50)} = \sqrt{17{,}500} = 132.3$ → FS = 250/132.3 = 1.89

25.4 Columns

Problem 18 – Euler load for different ends

A 50 mm solid steel column is 2.5 m long. Find the Euler load for both ends hinged, fixed–free and fixed–fixed.

$I = \pi \times 50^4/64 = 3.068 \times 10^5$ mm⁴; $P_e = \pi^2 EI/L^2 = 9.87 \times 2 \times 10^5 \times 3.068 \times 10^5/2500^2 = 96.9$ kN (hinged)

Fixed–free ($L_e = 2L$): 96.9/4 = 24.2 kN; fixed–fixed ($L_e = L/2$): 96.9 × 4 = 387.6 kN

Problem 19 – Slenderness ratio

For the column in Problem 18 (hinged), find k and λ.

$k = d/4 = 12.5$ mm; $\lambda = 2500/12.5 = 200$ (long column – Euler valid)

Problem 20 – Rankine load

For the same column take $\sigma_c = 320$ MPa. Find the Rankine load.

$P_c = 320 \times 1963.5 = 628.3$ kN; $1/P_R = 1/628.3 + 1/96.9$ → $P_R = 83.9$ kN

- Check with $a = \sigma_c/(\pi^2 E) = 1/6168$: $P_R = 628.3/(1 + 200^2/6168) = 628.3/7.485 = 83.9$ kN ✓

Problem 21 – Effect of doubling dimensions

What happens to the Euler load if (a) length is doubled (b) diameter is doubled?

(a) $P \propto 1/L^2$ → one-fourth; (b) $P \propto I \propto d^4$ → 16 times

25.5 Cylinders

Problem 22 – Thin cylinder stresses

A cylinder of 1 m diameter and 10 mm thickness carries internal pressure 2 MPa. Find hoop and longitudinal stresses and maximum in-plane shear.

$\sigma_c = pd/2t = 2 \times 1000/20 = 100$ MPa; $\sigma_l = 50$ MPa; $\tau = (100 - 50)/2 = 25$ MPa

Problem 23 – Thickness with joint efficiency

A boiler shell 1.5 m diameter works at 1.5 MPa; allowable stress 120 MPa; longitudinal joint efficiency 80%. Find thickness.

$t = pd/(2\sigma\ \eta_l) = 1.5 \times 1500/(2 \times 120 \times 0.8) = 11.7\ mm \rightarrow 12\ mm$

Problem 24 – Thin sphere

A spherical vessel of 1 m diameter holds gas at 1.2 MPa; allowable stress 60 MPa. Find thickness.

$t = pd/(4\sigma) = 1.2 \times 1000/240 = 5\ mm$

Problem 25 – Volume change of thin cylinder

For Problem 22 (L = 3 m, μ = 0.3), find the change in volume.

$e_v = (pd/4tE)(5 - 4\mu) = (2 \times 1000/(40 \times 2 \times 10^5))(3.8) = 2.5 \times 10^{-4} \times 3.8 = 9.5 \times 10^{-4}$

$V = \pi/4 \times 1^2 \times 3 = 2.356\ m^3 \rightarrow \Delta V = 2.24 \times 10^{-3}\ m^3 = 2.24\ litres$

Problem 26 – Thick cylinder

A thick cylinder (r_1 = 100 mm, r_2 = 150 mm) has internal pressure 30 MPa. Find hoop stresses at inner and outer surfaces.

$\sigma_{c,inner} = p(r_2^2 + r_1^2)/(r_2^2 - r_1^2) = 30 \times 32{,}500/12{,}500 = 78\ MPa$

$\sigma_{c,outer} = 2pr_1^2/(r_2^2 - r_1^2) = 60 \times 10{,}000/12{,}500 = 48\ MPa$

CHAPTER 25B

Solved Numericals – Set 2 (SFD/BMD, Bending, Shear, Deflection, Torsion, Springs)

SSC-style problems solved step by step (no options).

25B.1 Shear Force and Bending Moment

Problem 1 – SSB with UDL

A simply supported beam of 6 m span carries 10 kN/m over the whole span. Find reactions and maximum BM.

R_A = R_B = 30 kN; M_max = wL²/8 = 10 × 36/8 = 45 kN·m at mid-span

Problem 2 – SSB with eccentric point load

A 6 m SSB carries 30 kN at 2 m from A. Find reactions and maximum BM.

R_A = 30 × 4/6 = 20 kN; R_B = 10 kN; M_max = Wab/L = 30 × 2 × 4/6 = 40 kN·m under the load

Problem 3 – Cantilever with UDL and end load

A 3 m cantilever carries 5 kN/m over its length and 10 kN at the free end. Find SF and BM at the fixed end.

SF = 5 × 3 + 10 = 25 kN; BM = 5 × 3²/2 + 10 × 3 = 22.5 + 30 = 52.5 kN·m (hogging)

Problem 4 – UDL on part of the span

An 8 m SSB carries 4 kN/m over the left 4 m. Find the maximum BM and its location.

Load = 16 kN at 2 m from A → R_A = 16 × 6/8 = 12 kN, R_B = 4 kN

SF = 12 − 4x = 0 → x = 3 m; M_max = 12 × 3 − 4 × 3²/2 = 36 − 18 = 18 kN·m

Problem 5 – Overhanging beam

A beam is supported at A and B (6 m apart) and overhangs 2 m beyond B, carrying 10 kN at the free end. Find reactions and BM at B.

ΣM_A = 0: R_B × 6 = 10 × 8 → R_B = 13.33 kN; R_A = 10 − 13.33 = −3.33 kN (downward − A must be held down)

M_B = −10 × 2 = −20 kN·m (hogging); BM varies linearly from 0 at A to −20 at B

Problem 6 – Fixed beam and propped cantilever

(a) A fixed beam 6 m long carries 12 kN/m. Find end and centre moments. (b) A propped cantilever 5 m long carries 8 kN/m. Find the prop reaction.

(a) M_end = wL²/12 = 36 kN·m (hogging); M_centre = wL²/24 = 18 kN·m (sagging)

(b) R_prop = 3wL/8 = 3 × 8 × 5/8 = 15 kN; fixed-end reaction = 25 kN

25B.2 Bending and Shear Stresses

Problem 7 – Bending stress in a rectangular beam

A beam 100 mm wide and 200 mm deep carries M = 20 kN·m. Find the maximum bending stress.

$Z = bd^2/6 = 100 \times 200^2/6 = 6.667 \times 10^5$ mm³; $\sigma = M/Z = 2 \times 10^7/6.667 \times 10^5 = 30$ MPa

Problem 8 – Moment of resistance of a circular section

Find the moment of resistance of a 100 mm diameter beam with σ = 120 MPa.

$M = \sigma \pi d^3/32 = 120 \times 98{,}175 = 11.78 \times 10^6$ N·mm = 11.78 kN·m

Problem 9 – Radius of curvature

For the beam of Problem 7 (E = 200 GPa), find the radius of curvature.

$I = bd^3/12 = 6.667 \times 10^7$ mm⁴; $R = EI/M = 2 \times 10^5 \times 6.667 \times 10^7/2 \times 10^7 = 6.67 \times 10^5$ mm = 667 m

Problem 10 – Shear stress in a rectangular section

The beam of Problem 7 carries a shear force of 40 kN. Find average and maximum shear stress.

$\tau_{avg} = 40{,}000/(100 \times 200) = 2$ MPa; $\tau_{max} = 1.5 \times 2 = 3$ MPa at the neutral axis

Problem 11 – Shear stress in a circular section

A 100 mm diameter beam carries a shear force of 50 kN. Find maximum shear stress.

$\tau_{avg} = 50{,}000/7854 = 6.37$ MPa; $\tau_{max} = (4/3) \times 6.37 = 8.49$ MPa

Problem 12 – Strongest beam from a log

Find the dimensions of the strongest rectangular beam that can be cut from a log of 300 mm diameter.

$b = D/\sqrt{3} = 173.2$ mm; $d = \sqrt{(2/3)}\ D = 244.9$ mm ($d/b = \sqrt{2}$)

Problem 13 – Eccentric load on a short column

A 300 mm × 200 mm column carries 600 kN at 40 mm eccentricity along the 300 mm side. Find extreme stresses.

$\sigma = (P/A)(1 \pm 6e/d) = (600{,}000/60{,}000)(1 \pm 6 \times 40/300) = 10 \times (1 \pm 0.8) = 18$ MPa and 2 MPa (both compressive; $e < d/6 = 50$ mm)

25B.3 Deflection

Problem 14 – Cantilever end load

A cantilever 2 m long ($EI = 1.6 \times 10^{12}$ N·mm²) carries 10 kN at the free end. Find the deflection and slope at the free end.

$\delta = WL^3/(3EI) = 10^4 \times 8 \times 10^9/(4.8 \times 10^{12}) = 16.7$ mm; $\theta = WL^2/(2EI) = 10^4 \times 4 \times 10^6/(3.2 \times 10^{12}) = 0.0125$ rad

Problem 15 – SSB with UDL

A 4 m SSB ($EI = 1.6 \times 10^{12}$ N·mm²) carries 10 kN/m. Find the central deflection.

$\delta = 5wL^4/(384EI) = 5 \times 10 \times 4000^4/(384 \times 1.6 \times 10^{12}) = 1.28 \times 10^{16}/6.144 \times 10^{14} = 20.8$ mm

Problem 16 – SSB with central load (same total load)

The same beam carries the total load 40 kN at the centre instead. Find the deflection and compare.

$\delta = WL^3/(48EI) = 4 \times 10^4 \times 6.4 \times 10^{10}/(7.68 \times 10^{13}) = 33.3$ mm; ratio = 20.8/33.3 = 0.625 = 5/8

Problem 17 – Effect of depth on deflection

How does the deflection change if the depth of a rectangular beam is doubled, other things remaining same?

$I \propto d^3 \rightarrow$ I becomes 8 times → deflection becomes 1/8; bending stress becomes 1/4 ($Z \propto d^2$)

Problem 18 – Fixed vs simply supported

Compare central deflections of fixed and SSB under UDL.

Fixed: $wL^4/(384EI)$; SSB: $5wL^4/(384EI)$ → fixed beam deflects 1/5 as much

25B.4 Torsion and Springs

Problem 19 – Shaft diameter for power

A solid shaft transmits 100 kW at 300 rpm; allowable shear stress 60 MPa. Find the diameter.

$T = 60P/(2\pi N) = 60 \times 10^5/(2\pi \times 300) = 3183$ N·m

$d^3 = 16T/(\pi\tau) = 16 \times 3.183 \times 10^6/(\pi \times 60) = 2.70 \times 10^5 \rightarrow d = 64.6$ mm, say 65 mm

Problem 20 – Angle of twist

For the 65 mm shaft of Problem 19 (L = 2 m, G = 80 GPa) find the angle of twist.

$J = \pi \times 65^4/32 = 1.752 \times 10^6$ mm⁴; $\theta = TL/(GJ) = 3.183 \times 10^6 \times 2000/(8 \times 10^4 \times 1.752 \times 10^6) = 0.0454$ rad = 2.6°

Problem 21 – Hollow vs solid of equal weight

Compare the strength of a hollow shaft (d = D/2) with a solid shaft of 100 mm diameter having the same cross-sectional area.

Area: $0.75\ D^2 = 100^2 \rightarrow D = 115.5$ mm, d = 57.7 mm

Strength ratio = $[(D^4 - d^4)/D]/d_s^3 = 0.9375 \times 115.5^3/10^6 = 1.44$ → hollow is 44% stronger

Problem 22 – Close-coiled helical spring

A spring has d = 10 mm, D = 80 mm, n = 10 active turns, G = 80 GPa, load 500 N. Find deflection, stiffness and shear stress (with Wahl factor).

$\delta = 8WD^3n/(Gd^4) = 8 \times 500 \times 512{,}000 \times 10/(8 \times 10^4 \times 10^4) = 25.6$ mm; k = 500/25.6 = 19.5 N/mm

$\tau = 8WD/(\pi d^3) = 8 \times 500 \times 80/(\pi \times 1000) = 101.9$ MPa; C = 8 → K = (31/28) + 0.615/8 = 1.184 → τ = 120.6 MPa

Problem 23 – Springs in series and parallel

Two springs of stiffness 20 and 30 N/mm are combined. Find the equivalent stiffness in series and in parallel.

Series: $k = 20 \times 30/50 = 12$ N/mm; parallel: k = 50 N/mm

Problem 24 – Leaf spring

A semi-elliptic leaf spring (span 1 m, 8 leaves, 60 mm wide, 8 mm thick) carries 5 kN at the centre. Find the maximum stress and deflection.

$\sigma = 3WL/(2nbt^2) = 3 \times 5000 \times 1000/(2 \times 8 \times 60 \times 64) = 244$ MPa

$\delta = 3WL^3/(8Enbt^3) = 3 \times 5000 \times 10^9/(8 \times 2 \times 10^5 \times 8 \times 60 \times 512) = 38.1$ mm

CHAPTER 25C

Solved Numericals – Set 3 (Casting, Welding, Forming, Machining)

SSC-style production problems solved step by step (no options).

25C.1 Casting

Problem 1 – Shrinkage allowance

A grey cast iron casting is to be 500 mm long. Find the pattern length (shrinkage 10 mm/m).

```
Allowance = 10 × 0.5 = 5 mm → pattern = 505 mm
```

Problem 2 – Chvorinov's rule

A 100 mm cube solidifies in 5 min. Find the solidification time of a sphere of the same volume and same material/mould.

```
Cube: V/A = a/6 = 16.67 mm; sphere: r = (3V/4π)^(1/3) = 62.04 mm → V/A = r/3 = 20.68 mm
t = 5 × (20.68/16.67)² = 5 × 1.539 = 7.7 min (sphere freezes slower)
```

Problem 3 – Velocity at sprue base and filling time

The effective sprue height is 200 mm, gate area 200 mm², mould volume 10^6 mm³ (top gating). Find gate velocity and filling time.

```
V = √(2gh) = √(2 × 9810 × 200) = 1981 mm/s ≈ 1.98 m/s
t = V_mould/(A_g V) = 10⁶/(200 × 1981) = 2.52 s
```

Problem 4 – Sprue taper

The sprue bottom (head 250 mm) has area 100 mm²; the sprue top is at a head of 50 mm. Find the top area to avoid aspiration.

```
A_top/A_bottom = √(h_bottom/h_top) = √(250/50) = 2.236 → A_top = 223.6 mm²
```

Problem 5 – Centrifugal casting speed

Find the rotational speed for a pipe of inner diameter 200 mm with G-factor 60.

```
ω²r = 60g → ω = √(60 × 9.81/0.1) = 76.7 rad/s → N = 732 rpm
```

25C.2 Welding

Problem 6 – Heat input

Arc welding at 25 V, 200 A and travel speed 5 mm/s with 80% efficiency. Find the heat input per mm.

```
H = ηVI/v = 0.8 × 25 × 200/5 = 800 J/mm
```

Problem 7 – Resistance spot welding

Current 10,000 A, resistance 100 μΩ, time 0.1 s. Find the heat generated.

```
H = I²Rt = 10⁸ × 10⁻⁴ × 0.1 = 1000 J
```

Problem 8 – Fillet weld strength

A 10 mm fillet weld 100 mm long (two parallel welds) has allowable shear stress 80 MPa. Find the load capacity.

```
Throat = 0.707 × 10 = 7.07 mm; P = 2 × 7.07 × 100 × 80 = 113 kN
```

Problem 9 – Arc length and voltage

The arc characteristic is V = 20 + 4l (l in mm). Find the voltage for a 3 mm arc.

```
V = 20 + 4 × 3 = 32 V
```

25C.3 Metal Forming

Problem 10 – Maximum draft in rolling

Roll radius 300 mm, μ = 0.1. Find the maximum possible draft and the bite angle.

```
Δh_max = μ²R = 0.01 × 300 = 3 mm; cosα = 1 − 3/600 = 0.995 → α = 5.73° (tanα ≈ 0.1 = μ ✓)
```

Problem 11 – Exit velocity in rolling

A strip 20 mm thick enters at 1 m/s and leaves 15 mm thick (width constant). Find the exit velocity.

```
h₀v₀ = h_f v_f → v_f = 20 × 1/15 = 1.33 m/s
```

Problem 12 – Blanking force and tool sizes

A 50 mm diameter blank is cut from 2 mm sheet (τ = 300 MPa); clearance 5% of t per side. Find force and tool sizes.

```
F = πDtτ = π × 50 × 2 × 300 = 94.2 kN
Clearance = 0.1 mm → die = 50 mm, punch = 50 − 0.2 = 49.8 mm
```

Problem 13 – Punching a hole

A 20 mm hole is punched in the same sheet. Find punch and die sizes.

```
Punch = 20 mm; die = 20 + 0.2 = 20.2 mm
```

Problem 14 – Deep drawing blank

Find the blank diameter for a cup of 50 mm diameter and 40 mm height. Can it be drawn in one stage?

```
D = √(d² + 4dh) = √(2500 + 8000) = 102.5 mm; drawing ratio = 102.5/50 = 2.05 > ≈ 2 →
needs two draws
```

Problem 15 – Extrusion pressure

A billet of 100 mm diameter is extruded to 50 mm (flow stress 100 MPa). Find ideal pressure and force.

```
R = (100/50)² = 4; p = σ ln R = 100 × 1.386 = 138.6 MPa; F = 138.6 × 7854 = 1.09 MN
```

Problem 16 – Wire drawing stress

Wire is drawn from 5 mm to 4 mm (flow stress 300 MPa). Find the ideal drawing stress. Is it feasible?

```
σ_d = 300 ln(25/16) = 300 × 0.446 = 133.9 MPa < 300 MPa → feasible (reduction = 36%)
```

25C.4 Metal Cutting

Problem 17 – Turning time and MRR

A 50 mm bar is turned at 500 rpm, feed 0.2 mm/rev, depth 2 mm, length 200 mm. Find V, time and MRR.

```
V = π × 50 × 500/1000 = 78.5 m/min; t = 200/(0.2 × 500) = 2 min
MRR = V f d = 78,540 × 0.2 × 2 = 31,416 mm³/min
```

Problem 18 – Merchant's analysis

Orthogonal cutting: α = 10°, uncut thickness 0.2 mm, chip thickness 0.5 mm, F_c = 1200 N, F_t = 500 N. Find shear angle, friction coefficient, shear force and shear strain.

```
r = 0.4; tanφ = 0.4 cos10°/(1 − 0.4 sin10°) = 0.3939/0.9305 = 0.4233 → φ = 22.9°
μ = (F_t + F_c tanα)/(F_c − F_t tanα) = (500 + 211.6)/(1200 − 88.2) = 0.640 → β = 32.6°
F_s = F_c cosφ − F_t sinφ = 1105.1 − 194.9 = 910 N
γ = cotφ + tan(φ − α) = 2.362 + 0.230 = 2.59
```

- Merchant's theory would predict φ = 45° + 5° − 16.3° = 33.7°; the difference shows the theory's limitations.

Problem 19 – Taylor's tool life

Tool life equation VT^0.25 = 300. Find the speed for 60 min life, and tool life at 150 m/min.

```
V = 300/60^0.25 = 300/2.783 = 107.8 m/min; T = (300/150)^4 = 16 min
```

Problem 20 – Taylor constants from two tests

Tool life is 60 min at 100 m/min and 12 min at 150 m/min. Find n and C.

```
n = ln(150/100)/ln(60/12) = 0.4055/1.6094 = 0.252; C = 100 × 60^0.252 = 280.6
```

Problem 21 – Percentage change in tool life

For n = 0.25, if cutting speed is increased by 20%, find the change in tool life.

```
T₂/T₁ = (V₁/V₂)^(1/n) = (1/1.2)⁴ = 0.482 → tool life falls by about 52%
```

Problem 22 – Drilling time

A 20 mm drill makes a 40 mm deep hole at 400 rpm, feed 0.2 mm/rev. Find time (approach 0.3D).

```
t = (40 + 6)/(0.2 × 400) = 0.575 min = 34.5 s
```

Problem 23 – Milling time

A 100 mm slab cutter with 10 teeth runs at 200 rpm, feed 0.1 mm/tooth, depth 5 mm, work length 300 mm. Find time.

```
f_m = 0.1 × 10 × 200 = 200 mm/min; approach = √(5 × 95) = 21.8 mm
t = (300 + 21.8)/200 = 1.61 min
```

Problem 24 – Indexing

Find the indexing for 36 and 52 divisions (plates with 18- and 39-hole circles).

```
36: 40/36 = 1 1/9 = 1 turn + 2 holes in 18-hole circle; 52: 40/52 = 10/13 = 30 holes in 39-hole circle
```

Problem 25 – Shaper cutting speed and time

Stroke 200 mm, 30 strokes/min, return/cutting time ratio 2/3; job width 150 mm, feed 1 mm/stroke. Find average cutting speed and time.

```
V = LN(1 + m)/1000 = 200 × 30 × 1.667/1000 = 10 m/min; strokes = 150 → time = 150/30 = 5 min
```

Problem 26 – Surface roughness from feed and nose radius

Feed 0.2 mm/rev, nose radius 0.8 mm. Find peak-to-valley roughness and approximate Ra.

```
h = f²/(8R) = 0.04/6.4 = 6.25 µm; Ra ≈ h/4 ≈ 1.56 µm
```

CHAPTER 25D

Solved Numericals – Set 4 (Metrology, Industrial Engineering, Mechanics)

Short problems testing one concept each (no options).

25D.1 Metrology

Problem 1 – Clearance from limits

Hole 50 +0.025/+0.000 mm; shaft 50 −0.009/−0.025 mm. Find maximum and minimum clearance and the type of fit.

Max clearance = 50.025 − 49.975 = 0.050 mm; min clearance (allowance) = 50.000 − 49.991 = 0.009 mm → clearance fit

Hole tolerance = 0.025 mm; shaft tolerance = 0.016 mm

Problem 2 – Interference fit

Hole 30 +0.021/0; shaft 30 +0.035/+0.022. Classify the fit.

Max interference = 30.035 − 30.000 = 0.035 mm; min interference = 30.022 − 30.021 = 0.001 mm → interference fit

Problem 3 – Standard tolerance

Find IT7 for a 40 mm shaft (diameter step 30–50 mm).

$D = \sqrt{(30 \times 50)} = 38.73$ mm; $i = 0.45 \sqrt[3]{38.73} + 0.001 \times 38.73 = 1.522 + 0.039 = 1.561$ µm

IT7 = 16i = 25 µm

Problem 4 – Vernier least count

A vernier has 50 divisions coinciding with 49 main scale divisions of 1 mm. Find the least count.

LC = 1 MSD − 1 VSD = 1 − 49/50 = 0.02 mm

Problem 5 – Micrometer reading

A micrometer (pitch 0.5 mm, 50 thimble divisions) shows 12.5 mm on the sleeve and 28 on the thimble. Find the reading.

LC = 0.5/50 = 0.01 mm; reading = 12.5 + 28 × 0.01 = 12.78 mm

Problem 6 – Sine bar

A 200 mm sine bar is used to set 30°. Find the slip gauge height.

$h = L \sin\theta = 200 \times 0.5 = 100$ mm

Problem 7 – Best wire size

Find the best wire size for an M20 × 2.5 thread (60°).

$d_w = (p/2) \sec 30° = 0.577 \times 2.5 = 1.443$ mm

Problem 8 – Optical flat

Five fringes are seen across a surface under light of wavelength 0.6 µm. Find the height difference.

$h = n\lambda/2 = 5 \times 0.3 = 1.5\ \mu m$

25D.2 Industrial Engineering

Problem 9 – Standard time

Observed time 2 min, rating 110%, allowances 15% of normal time. Find standard time.

Normal time = $2 \times 1.1 = 2.2$ min; standard time = $2.2 \times 1.15 = 2.53$ min

Problem 10 – Work sampling observations

A machine is idle 20% of the time ($p = 0.2$). How many observations are needed for ±5% absolute accuracy at 95% confidence?

$N = 4p(1 - p)/L^2 = 4 \times 0.2 \times 0.8/0.05^2 = 256$

Problem 11 – EOQ

Annual demand 12,000 units, ordering cost ₹200 per order, holding cost ₹12 per unit per year. Find EOQ, number of orders and minimum total variable cost.

$Q^* = \sqrt{(2 \times 12{,}000 \times 200/12)} = \sqrt{400{,}000} = 632$ units; orders = $12{,}000/632 \approx 19$ per year

Total cost = $\sqrt{(2DSH)} = \sqrt{(2 \times 12{,}000 \times 200 \times 12)} = ₹7{,}589$

Problem 12 – Reorder point

Daily demand 40 units, lead time 5 days, safety stock 60 units. Find ROP.

ROP = $40 \times 5 + 60 = 260$ units

Problem 13 – Exponential smoothing

Last forecast 100, actual demand 120, $\alpha = 0.2$. Find the next forecast.

$F = 100 + 0.2 \times (120 - 100) = 104$

Problem 14 – Break-even

Fixed cost ₹1,00,000; selling price ₹50; variable cost ₹30 per unit. Find the break-even quantity.

$Q_{BE} = 1{,}00{,}000/(50 - 30) = 5{,}000$ units

Problem 15 – Line balancing

Task times total 40 s; cycle time 10 s; 5 stations used. Find minimum stations and line efficiency.

Minimum stations = $40/10 = 4$; efficiency = $40/(5 \times 10) = 80\%$; balance delay = 20%

Problem 16 – PERT

An activity has $t_o = 4$, $t_m = 6$, $t_p = 14$ days. Find expected time and variance.

$t_e = (4 + 24 + 14)/6 = 7$ days; $\sigma^2 = ((14 - 4)/6)^2 = 2.78$

Problem 17 – Probability of completion

Critical path expected time 40 days, $\sigma = 3$ days. Find the probability of completion in 43 days.

$Z = (43 - 40)/3 = 1 \rightarrow$ probability ≈ 84%

Problem 18 – Control chart limits

20 samples of size 5 give $\bar{\bar{X}} = 50$, $\bar{R} = 4$ ($A_2 = 0.577$, $D_3 = 0$, $D_4 = 2.114$). Find limits.

$\bar{X}$ chart: $50 \pm 0.577 \times 4 = 52.31 / 47.69$; R chart: UCL = 8.46, LCL = 0

Problem 19 – Process capability

Specification 50 ± 0.06 mm; process σ = 0.015 mm. Find C_p.

```
C_p = 0.12/(6 × 0.015) = 1.33 (capable)
```

Problem 20 – Reliability

Three components each with reliability 0.9 are connected (a) in series (b) two in parallel. Find system reliability.

```
(a) 0.9³ = 0.729; (b) 1 − 0.1² = 0.99
```

Problem 21 – Johnson's rule

Jobs (M1, M2 times): A (5, 2), B (1, 6), C (9, 7), D (3, 8). Find the optimal sequence.

```
Smallest = B on M1 (1) → first; A on M2 (2) → last; D on M1 (3) → second; C → third → sequence B–D–C–A
```

Problem 22 – M/M/1 queue

Arrivals 8 per hour, service 10 per hour. Find utilisation, average number in system and time in system.

```
ρ = 0.8; L_s = λ/(μ − λ) = 4; W_s = 1/(μ − λ) = 0.5 h
```

25D.3 Engineering Mechanics

Problem 23 – Resultant of two forces

Two forces of 30 N and 40 N act at 90°. Find the resultant.

```
R = √(30² + 40²) = 50 N at tan⁻¹(40/30) = 53.1° to the 30 N force
```

Problem 24 – Lami's theorem

A 100 N weight hangs from two strings making 30° and 60° with the horizontal. Find the tensions.

```
Angles between forces: T₁ (30° string) and W: 120°; T₂ and W: 150°; T₁ and T₂: 90°
T₁/sin150° = T₂/sin120° = 100/sin90° → T₁ = 50 N, T₂ = 86.6 N
```

Problem 25 – Inclined plane

A 500 N block rests on a 30° incline with μ = 0.2. Find the force parallel to the plane to just move it up.

```
P = W(sin30° + μ cos30°) = 500(0.5 + 0.173) = 336.6 N
```

Problem 26 – Belt tension ratio

A flat belt has μ = 0.3 and angle of lap 180°. Find T_1/T_2.

```
T₁/T₂ = e^(0.3π) = e^0.942 = 2.57
```

Problem 27 – Truss check

A truss has 9 members and 6 joints. Is it perfect?

```
2j − 3 = 9 = m → perfect truss
```

Problem 28 – Projectile

A ball is projected at 20 m/s at 30°. Find range and maximum height (g = 9.81).

```
R = u² sin60°/g = 400 × 0.866/9.81 = 35.3 m; H = u² sin²30°/(2g) = 400 × 0.25/19.62 = 5.10 m
```

Problem 29 – Moment of inertia using parallel axis

Find I of a 100 × 200 mm rectangle about its base.

```
I = bd³/3 = 100 × 200³/3 = 2.667 × 10⁸ mm⁴ (= bd³/12 + A(d/2)²)
```

Problem 30 – Screw jack efficiency

A screw jack has helix angle 6° and friction angle 8°. Find efficiency; is it self-locking?

```
η = tan6°/tan14° = 0.1051/0.2493 = 42.2% < 50% → self-locking (φ > α)
```

CHAPTER 25E

Solved Numericals – Set 5 (Advanced SOM)

Slightly longer problems combining two or three concepts (no options).

25E.1 Bars and Energy

Problem 1 – Bar fixed at both ends with intermediate load

A bar of length 600 mm fixed at both ends carries an axial load of 60 kN at 200 mm from end A. Find reactions.

R_A = Pb/L = 60 × 400/600 = 40 kN (tension portion AC); R_B = Pa/L = 20 kN (compression portion CB)

Problem 2 – Composite bar heated

A steel rod (A = 500 mm², α = 12 × 10⁻⁶, E = 200 GPa) and a brass tube (A = 1000 mm², α = 19 × 10⁻⁶, E = 100 GPa) of equal length are rigidly joined and heated by 50 °C. Find stresses.

Equilibrium: σ_s × 500 = σ_b × 1000 → σ_s = 2σ_b (steel in tension, brass in compression)

Compatibility: σ_s/E_s + σ_b/E_b = (α_b − α_s)ΔT = 7 × 10⁻⁶ × 50 = 3.5 × 10⁻⁴

2σ_b/2 × 10⁵ + σ_b/10⁵ = 3.5 × 10⁻⁴ → σ_b = 17.5 MPa (comp.), σ_s = 35 MPa (tension)

Problem 3 – Bolt and sleeve

A steel bolt (A = 300 mm²) passes through a copper sleeve (A = 600 mm²); the nut is tightened by 1/4 turn with pitch 2 mm, length 400 mm. E_s = 200, E_c = 100 GPa. Find stresses.

Total deformation = 0.5 mm = δ_s + δ_c; force equal: σ_s × 300 = σ_c × 600 → σ_s = 2σ_c

0.5 = (2σ_c × 400/2 × 10⁵) + (σ_c × 400/10⁵) = σ_c(0.004 + 0.004) → σ_c = 62.5 MPa (comp.), σ_s = 125 MPa (tension)

Problem 4 – Strain energy comparison

Compare the strain energy stored in two bars of equal length under the same load P: bar A uniform area A, bar B area A for half length and 2A for the other half.

U_A = P²L/(2AE); U_B = (P²/2E)(L/2A + L/4A) = (3/4) × P²L/(2AE) → U_B = 0.75 U_A

Problem 5 – Proof resilience

Find the proof resilience of a 50 mm × 50 mm × 1 m bar with elastic limit 250 MPa (E = 200 GPa) and the modulus of resilience.

Modulus of resilience = σ²/2E = 250²/(4 × 10⁵) = 0.156 N·mm/mm³

Proof resilience = 0.156 × 2.5 × 10⁶ = 3.9 × 10⁵ N·mm = 390 J

25E.2 Stress Transformation

Problem 6 – Normal and shear on inclined plane (biaxial)

σx = 100 MPa (tension), σy = 50 MPa (compression). Find stresses on a plane at 30° to the y-axis (normal at 30° to x).

σ_n = (100 − 50)/2 + (150/2) cos60° = 25 + 37.5 = 62.5 MPa; τ = 75 sin60° = 65 MPa

Problem 7 – Principal strains

At a point $\sigma_1 = 120$ MPa, $\sigma_2 = 40$ MPa ($E = 200$ GPa, $\mu = 0.3$). Find principal strains.

$\varepsilon_1 = (120 - 0.3 \times 40)/2 \times 10^5 = 5.4 \times 10^{-4}$; $\varepsilon_2 = (40 - 36)/2 \times 10^5 = 2 \times 10^{-5}$

Problem 8 – Shaft with axial load and torque

A 50 mm shaft carries 100 kN axial tension and 1 kN·m torque. Find principal stresses and maximum shear.

$\sigma = 100{,}000/1963.5 = 50.9$ MPa; $\tau = 16 \times 10^6/(\pi \times 125{,}000) = 40.7$ MPa

$\sigma_{1,2} = 25.5 \pm \sqrt{(25.5^2 + 40.7^2)} = 25.5 \pm 48.0 \rightarrow 73.5$ MPa, -22.5 MPa; $\tau_{max} = 48.0$ MPa

Problem 9 – Shaft diameter by two theories

A shaft carries $M = 3$ kN·m and $T = 4$ kN·m; allowable $\tau = 50$ MPa (Tresca) and $\sigma = 100$ MPa (Rankine). Find diameter.

Tresca: $T_e = 5$ kN·m $\rightarrow d^3 = 16 \times 5 \times 10^6/(\pi \times 50) = 5.09 \times 10^5 \rightarrow d = 79.9$ mm

Rankine: $M_e = \frac{1}{2}(3 + 5) = 4$ kN·m $\rightarrow d^3 = 32 \times 4 \times 10^6/(\pi \times 100) = 4.07 \times 10^5 \rightarrow d = 74.1$ mm $\rightarrow$ choose 80 mm

25E.3 Beams

Problem 10 – Overhanging beam with UDL

A beam 8 m long rests on supports 6 m apart with 1 m overhang on each side and carries UDL 10 kN/m over its whole length. Find BM at supports and at centre.

Reactions = 40 kN each; $M_{support} = -10 \times 1^2/2 = -5$ kN·m

$M_{centre} = 40 \times 3 - 10 \times 4^2/2 = 120 - 80 = 40$ kN·m (sagging) $\rightarrow$ two points of contraflexure

Problem 11 – Section modulus of I-section

An I-section has flanges 150 × 20 mm and web 260 × 10 mm (overall depth 300 mm). Find I and Z.

$I = [150 \times 300^3 - 140 \times 260^3]/12 = [4.050 \times 10^9 - 2.461 \times 10^9]/12 = 1.325 \times 10^8$ mm⁴

$Z = I/150 = 8.83 \times 10^5$ mm³

Problem 12 – Moment capacity of the I-section

With $\sigma = 150$ MPa, find the safe BM for Problem 11, and compare with a rectangle of the same area (100 mm wide).

$M = 150 \times 8.83 \times 10^5 = 132.5$ kN·m

Area = $2 \times 3000 + 2600 = 8600$ mm² $\rightarrow$ rectangle 100 × 86 mm: $Z = 100 \times 86^2/6 = 1.233 \times 10^5$ mm³ $\rightarrow M = 18.5$ kN·m (I-section ≈ 7 times stronger)

Problem 13 – Shear stress in I-section web

The I-section of Problem 11 carries $V = 200$ kN. Find the maximum shear stress (at the neutral axis) and compare with the average web shear.

$A\bar{y}$ (above NA) $= 150 \times 20 \times 140 + 130 \times 10 \times 65 = 420{,}000 + 84{,}500 = 504{,}500$ mm³

$\tau_{max} = VA\bar{y}/(Ib) = 2 \times 10^5 \times 504{,}500/(1.325 \times 10^8 \times 10) = 76.2$ MPa

Approximate (V/overall depth × web thickness) $= 2 \times 10^5/3000 = 66.7$ MPa – the web carries nearly all the shear

Problem 14 – Macaulay's method

A 6 m SSB carries 30 kN at 2 m from A. Find the deflection under the load (EI = 2 × 10^4 kN·m^2).

$\delta = Wa^2b^2/(3EIL) = 30 \times 4 \times 16/(3 \times 2 \times 10^4 \times 6) = 1920/3.6 \times 10^5 = 5.33 \times 10^{-3}$ m = 5.33 mm

Problem 15 – Cantilever with partial load

A 3 m cantilever carries 20 kN at 2 m from the fixed end (EI = 10^4 kN·m^2). Find deflection at the free end.

$\delta_{load} = Wa^3/(3EI) = 20 \times 8/(3 \times 10^4) = 5.33$ mm; $\theta = Wa^2/(2EI) = 20 \times 4/(2 \times 10^4) = 0.004$ rad

$\delta_{free} = 5.33 + 0.004 \times 1000 = 9.33$ mm

25E.4 Torsion, Columns, Cylinders

Problem 16 – Stepped shaft twist

A shaft has 1 m of 60 mm diameter and 1 m of 40 mm diameter (G = 80 GPa) and carries 1 kN·m. Find total twist.

$J_1 = \pi \times 60^4/32 = 1.272 \times 10^6$; $J_2 = \pi \times 40^4/32 = 2.513 \times 10^5$ mm^4

$\theta = (T/G)(L_1/J_1 + L_2/J_2) = (10^6/8 \times 10^4)(1000/1.272 \times 10^6 + 1000/2.513 \times 10^5) = 12.5 \times (7.86 \times 10^{-4} + 3.98 \times 10^{-3}) = 0.0596$ rad = 3.4°

Problem 17 – Shaft fixed at both ends

A shaft fixed at both ends carries a torque of 900 N·m at 1 m from end A; total length 3 m (uniform). Find end torques.

$T_A = T \times b/L = 900 \times 2/3 = 600$ N·m; $T_B = 300$ N·m

Problem 18 – Column with rectangular section

A 2 m long column, both ends fixed, has section 40 × 60 mm (E = 200 GPa). Find the Euler load.

$I_{min} = 60 \times 40^3/12 = 3.2 \times 10^5$ mm^4; $L_e = 1$ m

$P = \pi^2 \times 2 \times 10^5 \times 3.2 \times 10^5/10^6 = 631.6$ kN (check crushing: $\sigma = 263$ MPa – near yield, so Euler may not govern)

Problem 19 – Hollow column

A hollow cast-iron column (D = 200 mm, d = 160 mm), 5 m long, hinged ends. E = 100 GPa. Find the Euler load and slenderness ratio.

$I = \pi(200^4 - 160^4)/64 = 4.637 \times 10^7$ mm^4; $A = 11{,}310$ mm^2; $k = \sqrt{(I/A)} = 64.0$ mm; $\lambda = 5000/64 = 78$

$P = \pi^2 \times 10^5 \times 4.637 \times 10^7/25 \times 10^6 = 1.83$ MN

Problem 20 – Change in diameter of thin cylinder

A cylinder (d = 800 mm, t = 10 mm) is under 2 MPa (E = 200 GPa, μ = 0.3). Find the change in diameter.

$e_c = (pd/4tE)(2 - \mu) = (2 \times 800/(40 \times 2 \times 10^5))(1.7) = 2 \times 10^{-4} \times 1.7 = 3.4 \times 10^{-4}$

$\Delta d = 3.4 \times 10^{-4} \times 800 = 0.272$ mm

CHAPTER 25F

Solved Numericals – Set 6 (Advanced Production and Industrial Engineering)

Longer production problems combining two or three concepts (no options).

25F.1 Casting and Welding

Problem 1 – Riser size by Chvorinov

A casting 200 × 100 × 50 mm needs a cylindrical riser with H = D whose solidification time is 1.25 times that of the casting. Find D.

Casting: $V = 10^6$ mm³, $A = 2(200 \times 100 + 200 \times 50 + 100 \times 50) = 70{,}000$ mm² → $M_c = 14.29$ mm

Riser (H = D): $V/A = (\pi D^3/4)/(1.5\pi D^2) = D/6$; $t \propto M^2$ → $M_r = \sqrt{1.25} \times 14.29 = 15.97$ mm → $D = 95.8$ mm

Problem 2 – Pouring time with bottom gating

A mould 300 mm high is filled through a bottom gate of area 400 mm² from a sprue with 300 mm effective head; mould plan area 40,000 mm². Find the filling time.

$t = (A_m/A_g) \times (2/\sqrt{(2g)}) \times (\sqrt{H} - \sqrt{(H - h)}) = (40{,}000/400) \times (2/\sqrt{19{,}620}) \times (\sqrt{300} - 0)$

$= 100 \times 0.01428 \times 17.32 = 24.7$ s

Problem 3 – Weld bead and electrode consumption

A butt weld 1 m long has a cross-section of 60 mm². Find the mass of weld metal (steel density 7.85 g/cm³) and the electrode needed if deposition efficiency is 65%.

Volume = 60 × 1000 = 60,000 mm³ = 60 cm³ → mass = 471 g; electrode = 471/0.65 = 725 g

Problem 4 – Melting efficiency

An arc of 30 V, 250 A with heat transfer efficiency 0.8 welds at 6 mm/s, depositing 50 mm³ of metal per second requiring 10 J/mm³ to melt. Find melting efficiency.

Heat to work = 0.8 × 30 × 250 = 6000 W; heat used for melting = 50 × 10 = 500 W → η_{melt} = 8.3%

25F.2 Forming and Press Work

Problem 5 – Rolling load

A strip 300 mm wide is rolled from 25 mm to 20 mm with rolls of 250 mm radius; mean flow stress 200 MPa. Estimate the roll force (no friction correction) and torque per roll.

$L = \sqrt{(R\Delta h)} = \sqrt{(250 \times 5)} = 35.4$ mm; $F = \bar{\sigma} w L = 200 \times 300 \times 35.4 = 2.12$ MN

Torque per roll ≈ $F \times L/2 = 2.12 \times 10^6 \times 0.0177 = 37.5$ kN·m

Problem 6 – Number of passes

A 40 mm slab is to be reduced to 25 mm with μ = 0.12 and R = 300 mm. Find the minimum number of passes.

```
Δh_max = μ²R = 0.0144 × 300 = 4.32 mm → passes ≥ 15/4.32 = 3.5 → 4 passes
```

Problem 7 – Blanking with shear on punch

A 100 mm blank is cut from 4 mm sheet (τ = 400 MPa); penetration at fracture 40%; shear on punch 2 mm. Find the maximum force without and with shear.

```
F = π × 100 × 4 × 400 = 502.7 kN; with shear: F' = F × (p t)/S = 502.7 × (0.4 × 4)/2 = 402 kN
```

Problem 8 – Work done in blanking

For Problem 7 (no shear), estimate the work done per stroke.

```
W = F × p t = 502.7 × 10³ × 0.4 × 4 × 10⁻³ = 804 J
```

Problem 9 – Bend allowance

A 3 mm sheet is bent through 90° with inside radius 6 mm (k = 0.5). Find the bend allowance.

```
BA = α(R + kt) = (π/2)(6 + 1.5) = 11.8 mm
```

Problem 10 – Forging load (open die)

A cylinder 100 mm diameter and 50 mm high is upset to 25 mm height (flow stress at the end 300 MPa, μ = 0.2). Estimate the final forging load.

```
Final d = 100√2 = 141.4 mm (volume constant), r = 70.7 mm, h = 25 mm
F = σ_f πr²(1 + 2μr/(3h)) = 300 × 15,708 × (1 + 2 × 0.2 × 70.7/75) = 4.71 MN × 1.377 = 6.49 MN
```

25F.3 Machining Economics

Problem 11 – Optimum tool life for minimum cost

Taylor: VT^0.25 = 250. Machine + labour cost ₹2/min, tool cost per edge ₹40, tool change time 4 min. Find the tool life and speed for minimum cost.

```
T_opt = (1/n − 1)(C_t/C_m + t_c) = 3 × (40/2 + 4) = 72 min
V_opt = 250/72^0.25 = 250/2.913 = 85.8 m/min
```

Problem 12 – Tool life for maximum production

For Problem 11, find the tool life and speed for maximum production rate.

```
T = (1/n − 1) t_c = 3 × 4 = 12 min; V = 250/12^0.25 = 250/1.861 = 134.3 m/min
```

Problem 13 – Cutting power and specific energy

In turning, F_c = 1500 N at V = 120 m/min with feed 0.25 mm/rev and depth 3 mm. Find power and specific cutting energy.

```
P = 1500 × 2 = 3 kW; MRR = 120,000 × 0.25 × 3/60 = 1500 mm³/s
u = 3000/1500 = 2 J/mm³
```

Problem 14 – Shaper quick return

In a crank and slotted lever mechanism, the crank rotates 220° during cutting and 140° during return. Find the quick return ratio.

```
Ratio = 220/140 = 1.57
```

Problem 15 – Taper turning by tailstock set-over

A job 300 mm long has a 150 mm long taper from 50 mm to 40 mm diameter. Find the set-over.

```
S = (D − d)L/(2l) = 10 × 300/(2 × 150) = 10 mm
```

Problem 16 – Change gears for thread cutting

Lead screw pitch 6 mm; required pitch 1.5 mm. Find the gear ratio.

```
Driver/driven = 1.5/6 = 1/4 → e.g. 20 T driver and 80 T driven
```

25F.4 Industrial Engineering

Problem 17 – EBQ

Annual demand 24,000; production rate 48,000 per year; set-up cost ₹500; holding cost ₹4/unit/year. Find the economic batch quantity.

```
Q = √(2DS/(H(1 − d/p))) = √(2 × 24,000 × 500/(4 × 0.5)) = √12,000,000 = 3464 units
```

Problem 18 – Quantity discount decision

D = 10,000 units/year, S = ₹100, H = 20% of price. Price ₹10 for Q < 1000, ₹9.5 for Q ≥ 1000. Find the best order quantity.

```
At ₹10: EOQ = √(2 × 10,000 × 100/2) = 1000 → not valid (falls in discount range); at ₹9.5: EOQ = √(2 × 10⁶/1.9) = 1026 (valid)
Total cost at Q = 1026: 95,000 + 10,000 × 100/1026 + 1026 × 1.9/2 = 95,000 + 975 + 975 = ₹96,950 → order 1026 units
```

Problem 19 – Halsey and Rowan bonus

Time allowed 10 h, time taken 8 h, rate ₹50/h. Find earnings under Halsey (50%) and Rowan plans.

```
Halsey: 8 × 50 + 0.5 × 2 × 50 = 400 + 50 = ₹450
Rowan: 400 + (2/10) × 8 × 50 = 400 + 80 = ₹480
```

Problem 20 – Depreciation

A machine costs ₹1,00,000, salvage ₹10,000 after 9 years. Find annual straight-line depreciation.

```
D = (1,00,000 − 10,000)/9 = ₹10,000 per year
```

Problem 21 – OEE

Availability 90%, performance 95%, quality 98%. Find OEE.

```
OEE = 0.90 × 0.95 × 0.98 = 83.8%
```

Problem 22 – CPM floats

Activity duration 5 days; ES = 4, LF = 15. Find total float. Successor ES = 10; find free float.

```
EF = 9; LS = 10; total float = 10 − 4 = 6 days; free float = 10 − 9 = 1 day
```

Problem 23 – Crashing cost slope

Normal: 10 days, ₹5000; crash: 7 days, ₹6500. Find the cost slope.

```
Slope = (6500 − 5000)/(10 − 7) = ₹500 per day
```

Problem 24 – p chart limits

Average fraction defective 0.04 with sample size 100. Find control limits.

```
σ = √(0.04 × 0.96/100) = 0.0196; UCL = 0.04 + 0.0588 = 0.0988; LCL = 0.04 − 0.0588 < 0 → 0
```

Problem 25 – c chart limits

Average 9 defects per casting. Find control limits.

```
UCL = 9 + 3 × 3 = 18; LCL = 9 − 9 = 0
```

Problem 26 – Reliability with exponential law

A component has constant failure rate 0.0002 per hour. Find reliability for 1000 h and MTBF.

```
R = e^(−0.2) = 0.819; MTBF = 1/λ = 5000 h
```

CHAPTER 25G

Solved Numericals – Set 7 (Quick One-Concept Drills)

Very short problems for speed practice – aim for 30–45 seconds each (no options).

25G.1 Mechanics and SOM Drills

Problem 1 – Equal forces

Two forces of 10 N each act at 120°. Find the resultant.

$R = 2P\cos(\theta/2) = 2 \times 10 \times \cos 60° = 10\ N$

Problem 2 – Friction angle

μ = 0.577. Find the angle of repose.

$\tan\varphi = 0.577 \rightarrow \varphi = 30°$

Problem 3 – Centroid of composite area

A T-section has flange 100 × 20 mm and web 20 × 80 mm (web below flange). Find ȳ from the bottom of the web.

$\bar{y} = (2000 \times 90 + 1600 \times 40)/3600 = (180{,}000 + 64{,}000)/3600 = 67.8\ mm$

Problem 4 – Polar MOI of square

Find I_z of a 100 mm square about its centroid.

$I_x = I_y = 100^4/12 = 8.33 \times 10^6\ mm^4 \rightarrow I_z = 1.67 \times 10^7\ mm^4$

Problem 5 – Strain from stress

Steel is stressed to 100 MPa. Find the strain.

$e = 100/2 \times 10^5 = 5 \times 10^{-4}$

Problem 6 – Bulk modulus from E and μ

E = 120 GPa, μ = 0.3. Find K.

$K = 120/(3 \times 0.4) = 100\ GPa$

Problem 7 – Max shear from principal stresses

σ_1 = 60 MPa, σ_2 = 20 MPa (plane stress). Find in-plane and absolute maximum shear.

In-plane = 20 MPa; absolute (σ_3 = 0) = 60/2 = 30 MPa

Problem 8 – Mohr's circle radius

σx = 50, σy = 10, τ = 15 MPa. Find R.

$R = \sqrt{(20^2 + 15^2)} = 25\ MPa$

Problem 9 – Max BM of SSB with two loads

A 6 m SSB carries 10 kN at 2 m and 4 m. Find maximum BM.

R = 10 kN each → M under loads = 10 × 2 = 20 kN·m (constant between loads)

Problem 10 – Section modulus ratio

Compare Z of a 100 × 200 beam placed with depth 200 vs depth 100.

$Z_1 = 100 \times 200^2/6$, $Z_2 = 200 \times 100^2/6 \rightarrow$ ratio 2:1

Problem 11 – Shape factor use

A rectangular beam yields at $M_y = 20$ kN·m. Find the plastic moment.

$M_p = 1.5 \times 20 = 30$ kN·m

Problem 12 – Deflection ratio

A cantilever's length is doubled under the same end load. Deflection becomes?

$\delta \propto L^3 \rightarrow$ 8 times

Problem 13 – Torque from power

A shaft transmits 20 kW at 1000 rpm. Find torque.

$T = 60 \times 20{,}000/(2\pi \times 1000) = 191$ N·m

Problem 14 – Shear stress in shaft

Find the shear stress in a 40 mm shaft carrying 191 N·m.

$\tau = 16 \times 191{,}000/(\pi \times 64{,}000) = 15.2$ MPa

Problem 15 – Spring cut in half

A spring of stiffness 10 N/mm is cut into two equal halves. Stiffness of each? Both in parallel?

Each 20 N/mm; parallel 40 N/mm

Problem 16 – Euler ratio

Ratio of Euler loads: both ends fixed vs one end fixed other free.

4 : 1/4 = 16 : 1

Problem 17 – Thin cylinder hoop stress

d = 500 mm, t = 5 mm, p = 1 MPa. Find σ_c and σ_l.

$\sigma_c = 50$ MPa; $\sigma_l = 25$ MPa

Problem 18 – Rotating ring

A steel ring ($\rho = 7800$) has rim speed 50 m/s. Find hoop stress.

$\sigma = \rho v^2 = 7800 \times 2500 = 19.5$ MPa

Problem 19 – Water thrust on dam

Find the water thrust per metre on a vertical face 10 m deep.

$P = \rho g h^2/2 = 1000 \times 9.81 \times 100/2 = 490.5$ kN/m acting at 3.33 m above base

Problem 20 – Riveted joint efficiency

Pitch 60 mm, rivet hole 20 mm (tearing governs). Find efficiency.

$\eta = (60 - 20)/60 = 66.7\%$

25G.2 Production Drills

Problem 21 – Lever rule

Find the percentage of pearlite in a 0.4% C steel at room temperature (ferrite ≈ 0% C).

```
Pearlite = 0.4/0.8 = 50%
```

Problem 22 – Draft allowance

A 200 mm deep pattern face needs 1° draft. Find the taper per side.

```
200 × tan1° = 3.5 mm
```

Problem 23 – Cutting speed

A 100 mm workpiece is turned at 300 rpm. Find V.

```
V = π × 100 × 300/1000 = 94.2 m/min
```

Problem 24 – Spindle speed

Required V = 30 m/min with a 20 mm drill. Find N.

```
N = 1000V/(πD) = 30,000/62.83 = 477 rpm
```

Problem 25 – Chip thickness ratio

Uncut thickness 0.25 mm, chip 0.75 mm. Find r and chip reduction coefficient.

```
r = 0.333; ζ = 3
```

Problem 26 – Taylor with n = 0.5

If speed is halved (n = 0.5), tool life becomes?

```
T ∝ V^(−1/n) = V^(−2) → 4 times
```

Problem 27 – Angular indexing

Index 27°.

```
27/9 = 3 full turns
```

Problem 28 – Vernier least count

1 MSD = 0.5 mm; 25 vernier divisions equal 24 main scale divisions. Find LC.

```
LC = 1 MSD − 1 VSD = 0.5 − (24 × 0.5/25) = 0.5 − 0.48 = 0.02 mm
```

Problem 29 – Number of IT grade steps

How does tolerance change from IT7 to IT9?

```
16i → 40i → 2.5 times
```

Problem 30 – Standard time with allowance on standard time

Normal time 5 min; allowance 20% of standard time. Find standard time.

```
ST = 5/(1 − 0.2) = 6.25 min (note: if allowance is 20% of normal time, ST = 6 min)
```

CHAPTER 26

Concept Recall Bank – Part A (Mechanics and SOM)

Short question → crisp answer. Cover the right column and test yourself (active recall, not MCQs).

R1. Engineering Mechanics

Question	Answer
Lami's theorem applies to?	Three concurrent coplanar forces in equilibrium
Resultant of two equal forces P at angle θ?	2P cos(θ/2)
Couple is what kind of vector?	Free vector
Varignon's theorem?	Moment of resultant = sum of moments of components
Reactions at a hinge / roller / fixed end?	2 / 1 / 3
Angle of repose equals?	Angle of friction
μ in terms of friction angle?	tanφ
Least force on horizontal plane applied at?	Angle φ to horizontal (P = W sinφ)
Belt tension ratio?	e^(μθ)
Self-locking screw efficiency?	Less than 50%
Perfect truss condition?	m = 2j − 3
Centroid of triangle from base?	h/3
Centroid of semicircle from diameter?	4r/3π
Centroid of solid hemisphere?	3r/8 from flat face
Centroid of solid cone?	h/4 from base
I of rectangle about centroid / base?	$bd^3/12$ / $bd^3/3$
I of triangle about centroid / base?	$bh^3/36$ / $bh^3/12$
I of circle / polar?	$\pi d^4/64$ / $\pi d^4/32$
Mass MOI of solid sphere?	$(2/5)MR^2$
Mass MOI of rod about end?	$ML^2/3$
Maximum range of projectile at?	45°
e for perfectly plastic impact?	0
D'Alembert's principle?	Inertia force makes dynamics a statics problem

R2. Stress, Strain and Properties

Question	Answer
Hooke's law?	σ = Eε within proportional limit
Poisson's ratio range?	0 to 0.5
μ of steel / rubber / cork?	≈ 0.3 / ≈ 0.5 / ≈ 0

Question	Answer
E–G relation?	$E = 2G(1 + \mu)$
E–K relation?	$E = 3K(1 - 2\mu)$
E in terms of K and G?	$9KG/(3K + G)$
Independent elastic constants (isotropic)?	2
Elongation of bar?	PL/AE
Elongation under self-weight?	$\rho gL^2/2E$ (= $WL/2AE$)
Tapering circular bar elongation?	$4PL/(\pi E d_1 d_2)$
Thermal stress (fully restrained)?	$E\alpha\Delta T$ – independent of length
Composite bar condition?	Equal strains
Suddenly applied load stress?	$2P/A$
Strain energy per unit volume?	$\sigma^2/2E$
Proof resilience?	Max strain energy up to elastic limit
Modulus of toughness?	Area under σ–ε curve up to fracture
Necking begins at?	Ultimate tensile strength
Proof stress for non-ferrous?	0.2% offset
Most ductile and malleable metal?	Gold
Creep stage used for design?	Secondary (steady state)
Endurance limit of steel ≈?	0.5 × UTS
Izod specimen supported as?	Cantilever
Charpy specimen supported as?	Simply supported beam
Vickers indenter?	Diamond pyramid 136°
Rockwell C indenter?	Diamond cone 120°
Brinell indenter?	10 mm steel/carbide ball
Stress concentration at small circular hole?	3
Shot peening improves?	Fatigue strength
Castigliano's theorem?	$\delta = \partial U/\partial P$

R3. Principal Stresses and Failure

Question	Answer
Max shear in uniaxial tension?	$\sigma/2$ on 45° planes
Shear on principal planes?	Zero
Angle between principal and max shear planes?	45°
Mohr's circle radius?	$\tau_{max} = \sqrt{((\sigma x - \sigma y)/2)^2 + \tau^2)}$
Mohr's circle for equal biaxial stress?	A point
Mohr's circle for pure shear?	Centred at origin
Theory for ductile materials (conservative)?	Maximum shear stress (Tresca)
Most accurate theory for ductile materials?	Distortion energy (von Mises)
Theory for brittle materials?	Maximum principal stress (Rankine)
Shear yield by von Mises?	$0.577\sigma_y$
Shear yield by Tresca?	$0.5\sigma_y$
Tresca locus shape?	Hexagon

Question	Answer
von Mises locus shape?	Ellipse
Equivalent torque?	$\sqrt{(M^2 + T^2)}$
Equivalent bending moment?	$\frac{1}{2}[M + \sqrt{(M^2 + T^2)}]$

R4. Beams

Question	Answer
dM/dx equals?	Shear force
BM maximum where?	SF = 0 or changes sign
BMD for UDL?	Parabola
BMD for UVL?	Cubic
SFD for UDL?	Linear
Point of contraflexure?	BM changes sign
Max BM SSB UDL / central load?	$wL^2/8$ / $WL/4$
Max BM cantilever UDL / end load?	$wL^2/2$ / WL
Fixed beam UDL end moment?	$wL^2/12$
Propped cantilever UDL prop reaction?	$3wL/8$
Bending equation?	$M/I = \sigma/y = E/R$
Neutral axis passes through?	Centroid
Z of rectangle / circle?	$bd^2/6$ / $\pi d^3/32$
Strongest beam from log d/b?	$\sqrt{2}$
Stiffest beam from log d/b?	$\sqrt{3}$
Shape factor rectangle / circle / I?	1.5 / 1.7 / ≈ 1.14
Middle-third rule?	$e \leq d/6$ for no tension
Core of circular section?	Diameter d/4
τ_{max}/τ_{avg} rectangle / circle?	1.5 / 4/3
Max shear stress location in triangle?	Mid-height
Shear centre of channel?	Outside web
Max stress in crane hook?	Inner fibre
Cantilever end deflection (point / UDL)?	$WL^3/3EI$ / $wL^4/8EI$
SSB central deflection (point / UDL)?	$WL^3/48EI$ / $5wL^4/384EI$
Fixed beam central deflection (point / UDL)?	$WL^3/192EI$ / $wL^4/384EI$
Conjugate beam: fixed end becomes?	Free end
Moment-area: change of slope?	Area of M/EI diagram

R5. Torsion, Springs, Columns, Cylinders

Question	Answer
Torsion equation?	$T/J = \tau/r = G\theta/L$
Max shear in solid shaft?	$16T/\pi d^3$
Power from torque?	$2\pi NT/60$
Shafts in series share?	Same torque
Shafts in parallel share?	Same angle of twist

Question	Answer
Strain energy in solid shaft?	$\tau^2V/4G$
Spring stiffness?	$Gd^4/(8D^3n)$
Springs in series?	$1/k = 1/k_1 + 1/k_2$
Wahl factor accounts for?	Direct shear and curvature
Leaf spring is a?	Beam of uniform strength
Euler load?	π^2EI/L_e^2
L_e fixed–free / fixed–fixed / fixed–hinged?	2L / L/2 / L/√2
Ratio of Euler loads (F–F : H–H : F–H : Fx–Fx)?	1 : 4 : 8 : 16
Slenderness ratio?	L_e/k_min
Rankine constant?	σ_c/π^2E
Hoop stress in thin cylinder?	pd/2t
Longitudinal stress?	pd/4t
Thin sphere stress?	pd/4t
Volumetric strain of thin cylinder?	$(pd/4tE)(5 - 4\mu)$
Max hoop stress in thick cylinder at?	Inner surface
Theory for thick cylinders?	Lamé's theory
Fillet weld throat?	0.707 × leg size
Joint efficiency?	Least strength/solid plate strength
Unwin's formula?	$d = 6\sqrt{t}$

CHAPTER 26B

Concept Recall Bank – Part B (Production Engineering)

Quick recall for materials, processes, machining, metrology and industrial engineering.

R6. Materials and Heat Treatment

Question	Answer
APF of BCC / FCC?	0.68 / 0.74
Coordination number BCC / FCC?	8 / 12
Austenite structure?	FCC
Ferrite structure?	BCC
Martensite structure?	BCT
Eutectoid point?	727 °C, 0.8% C – pearlite
Eutectic point?	1147 °C, 4.3% C – ledeburite
Peritectic temperature?	≈ 1495 °C
Max C in austenite?	2.14%
Carbon in cementite?	6.67%
Mild steel carbon?	0.15–0.3%
Tool steel carbon?	0.8–1.5%
40C8 means?	0.40% C, 0.8% Mn
AISI 1040 means?	Plain carbon steel with 0.40% C
Element for red hardness?	Tungsten
Element causing hot shortness?	Sulphur
Element causing cold shortness?	Phosphorus
Element reducing hardenability?	Cobalt
18-4-1 HSS?	W 18%, Cr 4%, V 1%
18-8 stainless?	Cr 18%, Ni 8% (austenitic)
Hadfield steel?	12–14% Mn, work-hardening
Graphite form in grey CI?	Flakes
SG iron made by adding?	Mg or Ce
Malleable iron from?	Annealing white CI
Brass / bronze?	Cu–Zn / Cu–Sn
Duralumin used in?	Aircraft
Annealing cooling?	Furnace
Normalising cooling?	Still air
Tempering purpose?	Reduce brittleness, increase toughness
Austempering gives?	Bainite
Hardenability test?	Jominy end-quench

Question	Answer
Nitriding temperature?	500–550 °C, no quench
Case depth in induction hardening vs frequency?	Decreases as frequency increases

R7. Casting and Welding

Question	Answer
Negative pattern allowance?	Shake (rapping)
Shrinkage allowance CI / steel?	≈ 1% / ≈ 2%
Pattern for symmetrical large castings?	Sweep
Pattern for mass production of small castings?	Match-plate
Binder in green sand?	Clay (bentonite)
CO_2 process binder?	Sodium silicate
Support for cores?	Chaplets
Chvorinov's rule?	$t = B(V/A)^2$
Riser should solidify?	Last
Most efficient riser shape?	Sphere
Why is sprue tapered?	Prevent aspiration
Turbine blades / jewellery cast by?	Investment casting
Pipes without cores?	True centrifugal casting
Hot chamber die casting metals?	Zn, Sn, Pb (low melting)
Cold chamber die casting metals?	Al, brass
Furnace for cast iron?	Cupola
Defect from moisture and low permeability?	Blowholes
Two streams not fused?	Cold shut
Hot tear due to?	Poor collapsibility
Flame for brass?	Oxidising
Flame for most metals?	Neutral
Acetylene dissolved in?	Acetone
Deeper penetration polarity?	DCSP (electrode negative)
TIG electrode?	Non-consumable tungsten
SAW position?	Flat only
Heat in resistance welding?	I^2Rt
Rail welding?	Thermit / flash butt
Brazing vs soldering temperature?	Above / below 450 °C
Welding in vacuum?	Electron beam

R8. Forming

Question	Answer
Hot vs cold working decided by?	Recrystallisation temperature
Lead at room temperature is?	Hot worked
Maximum draft in rolling?	μ^2R
Mill for thin hard strips?	Cluster (Sendzimir)

Question	Answer
Coining is?	Closed-die cold forging
Upsetting does?	Increases cross-section
Toothpaste tubes made by?	Impact extrusion
Less force extrusion?	Indirect (backward)
Max reduction in wire drawing (ideal)?	≈ 63%
Chevron cracks in?	Extrusion/drawing (centre burst)
Blanking – product is?	Removed piece
Punching – product is?	Sheet with hole
Clearance in blanking applied to?	Punch
Clearance in punching applied to?	Die
Punching force?	L t τ
Deep-drawn blank diameter?	$\sqrt{(d^2 + 4dh)}$
Limiting drawing ratio?	≈ 2
Wrinkling cause?	Low blank-holder force
Several operations at successive stations?	Progressive die
Thread rolling advantage?	Stronger threads (grain flow)

R9. Machining and Machine Tools

Question	Answer
Tool signature first term (ASA)?	Back rake
Rake for carbides?	Negative
Continuous chip conditions?	Ductile, high speed, positive rake
Chip in cast iron?	Discontinuous
BUE occurs at?	Low speed, ductile material
Chip thickness ratio?	$t/t_c < 1$
Merchant's relation?	$2\varphi + \beta - \alpha = 90°$
Largest cutting force?	Tangential (cutting) force
Most heat goes to?	Chip
Taylor's equation?	$VT^n = C$
n for HSS / carbide?	≈ 0.125 / ≈ 0.25
Crater wear location?	Rake face
Tool-life criterion?	Flank wear land (VB)
Hardest tool material?	Diamond
Tool for hardened steel?	CBN
Diamond unsuitable for?	Ferrous metals
Lead screw used for?	Thread cutting only
Long tapers on lathe?	Tailstock set-over
Short steep tapers?	Compound rest
Twist drill point angle?	118°
Hole finishing with fixed tool?	Reaming
Shaper mechanism?	Quick return (crank and slotted lever)

Question	Answer
Planer: what reciprocates?	Work table
Simple indexing?	40/N
Climb milling needs?	Backlash eliminator
Fastest gear generating?	Hobbing
Internal gears cut by?	Gear shaper
Hard work needs wheel grade?	Soft
Abrasive for cast iron?	Silicon carbide
Centreless grinding uses?	Regulating wheel
Cylinder bore finishing?	Honing
Gauge blocks finished by?	Lapping
EDM dielectric?	Kerosene / deionised water
ECM principle?	Faraday's laws (anodic dissolution)
USM frequency?	≈ 20 kHz
Machining glass/ceramics?	USM, AJM
EBM needs?	Vacuum
Aircraft panel thinning?	Chemical milling

R10. Metrology, CNC and Industrial Engineering

Question	Answer
Preferred fit system?	Hole basis
H7/g6 fit?	Clearance
H7/k6 fit?	Transition
H7/p6 fit?	Interference
Number of IT grades?	18 (IT01 to IT16)
Tolerance unit?	$0.45\sqrt[3]{D} + 0.001D$ µm
Taylor's principle – GO gauge checks?	All features (maximum material)
Wear allowance on?	GO gauge
Vernier least count (50 div)?	0.02 mm
Micrometer least count?	0.01 mm
Instrument obeying Abbe's principle?	Micrometer
Sine bar used up to?	≈ 45°
Best wire size (60° thread)?	0.577p
Optical flat fringe?	λ/2
Ra definition?	Arithmetic mean deviation (CLA)
G02 / G03?	Circular CW / CCW
M03 / M06 / M30?	Spindle CW / tool change / end and rewind
Z-axis in CNC?	Along spindle
Jig vs fixture?	Jig guides tool; fixture only holds
3-2-1 principle restricts?	9 DOF by locators
Standard time?	Normal time + allowances
Therbligs by?	Gilbreth

Question	Answer
Work sampling by?	Tippett
MTM unit TMU?	0.036 s
EOQ?	√(2DS/H)
ABC analysis basis?	Annual consumption value
VED analysis for?	Spare parts (criticality)
Chart for defects per unit?	c chart
Chart for fraction defective?	p chart
PERT time estimates?	Three (beta distribution)
CPM nature?	Deterministic
Critical path float?	Zero
Johnson's rule?	n jobs, 2 machines
SPT rule minimises?	Mean flow time
Product layout for?	Mass production
Fixed-position layout for?	Ships, aircraft
Six Sigma defect level?	3.4 per million
Fishbone diagram by?	Ishikawa

CHAPTER 26C

Concept Recall Bank – Part C (Special Topics)

Quick recall for NDT, composites, special stresses, presses, welding practice, corrosion, maintenance and safety.

R11. NDT, Composites and Polymers

Question	Answer
NDT only for ferromagnetic materials?	Magnetic particle test
NDT for surface cracks in any non-porous material?	Dye penetrant test
Internal porosity in castings/welds?	Radiography
Thickness measurement from one side?	Ultrasonic testing
Gamma-ray sources?	Co-60, Ir-192
Conductive tubes inspection?	Eddy current testing
Bullet-proof vests fibre?	Kevlar (aramid)
Cemented carbide is which composite?	Particulate (cermet)
Rule of mixtures (longitudinal)?	$E_c = E_f V_f + E_m V_m$
Natural composite examples?	Wood, bone
Aircraft brake discs?	Carbon–carbon composite
Polymer with lowest friction?	PTFE (Teflon)
Bakelite is?	Thermoset (phenol-formaldehyde)
Rubber vulcanised with?	Sulphur
Plasticiser used in?	PVC
Nylon formed by?	Condensation polymerisation
Polyethylene formed by?	Addition polymerisation
Glass structure?	Amorphous
Refractory for basic slag?	Basic (magnesite, dolomite)

R12. Special Stresses and Presses

Question	Answer
Hoop stress in rotating ring?	ρv^2
Effect of small hole in rotating disc?	Doubles max stress
Water thrust on vertical face?	$\rho g h^2/2$ at $h/3$
No tension in dam base?	Resultant within middle third
Rankine k_a?	$(1 - \sin\varphi)/(1 + \sin\varphi)$
Max stress in curved beam?	Inner fibre
Fatigue notch factor?	$1 + q(K_t - 1)$
Subsurface maximum shear causes?	Pitting (contact fatigue)

Question	Answer
Press for deep drawing?	Hydraulic (double action)
Press for coining?	Knuckle-joint
Shut height?	Bed to slide at BDC (adjustment up)
Die doing blanking + piercing at one station?	Compound die
Die doing operations at successive stations?	Progressive die
Cutting + forming at one station?	Combination die
Pilots used in?	Progressive dies
Stripper function?	Removes strip from punch
Large dish ends in small numbers?	Explosive forming
Crimping tubes with magnetic field?	Electromagnetic forming
Long channel sections?	Roll forming

R13. Welding Practice and Corrosion

Question	Answer
Fillet weld symbol?	Triangle
Symbol below reference line means?	Weld on arrow side
Field weld symbol?	Flag
Edge preparation for thick plates?	Double-V, U or J
Residual stress in weld metal?	Tensile
Distortion control sequence?	Back-step / skip welding
Aluminium welding current (TIG)?	AC
Cast iron welding electrode?	Nickel-based
Cold cracking prevented by?	Preheat + low-hydrogen electrodes
Gas cutting unsuitable for?	Stainless steel, aluminium, cast iron
Back gouging process?	Air-carbon arc gouging
Most active metal in galvanic series?	Magnesium
Sacrificial anodes?	Zinc, magnesium, aluminium
Season cracking of brass?	SCC in ammonia
Zinc lost from brass?	Dezincification
Caustic embrittlement occurs in?	Boilers
Worst galvanic area ratio?	Small anode, large cathode
Rusting requires?	Oxygen and moisture
Age hardening steps?	Solution treat → quench → age
Machine tool bed material?	Grey cast iron (damping)
Ball bearing steel?	High-carbon chromium (52100)
Turbine blade material?	Nickel superalloy

R14. Maintenance, Safety and Costing

Question	Answer
Maintenance based on condition monitoring?	Predictive
Fixed-interval maintenance?	Preventive

Question	Answer
OEE?	Availability × performance × quality
Main cause of accidents (Heinrich)?	Unsafe acts
Heinrich ratio?	1 : 29 : 300
Fire on oil – avoid?	Water jet
Electrical fire extinguisher?	CO_2 / dry powder
Class D fire?	Metals
Value?	Function/cost
Value engineering pioneer?	L. D. Miles
Prime cost?	Direct material + labour + expenses
Halsey bonus share?	50% of time saved
Straight-line depreciation?	(P − S)/n
Noise limit for 8 h (India)?	90 dB(A)

CHAPTER 27

Paper-II Model Answers

Descriptive answers in the length expected for SSC JE Paper-II (conventional) – learn the structure: definition → explanation/sketch → equation → application.

P1. Strength of Materials

✎ **DESCRIPTIVE Q:** Derive the relation between E, G and μ.

Consider a square element ABCD of side a under pure shear stress τ. The diagonal AC elongates and BD shortens. Pure shear is equivalent to a tensile stress τ along one diagonal and a compressive stress τ along the other. The strain along AC is therefore $e = \tau/E + \mu\tau/E = (\tau/E)(1 + \mu)$. Geometrically, the shear strain $\varphi = \tau/G$ distorts the square so that corner C moves horizontally by $a\varphi$; the extension of diagonal AC (length $a\sqrt{2}$) is $a\varphi \cos 45° = a\varphi/\sqrt{2}$, so the diagonal strain = $(a\varphi/\sqrt{2})/(a\sqrt{2}) = \varphi/2 = \tau/(2G)$. Equating the two expressions for the diagonal strain: $(\tau/E)(1 + \mu) = \tau/(2G)$, which gives $E = 2G(1 + \mu)$. Similarly, by considering a cube under equal stresses σ on all faces, volumetric strain = $3(\sigma/E)(1 - 2\mu) = \sigma/K$, giving $E = 3K(1 - 2\mu)$. Eliminating μ between these yields $E = 9KG/(3K + G)$. Since μ lies between 0 and 0.5, E lies between 2G and 3G.

✎ **DESCRIPTIVE Q:** Explain the theory of pure torsion and state why hollow shafts are preferred to solid shafts.

In pure torsion a shaft carries a constant torque with no bending moment or axial load. Assuming that cross-sections remain plane and radii remain straight, the shear strain at radius r is $r\theta/L$, so shear stress $\tau = G\theta r/L$ increases linearly from zero at the axis to a maximum at the surface. Integrating the moment of the shear stresses over the section gives $T = G\theta J/L$, leading to the torsion equation $T/J = \tau/r = G\theta/L$. Because the material near the centre is lightly stressed and has a small lever arm, it contributes little to torque capacity. A hollow shaft removes this inefficiently used material, so for the same weight it has a larger polar moment of inertia and polar modulus: it can transmit more torque and is stiffer (smaller twist). For example, a hollow shaft with inner diameter half the outer has about 94% of the strength of a solid shaft of the same outer diameter but only 75% of its weight; for equal weight it is about 44% stronger. Hollow shafts also allow passage of oil or cables and have higher natural frequency. Their disadvantages are higher manufacturing cost and larger outside diameter for the same torque, so they are used where weight matters – aircraft, propeller shafts, machine tool spindles.

✎ **DESCRIPTIVE Q:** What are the assumptions in Euler's column theory? State its limitations and how Rankine's formula overcomes them.

Euler's theory assumes that the column is initially perfectly straight and of uniform cross-section; the load is exactly axial; the material is homogeneous, isotropic and obeys Hooke's law; the column is long compared with its lateral dimensions so that it fails by buckling only; self-weight is neglected; and end conditions are ideal (pinned, fixed or free). It gives $P = \pi^2 EI/L_e^2$. Its limitations: it ignores direct compressive (crushing) stress, so for short and medium columns it predicts loads far higher than the crushing load; it is valid only when the critical stress $\pi^2 E/\lambda^2$ is below the proportional limit, i.e. for large slenderness ratios (above

about 80–100 for mild steel); and real columns have initial crookedness, eccentric loading and imperfect end fixity, which reduce the load. Rankine's empirical formula combines crushing and buckling: $1/P_R = 1/P_c + 1/P_e$, or $P_R = \sigma_c A/(1 + a\lambda^2)$. For short columns P_e is very large and P_R approaches the crushing load; for long columns P_e is small compared with P_c and P_R approaches the Euler load. Thus a single formula covers the whole range of slenderness, with the constant a determined from tests for each material.

✎ **DESCRIPTIVE Q:** Explain the concept of the point of contraflexure with an example.

A point of contraflexure (or inflexion) is a point in a beam where the bending moment changes sign, from sagging to hogging, passing through zero. At this point the curvature of the elastic curve changes direction. It occurs only in beams where both sagging and hogging moments exist – overhanging beams, fixed beams, propped cantilevers and continuous beams – and not in simply supported beams with downward loads or in cantilevers. For example, a fixed beam carrying a uniformly distributed load w over span L has hogging moments $wL^2/12$ at the supports and a sagging moment $wL^2/24$ at the centre; setting the BM $M(x) = wLx/2 - wx^2/2 - wL^2/12 = 0$ gives $x = 0.211L$ from each end, which are the points of contraflexure. Similarly, an overhanging beam with loads on the overhang has a point of contraflexure between the supports. These points are significant because the bending stress is zero there: they are suitable locations for joints or splices in continuous beams, and in reinforced concrete the main steel must be shifted from the bottom to the top across them.

P2. Production Engineering

✎ **DESCRIPTIVE Q:** Explain the iron–carbon equilibrium diagram with its important reactions.

The iron–iron carbide diagram shows the phases present in steels and cast irons at different carbon contents (0–6.67%) and temperatures under slow (equilibrium) cooling. Pure iron exists as δ-ferrite (BCC) above 1394 °C, austenite γ (FCC) between 1394 and 912 °C, and α-ferrite (BCC) below 912 °C. Austenite dissolves up to 2.14% carbon at 1147 °C; ferrite only 0.022% at 727 °C. Cementite (Fe_3C, 6.67% C) is a hard, brittle compound. Three invariant reactions occur: the peritectic at about 1495 °C and 0.17% C (δ + liquid → γ); the eutectic at 1147 °C and 4.3% C, where liquid transforms to ledeburite (γ + Fe_3C); and the eutectoid at 727 °C and 0.8% C, where austenite transforms to pearlite, a lamellar mixture of ferrite and cementite. Steels (below 2.14% C) are classified as hypo-eutectoid (ferrite + pearlite), eutectoid (pearlite) and hyper-eutectoid (pearlite + cementite); cast irons contain 2.14–6.67% C. The lines A_1 (727 °C), A_3 and A_{cm} define the critical temperatures used in heat treatment – for example, hypo-eutectoid steels are hardened from about 30–50 °C above A_3. The diagram does not show non-equilibrium products such as martensite and bainite, which are described by TTT diagrams.

✎ **DESCRIPTIVE Q:** Describe the different types of chips formed in metal cutting.

(1) Continuous chips are long ribbons with a smooth shiny under-surface, formed when machining ductile materials (mild steel, copper, aluminium) at high cutting speed, with a large positive rake angle, small feed and depth, sharp tool and efficient cutting fluid. The material deforms plastically along the shear plane without fracture. They give good surface finish and low power consumption, but they are hazardous and difficult to dispose of, so chip breakers are used. (2) Continuous chips with built-up edge form when ductile materials are cut at low speeds with high friction at the tool–chip interface: work material welds to the rake face and builds up, periodically breaking off and sticking to the chip and machined surface, giving poor finish and altering the tool geometry; higher speed, larger rake, better lubrication and polished tools reduce it. (3) Discontinuous (segmental) chips are small separate segments formed when

machining brittle materials such as cast iron, brass and bronze, or ductile materials at very low speed with small rake and large feed; they are easy to handle and the finish is fair in brittle materials, but force fluctuations can cause vibration. (4) Serrated or saw-tooth chips form with low-conductivity, hard materials (titanium alloys, hardened steels) at high speeds due to adiabatic shear bands.

✎ **DESCRIPTIVE Q:** Explain the gating system of a sand mould and the functions of each element.

The gating system leads molten metal from the ladle into the mould cavity in a controlled way. The pouring basin (cup) receives the metal, reduces its momentum and helps keep slag back; a skim core or strainer may be included. The sprue is a vertical passage leading metal down to the parting level; it is tapered (larger at the top) so that, as the stream accelerates, it stays in contact with the walls, avoiding aspiration of air and gas into the metal. The sprue well at its base reduces the velocity and turbulence where the flow turns horizontal. The runner carries metal horizontally, usually in the drag or cope, and is extended beyond the last gate to trap the first cold, dirty metal; skim bobs trap slag. The ingates connect the runner to the cavity – top, parting-line, bottom or step gates – and control the rate, direction and distribution of metal entry. The gating ratio (sprue : runner : gate area) determines whether the system is pressurised (choke at gates, used for ferrous metals) or unpressurised (choke at sprue, used for aluminium and magnesium to reduce turbulence and oxidation). Risers, though not part of the gating proper, are connected to supply feed metal as the casting shrinks. A good gating system fills the mould quickly without turbulence, erosion or cold shuts, removes slag and promotes directional solidification towards the risers.

✎ **DESCRIPTIVE Q:** Explain the various methods of taper turning on a lathe.

(1) Form tool method: a broad-nosed tool with its cutting edge set at the half-taper angle is fed straight in; suitable only for short tapers (chamfers) because long edges cause chatter. (2) Compound rest (swivelling) method: the compound rest is swivelled to the half-taper angle α ($\tan\alpha = (D - d)/2L$) and the tool is fed by hand along it; suitable for short, steep tapers, both external and internal, because the travel of the compound slide is limited. (3) Tailstock set-over method: the tailstock body is offset laterally so that the axis of the job held between centres is inclined to the lathe axis; the tool is fed by the carriage parallel to the bed. The set-over $S = (D - d)L/(2l)$, where L is the total job length and l the taper length. It is used for long, gentle external tapers but not for internal or steep tapers, and the centres do not bear properly. (4) Taper-turning attachment: a guide bar at the back of the lathe set at the half-taper angle guides the cross-slide (disconnected from its screw) as the carriage moves, so the tool follows the taper; it gives accurate long tapers, internal and external, with the work on centres or in a chuck. (5) Combined feed method: the tool is moved simultaneously by longitudinal and cross feeds. CNC lathes produce tapers directly by linear interpolation.

✎ **DESCRIPTIVE Q:** Distinguish between jigs and fixtures and state the principles of their design.

A jig is a device that holds and locates the workpiece and also guides the cutting tool to the correct position, usually through hardened drill bushes; jigs are used for drilling, reaming and tapping, and they are generally lighter and are not clamped to the machine table. A fixture holds and locates the work but does not guide the tool; the tool position is set by setting blocks and feeler gauges, and the fixture is firmly bolted to the machine table. Fixtures are used in milling, turning, grinding, welding, assembly and inspection. Design principles common to both: locate the workpiece positively using the 3-2-1 principle on machined datum surfaces, avoiding redundant location; provide rigid support against cutting forces, with clamps acting directly over supports and not on unsupported areas, so the work does not distort; ensure quick loading

and unloading with simple, fast-acting clamps (cam, toggle, quick-release); make the design fool-proof so the part can be loaded only in the correct orientation; provide chip clearance and coolant drainage, and relief at corners for burrs; use hardened, replaceable wear parts (bushes, locators); keep the device light but rigid, and safe to operate. Using jigs and fixtures reduces marking out, ensures interchangeability, permits less skilled labour and increases productivity in batch and mass production.

CHAPTER 27B

Paper-II Model Answers – Set 2

More conventional-style answers; practise writing each in 10–12 minutes with a neat sketch.

P3. Strength of Materials

✎ **DESCRIPTIVE Q:** Derive the expression for strain energy stored in a body due to shear and state the strain energy in a solid shaft under torsion.

Consider a rectangular block of height h, area A of the top face, fixed at the bottom and subjected to a shear force P on the top face applied gradually. The top face moves through a distance δ = φh, where the shear strain φ = τ/G. Work done = ½Pδ = ½(τA)(φh) = ½τφ(Ah) = (τ²/2G) × V, where V = Ah is the volume. This work is stored as shear strain energy, so the energy per unit volume is τ²/(2G). In a solid circular shaft under torque, the shear stress varies linearly from zero at the centre to τ_max at the surface: τ_r = τ_max r/R. The energy in an elemental ring of radius r, thickness dr and length L is (τ_r²/2G) × 2πr dr L. Integrating from 0 to R: U = (τ_max²/2GR²) × 2πL ∫r³ dr = (τ_max²/2GR²) × 2πL × R⁴/4 = (τ_max²/4G) × πR²L = (τ_max²/4G) × V. Thus a solid shaft stores only half the energy it would if the whole volume were stressed to τ_max. For a hollow shaft U = (τ²/4G) × [(D² + d²)/D²] × V, which is higher per unit volume – another reason hollow shafts are efficient for absorbing energy.

✎ **DESCRIPTIVE Q:** Explain the different methods of finding the slope and deflection of beams.

(1) Double integration: the differential equation EI d²y/dx² = M is integrated twice; constants are found from boundary conditions (y = 0 at supports, dy/dx = 0 at fixed ends). It is direct for simple loading but needs separate equations for each segment when loads are discontinuous. (2) Macaulay's method: a single BM expression valid for the whole beam is written using Macaulay brackets ⟨x − a⟩ that are taken as zero when negative; integration is done without opening the brackets, so only two constants are needed. It is convenient for several point loads. (3) Moment-area method: Mohr's theorems state that the change of slope between two points equals the area of the M/EI diagram between them, and the tangential deviation equals the moment of that area; it is quick for cantilevers and symmetrically loaded beams. (4) Conjugate beam method: the real beam's M/EI diagram is applied as load on an imaginary conjugate beam with modified supports (fixed ↔ free, simple support unchanged); the shear and BM in the conjugate beam give the slope and deflection of the real beam. (5) Energy methods – Castigliano's theorem and the unit-load (virtual work) method – are useful for curved members, frames and beams of varying section. (6) Superposition of standard results from tables for combined loads.

✎ **DESCRIPTIVE Q:** Explain the concept of a beam of uniform strength with an example.

A beam of uniform strength is one in which the maximum bending stress at every cross-section is the same and equal to the allowable stress. Since σ = M/Z, the section modulus must vary along the length in proportion to the bending moment. This makes the most economical use of material and reduces weight. For example, a cantilever carrying a point load W at the free end has M = Wx at distance x from the free end. If the depth d is kept constant, Z = bd²/6 ∝ b, so the width must vary linearly with x: b = 6Wx/(σd²) – the plan is triangular. If the width is

constant, d must vary as √x – a parabolic profile. For a simply supported beam with a central load and constant width, the depth varies as √x from each support. A laminated leaf spring is a practical beam of uniform strength: a triangular plate of uniform thickness (constant stress) is cut into strips and stacked, giving leaves of decreasing length. Other examples are tapered cantilever brackets, girders with cover plates added near mid-span, and fish-bellied beams. In practice a minimum section is kept near supports to carry shear.

P4. Production Engineering

✎ **DESCRIPTIVE Q:** Explain the principle and applications of electrochemical machining (ECM).

ECM is based on Faraday's laws of electrolysis – reverse electroplating. The workpiece is made the anode and a shaped tool the cathode of a DC circuit (low voltage 5–30 V, very high current density 10–100 A/cm²); a conductive electrolyte, usually sodium chloride or sodium nitrate solution, is pumped at high velocity through the small gap (0.1–0.6 mm) between them. Metal atoms at the anode surface dissolve as ions, forming hydroxides that are flushed away as sludge, while hydrogen gas is evolved at the cathode; the tool itself is not eroded. As the tool is fed towards the work at constant rate, an equilibrium gap is established and the work takes the mirror shape of the tool. The material removal rate depends on current and on the electrochemical equivalent of the metal, not on its hardness: MRR = AI/(ρnF). Advantages: any conductive material regardless of hardness can be machined; there is no tool wear; complex three-dimensional shapes are produced in one pass; no residual stresses, burrs or heat-affected zone; good surface finish. Limitations: high equipment cost, large power requirement, corrosion and disposal problems with electrolyte, inability to produce sharp internal corners, and only conductive materials. Applications: turbine blades and blisks, die cavities, rifling of gun barrels, deburring (electrochemical deburring), drilling small and deep holes, and electrochemical grinding of carbide tools.

✎ **DESCRIPTIVE Q:** Explain Taylor's tool life equation and the factors affecting tool life.

Tool life is the time a tool cuts satisfactorily before it must be reground or replaced, usually defined by a limiting flank wear land (e.g. VB = 0.3 mm) or by crater depth, failure or deterioration of finish. F. W. Taylor found experimentally that $VT^n = C$, where V is the cutting speed, T the tool life in minutes, n an exponent depending mainly on the tool material (about 0.1–0.15 for HSS, 0.2–0.4 for carbides and 0.5–0.7 for ceramics) and C a constant equal to the cutting speed for one-minute life. On a log–log plot the relation is a straight line of slope −n. The extended form VTⁿf^a d^b = K shows that feed and depth of cut also reduce tool life, but less than speed, and depth least of all. Factors affecting tool life: cutting speed (most important, since it controls temperature); feed and depth of cut; tool material and its hot hardness and wear resistance; tool geometry – rake, clearance, nose radius, approach angle; work material – hardness, abrasive inclusions, work hardening; cutting fluid; rigidity of machine and set-up (vibration and chatter); and type of cut (continuous or interrupted). Tool life is used to select economical cutting speeds: the speed for minimum cost or maximum production rate is obtained from Taylor's equation together with tool and machine costs.

✎ **DESCRIPTIVE Q:** Explain the various methods of control charts used in statistical quality control.

Control charts, introduced by Shewhart, plot a quality characteristic of samples taken at intervals against control limits normally placed at ±3σ from the centre line, so that, when only chance causes act, 99.73% of points fall within them. A point outside the limits, or a non-random pattern (runs, trends), indicates an assignable cause to be investigated. Charts for

variables (measurable characteristics such as diameter or weight): the $\bar{X}$ chart monitors the process mean, with limits $\bar{\bar{X}} \pm A_2\bar{R}$; the R chart monitors variability using $D_3\bar{R}$ and $D_4\bar{R}$; the s chart uses standard deviation for larger samples. They are used together. Charts for attributes (counted characteristics): the p chart plots fraction defective, limits $\bar{p} \pm 3\sqrt{(\bar{p}(1-\bar{p})/n)}$, and can handle variable sample sizes; the np chart plots number defective for constant n; the c chart plots number of defects per unit (Poisson distribution), limits $\bar{c} \pm 3\sqrt{\bar{c}}$, used for defects on castings, cloth, or painted surfaces; the u chart plots defects per unit for varying unit sizes. Variable charts give more information per sample and detect shifts earlier; attribute charts are simpler and cover many characteristics at once. Control charts help to achieve and maintain statistical control, estimate process capability and reduce scrap and inspection costs.

✎ **DESCRIPTIVE Q:** Explain the types of plant layout and their suitability.

Plant layout is the physical arrangement of machines, work centres, storage and services to allow smooth flow of materials at minimum cost. (1) Product or line layout arranges machines in the sequence of operations required by a product, as in automobile assembly lines and bottling plants. It suits mass production of standardised products with stable demand; material handling and work-in-process are minimal, control is simple and throughput high, but it is inflexible, requires high investment, and a breakdown of one machine stops the line; line balancing is essential. (2) Process or functional layout groups similar machines together – lathes in one section, milling machines in another – as in job shops, hospitals and tool rooms. It handles a variety of products in small batches, offers flexibility and better machine utilisation, and breakdowns affect only a part, but material handling, WIP and throughput time are high and production control is complex. (3) Fixed-position layout keeps the product stationary while workers, materials and equipment move to it – ship building, aircraft and locomotive assembly, large turbines, construction – used when the product is very large or heavy. (4) Cellular or group-technology layout forms cells of dissimilar machines to process families of similar parts, combining the efficiency of product layout with the flexibility of process layout for medium variety and volume. (5) Combination layouts mix these types.

CHAPTER 27C

Paper-II Model Answers – Set 3

Further descriptive answers covering topics that frequently appear in conventional papers.

P5. Mechanics and SOM

✎ **DESCRIPTIVE Q:** State the laws of dry friction and explain the angle of friction and angle of repose.

The laws of static (Coulomb) friction are: (1) the frictional force always acts opposite to the direction in which the body tends to move; (2) up to the limiting value, the friction force is exactly equal to the applied force and the body stays at rest (friction is self-adjusting); (3) the limiting friction bears a constant ratio to the normal reaction, F = μN; (4) limiting friction is independent of the apparent area of contact; (5) it depends on the nature and condition of the surfaces. For kinetic friction, the friction force is slightly less than limiting friction, roughly constant for moderate speeds, and the ratio F_k/N = μ_k is less than μ_s. The angle of friction φ is the angle that the resultant of the limiting friction force and normal reaction makes with the normal: tanφ = F/N = μ. If a body is placed on an inclined plane and the inclination is slowly increased, the body is on the point of sliding when the component of weight along the plane W sinα equals the limiting friction μW cosα, i.e. tanα = μ. This limiting angle α is the angle of repose, and it equals the angle of friction. Rotating the resultant reaction about the normal generates the cone of friction of semi-vertex angle φ; a body stays at rest as long as the resultant force lies within the cone.

✎ **DESCRIPTIVE Q:** Explain the method of joints and the method of sections for analysing trusses.

A truss is analysed on the assumptions that members are joined by frictionless pins, loads act only at joints and self-weight is neglected, so each member is a two-force member in pure tension or compression. In the method of joints, the support reactions are first found from the equilibrium of the whole truss. Then a joint at which not more than two unknown member forces meet is isolated; since the forces are concurrent, two equations ΣFx = 0 and ΣFy = 0 give the two unknowns. Proceeding from joint to joint, all member forces are found. Forces directed away from a joint indicate tension, towards it compression. It is systematic but forces in a central member may require analysing many joints first, and errors accumulate. In the method of sections, an imaginary section is passed through the truss cutting not more than three members whose forces are unknown and which are not all concurrent or parallel. One part of the truss is taken as a free body; three equations ΣFx = 0, ΣFy = 0 and ΣM = 0 are applied, taking moments about the point where two unknown forces intersect to find the third directly. It is quick for finding forces in a few particular members. Zero-force members can be identified first by inspection to simplify both methods.

✎ **DESCRIPTIVE Q:** Explain the strain energy method (Castigliano's theorem) for finding deflections.

Castigliano's first theorem states that, for a linearly elastic structure, the partial derivative of the total strain energy with respect to an external load equals the deflection of the point of application of that load in the direction of the load: δ = ∂U/∂P. Similarly, the partial derivative

with respect to an applied couple gives the rotation: $\theta = \partial U/\partial M$. For a beam, $U = \int M^2 dx/(2EI)$ (shear energy usually neglected), so $\delta = \int (M/EI)(\partial M/\partial P)dx$. To find the deflection at a point where no load acts, a fictitious load Q is applied there, the expression is differentiated with respect to Q, and Q is then set to zero. For example, for a cantilever with end load W, $M = -Wx$ and $\partial M/\partial W = -x$, so $\delta = (1/EI)\int_0^L Wx^2\, dx = WL^3/(3EI)$. The method is particularly useful for curved bars, rings, frames, springs and beams of varying cross-section, and its companion theorem of least work ($\partial U/\partial R = 0$ for a redundant reaction) is used to analyse statically indeterminate structures such as propped cantilevers and fixed beams.

P6. Production Engineering

✎ **DESCRIPTIVE Q:** Explain the working of a cupola furnace.

The cupola is a vertical cylindrical steel shell lined with refractory bricks, supported on legs, with a drop-bottom door at its base. Air from a blower enters a wind box and passes into the furnace through tuyeres located a little above the bottom. Before the blast, a coke bed is laid and ignited up to a height above the tuyeres; then alternate layers of charge – pig iron, foundry return scrap and steel scrap, coke as fuel (metal-to-coke ratio about 8–10:1) and limestone as flux – are charged from the charging door near the top. The furnace has zones: the well or crucible (below the tuyeres) where molten metal collects; the combustion (oxidising) zone just above the tuyeres where coke burns to CO_2, releasing heat and reaching about 1500–1800 °C; the reducing zone where CO_2 reacts with hot coke to form CO, protecting the metal from oxidation; the melting zone where iron melts and picks up carbon; the preheating zone where the descending charge is heated by rising gases; and the stack through which gases leave. Molten iron is tapped from the tap hole at intervals, and slag (formed by limestone combining with ash, sand and oxides) is removed from the slag hole at a higher level. Cupolas are simple, cheap and continuous for melting grey cast iron, though temperature and composition control are less precise than in induction furnaces, and emissions are high.

✎ **DESCRIPTIVE Q:** Compare casting, forging and machining as manufacturing processes.

Casting shapes a part by pouring liquid metal into a mould; it can produce very large, complex shapes with internal cavities, from any castable alloy, at low tooling cost for sand moulds, but the structure is coarse, may contain porosity and shrinkage defects, mechanical properties (particularly ductility and fatigue strength) are lower, and dimensional accuracy and finish are moderate except in precision processes. Forging shapes heated (or cold) metal by compressive plastic deformation; the grain flow follows the part contour and porosity is closed, giving superior strength, toughness and fatigue resistance – hence crankshafts, connecting rods, gears and hand tools are forged. Its limitations are the need for ductile materials, limited complexity (no internal cavities), high die costs and the need for machining of critical surfaces. Machining removes material with cutting tools to obtain precise dimensions, close tolerances and good surface finish on parts made by casting, forging or from bar stock; it is flexible and suits small quantities, but wastes material as chips, is slower, cuts across grain flow and becomes costly for hard materials. In practice the three are combined: a part is cast or forged near to shape and then machined on functional surfaces, the choice depending on shape, size, quantity, material and required properties.

✎ **DESCRIPTIVE Q:** Explain the principle of ultrasonic machining with its applications.

Ultrasonic machining removes material from hard and brittle workpieces by the hammering action of abrasive grains. A transducer (magnetostrictive or piezoelectric) converts high-frequency electrical energy (about 20 kHz) into mechanical vibrations, which are amplified

by a tapered horn (concentrator) to an amplitude of about 10–50 μm at the tool tip. The tool, made of a soft but tough material such as mild steel, brass or stainless steel and shaped as the required cavity, is held close to the workpiece under a light static feed force. A slurry of fine abrasive particles (boron carbide, silicon carbide or alumina in water) is circulated through the gap. The vibrating tool drives the abrasive grains against the work surface many thousands of times per second, chipping out minute particles by brittle fracture; cavitation and slurry flow assist removal. The tool wears slowly, so the work takes the negative shape of the tool. USM can machine both conductive and non-conductive materials, produces no heat-affected zone or chemical change and gives good accuracy, but the MRR is low, tool wear is significant and it is ineffective on soft ductile metals. Applications: drilling holes and cavities of any shape in glass, ceramics, quartz, precious stones, carbides and hardened steels; making dies, wire-drawing dies and coining tools; engraving; and machining semiconductor wafers.

✎ **DESCRIPTIVE Q:** Explain work sampling and its advantages over stop-watch time study.

Work sampling (activity sampling or ratio-delay study), developed by L. H. C. Tippett, estimates the proportion of time that workers or machines spend on various activities by making a large number of instantaneous observations at random times, rather than continuous timing. It is based on the laws of probability: if an activity is observed in p fraction of the observations, it occupies approximately p fraction of the total time. The number of observations required for a given accuracy L at 95% confidence is $N = 4p(1 - p)/L^2$ (absolute accuracy). The procedure involves defining activities, making a preliminary study to estimate p, determining N, preparing a random schedule of observation times, recording observations, checking with control limits and computing the results; combined with output and rating data it can set standard times. Advantages over stop-watch study: suitable for long-cycle, non-repetitive and group work (maintenance, office work, material handling); one observer can cover many workers and machines; it needs no stop-watch or continuous presence, causes less fatigue and less operator resentment; observations can be interrupted without bias; and it is cheaper. Its disadvantages are that it is less suitable for short-cycle repetitive work, gives less detailed element breakdown, and requires a large number of observations for accuracy.

CHAPTER 28

High-Yield Topic Map

A chapter-by-chapter checklist of what SSC JE papers repeatedly test. Tick each line only when you can explain it without notes.

M1. Engineering Mechanics and SOM

Concept	What is usually asked
Force systems	Resultant of two forces, Lami's theorem, Varignon's theorem, couples
Friction	Angle of repose, inclined plane forces, belt tension ratio, screw jack efficiency
Trusses	m = 2j − 3, zero-force members, method of sections
Centroid and MOI	Standard centroids; I of rectangle, circle, triangle; parallel/perpendicular axis theorems
Stress–strain	Hooke's law, elastic constants relations, Poisson's ratio values, stress–strain curve points
Bars	Stepped, tapering, self-weight, composite bars, thermal stress
Strain energy	Gradual, sudden and impact loads; resilience; Castigliano
Material testing	Hardness tests (BHN, VHN, Rockwell), Izod/Charpy, fatigue, creep
Principal stresses	Formulae, Mohr's circle centre/radius, special cases
Failure theories	Which theory for which material, loci shapes, shear yield values
SFD/BMD	Relations dM/dx = F, standard maximum BMs, shapes, contraflexure
Bending	M/I = σ/y = E/R, section moduli, strongest beam, shape factor, middle third
Shear stress	τ_max/τ_avg ratios, distribution shapes, shear centre
Deflection	Standard slope/deflection table, methods (Macaulay, moment area, conjugate beam)
Torsion	T/J = τ/r = Gθ/L, power, hollow vs solid, series/parallel shafts, equivalent torque
Springs	Stiffness, series/parallel, Wahl factor, leaf springs
Columns	Euler loads and effective lengths, slenderness, Rankine formula
Cylinders	Hoop/longitudinal stresses, volumetric strain, spheres, Lamé's theory
Joints	Riveted joint efficiency, fillet weld throat, keys and couplings

M2. Production Engineering

Concept	What is usually asked
Crystal structures	APF, coordination numbers, allotropy of iron
Steels	Carbon ranges and uses, designations, alloying element effects, stainless and tool steels
Cast irons and non-ferrous alloys	Grey, white, malleable, SG; brass, bronze, duralumin, babbitt
Fe–C diagram	Invariant reactions, phases, critical temperatures
Heat treatment	Annealing vs normalising, hardening, tempering, austempering, case hardening methods
Casting	Pattern types and allowances, sand properties, gating and risering, Chvorinov's rule
Special casting	Investment, die (hot/cold chamber), centrifugal, shell, continuous
Casting defects	Causes and remedies of blowholes, misrun, cold shut, hot tear, shift
Welding	Flames, arc processes, polarity, resistance welding heat, thermit, brazing/soldering, defects
Forming	Hot vs cold working, rolling draft, forging types, extrusion and drawing, sheet-metal operations and forces
Metal cutting	Tool signature, chip types, Merchant's circle, Taylor's equation, tool materials
Machine tools	Lathe parts and tapers, drilling operations, shaper quick return, milling (up/down, indexing), gear cutting
Finishing and NTM	Grinding wheel marking, honing/lapping, EDM, ECM, USM, AJM, LBM, EBM
Metrology	Limits and fits, IT grades, gauges and Taylor's principle, vernier/micrometer, surface roughness
CNC and jigs	G and M codes, axes, point-to-point vs contouring, 3-2-1 principle, jig vs fixture
P/M and plastics	P/M steps and products, moulding processes for thermoplastics and thermosets
Industrial engineering	Work study, standard time, layouts, EOQ, ABC, control charts, PERT/CPM, forecasting

M3. How to Use This Map in the Last Two Weeks

- Week 1: for each row above, revise the chapter section, then answer the matching rows of the Concept Recall Banks without looking.
- Week 2: solve one Solved Numericals set per day under time pressure (about 1 minute per short problem), and write two Paper-II answers per day.
- Final three days: Formula Appendix, PYQ-Trend Appendix and the Last-Minute Revision Sheet only.
- Mark any row you cannot explain in 30 seconds and revisit it the next morning – spaced repetition beats re-reading.

CHAPTER 28B

Abridged Data Tables

Standard textbook values (rounded) for quick numericals. Exam papers supply the data they need – use these for practice only.

T1. Typical Mechanical Properties

Material	E (GPa)	G (GPa)	μ	Yield (MPa)	UTS (MPa)	Density (kg/m³)
Mild steel	200–210	80	0.30	250	410–450	7850
Medium-carbon steel (C45)	210	80	0.29	350	600–700	7850
Stainless steel 304	193	75	0.29	215	505	8000
Grey cast iron	100–150	40–60	0.25	–	150–300 (tension)	7200
SG iron	170	65	0.28	300–400	450–700	7100
Aluminium alloy (6061-T6)	70	26	0.33	275	310	2700
Copper	110–120	45	0.34	70 (annealed)	220	8900
Brass (70/30)	100–110	40	0.33	100–400	300–550	8500
Titanium alloy (Ti-6Al-4V)	114	44	0.34	880	950	4430
Concrete (M20)	22–25	–	0.15	–	20 (compression)	2400
Timber (along grain)	10–12	–	–	–	40–100	500–800
Rubber	0.01–0.1	–	0.49	–	15–25	1100

T2. Coefficients of Thermal Expansion (× 10^{-6} per °C)

Material	α	Material	α
Invar	1.2	Copper	17
Glass (soda-lime)	9	Brass	19
Concrete	10–12	Aluminium	23
Steel	11–12	Zinc	30
Cast iron	10–11	Lead	29

T3. Standard Section Properties

Section	Area	I (centroidal)	Z	Radius of gyration k
Rectangle b × d	bd	$bd^3/12$	$bd^2/6$	$d/\sqrt{12} = 0.289d$
Solid circle d	$\pi d^2/4$	$\pi d^4/64$	$\pi d^3/32$	$d/4$
Hollow circle D, d	$\pi(D^2 - d^2)/4$	$\pi(D^4 - d^4)/64$	$\pi(D^4 - d^4)/(32D)$	$\sqrt{(D^2 + d^2)}/4$
Triangle b × h	$bh/2$	$bh^3/36$	$bh^2/24$ (top)	$h/\sqrt{18}$
Thin tube (mean radius r, thickness t)	$2\pi rt$	$\pi r^3 t$	$\pi r^2 t$	$r/\sqrt{2}$

T4. Hardness Conversion (approximate, steels)

Brinell HB	Vickers HV	Rockwell HRC	UTS approx. (MPa)
120	120	– (HRB 67)	410
200	210	15	690
300	315	32	1030
400	420	43	1380
500	530	51	1720
600 (carbide ball)	650	58	–

T5. Fundamental Tolerance (IT) Values for Common Sizes (µm)

Diameter step (mm)	IT6	IT7	IT8	IT9	IT11
6–10	9	15	22	36	90
10–18	11	18	27	43	110
18–30	13	21	33	52	130
30–50	16	25	39	62	160
50–80	19	30	46	74	190
80–120	22	35	54	87	220

T6. Typical Cutting Speeds (HSS tool, turning, m/min)

Work material	HSS	Carbide
Mild steel	25–35	100–200
Medium-carbon steel	20–30	80–150
Grey cast iron	15–25	60–120
Brass	60–90	150–300
Aluminium	75–150	300–600
Stainless steel	10–20	60–120

T7. Control Chart Factors

n	A_2	D_3	D_4	d_2
2	1.880	0	3.267	1.128
3	1.023	0	2.574	1.693
4	0.729	0	2.282	2.059

n	A_2	D_3	D_4	d_2
5	0.577	0	2.114	2.326
6	0.483	0	2.004	2.534
7	0.419	0.076	1.924	2.704

CHAPTER 29

Last-Minute Revision Sheet

29.1 Must-Know Values

Quantity	Value
E of steel / CI / Al / Cu	200–210 / 100–150 / 70 / 110–120 GPa
G of steel	≈ 80 GPa
μ of steel / rubber / cork / concrete	≈ 0.3 / 0.5 / 0 / 0.1–0.2
α of steel / Al / Cu	12 / 23 / 17 × 10^{-6} per °C
Density of steel	7850 kg/m^3
Stress concentration (small hole)	3
Endurance limit of steel	≈ 0.5 UTS
Shape factor rectangle / circle / I	1.5 / 1.7 / 1.12–1.15
Eutectoid / eutectic / peritectic	727 °C (0.8% C) / 1147 °C (4.3% C) / ≈ 1495 °C
α → γ / γ → δ / Curie	912 °C / 1394 °C / 768 °C
APF BCC / FCC / HCP	0.68 / 0.74 / 0.74
Shrinkage allowance CI / steel	1% / 2%
Neutral flame temperature	≈ 3200 °C
Brazing / soldering	> 450 °C / < 450 °C
Twist drill point angle	118°
Indexing ratio	40 : 1
Taylor n: HSS / carbide / ceramic	0.125 / 0.25 / 0.5–0.7
Vernier / micrometer LC	0.02 / 0.01 mm
Best wire size (60°)	0.577p
TMU	0.036 s
Control limits	±3σ (99.73%)
Six Sigma	3.4 DPMO

29.2 Formula Flash Card

$E = 2G(1 + \mu) = 3K(1 - 2\mu) = 9KG/(3K + G)$

$\delta = PL/AE$; self-weight $\rho g L^2/2E$; taper $4PL/(\pi E d_1 d_2)$; thermal $\sigma = E\alpha\Delta T$

Sudden load $\sigma = 2P/A$; impact $\sigma = (W/A)[1 + \sqrt{(1 + 2AEh/WL)}]$; $U = \sigma^2 V/2E$

$\sigma_{1,2} = (\sigma x + \sigma y)/2 \pm \sqrt{(((\sigma x - \sigma y)/2)^2 + \tau^2)}$; $\tau_{max} = (\sigma_1 - \sigma_2)/2$

$T_e = \sqrt{(M^2 + T^2)}$; $M_e = \frac{1}{2}[M + \sqrt{(M^2 + T^2)}]$

$M/I = \sigma/y = E/R$; $Z_{rect} = bd^2/6$; $Z_{circle} = \pi d^3/32$

$\tau = VA\bar{y}/(Ib)$; $\tau_{max} = 1.5\tau_{avg}$ (rect), 4/3 τ_{avg} (circle)

Deflections: $WL^3/3EI$, $wL^4/8EI$ (cantilever); $WL^3/48EI$, $5wL^4/384EI$ (SSB); $WL^3/192EI$, $wL^4/384EI$ (fixed)

$T/J = \tau/r = G\theta/L$; $\tau = 16T/\pi d^3$; $P = 2\pi NT/60$

Spring: $\delta = 8WD^3n/(Gd^4)$; $k = Gd^4/(8D^3n)$
Euler $P = \pi^2EI/L_e^2$; Rankine $P = \sigma_{cA}/(1 + a\lambda^2)$
Thin cylinder $\sigma_c = pd/2t$, $\sigma_l = pd/4t$; sphere $pd/4t$; $e_v = (pd/4tE)(5 - 4\mu)$
Chvorinov $t = B(V/A)^2$; sprue $A_1/A_2 = \sqrt{(h_2/h_1)}$
Resistance welding $H = I^2Rt$; arc heat input $\eta VI/v$
Rolling $\Delta h_{max} = \mu^2R$; blanking $F = \pi Dt\tau$; deep-draw blank $D = \sqrt{(d^2 + 4dh)}$
$\tan\varphi = r\cos\alpha/(1 - r\sin\alpha)$; $2\varphi + \beta - \alpha = 90°$; $VT^n = C$
$V = \pi DN/1000$; $t_{turning} = L/(fN)$; roughness $f^2/8R$
Indexing $40/N$; angular $\theta/9$
$i = 0.45\sqrt[3]{D} + 0.001D$; sine bar $\sin\theta = h/L$
EOQ $= \sqrt{(2DS/H)}$; standard time = normal time × (1 + allowance)
PERT $t_e = (t_o + 4t_m + t_p)/6$; $\sigma^2 = ((t_p - t_o)/6)^2$

29.3 Comparison Quick-Hits

Topic	Remember
Ductility vs malleability	Tension (wires) vs compression (sheets)
Tresca vs von Mises	Hexagon, conservative vs ellipse, accurate
Izod vs Charpy	Cantilever vs simply supported
Annealing vs normalising	Furnace cool (softer) vs air cool (finer, stronger)
Hot vs cold chamber die casting	Zn/Pb/Sn vs Al/brass
Misrun vs cold shut	Not filled vs not fused
DCSP vs DCRP	Deep penetration vs thin sheets/Al cleaning
TIG vs MIG	Non-consumable W vs consumable wire
Blanking vs punching	Piece is product vs hole is product
Direct vs indirect extrusion	More vs less friction
Up vs down milling	Lifts work, backlash-safe vs presses work, better finish
Hobbing vs gear shaping	Fastest, no internal gears vs internal gears possible
EDM vs ECM	Sparks, tool wears vs electrolysis, no tool wear
Jig vs fixture	Guides tool vs only holds
PERT vs CPM	Probabilistic, event vs deterministic, activity
Product vs process layout	Mass vs job/batch production

29.4 Traps Toppers Avoid

- Use effective length (not actual length) and least radius of gyration in column formulas.
- In thin cylinders, hoop stress uses longitudinal joint efficiency and longitudinal stress uses circumferential joint efficiency.
- Maximum shear stress in a triangle section is at mid-height, not at the neutral axis.
- Springs in series: reciprocals of stiffness add (softer combination).
- Shake allowance is negative; all other pattern allowances are positive.
- Normalised steel is harder than annealed steel – not softer.
- Clearance goes on the punch in blanking and on the die in punching.
- Chip thickness ratio is less than 1; the chip is thicker than the uncut layer.
- Lead screw is used only for thread cutting; feed rod for ordinary turning.

- Hard materials need soft-grade grinding wheels.
- GO gauge gets wear allowance; NO-GO gauge does not.
- EOQ varies as square root of demand – doubling demand raises EOQ by 41%, not 100%.

29.5 Final 10-Minute Checklist

- Mechanics: Lami, friction angle, truss condition, centroids and MOI.
- SOM: elastic constants, bars, thermal and impact stresses, Mohr's circle, failure theories, BM/deflection tables, bending/shear stress, torsion, springs, Euler, cylinders.
- Materials: steels, alloying effects, cast irons, Fe–C diagram, heat treatments, case hardening.
- Processes: casting allowances and defects, special casting, welding processes and polarity, forming calculations, press work.
- Machining: Merchant, Taylor, tool materials, lathe/milling/drilling facts, grinding wheel marking, NTM processes.
- Metrology and IE: fits and tolerances, gauges, instruments, CNC codes, jigs, work study, EOQ, control charts, PERT/CPM.

APPENDIX

Appendix A – One-Place Formula Sheet

Every formula box from the theory chapters, collected chapter by chapter for the final revision. Read it with a pen and rewrite any formula you hesitate on.

Engineering Mechanics – Forces, Equilibrium and Friction

Parallelogram law: $R = \sqrt{(P^2 + Q^2 + 2PQ\cos\theta)}$; $\tan\alpha = Q\sin\theta/(P + Q\cos\theta)$

Resolution: $R = \sqrt{((\Sigma Fx)^2 + (\Sigma Fy)^2)}$; $\tan\alpha = \Sigma Fy/\Sigma Fx$

Moment of a force $M = F \times$ perpendicular distance; Varignon's theorem: moment of resultant = sum of moments of components

Couple: two equal, opposite, parallel non-collinear forces; moment = $F \times$ arm, same about every point (free vector)

Coplanar equilibrium: $\Sigma Fx = 0$, $\Sigma Fy = 0$, $\Sigma M = 0$ (three equations)

Lami's theorem (three concurrent forces in equilibrium): $P/\sin\alpha = Q/\sin\beta = R/\sin\gamma$ (α, β, γ = angles opposite each force)

Limiting friction $F = \mu N$; angle of friction φ: $\tan\varphi = \mu$

Angle of repose α = angle of friction φ (for a body on an inclined plane just on the point of sliding)

Cone of friction: semi-vertex angle = φ

Force to pull up an incline (force parallel to plane): $P = W(\sin\alpha + \mu\cos\alpha)$; to just prevent sliding down: $P = W(\sin\alpha - \mu\cos\alpha)$

Least force to drag a body on a horizontal plane: $P_{min} = W\sin\varphi$, applied at angle φ to the horizontal

Ladder problems: $\Sigma Fx = 0$, $\Sigma Fy = 0$, ΣM about foot $= 0$

Belt friction: $T_1/T_2 = e^{(\mu\theta)}$ (flat belt); V-belt: $T_1/T_2 = e^{(\mu\theta \operatorname{cosec}\beta)}$ (β = half groove angle)

Screw jack: effort $P = W\tan(\alpha + \varphi)$; efficiency $\eta = \tan\alpha/\tan(\alpha + \varphi)$; max efficiency $= (1 - \sin\varphi)/(1 + \sin\varphi)$; self-locking if $\varphi > \alpha$ ($\eta < 50\%$)

Perfect (just rigid) truss: $m = 2j - 3$; deficient: $m < 2j - 3$; redundant: $m > 2j - 3$ (m = members, j = joints)

Parallel axis theorem: $I = I_G + A h^2$; perpendicular axis theorem (plane areas): $I_z = I_x + I_y$

Radius of gyration $k = \sqrt{(I/A)}$

Equations of motion: $v = u + at$; $s = ut + \frac{1}{2}at^2$; $v^2 = u^2 + 2as$

Projectile: range $R = u^2\sin 2\theta/g$ (max at 45°: u^2/g); max height $H = u^2\sin^2\theta/(2g)$; time of flight $T = 2u\sin\theta/g$

Work–energy: $W = \Delta KE$; impulse–momentum: $F \cdot t = m\Delta v$

Coefficient of restitution $e = (v_2 - v_1)/(u_1 - u_2)$; $e = 1$ perfectly elastic, $e = 0$ perfectly plastic

Circular motion: centripetal acceleration $v^2/r = \omega^2 r$; banking angle $\tan\theta = v^2/(rg)$

Simple pendulum $T = 2\pi\sqrt{(L/g)}$; spring–mass $T = 2\pi\sqrt{(m/k)}$

Simple Stresses and Strains

σ = P/A; e = δL/L; τ = P/A (shear); bearing stress $\sigma_b = P/(d\,t)$

Hooke's law: σ = E e (within proportional limit); τ = G φ; $\sigma_v = K e_v$

Elongation of a bar: $\delta L = PL/(AE)$

Poisson's ratio μ = − lateral strain/longitudinal strain (0 ≤ μ ≤ 0.5)

$E = 2G(1 + \mu)$; $E = 3K(1 - 2\mu)$; $E = 9KG/(3K + G)$; $\mu = (3K - 2G)/(6K + 2G)$

Volumetric strain of a rectangular bar under axial load: $e_v = e(1 - 2\mu)$; for three-dimensional stresses: $e_v = (\sigma x + \sigma y + \sigma z)(1 - 2\mu)/E$

Volumetric strain of a cylindrical rod: $e_v = e_L - 2e_d$ (e_d = magnitude of diametral strain); of a sphere: $e_v = 3 \times$ diametral strain

Stepped bar: $\delta L = (P/E)\,\Sigma(L_i/A_i)$

Uniformly tapering circular bar (diameters d_1, d_2): $\delta L = 4PL/(\pi E d_1 d_2)$

Tapering rectangular bar (thickness t, widths a, b): $\delta L = PL \ln(a/b)/(E\,t\,(a - b))$

Bar hanging under self-weight: $\delta L = \rho g L^2/(2E) = WL/(2AE)$ (half that of the same weight applied at the end)

Bar of uniform strength (constant stress σ under self weight + load): $A_2 = A_1\, e^{(\rho g L/\sigma)}$

Conical bar under self-weight: $\delta L = \rho g L^2/(6E)$

Composite bar: $P = \sigma_1 A_1 + \sigma_2 A_2$; $\sigma_1/E_1 = \sigma_2/E_2 \rightarrow \sigma_1 = (E_1/E_2)\,\sigma_2$ (modular ratio $m = E_1/E_2$)

Free thermal expansion $\delta = \alpha L \Delta T$ (no stress)

Fully restrained bar: thermal stress $\sigma = E\alpha\Delta T$ (compressive on heating, tensile on cooling) — independent of length and area

Partially restrained (gap δ allowed): $\sigma = E(\alpha L \Delta T - \delta)/L$

Composite bar of two materials heated ($\alpha_1 > \alpha_2$): $\sigma_1 A_1 = \sigma_2 A_2$ (equilibrium); $\sigma_1/E_1 + \sigma_2/E_2 = (\alpha_1 - \alpha_2)\Delta T$ (compatibility) → material with higher α in compression

Tapered circular bar ($d_1 > d_2$) fixed at both ends and heated: $P = E\alpha\Delta T\,\pi\, d_1 d_2/4$; σ_{max} (at small end) $= E\alpha\Delta T\, d_1/d_2$

Strain energy (axial) $U = \sigma^2 V/(2E) = P^2 L/(2AE)$; strain energy density $= \sigma^2/(2E)$

Proof resilience = $\sigma_p^2 V/(2E)$ (σ_p = proof/elastic limit stress); modulus of resilience = $\sigma_p^2/(2E)$

Gradually applied load: σ = P/A; suddenly applied load: σ = 2P/A (twice); impact (falling weight W from height h): $\sigma = (W/A)[1 + \sqrt{(1 + 2AEh/(WL))}]$

If h □ δ: $\sigma \approx \sqrt{(2WhE/(AL))}$

Shear strain energy $U = \tau^2 V/(2G)$; torsion of solid shaft $U = (\tau^2/(4G))\,V$; hollow shaft $U = (\tau^2/(4G))\,(D^2 + d^2)/D^2 \times V$

Bending: $U = \int M^2\,dx/(2EI)$; beam of uniform strength stores more energy

Mechanical Properties and Material Testing

Percentage elongation = $(L_f - L_0)/L_0 \times 100$; percentage reduction in area = $(A_0 - A_f)/A_0 \times 100$

Factor of safety = ultimate (or yield) stress/working (allowable) stress

True stress $\sigma_t = \sigma(1 + e)$; true strain $\varepsilon = \ln(1 + e)$; Hollomon: $\sigma_t = K\,\varepsilon^n$ (n = strain-hardening exponent = true strain at UTS)

BHN = $2P/[\pi D(D - \sqrt{(D^2 - d^2)})]$ (P in kgf, D ball diameter, d impression diameter, mm)

VHN = $1.854\ P/d^2$ (d = mean diagonal); approximate: UTS (MPa) ≈ 3.45 × BHN for steels

Soderberg: $\sigma_a/\sigma_e + \sigma_m/\sigma_y = 1/N$; Goodman: $\sigma_a/\sigma_e + \sigma_m/\sigma_u = 1/N$; Gerber: $\sigma_a/\sigma_e + N(\sigma_m/\sigma_u)^2 = 1/N$

Stress concentration factor $K_t = \sigma_{max}/\sigma_{nominal}$; for a small circular hole in a wide plate $K_t = 3$; elliptical hole $K_t = 1 + 2a/b$

Non-Destructive Testing, Composites, Ceramics and Polymers

Rule of mixtures (longitudinal, iso-strain): $E_c = E_f V_f + E_m V_m$; $\sigma_c = \sigma_f V_f + \sigma_m V_m$

Transverse (iso-stress): $1/E_c = V_f/E_f + V_m/E_m$

Principal Stresses, Mohr's Circle and Theories of Failure

$\sigma_n = \sigma \cos^2\theta$; $\tau = (\sigma/2) \sin2\theta$; $\tau_{max} = \sigma/2$ at $\theta = 45°$

Two perpendicular direct stresses σx, σy: $\sigma_n = (\sigma x + \sigma y)/2 + ((\sigma x - \sigma y)/2) \cos2\theta$; $\tau = ((\sigma x - \sigma y)/2) \sin2\theta$

General plane stress (σx, σy, τxy): $\sigma_n = (\sigma x + \sigma y)/2 + ((\sigma x - \sigma y)/2) \cos2\theta + \tau xy \sin2\theta$; $\tau = ((\sigma x - \sigma y)/2) \sin2\theta - \tau xy \cos2\theta$

$\sigma_{1,2} = (\sigma x + \sigma y)/2 \pm \sqrt{[((\sigma x - \sigma y)/2)^2 + \tau xy^2]}$

Principal plane: $\tan2\theta_p = 2\tau xy/(\sigma x - \sigma y)$

Maximum in-plane shear $\tau_{max} = \sqrt{[((\sigma x - \sigma y)/2)^2 + \tau xy^2]} = (\sigma_1 - \sigma_2)/2$; normal stress on that plane $= (\sigma x + \sigma y)/2$

Absolute maximum shear (3-D): $\tau_{abs} = (\sigma_{max} - \sigma_{min})/2$ — if σ_1 and σ_2 have the same sign, $\tau_{abs} = \sigma_1/2$ (taking $\sigma_3 = 0$)

Principal strains: $\varepsilon_1 = (\sigma_1 - \mu\sigma_2)/E$; $\varepsilon_2 = (\sigma_2 - \mu\sigma_1)/E$ (plane stress)

Equivalent bending moment $M_e = \frac{1}{2}[M + \sqrt{(M^2 + T^2)}]$; equivalent torque $T_e = \sqrt{(M^2 + T^2)}$

Centre $C = ((\sigma x + \sigma y)/2, 0)$; radius $R = \sqrt{[((\sigma x - \sigma y)/2)^2 + \tau xy^2]} = \tau_{max}$

$\sigma_1 = C + R$; $\sigma_2 = C - R$; angle on circle = 2 × angle on element

Shear Force and Bending Moment Diagrams

$dF/dx = -w$ (rate of change of SF = intensity of load); $dM/dx = F$ (rate of change of BM = SF)

BM is maximum (or minimum) where SF = 0 or changes sign

Change in BM between two sections = area of SFD between them; change in SF = area of load diagram

Bending Stresses in Beams

Bending equation: $M/I = \sigma/y = E/R$

Bending stress $\sigma = M y/I$; maximum $\sigma = M/Z$; section modulus $Z = I/y_{max}$

Moment of resistance $M = \sigma_{allow} Z$; flexural rigidity $= EI$; curvature $1/R = M/(EI)$

Cantilever with point load, constant depth: width $b \propto x$ (triangular plan); constant width: depth $d \propto \sqrt{x}$

SSB central load, constant width: $d \propto \sqrt{x}$ from each support

Composite beam: $M = M_1 + M_2 = \sigma_1 Z_1 + \sigma_2 Z_2$ with $\sigma_1/\sigma_2 = E_1/E_2$ at equal distance from NA

Eccentric axial load P at eccentricity e: $\sigma = P/A \pm M/Z = P/A \pm P\,e/Z$

No tension condition (rectangle $b \times d$, eccentricity along d): $e \le d/6$ — middle-third rule

Circular section: $e \le d/8$ — load within a central circle of diameter d/4 (middle-quarter rule)

Hollow circle: $e \le (D^2 + d^2)/(8D)$

Core (kern) of rectangular section: a rhombus with diagonals b/3 and d/3

Chimney/dam base stresses: $\sigma_{max,min} = W/A\,(1 \pm 6e/b)$

Plastic moment $M_p = \sigma_y Z_p$; plastic section modulus $Z_p = (A/2)(\bar{y}_1 + \bar{y}_2)$

Shape factor $S = Z_p/Z = M_p/M_y$

Curved beam: $\sigma = (M/(A\,e)) \times (y/(R_n - y))$ where $e = \bar{R} - R_n$ (shift of NA)

Shear Stresses in Beams and Deflection of Beams

$\tau = V A \bar{y}/(I b)$ (V = shear force, $A\bar{y}$ = first moment of area above the layer about NA, b = width at the layer)

Differential equation of the elastic curve: $EI\, d^2y/dx^2 = M$; $EI\, d^3y/dx^3 = F$; $EI\, d^4y/dx^4 = -w$

Torsion of Shafts and Springs

Torsion equation: $T/J = \tau/r = G\theta/L$

Polar moment of inertia: solid $J = \pi d^4/32$; hollow $J = \pi(D^4 - d^4)/32$

Maximum shear stress: solid $\tau = 16T/(\pi d^3)$; hollow $\tau = 16TD/[\pi(D^4 - d^4)]$

Polar modulus $Z_p = J/r_{max}$: solid $\pi d^3/16$; hollow $\pi(D^4 - d^4)/(16D)$

Power transmitted $P = 2\pi NT/60$ (W, with T in N·m, N in rpm)

Torsional rigidity = GJ; torsional stiffness $k_t = GJ/L$; angle of twist $\theta = TL/(GJ)$

Equivalent torque (Tresca): $T_e = \sqrt{(M^2 + T^2)} \rightarrow \tau_{max} = 16T_e/(\pi d^3)$

Equivalent bending moment (Rankine): $M_e = \frac{1}{2}[M + \sqrt{(M^2 + T^2)}] \rightarrow \sigma_{max} = 32M_e/(\pi d^3)$

With shock and fatigue factors (ASME code): $T_e = \sqrt{((K_m M)^2 + (K_t T)^2)}$

Shaft under axial load + torque: $\sigma = P/A$, $\tau = 16T/\pi d^3 \rightarrow$ principal stresses from formula

Shear stress $\tau = 8WD/(\pi d^3)$ (pure torsion); with Wahl factor $K = (4C - 1)/(4C - 4) + 0.615/C$: $\tau = K \times 8WD/(\pi d^3)$

Deflection $\delta = 8WD^3n/(Gd^4) = 64WR^3n/(Gd^4)$

Stiffness $k = W/\delta = Gd^4/(8D^3n)$

Strain energy $U = \frac{1}{2}W\delta$; springs in series: $1/k = 1/k_1 + 1/k_2$ (same load); springs in parallel: $k = k_1 + k_2$ (same deflection)

Cutting a spring into two halves doubles the stiffness of each half

Semi-elliptic leaf spring (span L, n leaves, width b, thickness t, central load W): $\sigma = 3WL/(2nbt^2)$; $\delta = 3WL^3/(8Enbt^3)$

Quarter-elliptic (cantilever) leaf spring: $\sigma = 6WL/(nbt^2)$; $\delta = 6WL^3/(Enbt^3)$

Proof load: load that straightens the spring; camber = δ at proof load

Columns and Struts

Slenderness ratio $\lambda = L_e/k_{min}$ (k_{min} = least radius of gyration = $\sqrt{(I_{min}/A)}$)

Euler crippling load $P_e = \pi^2 E I_{min}/L_e^2$

Euler critical stress $\sigma_e = \pi^2 E/\lambda^2$

Rankine: $1/P_R = 1/P_c + 1/P_e \rightarrow P_R = \sigma_c A/(1 + a \lambda^2)$

Rankine constant $a = \sigma_c/(\pi^2 E)$ ($a \approx 1/7500$ for mild steel, $1/1600$ for cast iron, $1/750$ for timber, for hinged ends)

Johnson's parabolic formula: $\sigma = \sigma_y[1 - \sigma_y \lambda^2/(4\pi^2 E)]$ (intermediate columns)

Straight-line formula: $P = \sigma_c A(1 - n \lambda)$

Secant formula (eccentric load e): $\sigma_{max} = (P/A)[1 + (e\, y_c/k^2)\, \sec((L_e/2k)\sqrt{(P/(EA))})]$

Perry–Robertson formula: allows for initial curvature; basis of IS code curves

Thin and Thick Cylinders and Spheres

Hoop (circumferential) stress $\sigma_c = pd/(2t)$

Longitudinal stress $\sigma_l = pd/(4t) = \sigma_c/2$

Maximum in-plane shear $\tau = (\sigma_c - \sigma_l)/2 = pd/(8t)$; absolute maximum shear (including radial ≈ 0) = $\sigma_c/2 = pd/(4t)$

With joint efficiencies: $\sigma_c = pd/(2t\eta_l)$ (longitudinal joint); $\sigma_l = pd/(4t\eta_c)$ (circumferential joint)

Hoop strain $e_c = (\sigma_c - \mu\sigma_l)/E = (pd/(4tE))(2 - \mu)$; longitudinal strain $e_l = (pd/(4tE))(1 - 2\mu)$

Volumetric strain $e_v = 2e_c + e_l = (pd/(4tE))(5 - 4\mu)$; change in volume $\delta V = e_v V$

Wire-wound cylinders: initial wire tension puts the cylinder in compression, increasing its pressure capacity

Stress (all directions in the wall) $\sigma = pd/(4t)$

Strain $e = (pd/(4tE))(1 - \mu)$; volumetric strain $e_v = 3e = (3pd/(4tE))(1 - \mu)$

Lamé's equations: $\sigma_r = a/r^2 - b$ (compressive radial stress, taken positive); $\sigma_c = a/r^2 + b$ (hoop tensile)

Internal pressure p only (radii r_1 inner, r_2 outer): $\sigma_{c,max}$ (at inner surface) = $p(r_2^2 + r_1^2)/(r_2^2 - r_1^2)$

Hoop stress at outer surface = $2p\, r_1^2/(r_2^2 - r_1^2)$; $\sigma_r = p$ at inner surface, 0 at outer surface

Maximum shear stress (inner surface) = $p\, r_2^2/(r_2^2 - r_1^2)$

Riveted, Welded and Bolted Joints (Strength Basics)

Tearing strength of plate (per pitch) $P_t = (p - d)\, t\, \sigma_t$

Shearing strength of rivets $P_s = n\, (\pi/4)\, d^2\, \tau$ (single shear); $= n \times 2 \times (\pi/4)d^2\, \tau$ (double shear; IS/boiler practice may use 1.75 or 1.875 instead of 2)

Crushing (bearing) strength $P_c = n\, d\, t\, \sigma_c$

Efficiency η = least of (P_t, P_s, P_c)/strength of solid plate $(p\, t\, \sigma_t)$

Efficiency of joint (tearing criterion) = $(p - d)/p$

Throat of fillet weld $t = 0.707\, s$; weld strength = $\tau \times 0.707\, s \times L$

Bolt in tension: $P = (\pi/4)\, d_c^2\, \sigma_t$ (d_c = core/minor diameter); bolt in shear: $P = (\pi/4)\, d^2\, \tau$

Eccentrically loaded bolted bracket: direct shear P/n + torsional shear $P\, e\, r_i/\Sigma r^2$

Rectangular sunk key: width $w = d/4$, thickness $= 2w/3$; shear: $T = w\, l\, \tau\, d/2$; crushing: $T = (t/2)\, l\, \sigma_c\, d/2$

Key equally strong in shear and crushing (square key): $\sigma_c = 2\tau$

Special Topics – Rotating Rings and Discs, Dams, Curved Bars and Contact Stresses

Thin rotating ring: hoop stress $\sigma = \rho\, v^2 = \rho\, \omega^2\, r^2$ (independent of cross-sectional area)

Maximum peripheral speed for allowable stress: $v_{max} = \sqrt{(\sigma/\rho)}$ (steel $\sigma = 100$ MPa → $v \approx 113$ m/s)

Solid disc of uniform thickness: $\sigma_r = \sigma_\theta = (3 + \mu)\rho\omega^2R^2/8$ at the centre (maximum)

Disc with central hole (inner radius small): $\sigma_{\theta,max}$ at hole ≈ $(3 + \mu)\rho\omega^2R^2/4$ — twice the solid-disc value

Rotating long cylinder (μ effects similar) and disc of uniform strength: thickness $t = t_0$ e^(−ρω²r²/(2σ))

Water pressure on vertical face of height h: $P = \rho g h^2/2$ per unit length, acting at h/3 above the base

Base stresses (width b, total vertical load W, eccentricity e): $\sigma_{max,min} = (W/b)(1 \pm 6e/b)$

Earth pressure (Rankine, active, dry cohesionless soil): p = k_a w h with k_a = (1 − sinφ)/(1 + sinφ); total P = k_a w h²/2 at h/3

Closed ring under diametral pull W: M at load = 0.182WR (= WR(1/2 − 1/π)); M at 90° = −0.137WR... (sign opposite)

Curved bar with small curvature (R/h > 10): use straight-beam formula with little error

Two spheres: contact radius a ∝ (F R)^(1/3); maximum pressure p_max = 1.5F/(πa²)

Two cylinders (line contact): half-width b = √(2F/(πL) × [(1 − μ₁²)/E₁ + (1 − μ₂²)/E₂]/(1/d₁ + 1/d₂)); p_max = 2F/(πbL)

Fatigue stress concentration K_f = 1 + q(K_t − 1)

Metal Casting – Patterns, Moulds and Gating

Gating ratio = sprue area : runner area : ingate area — pressurised (e.g. 1 : 0.75 : 0.5, choke at gates; ferrous) or unpressurised (e.g. 1 : 2 : 2 to 1 : 4 : 4, choke at sprue; aluminium, magnesium)

Sprue taper (avoid aspiration): A₁/A₂ = √(h₂/h₁) (continuity + Bernoulli; velocity at bottom V = √(2gh))

Filling time (top gate): t = V/(A_g √(2gH)); bottom gate: t = (A_m/A_g)(2/√(2g))(√H − √(H − h_m))

Chvorinov's rule: solidification time t_s = B (V/A)ⁿ (n ≈ 2) — modulus M = V/A

Riser design: riser modulus ≥ 1.2 × casting modulus (solidification time of riser ≈ 1.2–1.44 × casting)

Caine's method: freezing ratio X = (A/V)_casting/(A/V)_riser; most efficient riser shape is a sphere, cylinder with H = D commonly used

Special Casting Processes, Melting and Casting Defects

Rotational speed for G-factor GF: N = (30/π)√(2 g GF/D) (D = inside diameter)

Welding, Brazing and Soldering

Heat input per unit length H = η V I/v (v = welding speed); arc power = V × I

Arc characteristic (Ayrton): V = A + B·l (voltage increases linearly with arc length)

Heat generated in resistance welding H = I² R t (current most influential — heat ∝ I²)

Metal Forming – Rolling, Forging, Extrusion, Drawing and Sheet Metal

True strain ε = ln(L/L₀) = ln(A₀/A); flow stress σ_f = K εⁿ; average flow stress = K εⁿ/(n + 1)

Draft Δh = h₀ − h_f; maximum draft Δh_max = μ² R (R = roll radius)

Angle of bite (contact angle) α: cosα = 1 − Δh/(2R); condition for bite: μ ≥ tanα

Projected contact length L ≈ √(R Δh); roll force F ≈ σ_avg × w × √(RΔh); power per roll P = F (L/2) ω (approx.)

Volume constancy: h₀w₀v₀ = h_f w_f v_f → exit speed greater than entry speed

Forging load (open-die, cylinder): F = σ_f π r²(1 + 2μr/(3h))

Extrusion ratio R = A₀/A_f; extrusion pressure p = σ_f ln R (ideal) or k(a + b ln R) (Johnson)

Drawing stress (ideal) σ_d = σ_f ln(A₀/A_f); maximum reduction per pass (ideal, perfect plasticity) = 1 − 1/e ≈ 63%

Blanking/punching force F = L t τ_s (L = perimeter, t = thickness, τ_s = shear strength)

Clearance c ≈ (0.03 to 0.1) t per side (≈ 0.0032 t √τ for steel by empirical rule)

Blanking: die size = blank size, punch = blank size − 2c; punching: punch size = hole size, die = hole size + 2c

Shear on punch or die reduces peak force: F' = F_max × (p t)/S (p = fractional penetration at fracture, S = amount of shear, valid for S ≥ p t)

Deep drawing blank diameter (cup without flange): D = √(d² + 4dh); drawing ratio D/d ≤ ≈ 2 (limiting drawing ratio); percentage reduction (D − d)/D ≤ ≈ 50%

Bend allowance BA = α(R + k t) (α in radians; k ≈ 0.33 for R < 2t, 0.5 for R ≥ 2t); minimum bend radius ∝ t

Press tonnage = cutting force × factor of safety; work done = F × t × penetration fraction

Presses, Press Tools and High-Energy-Rate Forming

Flywheel energy released: E = ½I(ω₁² − ω₂²) = I ω² C_s (energy for one stroke = F × t × penetration)

Strip utilisation η = (area of blanks)/(area of strip consumed) × 100

Theory of Metal Cutting

Ideal surface roughness (sharp tool): h = f/(tan ψ + cot κ); with nose radius: h_max = f²/(8R)

Chip thickness ratio r = t/t_c = L_c/L = V_c/V (< 1); cutting ratio 1/r = chip reduction coefficient ζ

Shear angle: tan φ = r cos α/(1 − r sin α)

Shear strain γ = cot φ + tan(φ − α); shear strain rate = V cos α/(cos(φ − α) Δy)

Velocities: chip velocity V_c = V sinφ/cos(φ − α) = rV; shear velocity V_s = V cosα/cos(φ − α)

Merchant's circle: F = F_c sin α + F_t cos α (friction force); N = F_c cos α − F_t sin α; μ = tan β = F/N = (F_t + F_c tan α)/(F_c − F_t tan α)

Shear force F_s = F_c cos φ − F_t sin φ; normal on shear plane N_s = F_c sin φ + F_t cos φ; shear stress τ_s = F_s sin φ/(b t)

Merchant's first solution (minimum energy): 2φ + β − α = 90° → φ = 45° + α/2 − β/2

Lee–Shaffer: φ = 45° + α − β

Cutting power P = F_c V; specific cutting energy u = F_c/(b t) (J/mm³); material removal rate MRR = V f d (turning)

Taylor's tool life equation: V Tⁿ = C (V in m/min, T in min)

Extended Taylor: V Tⁿ f^a d^b = K (feed has more effect than depth on tool life)

Speed for minimum cost: T_opt = (1/n − 1)(C_t/C_m + t_c) (C_t = tool cost per edge, C_m = machine + labour cost per minute, t_c = tool change time); maximum production rate: T_opt = (1/n − 1) t_c

Machine Tools – Lathe, Drilling, Shaper, Planer, Milling and Others

Cutting speed V = πDN/1000 (m/min, D in mm, N in rpm)

Machining time (turning) t = L/(f N) per pass (L = length including approach and overtravel)

MRR (turning) = π D_avg d f N ≈ V f d

Taper: tan(α/2) = (D − d)/(2L); tailstock set-over S = (D − d) L_total/(2L_taper) (whole job length L_total)

Thread cutting: (driver teeth/driven teeth) = pitch of work/pitch of lead screw

Depth of cut in turning $d = (D_1 - D_2)/2$

Drilling time $t = (L + 0.3D)/(f\,N)$; MRR = $(\pi D^2/4)\,f\,N$; cutting speed at periphery $V = \pi DN/1000$

Tap drill size ≈ major diameter − pitch (metric threads)

Quick return ratio = time of cutting stroke/time of return stroke = $(360° - \theta)/\theta$ (θ = crank angle for return) — typically 3 : 2

Average cutting speed (shaper) $V = L\,N\,(1 + m)/1000$ m/min (m = return/cutting time ratio, N = strokes/min)

Machining time = $B/(f\,N)$ (B = width of job, f = feed per stroke)

Cutting speed $V = \pi DN/1000$; table feed $f_m = f_z \times Z \times N$ (mm/min)

Milling time $t = (L + \text{approach})/f_m$; approach (slab milling) = $\sqrt{d(D - d)}$; face milling (centred) = D/2 approx.

MRR = $w\,d\,f_m$; maximum uncut chip thickness (slab) $t_{max} = 2\,f_z\,\sqrt{(d/D)}$

Simple indexing: crank turns = 40/N (N = number of divisions); fractional turns using holes in index plate circles (Brown & Sharpe/Cincinnati plates)

Direct (rapid) indexing: 24-hole plate on spindle — divisions that are factors of 24

Angular indexing: crank turns = $\theta°/9$ (one crank turn = 9°)

Differential indexing for numbers not possible by simple indexing (e.g. 61, 127): gear train between spindle and index plate

Grinding, Finishing and Non-Traditional Machining

Wheel speed (surface) ≈ 30–35 m/s (vitrified); centreless through-feed rate $f = \pi\,d_r\,N_r \sin\alpha$

Grinding ratio G = volume of work removed/volume of wheel wear; specific energy much higher than turning

ECM MRR (volume/time) = $A\,I/(\rho\,n\,F)$ per unit (F = 96,500 C) — depends on current, not on hardness

EDM: spark energy $E = \frac{1}{2}CV^2$ (RC relaxation circuit); MRR ∝ energy × frequency; surface finish improves with lower energy per spark

Metrology – Limits, Fits, Gauges and Measuring Instruments

Standard tolerance unit i (μm) = $0.45\sqrt[3]{D} + 0.001D$ (D = geometric mean of diameter step, mm)

IT grades: IT5 = 7i, IT6 = 10i, IT7 = 16i, IT8 = 25i, IT9 = 40i, IT10 = 64i, IT11 = 100i (multiplying by ≈ 1.6 each grade from IT6)

Maximum clearance = upper limit of hole − lower limit of shaft; minimum clearance (allowance) = lower limit of hole − upper limit of shaft

Vernier least count = value of 1 main scale division/number of vernier divisions; micrometer LC = pitch/number of thimble divisions

Optical flat: height difference = $n\,\lambda/2$ (n = number of fringes)

Best wire size $d_w = (p/2)\sec(\theta/2)$; for 60° thread $d_w = 0.577p$

Ra (centre-line average, CLA) = $(1/L)\int|y|\,dx$ — most commonly used (μm)

Rq (RMS) = $\sqrt{((1/L)\int y^2\,dx)} \approx 1.11$ Ra for sinusoidal profile; Rz = average of five highest peaks and five deepest valleys (ten-point height); Rt / Rmax = peak-to-valley height

CNC Machines, Automation, Jigs and Fixtures

Stepper motor: step angle = 360°/steps per revolution; table movement per pulse (BLU) = lead of screw × step angle/360° (with gear ratio)

Number of pulses = distance/BLU; feed rate = pulse frequency × BLU

Industrial Engineering – Work Study, PPC, Inventory and Quality

Productivity = output/input; labour productivity = output/man-hours

Normal time = observed time × rating factor (%/100)

Standard time = normal time + allowances = normal time × (1 + allowance fraction)

Number of cycles for stop-watch study (95% confidence, ±5% accuracy): $N = [40\sqrt{(N\Sigma x^2 - (\Sigma x)^2)}/\Sigma x]^2$

Work sampling: number of observations $N = 4p(1 - p)/L^2$ (95% confidence, L = accuracy as a fraction)

Line efficiency = $\Sigma t_i/(n \times C)$; smoothness index = $\sqrt{(\Sigma(C - t_i)^2)}$

EOQ (Wilson/Harris): $Q^* = \sqrt{(2DS/H)}$ (D = annual demand, S = ordering cost per order, H = holding cost per unit per year)

At EOQ: annual ordering cost = annual holding cost; total cost = $\sqrt{(2DSH)}$; number of orders = D/Q*; cycle time = Q*/D

Economic production (batch) quantity with production rate p and usage d: $Q^* = \sqrt{(2DS/(H(1 - d/p)))}$

Reorder point ROP = demand during lead time + safety stock = d × L + SS

Safety stock = z σ_L (z for desired service level)

Quantity discount: compare total cost at EOQ and at price-break quantities

$\bar{X}$ chart limits: $\bar{\bar{X}} \pm A_2 \bar{R}$; R chart: UCL = $D_4 \bar{R}$, LCL = $D_3 \bar{R}$; control limits normally ±3σ (99.73% of points within)

p chart: $\bar{p} \pm 3\sqrt{(\bar{p}(1 - \bar{p})/n)}$; c chart: $\bar{c} \pm 3\sqrt{\bar{c}}$

Process capability: $C_p = (USL - LSL)/(6\sigma)$; $C_{pk} = \min[(USL - \mu)/3\sigma, (\mu - LSL)/3\sigma]$; $C_p \geq 1.33$ usually required

PERT expected time $t_e = (t_o + 4t_m + t_p)/6$; variance $\sigma^2 = ((t_p - t_o)/6)^2$

Project duration variance = sum of variances of critical activities; $Z = (T_s - T_e)/\sigma_{project}$

Total float = LS − ES = LF − EF; free float = ES of successor − EF of activity; independent float = ES(succ) − LF(pred) − duration

Critical path: longest path through the network; activities with zero total float

Crash cost slope = (crash cost − normal cost)/(normal time − crash time)

Break-even quantity $Q_{BE} = F/(p - v)$; margin of safety = (actual sales − BEP sales)/actual sales

M/M/1 queue: $L_s = \lambda/(\mu - \lambda)$; $W_q = \lambda/(\mu(\mu - \lambda))$

Maintenance, Safety, Value Engineering and Costing

OEE = availability × performance rate × quality rate (world class ≈ 85%)

Availability = MTBF/(MTBF + MTTR)

Frequency rate = (number of lost-time injuries × 10^6)/man-hours worked

Severity rate = (man-days lost × 10^6)/man-hours worked (in India per IS 3786; some texts use 1000 man-hours)

Halsey bonus = (time saved × rate)/2; Rowan bonus = (time saved/time allowed) × time taken × rate

APPENDIX

Appendix B – PYQ-Trend Concept Map

All PYQ-Trend boxes in one place, chapter-wise. These are the concepts SSC JE has tested repeatedly; revise this list in the last week.

Engineering Mechanics – Forces, Equilibrium and Friction

★ Parallelogram law and resultant of two equal forces (2P cos θ/2) – frequent; Varignon's theorem; couple is a free vector; principle of transmissibility.

★ Lami's theorem (very frequent); number of reactions at hinge/roller/fixed support; conditions of equilibrium; two-force member.

★ Angle of repose = angle of friction (very frequent); μ = tanφ; least pull on horizontal plane at angle φ; belt tension ratio e^(μθ); self-locking screw efficiency < 50%; laws of friction.

★ m = 2j − 3 condition (very frequent); zero-force member rules; method of sections uses at most three unknown members.

★ Centroid of triangle h/3, semicircle 4r/3π, cone h/4, hemisphere 3r/8 (very frequent); I of rectangle $bd^3/12$ and about base $bd^3/3$; triangle $bh^3/36$; parallel and perpendicular axis theorems; polar MOI of circle $\pi d^4/32$.

★ Maximum range at 45°; coefficient of restitution limits; D'Alembert's principle; banking of roads; conservation of momentum vs energy.

Simple Stresses and Strains

★ Hooke's law, units of stress; Poisson's ratio range 0–0.5 and value for steel/rubber/cork (very frequent); δL = PL/AE; Saint-Venant's principle.

★ E = 2G(1 + μ) and E = 3K(1 − 2μ) (very frequent); E = 9KG/(3K + G); number of independent constants for isotropic material = 2; incompressible material μ = 0.5.

★ Elongation due to self-weight $\rho g L^2/2E$ (very frequent); tapering bar $4PL/(\pi E d_1 d_2)$; composite bar equal strain; modular ratio; bar of uniform strength exponential profile.

★ Thermal stress EαΔT independent of length (very frequent); compound bar – higher-α material in compression on heating; steel and concrete have similar α; gap allowance.

★ Suddenly applied load gives double stress (very frequent); strain energy formula $\sigma^2 V/2E$; proof resilience vs modulus of resilience; impact stress; Castigliano's theorem; Maxwell's reciprocal theorem; strain energy of solid shaft $\tau^2 V/4G$.

Mechanical Properties and Material Testing

★ Sequence proportional limit → elastic limit → yield → UTS → fracture (very frequent); necking starts at UTS; 0.2% proof stress; cup-and-cone fracture of ductile materials; percentage elongation and reduction in area; factor of safety; torsion fracture planes of ductile vs brittle materials.

★ Ductility vs malleability (very frequent); gold most ductile and malleable; toughness = area under curve up to fracture; resilience within elastic limit; creep at high temperature; fatigue below UTS.

★ Vickers indenter 136° diamond pyramid (very frequent); Rockwell C uses 120° diamond cone and measures depth; Brinell uses ball and area; Shore scleroscope rebound; Mohs scale; UTS ≈ 3.45 BHN.

★ Izod (cantilever) vs Charpy (simply supported) (very frequent); endurance limit ≈ 0.5 UTS and S–N curve; Goodman/Soderberg lines; stress concentration factor 3 for a circular hole; creep stages – secondary used for design; shot peening improves fatigue; BCC metals ductile–brittle transition.

Non-Destructive Testing, Composites, Ceramics and Polymers

★ Magnetic particle test only for ferromagnetic materials (very frequent); dye penetrant for surface cracks only; radiography for internal porosity in castings and welds; ultrasonic for internal cracks and thickness; eddy current for conductive materials; Co-60 and Ir-192 gamma sources.

★ GFRP and CFRP uses; Kevlar for bullet-proof vests; cermets as particulate composites; rule of mixtures (frequent); wood and bone as natural composites; C–C composites for brake discs and rocket nozzles.

★ Bakelite is a thermoset; PTFE lowest coefficient of friction; vulcanisation of rubber with sulphur (very frequent); addition vs condensation polymerisation; glass is amorphous; acid/basic/neutral refractories; plasticisers in PVC; T_g.

Principal Stresses, Mohr's Circle and Theories of Failure

★ Maximum shear in uniaxial tension = σ/2 at 45° (very frequent); complementary shear stresses; pure shear equivalent to ±τ at 45°; invariance of σx + σy.

★ Principal stress formula and shear zero on principal planes (very frequent); principal and maximum shear planes 45° apart; $\tau_{max} = (\sigma_1 - \sigma_2)/2$; equivalent bending moment and torque; strain rosette.

★ Mohr's circle centre and radius (very frequent); circle reduces to a point for equal biaxial stresses; pure shear circle centred at origin; angle doubling; radius equals maximum shear stress.

★ Maximum shear stress theory for ductile and principal stress theory for brittle materials (very frequent); von Mises gives $\tau_y = 0.577\sigma_y$; shapes of failure loci (hexagon, ellipse, square, rhombus); Tresca conservative; hydrostatic stress causes no distortion.

Shear Force and Bending Moment Diagrams

★ dM/dx = F and max BM where SF = 0 (very frequent); SFD/BMD shapes for UDL (linear/parabolic) and UVL (parabolic/cubic); point of contraflexure definition; couple causes jump in BMD only.

★ Max BM: SSB UDL $wL^2/8$, central load WL/4, cantilever $wL^2/2$ (very frequent); fixed beam UDL $wL^2/12$ and $wL^2/24$; propped cantilever reaction 3wL/8 and 5W/16; UVL results $wL^2/(9\sqrt{3})$ and $wL^2/6$; contraflexure locations; SSB with couple has constant SF.

Bending Stresses in Beams

★ M/I = σ/y = E/R (very frequent); section modulus of rectangle $bd^2/6$ and circle $\pi d^3/32$; NA passes through centroid; bending stress zero at NA; strongest beam from log d/b = √2; I-section most economical; effect of doubling depth.

★ Beam of uniform strength (leaf spring example); flitched beams and modular ratio; equivalent section method.

★ Middle-third rule e ≤ d/6 (very frequent); middle-quarter rule for circular sections d/8; kern of rectangle is a rhombus; σ = P/A ± M/Z; masonry dams no tension.

★ Shape factor of rectangle 1.5, circle 1.7, I-section ≈ 1.14, diamond 2 (very frequent); plastic neutral axis divides area equally; plastic hinge.

★ Crane hook – curved beam, maximum stress at inner fibre, NA shifts towards centre of curvature (frequent); shear centre of channel outside the web; shear centre of angle at junction of legs.

Shear Stresses in Beams and Deflection of Beams

★ $\tau_{max} = 1.5\tau_{avg}$ for rectangle and 4/3 for circle (very frequent); maximum shear at NA (except triangle at h/2); I-section web carries shear and flanges carry bending; jump at web–flange junction; diamond 9/8.

★ $EI\,d^2y/dx^2 = M$ (very frequent); Macaulay's method for discontinuous loads; Mohr's moment-area theorems; conjugate beam supports conversion; boundary conditions.

★ Cantilever $WL^3/3EI$ and $wL^4/8EI$; SSB $WL^3/48EI$ and $5wL^4/384EI$ (very frequent); fixed beam $WL^3/192EI$ and $wL^4/384EI$; ratio of deflections UDL vs point load; effect of length and depth on deflection; slope at supports $wL^3/24EI$.

Torsion of Shafts and Springs

★ $T/J = \tau/r = G\theta/L$ (very frequent); $\tau = 16T/\pi d^3$; $P = 2\pi NT/60$; hollow shaft advantages; series vs parallel shafts; angle of twist TL/GJ; shear stress zero at centre.

★ Equivalent torque $\sqrt{(M^2 + T^2)}$ and equivalent BM $\frac{1}{2}(M + \sqrt{(M^2 + T^2)})$ (very frequent); ASME code factors; strain energy in torsion.

★ Spring stiffness $Gd^4/(8D^3n)$ and deflection $8WD^3n/Gd^4$ (very frequent); series and parallel springs; Wahl factor; spring index; cutting spring in half doubles stiffness.

★ Leaf spring is a beam of uniform strength; nipping; stress $3WL/2nbt^2$ and deflection $3WL^3/8Enbt^3$; semi-elliptic springs in vehicles.

Columns and Struts

★ Slenderness ratio uses least radius of gyration (very frequent); short columns crush, long columns buckle; buckling about axis of minimum I.

★ Euler load π^2EI/L_e^2 and effective lengths for four end conditions (very frequent); ratio 1 : 4 : 8 : 16; Euler valid for long columns only; critical stress π^2E/λ^2; $P \propto 1/L^2$.

★ Rankine formula combines crushing and buckling (very frequent); Rankine constant σ_c/π^2E; Johnson formula for intermediate columns; secant formula for eccentric loads.

Thin and Thick Cylinders and Spheres

★ $\sigma_c = pd/2t$ and $\sigma_l = pd/4t$ (very frequent); hoop stress twice longitudinal; volumetric strain $(pd/4tE)(5 - 4\mu)$; thin cylinder bursts along longitudinal seam; joint efficiencies.

★ Thin sphere stress $pd/4t$ (same as longitudinal stress of cylinder); volumetric strain $3pd(1 - \mu)/4tE$; sphere needs half the thickness.

★ Lamé's theory for thick cylinders; maximum hoop stress at inner surface (very frequent); compound cylinders by shrink fit; autofrettage; thick cylinder criterion $t > d/20$.

Riveted, Welded and Bolted Joints (Strength Basics)

★ Joint efficiency = least strength/solid plate strength (very frequent); Unwin's formula $d = 6\sqrt{t}$; double cover butt joint – rivets in double shear; caulking and fullering; diamond riveting most efficient for structures; margin 1.5d.

★ Throat thickness 0.707 × leg (very frequent); fillet weld strength 0.707 s L τ; welded vs riveted joints; butt weld strength uses plate thickness.

★ Woodruff key is semicircular and self-aligning; saddle key works by friction; Oldham coupling for parallel shafts with lateral offset; Hooke's joint for intersecting shafts; key width d/4; cotter taper 1 in 30; knuckle joint uses a pin.

Special Topics – Rotating Rings and Discs, Dams, Curved Bars and Contact Stresses

★ Hoop stress in rotating ring ρv^2 (frequent); limiting rim speed of flywheels; small hole doubles the stress in a rotating disc; disc of uniform strength.

★ Water thrust $\rho g h^2/2$ acting at h/3 (very frequent); no tension in dam base requires $e \le b/6$; Rankine active earth pressure coefficient; chimney wind load coefficient for circular sections.

★ Maximum stress in crane hook at inner fibre; neutral axis shifts towards centre of curvature; Winkler–Bach theory; curved-beam correction negligible for large R/h.

★ Hertz contact stress in bearings and gears; subsurface maximum shear causes pitting; $K_f = 1 + q(K_t - 1)$; notch sensitivity; reducing stress concentration with fillets and relief grooves; photoelasticity to find K_t.

Engineering Materials and Classification of Steels

★ APF of BCC 0.68 and FCC/HCP 0.74; coordination numbers 8 and 12 (very frequent); allotropic forms of iron and temperatures 912 °C and 1394 °C; Curie temperature 768 °C; plastic deformation by dislocation movement; fine grains increase strength (Hall–Petch).

★ Carbon ranges of low/medium/high carbon steel (very frequent); uses of each grade (rails, springs, tools); 40C8 designation; AISI 1040; killed vs rimmed steel; effect of carbon on properties; eutectoid 0.8% C.

★ Element effects – Ni toughness, Cr corrosion, W red hardness, Mn prevents hot shortness, S free cutting, P cold shortness, Co reduces hardenability (very frequent); 18-4-1 HSS composition; 18-8 stainless; Hadfield steel; invar; weld decay in stainless steel.

★ Grey CI flakes and good damping/machine beds; SG iron by Mg/Ce; malleable iron from white CI by annealing (very frequent); white CI hardest; chilled CI for rolls; carbon range 2–4%.

★ Brass = Cu + Zn, bronze = Cu + Sn (very frequent); gun metal composition; duralumin for aircraft; Monel Ni–Cu; babbitt for bearings; German silver has no silver; age hardening of duralumin; thermoplastic vs thermoset; bakelite thermoset; vulcanisation with sulphur.

Material Selection, Corrosion and Strengthening Mechanisms

★ Age hardening steps for duralumin (very frequent); Hall–Petch grain-size strengthening; recrystallisation restores ductility; strain hardening by cold work; over-ageing reduces strength.

★ Galvanic series – zinc/magnesium most active, used as sacrificial anodes (very frequent); rusting needs O_2 and water; season cracking of brass in ammonia; dezincification of brass; caustic embrittlement in boilers; pitting of stainless steel by chlorides; small anode–large cathode is worst.

★ Machine bed of grey CI because of damping (very frequent); pistons of aluminium alloy; bearings of babbitt/bronze; ball bearings of high-carbon chromium steel; turbine blades of nickel superalloys; rails high-carbon steel; heating elements nichrome; specific strength for aerospace.

Iron–Carbon Diagram and Heat Treatment

★ Eutectoid 727 °C, 0.8% C – pearlite; eutectic 1147 °C, 4.3% C – ledeburite; peritectic 1495 °C (very frequent); cementite 6.67% C; max solubility of C in austenite 2.14% and in ferrite 0.022%; martensite BCT not on diagram; lever rule.

★ TTT diagram nose and critical cooling rate; martensite diffusionless BCT; Jominy end-quench test for hardenability (very frequent); hardness depends on carbon, hardenability on alloying; sub-zero treatment for retained austenite.

★ Annealing (furnace cool) vs normalising (air cool) (very frequent); hardening by quenching gives martensite; tempering reduces brittleness; spheroidising for machinability of high-carbon steel; austempering gives bainite; martempering reduces distortion; quenching media severity order.

★ Nitriding needs no quenching and is done below A_1 (very frequent); carburising for low-carbon steels; flame and induction hardening for medium-carbon steels; cyaniding adds C and N; induction case depth decreases with frequency; pack carburising energiser $BaCO_3$.

Metal Casting – Patterns, Moulds and Gating

★ Pattern allowances – shake allowance is negative (very frequent); shrinkage allowance for cast iron 1% and steel 2%; match-plate patterns for mass production; sweep pattern for symmetrical shapes; draft for withdrawal; core prints.

★ Moulding sand properties – permeability, collapsibility, refractoriness (very frequent); bentonite clay binder; CO_2 process uses sodium silicate; chaplets support cores; sea coal for cast iron finish; grain fineness number; parting sand clay-free.

★ Chvorinov's rule $t \propto (V/A)^2$ (very frequent); riser should freeze last – placed at thickest section; sprue tapered to prevent aspiration; chills promote directional solidification; pressurised vs unpressurised gating; bottom gating minimises turbulence; spherical riser most efficient; three stages of shrinkage.

Special Casting Processes, Melting and Casting Defects

★ Investment casting for turbine blades and jewellery (very frequent); hot chamber die casting for zinc and cold chamber for aluminium; true centrifugal casting for pipes without core; shell moulding uses thermosetting resin; slush casting for hollow toys; continuous casting for billets; lost-foam uses polystyrene; die casting suits large quantities of non-ferrous parts.

★ Cupola for cast iron (very frequent); cupola zones; limestone as flux; induction furnace clean melting; crucible furnace for non-ferrous; LD converter uses oxygen.

★ Blowholes from moisture and low permeability; misrun vs cold shut (very frequent); hot tear from poor collapsibility; swell from soft ramming; scab and rat tail from sand expansion; core shift prevented by chaplets; radiography for internal porosity; dye penetrant for surface cracks.

Welding, Brazing and Soldering

★ Neutral flame for most metals, oxidising for brass, carburising for high-carbon steel (very frequent); neutral flame temperature ≈ 3200 °C; acetylene dissolved in acetone; cylinder colours (oxygen black, acetylene maroon); oxy-fuel cutting only for ferrous; leftward vs rightward welding.

★ DCSP gives deeper penetration – two-thirds heat at the work; DCRP for thin sheets and aluminium (cleaning action) (very frequent); TIG uses non-consumable tungsten; MIG consumable wire; SAW flat position, thick plates; functions of electrode coating; plasma arc highest temperature; heat input VI/v.

★ Resistance welding heat I^2Rt (very frequent); spot welding for car bodies; seam welding for leak-tight tanks; thermit welding for rails (aluminium + iron oxide); friction welding for dissimilar metals; electron beam welding in vacuum; flash butt welding for rails.

★ Brazing above 450 °C, soldering below 450 °C, base metal not melted (very frequent); brazing uses capillary action; soft solder Sn–Pb; borax flux; HAZ; arc blow in DC; undercut from excess current; carbon equivalent and weldability.

Welding Practice – Symbols, Joint Design, Metals and Residual Stresses

★ Weld symbol placement – below reference line for arrow side (frequent); triangle symbol for fillet weld; edge preparation by plate thickness; double-V reduces distortion and filler; field-weld flag.

★ Weld region left in residual tension (frequent); back-step and skip welding reduce distortion; presetting; stress-relieving heat treatment after welding; angular distortion in single-V welds.

★ Aluminium welded with AC TIG due to oxide film (very frequent); preheating for high-carbon steel and cast iron; nickel electrodes for cast iron; low-hydrogen electrodes prevent cold cracking; stainless steel weld decay; titanium needs full inert shielding.

★ Gas cutting not suitable for stainless steel and aluminium; plasma cutting for non-ferrous; air-carbon arc gouging for back gouging; kerf.

Metal Forming – Rolling, Forging, Extrusion, Drawing and Sheet Metal

★ Hot vs cold working defined by recrystallisation temperature (very frequent); lead worked at room temperature is hot working; cold working improves finish and strength by strain hardening; recrystallisation temperature ≈ 0.4 T_m.

★ Maximum draft μ^2R and angle of bite (very frequent); neutral point; four-high mill uses backup rolls; cluster mill for thin hard strips; thread rolling stronger than cut threads; volume constancy.

★ Drop forging for crankshafts and connecting rods; upsetting increases cross-section; fullering; coining is closed-die cold forming (very frequent); flash in closed-die forging; forging gives favourable grain flow; swaging.

★ Impact extrusion for collapsible toothpaste tubes (very frequent); indirect extrusion needs less force; wire drawing is a tensile process with maximum reduction 63% (ideal); centre burst in extrusion; aluminium sections by extrusion; extrusion ratio.

★ Blanking vs punching – which part is the product (very frequent); clearance applied on punch for blanking and on die for punching; punching force $L\ t\ \tau$; deep drawing blank diameter $\sqrt{(d^2 + 4dh)}$; limiting drawing ratio ≈ 2; springback; wrinkling and earing; progressive vs compound dies; spinning for utensils.

Presses, Press Tools and High-Energy-Rate Forming

★ Hydraulic press for deep drawing (full force over stroke); knuckle-joint press for coining; shut height definition; C-frame vs straight-side presses; double-action press for deep drawing.

★ Progressive vs compound vs combination dies (very frequent); stripper function; pilots in progressive dies; centre of pressure; compound die for washers; strip layout to reduce scrap.

★ Explosive forming for large parts; electromagnetic forming for conductive tubes (crimping); hydroforming for auto frames; roll forming for long profiles; superplastic forming; Guerin process uses rubber pad.

Theory of Metal Cutting

★ Tool signature order (very frequent); positive rake for soft materials and negative rake for carbides/hard materials; orthogonal vs oblique cutting; roughness $f^2/8R$ – larger nose radius and smaller feed improve finish.

★ Continuous chips from ductile materials at high speed; discontinuous chips in cast iron; BUE at low speed (very frequent); chip breakers; BUE reduced by higher speed and rake.

★ Chip thickness ratio and shear angle formula (very frequent); Merchant's relation $2\varphi + \beta - \alpha = 90°$; Merchant circle force components; F_c largest force; most heat goes into the chip; specific energy; MRR = V f d; assumptions of Merchant theory.

★ Taylor's equation $VT^n = C$ and typical n values (very frequent); flank wear criterion and crater wear on rake face; speed most influential on tool life; diffusion wear in carbides; machinability index base AISI 1112; tool life for minimum cost vs maximum production.

★ Order of hot hardness of tool materials (very frequent); diamond not used for ferrous metals (graphitisation/reaction with iron); CBN for hardened steels; HSS 18-4-1; carbide grades P, M, K; ceramics need negative rake; cast iron machined dry; soluble oils for cooling.

Machine Tools – Lathe, Drilling, Shaper, Planer, Milling and Others

★ Lead screw used only for thread cutting (very frequent); lathe specification (swing, distance between centres); tailstock set-over for long tapers; compound rest for short steep tapers; capstan vs turret; steady rest vs follower rest; four-jaw chuck for irregular work; knurling is forming not cutting; change gear ratio = work pitch/lead screw pitch.

★ Point angle of twist drill 118° (very frequent); radial drilling machine for large jobs; reaming finishes holes; counterboring vs countersinking; chisel edge produces thrust; trepanning; drilling time formula; tap drill size.

★ Quick return mechanism in shaper (crank and slotted lever) (very frequent); planer for large work – work reciprocates; slotter for internal keyways; clapper box; shaper specified by stroke length; quick return ratio.

★ Up vs down (climb) milling (very frequent); simple indexing 40/N; angular indexing θ/9; straddle and gang milling; universal milling machine for helical gears; table feed = f_z Z N; T-slot and Woodruff cutters; down milling needs backlash eliminator.

★ Hobbing is the fastest generating process and cannot cut internal gears (very frequent); gear shaper cuts internal gears; form milling uses 8 cutters per module; broaching for internal keyways and splines – mass production; gear shaving before hardening, grinding after; thread rolling strongest threads.

Grinding, Finishing and Non-Traditional Machining

★ Wheel marking order (very frequent); hard material needs soft wheel; SiC for cast iron and non-ferrous, Al_2O_3 for steels; vitrified bond most common; dressing vs truing; glazing vs loading; centreless grinding with regulating wheel; diamond for carbides.

★ Honing for cylinder bores (cross-hatch) (very frequent); lapping uses loose abrasive and gives best flatness (gauge blocks); superfinishing for bearing races; burnishing removes no material; order of finish casting < machining < grinding < honing < lapping.

★ EDM tool materials (graphite, copper) and dielectric (kerosene) (very frequent); ECM based on Faraday's laws, work is anode, no tool wear; USM for hard brittle non-conductors with 20 kHz vibration; AJM for glass/ceramics; LBM and EBM for micro-holes – EBM needs vacuum; PAM for cutting thick plates; chemical milling for aircraft panels; ECM MRR independent of hardness; wire EDM for dies.

Metrology – Limits, Fits, Gauges and Measuring Instruments

★ Hole-basis system preferred and why (very frequent); clearance/transition/interference fit examples (H7/g6, H7/k6, H7/p6); tolerance vs allowance; 18 IT grades; tolerance unit $0.45\sqrt[3]{D} + 0.001D$; fundamental deviation letters; unilateral vs bilateral tolerances; MMC.

★ Taylor's principle of gauging (very frequent); GO gauge checks maximum material limit and gets wear allowance; plug gauge for holes, snap gauge for shafts; slip gauges wringing; sine bar for angles below 45°; feeler gauge for gaps.

★ Vernier and micrometer least counts (very frequent); micrometer obeys Abbe's principle; optical flat for flatness λ/2 per fringe; three-wire method and best wire size 0.577p; CMM; accuracy vs precision; standard temperature 20 °C; comparators; gear tooth vernier.

★ Ra (CLA) definition and unit µm (very frequent); Talysurf stylus instrument; roughness vs waviness vs lay; N-grade numbers; Rq ≈ 1.11 Ra.

CNC Machines, Automation, Jigs and Fixtures

★ G00 rapid, G01 linear, G02 CW, G03 CCW (very frequent); M03 spindle CW, M06 tool change, M30 end and rewind; Z-axis along spindle; point-to-point vs contouring; open vs closed loop; ball screws in CNC; G90/G91; DNC; machining centre with ATC.

★ FMS components; group technology and Opitz coding; robot configurations – SCARA for assembly; SLA uses photopolymer, FDM uses thermoplastic filament (very frequent); transfer lines for mass production; CAPP types; CIM.

★ Jig guides the tool, fixture only holds (very frequent); 3-2-1 principle restricts 9 DOF by locators and 3 by clamps; 12 degrees of freedom; diamond pin avoids redundant location; slip bushes for successive operations; V-block for cylindrical work; setting block in milling fixtures.

Powder Metallurgy, Plastics Processing and Surface Treatments

★ P/M steps compaction → sintering below melting point (very frequent); atomisation most common for powder production; self-lubricating bearings and cemented carbides by P/M; tungsten filaments; HIP gives full density; green compact; infiltration and impregnation.

★ Compression moulding for thermosets (bakelite) and injection moulding for thermoplastics (very frequent); blow moulding for bottles; rotational moulding for water tanks; calendering for sheets; filament winding for pressure vessels; pultrusion; sink marks.

★ Galvanising is zinc coating; anodising for aluminium (work is anode) (very frequent); tin plate for food cans; sherardising zinc diffusion; calorising aluminium diffusion; sacrificial anodes of zinc/magnesium; PVD/CVD TiN coatings on tools.

Industrial Engineering – Work Study, PPC, Inventory and Quality

★ Work study = method study + work measurement (very frequent); flow process chart symbols; therbligs by Gilbreth; string diagram for distance; man–machine chart; SIMO chart uses micromotion study; SREDIM steps.

★ Standard time = normal time + allowances (very frequent); normal time = observed × rating; work sampling for non-repetitive jobs (Tippett); MTM and TMU = 0.036 s;

Westinghouse rating; work sampling sample size $4p(1-p)/L^2$.

★ Product layout for mass production, process layout for job shop, fixed-position for ships/aircraft (very frequent); line balancing efficiency $\Sigma t/(nC)$; cycle time = available time/output; CRAFT.

★ Routing vs scheduling vs dispatching (very frequent); exponential smoothing formula; Delphi method; Johnson's rule for two machines; SPT minimises mean flow time; MRP uses BOM; JIT and kanban; Six Sigma 3.4 DPMO; 5S; Gantt chart.

★ EOQ formula and ordering cost = holding cost at EOQ (very frequent); ABC analysis based on annual consumption value; VED for spare parts; reorder point; safety stock; EBQ with production rate; EOQ $\propto \sqrt{D}$.

★ $\bar{X}$ and R charts for variables, p and c charts for attributes (very frequent); c chart for defects per unit (Poisson); $\pm 3\sigma$ limits; producer's risk vs consumer's risk; OC curve; C_p formula; bath-tub curve; series vs parallel reliability; Ishikawa diagram; Deming PDCA; Pareto principle.

★ PERT three time estimates and beta distribution, CPM deterministic (very frequent); $t_e = (t_o + 4t_m + t_p)/6$; critical path has zero float; crashing in CPM; Vogel's approximation; Hungarian method for assignment; $m + n - 1$ allocations; break-even point; M/M/1 formulas; dummy activity.

Maintenance, Safety, Value Engineering and Costing

★ Preventive vs predictive maintenance (frequent); vibration analysis for condition monitoring; TPM and OEE; straight-line depreciation formula; replacement when average annual cost is minimum.

★ Fire classes and extinguishers (water not for oil and electrical fires) (very frequent); unsafe acts cause most accidents; Heinrich's ratio; frequency and severity rates; PPE for welding; Factories Act 1948; colour codes.

★ Value = function/cost (very frequent); prime cost = direct material + labour + expenses; machine-hour rate; Halsey and Rowan plans; fixed vs variable cost; noise limit 90 dB; anthropometry 5th–95th percentile.